Program overview

12-Sep-2023 13:38

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Code	Omschrijving	ECTS	p1 | p2 | p3 | p4 | p5
Master CS 2023	**Master Computer Science 2023**		
Track Software Technology 2023			
Common Core ST 2023			
CESE4025	Real-time Systems	5	
CS4015	Behaviour Change Support Systems	5	
CS4200-A	Compiler Construction	5	
CS4220	Machine Learning 1	5	
IN4150	Distributed Algorithms	6	
IN4152	3D Computer Graphics and Animation	5	
IN4191	Security and Cryptography	5	
IN4252	Web Science & Engineering	5	
IN4315	Software Architecture	5	
IN4344	Advanced Algorithms	5	
Specialistievakken start kwartaal 1 2023	**Specialisation courses start first period 2023**		
CESE4055	Ad hoc and Sensor Networks	5	
CS4070	Multivariate Data Analysis	5	
CS4200-A	Compiler Construction	5	
CS4340	Seminar Probabilistic Programming	5	
CS4365	Applied Image Processing	5	
CS4375	Artificial Intelligence Techniques	5	
EE4C06	Networking	5	
IN4049TU	Introduction to High Performance Computing	6	
IN4191	Security and Cryptography	5	
IN4252	Web Science & Engineering	5	
IN4307	Medical Visualization	5	
IN4344	Advanced Algorithms	5	
WM-ITAV-4010	Scientific Writing	2	
Specialistievakken start kwartaal 2 2023	**Specialisation courses start second period 2023**		
CESE4025	Real-time Systems	5	
CESE4045	High-performance data networking	5	
CS4015	Behaviour Change Support Systems	5	
CS4135	Software Verification	5	
CS4220	Machine Learning 1	5	
CS4270	Conversational Agents	5	
CS4400	Deep Reinforcement Learning	5	
IN4089	Data Visualization	5	
IN4150	Distributed Algorithms	6	
IN4302TU	Building Serious Games	5	
IN4341	Performance Analysis	5	
QIST4310	Fundamentals of Quantum Information	4	
Specialistievakken start kwartaal 3 2023	**Specialisation courses start third period 2023**		
AP3132	Advanced Digital Image Processing	6	
CESE4030	Embedded Systems Laboratory	5	
CESE4050	Measuring and Simulating the Internet	5	
CESE4060	Wireless IoT and Local Area Networks	5	
CS4090	Quantum Communication and Cryptography	5	
CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5	
CS4160	Blockchain Engineering	5	
CS4195	Modeling and Data Analysis in Complex Networks	5	
CS4210-A	Algorithms for Intelligent Decision Making	5	
CS4225	Educational Technologies	5	
CS4230	Machine Learning 2	5	
CS4235	Socio-Cognitive Engineering	5	
CS4240	Deep Learning	5	
CS4345	Seminar Formal Methods for Learned Systems	5	
CS4370	Testing and Validation for AI-intensive Systems	5	
CS4405	Analysis of Concurrent and Distributed Programs	5	
CS4410	Category Theory for Programmers	5	
CS4415	Sustainable Software Engineering	5	
		5	

CS4430	Network Security	
EE4740	Data Compression: Entropy and Sparsity Perspectives	5
IN4152	3D Computer Graphics and Animation	5
IN4253ET	"Hacking Lab"-Applied Security Analysis	5
IN4315	Software Architecture	5
IN4325	Information Retrieval	5
Specialistievakken start kwartaal 4 2023	**Specialisation courses start fourth period 2023**	
CESE4120	Smart Phone Sensing	5
CS4035	Cyber Data Analytics	5
CS4125	Seminar Research Methodology for Data Science	5
CS4145	Crowd Computing	5
CS4205	Evolutionary Algorithms	5
CS4210-B	Intelligent Decision Making Project	5
CS4245	Seminar Computer Vision by Deep Learning	5
CS4280	Language-Based Software Security	5
CS4290	Seminar on Distributed Machine Learning Systems	5
CS4295	Release Engineering for Machine Learning Applications	5
CS4350	Machine Learning for Graph Data	5
CS4360	Natural Language Processing	5
EE4715	Array Processing	5
ET4030	Error Correcting Codes	4
IN4255	Geometric Data Processing	5
IN4331	Web-scale Data Management	5
Track Data Science & Technology 2023		
Common Core DST 2023		
CS4035	Cyber Data Analytics	5
CS4220	Machine Learning 1	5
CS4375	Artificial Intelligence Techniques	5
IN4089	Data Visualization	5
IN4252	Web Science & Engineering	5
IN4315	Software Architecture	5
IN4344	Advanced Algorithms	5
Specialistievakken start kwartaal 1 2023	**Specialisation courses start first period 2023**	
CESE4055	Ad hoc and Sensor Networks	5
CS4070	Multivariate Data Analysis	5
CS4200-A	Compiler Construction	5
CS4340	Seminar Probabilistic Programming	5
CS4365	Applied Image Processing	5
CS4375	Artificial Intelligence Techniques	5
EE4C06	Networking	5
IN4049TU	Introduction to High Performance Computing	6
IN4191	Security and Cryptography	5
IN4252	Web Science & Engineering	5
IN4307	Medical Visualization	5
IN4344	Advanced Algorithms	5
WM-ITAV-4010	Scientific Writing	2
Specialistievakken start kwartaal 2 2023	**Specialisation courses start second period 2023**	
CESE4025	Real-time Systems	5
CESE4045	High-performance data networking	5
CS4015	Behaviour Change Support Systems	5
CS4135	Software Verification	5
CS4220	Machine Learning 1	5
CS4270	Conversational Agents	5
CS4400	Deep Reinforcement Learning	5
IN4089	Data Visualization	5
IN4150	Distributed Algorithms	6
IN4302TU	Building Serious Games	5
IN4341	Performance Analysis	5
QIST4310	Fundamentals of Quantum Information	4
Specialistievakken start kwartaal 3 2023	**Specialisation courses start third period 2023**	
AP3132	Advanced Digital Image Processing	6
CESE4030	Embedded Systems Laboratory	5
CESE4050	Measuring and Simulating the Internet	5
CESE4060	Wireless IoT and Local Area Networks	5
CS4090	Quantum Communication and Cryptography	5
CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5
CS4160	Blockchain Engineering	5
CS4195	Modeling and Data Analysis in Complex Networks	5
CS4210-A	Algorithms for Intelligent Decision Making	5
CS4225	Educational Technologies	5
CS4230	Machine Learning 2	5
		5

CS4235	Socio-Cognitive Engineering	
CS4240	Deep Learning	5
CS4345	Seminar Formal Methods for Learned Systems	5
CS4370	Testing and Validation for AI-intensive Systems	5
CS4405	Analysis of Concurrent and Distributed Programs	5
CS4410	Category Theory for Programmers	5
CS4415	Sustainable Software Engineering	5
CS4430	Network Security	5
EE4740	Data Compression: Entropy and Sparsity Perspectives	5
IN4152	3D Computer Graphics and Animation	5
IN4253ET	"Hacking Lab"-Applied Security Analysis	5
IN4315	Software Architecture	5
IN4325	Information Retrieval	5
Specialistievakken start kwartaal 4 2023	**Specialisation courses start fourth period 2023**	
CESE4120	Smart Phone Sensing	5
CS4035	Cyber Data Analytics	5
CS4125	Seminar Research Methodology for Data Science	5
CS4145	Crowd Computing	5
CS4205	Evolutionary Algorithms	5
CS4210-B	Intelligent Decision Making Project	5
CS4245	Seminar Computer Vision by Deep Learning	5
CS4280	Language-Based Software Security	5
CS4290	Seminar on Distributed Machine Learning Systems	5
CS4295	Release Engineering for Machine Learning Applications	5
CS4350	Machine Learning for Graph Data	5
CS4360	Natural Language Processing	5
EE4715	Array Processing	5
ET4030	Error Correcting Codes	4
IN4255	Geometric Data Processing	5
IN4331	Web-scale Data Management	5
Track Artificial Intelligence Technology 2023		
Common Core AIT 2023		
CS4205	Evolutionary Algorithms	5
CS4210-A	Algorithms for Intelligent Decision Making	5
CS4220	Machine Learning 1	5
CS4240	Deep Learning	5
CS4270	Conversational Agents	5
CS4375	Artificial Intelligence Techniques	5
IN4315	Software Architecture	5
IN4325	Information Retrieval	5
Specialistievakken start kwartaal 1 2023	**Specialisation courses start first period 2023**	
CESE4055	Ad hoc and Sensor Networks	5
CS4070	Multivariate Data Analysis	5
CS4200-A	Compiler Construction	5
CS4340	Seminar Probabilistic Programming	5
CS4365	Applied Image Processing	5
CS4375	Artificial Intelligence Techniques	5
EE4C06	Networking	5
IN4049TU	Introduction to High Performance Computing	6
IN4191	Security and Cryptography	5
IN4252	Web Science & Engineering	5
IN4307	Medical Visualization	5
IN4344	Advanced Algorithms	5
WM-ITAV-4010	Scientific Writing	2
Specialistievakken start kwartaal 2 2023	**Specialisation courses start second period 2023**	
CESE4025	Real-time Systems	5
CESE4045	High-performance data networking	5
CS4015	Behaviour Change Support Systems	5
CS4135	Software Verification	5
CS4220	Machine Learning 1	5
CS4270	Conversational Agents	5
CS4400	Deep Reinforcement Learning	5
IN4089	Data Visualization	5
IN4150	Distributed Algorithms	6
IN4302TU	Building Serious Games	5
IN4341	Performance Analysis	5
QIST4310	Fundamentals of Quantum Information	4
Specialistievakken start kwartaal 3 2023	**Specialisation courses start third period 2023**	
AP3132	Advanced Digital Image Processing	6
CESE4030	Embedded Systems Laboratory	5
CESE4050	Measuring and Simulating the Internet	5
		5

CESE4060	Wireless IoT and Local Area Networks	
CS4090	Quantum Communication and Cryptography	5
CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5
CS4160	Blockchain Engineering	5
CS4195	Modeling and Data Analysis in Complex Networks	5
CS4210-A	Algorithms for Intelligent Decision Making	5
CS4225	Educational Technologies	5
CS4230	Machine Learning 2	5
CS4235	Socio-Cognitive Engineering	5
CS4240	Deep Learning	5
CS4345	Seminar Formal Methods for Learned Systems	5
CS4370	Testing and Validation for AI-intensive Systems	5
CS4405	Analysis of Concurrent and Distributed Programs	5
CS4410	Category Theory for Programmers	5
CS4415	Sustainable Software Engineering	5
CS4430	Network Security	5
EE4740	Data Compression: Entropy and Sparsity Perspectives	5
IN4152	3D Computer Graphics and Animation	5
IN4253ET	"Hacking Lab"-Applied Security Analysis	5
IN4315	Software Architecture	5
IN4325	Information Retrieval	5
Specialistievakken start kwartaal 4 2023	**Specialisation courses start fourth period 2023**	
CESE4120	Smart Phone Sensing	5
CS4035	Cyber Data Analytics	5
CS4125	Seminar Research Methodology for Data Science	5
CS4145	Crowd Computing	5
CS4205	Evolutionary Algorithms	5
CS4210-B	Intelligent Decision Making Project	5
CS4245	Seminar Computer Vision by Deep Learning	5
CS4280	Language-Based Software Security	5
CS4290	Seminar on Distributed Machine Learning Systems	5
CS4295	Release Engineering for Machine Learning Applications	5
CS4350	Machine Learning for Graph Data	5
CS4360	Natural Language Processing	5
EE4715	Array Processing	5
ET4030	Error Correcting Codes	4
IN4255	Geometric Data Processing	5
IN4331	Web-scale Data Management	5
Seminar Courses CS & Literature Survey 2023		
CESE4065	Advanced Practical I.o.T. and Seminar	5
CS4120	Seminar Science and Methods in Cyber security	5
CS4125	Seminar Research Methodology for Data Science	5
CS4130	Seminar Programming Languages	5
CS4165	Seminar Social Signal Processing	5
CS4210-B	Intelligent Decision Making Project	5
CS4245	Seminar Computer Vision by Deep Learning	5
CS4290	Seminar on Distributed Machine Learning Systems	5
CS4340	Seminar Probabilistic Programming	5
CS4345	Seminar Formal Methods for Learned Systems	5
IN4306	Literature Survey	10
IN4310	Seminar Computer Graphics	5
IN4314	Seminar Selected Topics in Multimedia Computing	5
IN4326	Seminar Web Information Systems	5
IN4334	Analytics and Machine Learning for Software Engineering	5
Free Elective Space 2023	**Suggested Free Elective Space 2023**	
Quantum Computing 2023		
AP3432	Quantum Hardware 1 - Theoretical Concepts	4
AP3442	Quantum Hardware 2 - Experimental State of the Art	4
CS4090	Quantum Communication and Cryptography	5
QIST4310	Fundamentals of Quantum Information	4
QIST4400	Quantum Computer Architecture	5
Language Courses & Skills 2023		
TPM018A	English Grammar for the University	2
TPM303A	Intermediate Writing in English for the University	2
TPM304A	Advanced Writing in English for the University	2
TPM305A	Writing a Masters Thesis in English	2
WM1115TU	Dutch Elementary 1	3
WM1116TU	Dutch Elementary 2	3
WM1117TU	Dutch Intermediate 1	3
WM1135TU	Advanced English for the University	3
Projects 2023		
IFEEMCS520200	Interdisciplinary Advanced Artificial Intelligence Project	15
TUD4040	Joint Interdisciplinary Project	15
Thesis Project 2023		

IN5000	Final Project	45
Research Groups 2023		
Algorithmics 2023		
CS4205	Evolutionary Algorithms	5
CS4210-A	Algorithms for Intelligent Decision Making	5
CS4210-B	Intelligent Decision Making Project	5
CS4340	Seminar Probabilistic Programming	5
CS4345	Seminar Formal Methods for Learned Systems	5
CS4400	Deep Reinforcement Learning	5
IN4344	Advanced Algorithms	5
Computer Graphics and Visualisation 2023		
CS4365	Applied Image Processing	5
IN4089	Data Visualization	5
IN4152	3D Computer Graphics and Animation	5
IN4255	Geometric Data Processing	5
IN4302TU	Building Serious Games	5
IN4307	Medical Visualization	5
IN4310	Seminar Computer Graphics	5
Cyber Security 2023		
CS4035	Cyber Data Analytics	5
CS4090	Quantum Communication and Cryptography	5
CS4120	Seminar Science and Methods in Cyber security	5
CS4150	Systems Security	5
CS4380	Privacy Enhancing Technologies	5
CS4430	Network Security	5
IN4191	Security and Cryptography	5
IN4253ET	"Hacking Lab"-Applied Security Analysis	5
Cyber Security/SERG 2023		
CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5
Distributed Systems 2023		
CS4160	Blockchain Engineering	5
CS4290	Seminar on Distributed Machine Learning Systems	5
IN4150	Distributed Algorithms	6
Embedded and Networked Systems 2023		
CESE4025	Real-time Systems	5
CESE4030	Embedded Systems Laboratory	5
CESE4045	High-performance data networking	5
CESE4050	Measuring and Simulating the Internet	5
CESE4055	Ad hoc and Sensor Networks	5
CESE4060	Wireless IoT and Local Area Networks	5
CESE4065	Advanced Practical I.o.T. and Seminar	5
CESE4110	Visible Light Communication and Sensing	5
CESE4120	Smart Phone Sensing	5
Interactive Intelligence 2023		
CS4015	Behaviour Change Support Systems	5
CS4125	Seminar Research Methodology for Data Science	5
CS4165	Seminar Social Signal Processing	5
CS4235	Socio-Cognitive Engineering	5
CS4270	Conversational Agents	5
CS4375	Artificial Intelligence Techniques	5
Multimedia Computing 2023		
CS4195	Modeling and Data Analysis in Complex Networks	5
CS4350	Machine Learning for Graph Data	5
CS4370	Testing and Validation for AI-intensive Systems	5
IN4314	Seminar Selected Topics in Multimedia Computing	5
Pattern Recognition & Bioinformatics 2023		
CS4070	Multivariate Data Analysis	5
CS4176	Algorithms for network-based bioinformatics	5
CS4220	Machine Learning 1	5
CS4230	Machine Learning 2	5
CS4240	Deep Learning	5
CS4245	Seminar Computer Vision by Deep Learning	5
CS4250	Selected Topics in Molecular Biology	5
CS4255	Algorithms for sequence-based Bioinformatics	5
CS4260	Machine Learning in Bioinformatics	5
CS4329	Recent topics in bioinformatics	5
Programming Languages 2023		
CS4130	Seminar Programming Languages	5
CS4135	Software Verification	5
CS4200-A	Compiler Construction	5
CS4280	Language-Based Software Security	5
CS4405	Analysis of Concurrent and Distributed Programs	5
CS4410	Category Theory for Programmers	5
Software Engineering 2023		

CS4295	Release Engineering for Machine Learning Applications	5
CS4415	Sustainable Software Engineering	5
IN4315	Software Architecture	5
IN4334	Analytics and Machine Learning for Software Engineering	5

Web Information Systems 2023

CS4145	Crowd Computing	5
CS4225	Educational Technologies	5
CS4360	Natural Language Processing	5
IN4252	Web Science & Engineering	5
IN4325	Information Retrieval	5
IN4326	Seminar Web Information Systems	5
IN4331	Web-scale Data Management	5

QCE/ Network Architectures and Services 2023

EE4396	Mobile Networks	5
ET4034	Telecom Business Architectures and Models	4
IN4341	Performance Analysis	5

Special Programmes 2023

Information Architecture 2023

IN4252	Web Science & Engineering	5
IN4325	Information Retrieval	5
IN4331	Web-scale Data Management	5
SEN1121	Complex Systems Engineering	5
SEN1141	Managing Multi-actor Decision-making	5
SEN1611	I&C Architecture Design	5
SEN1622	I&C Service Design	5

Bioinformatics 2023

DST & AIT Common Core courses 2023

Compulsory Bioinformatics courses 2023

CS4176	Algorithms for network-based bioinformatics	5
CS4250	Selected Topics in Molecular Biology	5
CS4255	Algorithms for sequence-based Bioinformatics	5
CS4260	Machine Learning in Bioinformatics	5
CS4329	Recent topics in bioinformatics	5

Specialization courses 2023

Bioinformatics specialization Courses Q1 2023

CS4070	Multivariate Data Analysis	5
CS4375	Artificial Intelligence Techniques	5
EE4C06	Networking	5
IN4049TU	Introduction to High Performance Computing	6
IN4252	Web Science & Engineering	5
IN4307	Medical Visualization	5
IN4344	Advanced Algorithms	5

Bioinformatics specialization Courses Q2 2023

CS4220	Machine Learning 1	5
IN4089	Data Visualization	5
IN4150	Distributed Algorithms	6

Bioinformatics specialization Courses Q3 2023

CS4195	Modeling and Data Analysis in Complex Networks	5
CS4230	Machine Learning 2	5
CS4240	Deep Learning	5
IN4315	Software Architecture	5
IN4325	Information Retrieval	5

Bioinformatics specialization Courses Q4 2023

CS4205	Evolutionary Algorithms	5
CS4245	Seminar Computer Vision by Deep Learning	5
CS4290	Seminar on Distributed Machine Learning Systems	5
IN4331	Web-scale Data Management	5

Literature study 2023

IN4306	Literature Survey	10

Free Electives 2023

Thesis project 2023

IN5000	Final Project	45

Cyber Security 2023

Common Core Courses Special Programme Cyber Security (7 courses) 2023

CS4035	Cyber Data Analytics	5
CS4150	Systems Security	5
CS4430	Network Security	5
IN4191	Security and Cryptography	5
TPM027A	Cyber Risk Management	5

Technical Electives (choose at least 3 courses) 2023

CS4090	Quantum Communication and Cryptography	5
CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5
CS4160	Blockchain Engineering	5

CS4280	Language-Based Software Security	5
CS4380	Privacy Enhancing Technologies	5
IN4253ET	"Hacking Lab"-Applied Security Analysis	5
QIST4310	Fundamentals of Quantum Information	4

Socio-Technical Electives (choose at least 3 courses) 2023

TPM010A	Cyber Crime Science	5
TPM020B	Economics of Cybersecurity	5
TPM025A	User-Centred Security	5
TPM030A	Introduction to Cloud as Infrastructure: The effects of the new business of computing on practice	5

Required Courses for CS Graduation 2023

CS4120	Seminar Science and Methods in Cyber security	5
IN5000	Final Project	45

Free Electives (Prerequisites and other Courses) 2023

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Master CS 2023	
Contact for students	MASTER COORDINATORS DATA SCIENCE & TECHNOLOGY Dr. L. Miranda da Cruz MASTER COORDINATORS SOFTWARE TECHNOLOGY Dr. C. Bach Poulsen Dr.ir. R. Bidarra MASTER COORDINATORS ARTIFICIAL INTELLIGENCE TECHNOLOGY Dr. J.C. van Gemert E-mailaddress: Mastercoordinator-CS@tudelft.nl
Introduction 1	The Computer Science programme consists of three tracks and three optional Special Programmes, it is a two-year programme (120 EC), and it is taught in English. The tracks: Data Science and Technology (DST), Software Technology (ST) and Artificial Intelligence Technology (AIT). DATA SCIENCE & TECHNOLOGY is meant for students who want to develop and use software that give meaning to data in order to support experts from various application domains (such as economy, medicine and marketing) to better understand the information that they possess by means of data science techniques. In the SOFTWARE TECHNOLOGY track the engineering of complex software systems takes on a central role. In this track, you will learn how to integrate them in real-world information-processing systems. Illustrative topics include distributed, multimedia, knowledge and secure processing systems, web and software engineering, visualisation and interaction. ARTIFICIAL INTELLIGENCE TECHNOLOGY is meant for students with a keen interest in AI and who want to focus on the development and engineering of systems using AI to solve problems from a variety of application domains. Students may opt for a Special Programme in Bioinformatics, Cyber Security or Information Architecture.

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Track Software Technology 2023

Introduction 1

In the Software Technology track of the Master of Computer Science program, the engineering of complex software systems takes on a central role. In this track, you will acquire knowledge and skills to design, develop and implement efficient algorithms, large-scale data structures and complex architectures. Furthermore, you will learn how to integrate them in real-world information-processing systems. Illustrative topics include distributed, multimedia, knowledge and secure processing, web and software engineering, visualization and interaction.

Software technology has a major impact on the economies of industrialised countries. Information-processing systems provide the backbone for almost all administrative and logistic operations within commercial production, business and public administration. Without software, high-tech systems are useless, whether large (e.g., a self-driving car or a robot for computer-assisted surgery) or small (e.g., a smartphone or an ordinary electronic card payment system). Moreover, software permeates all aspects of our life in society, ranging from Internet-based services like e-mail, online games, social networks and cloud computing, to large-scale scientific computing systems, traffic control systems and wireless sensor systems. All these systems need to have effective algorithms for their correct operation, good performance, high reliability and a well-thought-out architecture to make them easy to build and to maintain. Such essential features very well describe the specific focus of the Software Technology track.

THE CURRICULUM
1. An Individual Exam Programma (IEP) in this track consists of
a. a common core,
b. courses offered by the faculty EEMCS,
c. a seminar offered by the programme CS or a Literature survey (IN4306)
d. free electives,
e. a thesis project (IN5000 Final project) worth 45 credits and
f. if required, homologation.

The IEP must be drawn up in agreement with the thesis coordinator of the research group in which the student wishes to carry out his or her thesis project. The thesis coordinator is a member of the scientific staff of that research group.

The seminar of the research group in which the thesis is performed or the Literature Study (IN4306) is part of said IEP.

Free elective courses
- the number of credits spent on free electives in said IEP is no higher than 25 credits,
- the number of credits spent on homologation in said IEP is no higher than 15 credits,
- at least 40 credits of the courses in the IEP (notwithstanding the thesis project) should be computer Science courses. A list of these courses is published annually in the digital study guide.

Free electives - language course list:
Up to 3 credits may be spent on language courses. These may only be chosen if required. Placement tests showing the necessity to take one or more of these courses must be taken and submitted to the master coordinator.

TPM018A English Grammar for the University 2 EC
TPM303A Intermediate Writing in English for the University 2 EC
TPM304A Advanced Writing in English for the University 2 EC
TPM305A Writing a Masters Thesis in English 2 EC
WM1115TU Dutch Elementary 1 3 EC
WM1116TU Dutch Elementary 2 3 EC
WM1117TU Dutch intermediate 1 3 EC
WM1135TU Advanced English for the University 3 EC

The free elective space may also be used for an extra project:
TUD4040 Joint Interdisciplinary Project (JIP) 15 EC
IFEEMCS520200 Interdisciplinary Advanced Artificial Intelligence Project 15 EC

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Common Core ST 2023	
Introduction 1	Common Core MSc CS - ST (at least 25 EC): Choose 5 out of 10. Students follow the masters programme of the year in which they have started their programme. E.g. students who started their masters programme in 2022-2023 follow the Teaching and Examination Regulations (TER) of 2022-2023.

CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	- basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	The course intends to bring the student into the position to: - Explain the fundamental concepts and terminology of real-time systems - Construct task schedules using different scheduling policies under a given set of realistic system constraints - Analyze the timing behavior of a system for a given system model and scheduling policy - Discuss advantages and disadvantages of different scheduling policies for a given platform or system - Discuss the effect of hardware and software interferences on the timing behavior of a given system - Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system - Derive (reverse engineer) the system specification from a given implementation (in the lab) - Evaluate the scheduling overheads of a given implementation (in the lab) - Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	Written exam (grade) + lab work; the exam has a resit Repair option for the Individual Assignment = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4015	Behaviour Change Support Systems	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	none	
Course Contents	Behavior change support systems (BCSS) are computer-based systems that support individuals to form, alter or reinforce cognitions, attitudes or behaviors without using coercion or deception. They can serve individuals throughout the various stages of a change process, such as awareness development, contemplation, action strategy development, development of new behaviors, and maintaining these new behaviors. Virtual healthcare coaches, negotiation support systems, and applications that provide individuals with personalised financial guidance are three examples of these systems. To establish, modify or maintain change BCSS can deploy computerized persuasive strategies (e.g. reducing effort to establish target behavior, or argumentation reflection strategies, and also algorithmic nudge, including data-, expert-, and theory-driven), simulations (e.g. serious gaming, virtual reality), relational software agents (e.g. ePartners, virtual coaches), and personalization based on longitudinal user data. BCSSs are found in many domains, including education, sales, negotiation, management, and particularly in the health domain. The main topics are: Understanding the behaviour; strategies for behaviour change, action funnel, human decision making Identifying change support strategies; Behaviour change wheel, product vision, target outcome, target user, population diversity Conceptual design of BCSS; Behaviour plan, decision-making environment, story-editing technique Behaviour change and technology acceptance; Motivation, Trans-theoretical model, Theory of reasoned action, theory of planned behaviour, Technology acceptance model, Unified Theory of Acceptance and Use of Technology, technology adaption lifecycle, Fogg behaviour model, Self-Determination theory Persuasive technology; functional triad, levels of persuasion, Nudge Theory, principles of influence, triggers Algorithmic nudge/Persuasion and support Algorithms; concept, key characteristics, and feasibility Learning principles; Behaviouristic learning principles, cognitive learning principles, constructivism based learning principles, social cognitive (learning) theory Communication and framing; Behavioural economic principles, elaboration likelihood model (ELM), health belief model, protection motivation theory, persuasive communication Adherence and Gamification; Implementation Intentions, system 1 and system 2, treatment adherence, gamification Application domains eHealth and Cyber security; Self-monitoring, self-management, virtual health agents, virtual reality therapy, internet cognitive behavioural therapy (iCBT)	
Study Goals	This course provides students the opportunity to develop and demonstrate understanding, knowledge and competence in the field of BCSS. The learning outcomes for the course are that students will be able to: 1. Apply psychological theories, principles, and concepts in the design of BCSS systems. 2. Apply BCSS practices, principles and techniques. 3. Construct an implementation of an algorithmic nudge/boost/persuasion for a specific real-world application domain and objective 4. Design a BCSS system using BCSS principles, concepts, theories, and design methods.	
Education Method	The course uses a flipped-classroom approach, where students watch the pre-recorded lectures in their own time and complete online quizzes. In the on-campus lectures, students and teachers discuss small challenges related to the lecture topic. In the pre-recorded video material, theories, principles and methods are presented, discussed and illustrated with examples from the field. The video material is supported by online self-tests. In their own time, students work in small groups on coursework assignments to develop a product design for a BCSS with a computer simulation of supporting algorithmic nudge/boost/persuasion. In the on-campus session, student groups present the progress of their coursework and receive formative feedback. The presentations are done three times, following topic as the project progress over time: 1. The targeted behaviour 2. Behaviour plan 3. Conceptual design and the algorithmic nudge/boost/persuasion	
Literature and Study Materials	Available on brightspace	
Books	Wendel, S. (2020). Designing for Behavior Change: Applying Psychology and Behavioral Economics. O'Reilly Media, Inc,. or the older edition Wendel, S. (2013). Designing for Behavior Change: Applying Psychology and Behavioral Economics. " O'Reilly Media, Inc.	
Assessment	The final grade of the course consists of the following components: Computerised examination (or oral exam) (weighting 60%) If the expected number of students registering for the exam or resit is small, the teacher might decide to replace the computerized examination with an oral examination. Design BCSS Coursework Project (resulting in presentation slides and demonstration of computer simulation of a supporting algorithmic nudge/boost/persuasion, including question and answer round where individual group members are assessed on the coursework) (weighting 40%) Final grade calculation: 0.6 * Computerised examination (or oral exam) + 0.4 * Design BCSS Coursework Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Computerised examination (or oral exam): Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Design BCSS Coursework Project: Resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	3	
Co-Instructor	M.L. Tielman	

CS4200-A	Compiler Construction	5
Responsible Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	- programming (required) - software engineering (recommended) - functional programming (recommended) - formal languages and automata (recommended)	
Course Contents	Compilers translate the source code of programs in a high-level programming language into executable (virtual) machine code. In this course you will study the architecture of compilers, and important concepts and techniques underlying the components of that architecture. You will study these techniques by applying them in practice to implement your own compiler for a small language. The course covers the following topics: - Syntax and parsing - Static semantics and type checking - Virtual machines, assemby code, byte code - Interpretation vs. compilation - Code generation - Memory management, garbage collection - Data-flow analysis - Optimization	
Study Goals	At the end of this course, you will be able to: - Explain the architecture of a compiler and the role of each component of that architecture - Explain the algorithms and techniques for the implementation of compiler components and apply these techniques to examples - Implement a code generator that translates source language abstract syntax trees to target language instructions - Implement transformations on abstract syntax terms to simplify programs - Implement simple type checkers - Implement simple data-flow analyzers - Integrate these components into a working compiler	
Education Method	Attending the course involves: - Attending weekly lectures - Attending weekly lab session (which may start with a group tutorial) - Reading lecture material and papers - Making homework assignments (building a compiler)	
Literature and Study Materials	Required reading: - We will use selected chapters from the book "Essentials of Compilation" by Jeremy Siek, available online: https://github.com/IUCompilerCourse/Essentials-of-Compilation - Lecture slides and selected papers from the literature Optional reading: - We will use Scala to implement our compilers. The following book is optional reading which may be helpful in getting up to speed with programming in Scala: https://www.artima.com/shop/programming_in_scala_5ed (Additional) reading materials will be made available on Brightspace. We will use WebLab (https://weblab.tudelft.nl/cs4200/) for the submission of homework assignments.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Homework assignments (weighting 50%) Final grade calculation: 0.5 * Written exam + 0.5 * Homework assignments	

	A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Permitted Materials during Tests	No materials are permitted during the exam.
Co-Instructor	Dr. S.S. Chakraborty

CS4220	**Machine Learning 1**	**5**
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4150	Distributed Algorithms	6
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for network overlays, message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, broadcast, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	Understand the fundamental problems that distributed systems face Understand the most important distributed algorithms that solve these problems Reason about the execution of distributed algorithms Program distributed algorithms Select and summarize relevant literature on distributed algorithms	
Education Method	Lab work executed in groups of two students	
Literature and Study Materials	Lecture notes and lecture slides (available on Brightspace)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 75%). In case in-person, on-campus written exams are not possible because of unforeseen circumstances or measures, the written exam will be replaced by individual online oral exams. Paper summary (weighting 25%) Final grade calculation: 0.75 * Written exam + 0.25 * Paper summary A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper summary: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None except the list of algorithms	
Special Information	Email: in4150-ewi@tudelft.nl. Please send any emails related to the course to this address.	
Remarks	Lab work is 40 hrs.	

IN4152	3D Computer Graphics and Animation	5
Responsible Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. R. Marroquim	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Students that haven't followed any previous Computer Graphics courses (like CSE2215) will be able to participate, but might have to invest some more time to catch up in the first lectures.	
Course Contents	Have you ever wondered how Toy Story was made, why the game Last of Us 2 looks so beautiful, or have you ever wanted to create your own graphics application or game? Then you should consider following this course. If not, then you should still follow it... maybe, you will become interested! In this course, you will get a good idea of Computer Graphics in general. The topic is of very high relevance for the industry and the research community and has numerous applications in different domains, such as scientific visualization, video games, simulators, special effects, animated movies and many more. Here, you will learn about basic algorithms, as well as modern techniques. We will address several topics: the principles of image synthesis, object representations, geometric and hierarchical transformations, graphics cards and the graphics pipeline, realistic rendering (including global illumination and effects, such as reflections), expressive rendering, physics simulations, rendering control (including previsualization systems used by professionals in the movie industry), and perceptual rendering, which relies on properties of the human visual system to enhance the quality of the images. Besides course sessions on the theory of Computer Graphics, some of the algorithms will also be reproduced in practice, and deepened during the final project.	
Study Goals	The course teaches computer graphics techniques on an advanced level. After the course the student is able to classify the different modeling, shading, and display techniques. The student can reproduce the basic mathematical and algorithmic notions associated with these concepts, can comment on the weak and strong points of these techniques, and can apply the core concepts within a graphics program in practice.	
Education Method	lectures, instructions, research papers, lab work	
Literature and Study Materials	Research Papers in domain of selected topics, lecture sheets, online sources, optional books (see below)	
Books	Fundamentals of Computer Graphics by Shirley et al. - CRC Press Real-time Rendering by Tomas Akenine-Möller, Eric Haines, Naty Hoffman - Peters, Wellesley Real-Time Shadows by Elmar Eisemann, Michael Schwarz, Ulf Assarsson, Michael Wimmer - Taylor & Francis Computer Graphics. Principles and Practice by James D. Foley, Andries VanDam, Steven K. Feiner - Addison Wesley Advanced Material: Physically Based Rendering, From Theory to Implementation by Matt Pharr, Wenzel Jakob and Greg Humphreys - MIT Press	
Assessment	The final grade of the course consists of the following components (details of both elements will be presented during the lecture): Project (weighting 60%). The project grade is the result of a project and its presentation that is building upon the assignments that are handed out (roughly) weekly during the duration of this course. Paper (weighting 40%). The paper grade is the result of the presentation of a scientific paper and the development of an associated practical implementation. Final grade calculation: 0.6 * Project + 0.4 * Paper A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Paper: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4191	Security and Cryptography	5
Responsible Instructor	Z. Erkin	
Instructor	J.A. Martinez Castaneda	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	UT-201500042 Privacy Enhancing Technologies (Q4)	
Expected prior knowledge	Basic understanding on the following is suggested. -Probability and statistics -Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Learning outcomes: The goal is to make students familiar with the basic concepts applied cryptography, including classical cryptography and modern secret key and public key cryptography. In particular, the students will acquire A sound understanding of the notion of security An understanding of the confidentiality, integrity and authenticity needs of the society Understand the role of cryptographic primitives including the differences between symmetric and asymmetric cryptography, the role of hash functions, digital signatures and PKI Understand the advanced topics in cryptography needed for the modern society with untrustworthy entities Among others things, the following topics are covered: -Classical systems -Information theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures, -*Secret Sharing. (if time permits) -*Zero Knowledge proofs (if time permits)	
Study Goals	It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security, as well as is familiar with present applications. Through assignments, students are expected to have the chance to work on the topics covered in the lectures.	
Education Method	Lectures, assignments and weekly exercises. Attention: This course requires full effort of 140 hours. Even more, if you lack the background (probability and modular arithmetic) Planned Workload: Lectures: 28 x 45minutes sessions, total 22 hours Practice session: 7 x 90 min. total 12 hours Assignments: 3 x 20 hour, total 60 hours Weekly study: 7 x 4 hours, total 28 hours Exam preparation: 20 hours Exam: 3 hours	
Literature and Study Materials	-Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition -Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) Handouts of lectures	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%) Mandatory assignments (weighting 30%) Final grade calculation: 0.7 * Written exam + 0.3 * Mandatory assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mandatory assignments: re-submit deliverable	

	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	In case of in person examination at campus: The exam is closed book. A cheat sheet of size A4, hand written is allowed for the written exam. Name and student number has to be present on each side. In case of remote exam from home: open book: textbook, slides and self-made notes only. randomised/customised exam.
Permitted Materials during Tests	Only non-scientific calculators.

IN4252	Web Science & Engineering	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	J. Yang	
Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The expected entry level is (equivalent to) standard bachelor-level computer science.	
Course Contents	The main subject of the course is the Web, and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g. Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan / research paper draft in collaboration with others in a small group to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies, and verbally present and defend this plan / draft.	
Education Method	The education methods include: - Lectures, before which and after which students study material by themselves, to get an understanding of the relevant material; - Small assignments and hands-on exercises, to apply the understanding of relevant material; - One large assignment, with a number of feedback moments, to learn how to write a web science paper / research project plan. Lectures will not be in each week in the class period (1+2): in between lectures there is time reserved for studying before and after lectures, for small assignments and exercises, and for writing the large assignment paper. The writing of the large assignment paper / project plan happens throughout the class period (1+2) to enable frequent feedback.	
Literature and Study Materials	Will be provided in class, depending on the topics chosen for the assignments and final paper.	
Assessment	Assessments are based on two components: the "small assignments" (accompanying the lectures) which provide a total of 20% of the grade (these are a mix of literature and programming homework), and the "final paper" which provides 80% of the final grade. To pass, each of the components ("small assignments" and "final paper") need to be graded with a minimum of 5.0. The final grade must be at least a 5.8. If these criteria are not met, the "small assignments" and the "final paper" can be repaired provided that the to-be-repaired component was at least graded with a 4 in accordance with OER/TER article 17. This repair is not a resit, and is limited to a maximum of 6.0 as a result. Components that have been graded with less than a 4.0 cannot be repaired. If the course is still not passed after the repair or cannot be repaired, all components need to be redone if the course is retaken in a later year. Both the small assignments and the final paper are group works of groups of preferably 2 members.	
Special Information	Students are asked to register/enrol on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	The expected workload is 5ects and that is principally distributed uniformly over the two quarters.	

IN4315	Software Architecture	5
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software engineering	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four, in which the actual architectures of existing open source systems are analyzed in considerable detail. These systems are taken from github, and student teams are challenged to actually contribute to the systems under analysis in the course.	
Study Goals	Bring students into the position that they can (1) explain the key architectural concepts and methods for modeling software architectures; (2) apply viewpoints and perspectives to model software architectures; (3) discuss the benefits of architecting and the role of the software architect; (4) evaluate and validate software architectures; (5) explain and discuss the concepts of component-based and plugin architectures, service-oriented architectures, and software product lines; (6) explain and recognize technical debt and have an understanding of possible refactorings.	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books; Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; and Coplien and Bjørnvig, "Lean Architecture", Wiley, 2010. Additional reading material will be announced in the lectures.	
Assessment	Grades will be based on lab assignment including essay writing, coding, (video) presentation, peer reviewing, participation. The final grade of the course consists of the following components: Essays (weighting 60%) Coding (weighting 20%) Presentation (weighting 20%) Final grade calculation: 0.6 * Essays + 0.2 * Coding + 0.2 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Essays: re-submit deliverable Coding: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Course web site: https://se.ewi.tudelft.nl/delftswa/index.html	
Co-Instructor	M. Finavaro Aniche	

IN4344	Advanced Algorithms	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic courses in Algorithmics and Complexity Theory	
Course Contents	The course is on solving (abstract models of) complex real-world problems, with a focus on solving intractable problems exactly. The course consists of two main parts: * modeling and solving using linear programming * exact algorithms using search trees, dynamic programming, and/or decision diagrams	
Study Goals	By the end of this course, students will have knowledge of and experience with the following advanced algorithmic techniques: (Part 1:) - linear programming (LP) and LP modelling - duality and simplex algorithm - integer LP and integer modelling - LP relaxation and branch and bound (Part 2:) - complete and bounded search trees - tree decomposition - dynamic programming - preprocessing - decision diagrams Furthermore, they will be able to * recognize situations where they can use these techniques and adapt them to different applications. construct new algorithms that are similar to these techniques. construct proofs that are similar to a selected set of proofs from the material. analyze the run time of algorithms. implement an algorithm that is given in pseudo-code. experimentally evaluate the quality and the runtime of an algorithm on a set of instances.	
Education Method	Interactive lectures (usually 2x2 per week), optional homework exercises, programming assignments (2 hours lab support per week) The expected workload is 30% studying the written material and the recorded lectures, and participating in the interactive sessions 30% making the homework exercises and preparation for the exams 40% working on the programming assignments	
Literature and Study Materials	Part 1 of the course will be mainly based on chapters 1-8, some of 9-11 (most of 9-11 is assumed to be known), and 12-14 of the syllabus "Optimization" by Karen Aardal, Leo van Iersel and Remie Janssen, which can be ordered via https://www.webedu.nl/bestellen/tudelft/ Part 2 will use chapter 10 of the following textbook: J. Kleinberg and E. Tardos, Algorithm Design, Pearson Education, 2006. ISBN 0-321-37291-3 As well as some supplemental study material, which will be provided via BrightSpace.	
Assessment	The final mark depends on the marks obtained for (a) written reports and evaluation of programming assignments, PA (weight 40%) (b) a written exam, EX (2 parts, weight 60%) Each programming assignment is graded on a scale from 0 to 10. The final mark for the programming assignment (PA) is the average of the mark obtained for the assignments. Programming exercises can be completed by 2 students working together. If the average grade for the four programming assignments (PA) is between 4.0 and 6.0 (excluding 6.0), a repair is possible by making one extra assignment to replace the lowest grade of the four assignments. This leads to a repaired PA score, maximized at 6. (E.g. if grades were 3.5, 4, 4, 4.5, the 3.5 can be replaced. If the grade for the replacement assignment is a 7.5, the repaired PA score is a 5. If grades were 5, 5.5, 5.5, 6, the 5 can be replaced. If the replacement is a 9, the repaired PA score is not 6.25 but 6.) The exam consists of two parts. Each part will be examined after the lectures about that part have been delivered and will be graded on a scale from 1 to 10. The final mark for the exam (EX) is the average of the marks for the parts. Each exam part contains one or more challenging assignments; during each part of the course two representative homework assignments will be made available to prepare for this. There is a resit for the exam where any of the two parts can be redone. The result for a part after the resits is determined by the maximum score obtained for the original exam and the resit. The final mark for the course is determined as follows: - if the PA and the EX mark are above 5, the final mark is the weighted average of these three marks: 60% EX, 40% PA - if at least one of PA, EX is less than or equal to 5, the final mark is the minimum of the results obtained for PA or EX. Partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	For content questions / of general interest, please use StackOverflow. For personal questions, please use the course email address: aa-cs-ewi@tudelft.nl	
Tags	Algorithmics Artificial intelligence Mathematics	

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CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	

Course Contents

IMPORTANT NOTICE

Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students.
The above applies to all the fields in this coursebase for this course.

The course will be offered in PHYSICAL format.

Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure.

Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network.

The contents of the course are as follows:

Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5)
o Definition of ad-hoc networks
o Comparison with infrastructure based systems
o Typical applications
o Advantages and challenges
o Radio technologies for ad-hoc networks
o Wi-Fi, Zigbee, Bluetooth

Modelling ad-hoc networks
o Propagation models
o Topology models based on graph theory
o Degree and hopcount
o Connectivity theorems

MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook)
o Introduction to MAC protocols
o Issues and design goals
o Classification
o Directional, muli-channel MAC protocols
o Energy efficiency in MAC protocols
o Quality of service

Self organisation and Routing (Chapter 7, 8, 11 of the textbook)
o Flooding
o Node discovery, neighbour discovery
o Route establishment
o Topology maintenance, localisation
o Proactive, reactive and hybrid routing
o Typical protocols
o Energy efficiency in routing
o Broadcast and multicast
o Effects of mobility on connectivity and capacity
o Effect of nodes joining and leaving the network

Advanced issues in ad hoc networks
o Wireless sensor networks (Chapter 12 of the textbook and papers)
o Cooperation (Reference papers)
o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET)
o Energy Harvesting

Project presentations by students

Study Goals

By the end of this course students should be able to:
- Model the ad-hoc networks using Graphs.
- Describe the working principles of medium access control protocols for ad-hoc networks
- Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks
- Choose various components to form a coherent ad hoc networking architecture
- Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks
- Assess the suitability of ad-hoc networks for different communication needs and scenarios

Education Method

The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.

Literature and Study Materials

1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004.

2. Lecture notes consisting of slides presented at the lectures (Slides are only teaching aid and they are not substitute for

	textbooks, research papers, etc). 3. Some recent journal papers 4. Optional Reference Books 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms) 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar , C Puttamadappa , and T. G Basavaraju, Auerbach Publications, 2008. This book is avaliable online in the library. 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is avaliable online in the library. 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer. 5. OPNET/ns-2 web pages, tutorials and video lectures
Books	Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.
Assessment	1. There will be totally three written/oral tests/examinations for this course (including the final exam). 2. The students will carry out a project in a group and submit a short report. 3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis. Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class. Breakup: 2 Tests + Final Exam = 55% Project 40% Self assessment + Reflection 3% Activities on Feedback Fruits or any Online platform 2% Resit: written/oral exam [+ carry over from projects]. == Changes due to COVID19 in 2021 is below; in case there is a similar situation, we use the following methods. == There are three chapters Chapter 1: Network modelling Chapter 2: MAC protocol Chapter 3: Routing + Lab: Simple experiments using laptops (individual) + simulations (group-wise) Marks Breakup -------------- Homework/Assignments/Group works Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks Part B1: Individual experiment + report Part B2: Group-wise simulations + report + demo Part B1 + Part B2 - 50 Marks Part C: Self-assessment + peer activities -- 5 marks Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit. == (More information will be given in the first class) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Permitted Materials during Tests	Different conditions for different test/exams. Conditions will be informed 1 week before the exams/test.

CS4070	Multivariate Data Analysis	5
Responsible Instructor	Dr. J. Söhl	
Responsible Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Required: * Introduction Probability Theory and Statistics: see for instance A Modern Introduction to Probability and Statistics Understanding Why and How Series: Springer Texts in Statistics Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover ISBN: 1-85233-896-2 * Basic calculus * Linear Algebra: matrix multiplication, the inverse of a matrix, the transpose of a matrix, least square solution. see: David C. Lay: Linear Algebra and Its Applications ISBN-10: 0321385179 ISBN-13: 9780321385178 ©2012 Pearson)	
Course Contents	PART I: Stochastic models will be developed on the basis of probability theory. Probability theory describes the behavior of certain phenomena in terms of how likely it is that certain values will occur. Central features of the models will be discussed are random variables, probability density functions, and the expected value operator. In describing random processes and signals, the correlation function and conditional probabilities play a central role. It addresses the following subjects: 1. Random variables. Matlab exercise on estimation of PDF, expected value and variance. 2. Refresher correlation. Calculating with correlation functions. 3. Random processes, correlation function, stationarity, wide sense stationarity, estimation of correlation function (Matlab exercise). 4. Random signal processing, power spectral density function, white noise. 5. AR processes, linear prediction: theory and computer exercise. 6. Markov chains. PART II: A course in advanced statistics about linear models, Bayesian inference, classification problems, Gaussian processes and Markov Chain Monte Carlo.	
Study Goals	PART I: 1. Probability Theory - Conditional) probabilities, the law of total probability, and Bayes rule. - Solve probability problems that require the use of axioms of probability. 2. Definition and Description of Random Variables and Processes PDF, PMF, CDF, Covariance, Correlation- Determine if a given PDF, PMF, CDF, variance, (auto/cross-)correlation(-function), (auto/cross-)covariance(-function), power spectral density complies with (theoretical and analytical) requirements. - Convert the description of a probabilistic problem into a probabilistic model using PDF, PMF, or CDF. 3. PDF/PMF and Expected Value Calculate the various forms of expected value of (combinations of) random variables and random processes - For a given (amplitude continuous/discrete and time continuous/discrete) probability model calculate the following probabilistic (marginal, joint and conditional) characterizations: PDF, PMF, CDF, probability of an event, expected value, variance, covariance, correlation, correlation coefficient, auto/crosscorrelation function, auto/crosscovariance function, (cross) power spectral density. - Calculate the PDF, PMF, expected value and variance of a derived random variable. 4. Properties of Random Processes - Independence, orthogonality, uncorrelated, whiteness, IID- Determine if random variables/processes have the following properties: independent, orthogonal, uncorrelated, white, Poisson, Gaussian, Bernoulli, Markov, IID, stationary, WSS, ergodic. - Calculate the expected value, variance, auto/crosscorrelation(function), auto/crosscovariance(function), power spectral density of a linear combination of random variables and of a linearly filtered (WSS, amplitude discrete/continuous, time discrete/continuous) random process. 5. Large NumbersCentral limit theorem, law of large numbers - Solve problems that require the use of the central limit theorem in an engineering context - Explain the law of the large numbers in an engineering context. 6. Statistical Estimators - Estimated mean, variance, and correlation function - Given a set of outcomes, sample functions or realizations, calculate estimators for expected value, variance, and (auto-)correlation function. 7. Application to Engineering Problems and Simulations - Select and translate a simple electrical engineering or computer science problem into mathematical probability model. The emphasis is on problems in signal and image processing, telecommunication, and media and knowledge technology. The class of probability models encompasses the following random variables/processes: Bernoulli, exponential, binomial, Poisson, Gaussian, uniform. - Justify and reflect on the approach taken in calculating or simulating (MatLab) the following probabilistic properties: PDF,	

	PMF, expected value, variance, autocorrelation function, autocovariance function. PART II: By the end of the course, you should be able to: - compute properties of frequentist and Bayesian estimators in the linear model - derive posterior distributions and their properties - apply hyperparameter optimization in Bayesian models - apply computational methods for posterior approximation - derive properties of frequentist and Bayesian classification methods - derive properties of MCMC methods for sampling from the posterior distributions - derive properties of Gaussian process regression
Education Method	PART I: Lectures, working groups (problem solving), laboratory work (a Matlab exercise) Workload is around 15 hours for attending lectures, 5 hours of reading study material and preparing lectures, 15 hours for the lab course, 20 hours for preparing the exam, 3 hours for the exam, and 8 hours for a final report (66 hours in total). PART II: Classes and weekly exercises.
Books	PART I: R.D. Yates and D.J. Goodman, "Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers", ISBN 0-471-17837-3, John Wiley and Sons, New York, 2005, Second Edition. PART II: Simon Rogers and Mark Girolami "A first coruse in machine learning, 2nd edition" Chapman & Hall It is planned to cover chapters 1--4, 8 and 9 from this book. The precise material will be determined during the course.
Assessment	The final grade of the course consists of the following components: Part I (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Part II (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Final grade calculation: 0.5 * Part I + 0.5 * Part II A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Part I written exam: Resit (written or oral exam, depending on the number of students) Part II written exam: Resit (written or oral exam, depending on the number of students) In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Part I lab assignments: Re-submit deliverable Part II lab assignments: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	PART I: Exam of 3 hours. PART II: Exam of 3 hours.
Permitted Materials during Tests	PART I: Self made notes on a two-sided written A4 sheet. Calculator. PART II: none
Remarks	PART II: This course is particularly interesting for students that are interested in statistical exploratory and quantitative techniques to analyse multivariate data.

CS4200-A	Compiler Construction	5
Responsible Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	- programming (required) - software engineering (recommended) - functional programming (recommended) - formal languages and automata (recommended)	
Course Contents	Compilers translate the source code of programs in a high-level programming language into executable (virtual) machine code. In this course you will study the architecture of compilers, and important concepts and techniques underlying the components of that architecture. You will study these techniques by applying them in practice to implement your own compiler for a small language. The course covers the following topics: - Syntax and parsing - Static semantics and type checking - Virtual machines, assemby code, byte code - Interpretation vs. compilation - Code generation - Memory management, garbage collection - Data-flow analysis - Optimization	
Study Goals	At the end of this course, you will be able to: - Explain the architecture of a compiler and the role of each component of that architecture - Explain the algorithms and techniques for the implementation of compiler components and apply these techniques to examples - Implement a code generator that translates source language abstract syntax trees to target language instructions - Implement transformations on abstract syntax terms to simplify programs - Implement simple type checkers - Implement simple data-flow analyzers - Integrate these components into a working compiler	
Education Method	Attending the course involves: - Attending weekly lectures - Attending weekly lab session (which may start with a group tutorial) - Reading lecture material and papers - Making homework assignments (building a compiler)	
Literature and Study Materials	Required reading: - We will use selected chapters from the book "Essentials of Compilation" by Jeremy Siek, available online: https://github.com/IUCompilerCourse/Essentials-of-Compilation - Lecture slides and selected papers from the literature Optional reading: - We will use Scala to implement our compilers. The following book is optional reading which may be helpful in getting up to speed with programming in Scala: https://www.artima.com/shop/programming_in_scala_5ed (Additional) reading materials will be made available on Brightspace. We will use WebLab (https://weblab.tudelft.nl/cs4200/) for the submission of homework assignments.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Homework assignments (weighting 50%) Final grade calculation: 0.5 * Written exam + 0.5 * Homework assignments	

	A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Permitted Materials during Tests	No materials are permitted during the exam.
Co-Instructor	Dr. S.S. Chakraborty

CS4340	**Seminar Probabilistic Programming**	**5**
Responsible Instructor	S. Dumani	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Probabilistic reasoning is one of the core aspects of artificial intelligence. This course will look into probabilistic programming, the most powerful paradigm of probabilistic reasoning, which specifies probabilistic models as computer programs. Because of the representational power of programming languages, any computable process can be modelled as a probabilistic program. The course covers the topics of probabilistic modelling and inference, but the main focus will be on developing a working probabilistic programming language. The topics include: Probabilistic modelling techniques, Importance sampling, Particle filters, Metropolis-Hastings algorithm, Hamiltonian Monte Carlo and automated differentiation, Variational inference, Discrete and logic probabilistic programming languages, Probabilistic circuits, Compiled/Amortised inference, Deep generative models	
Study Goals	By the end of the course, you will be able to explain the standard probabilistic inference procedure, solve real-world problems using probabilistic programs, identify the strengths and weaknesses in recent probabilistic programming literature, and design a research proposal to address an open problem in the field	
Education Method	This is a seminar course: we will meet twice a week to discuss a research paper. The paper should be read before the lecture and the participants need to submit a short summary (max 0.5 - 1 page). The students are expected to study every discussed paper (but not to the same amount of detail), prepare a presentation and lead a discussion for a few papers, and actively participate in discussions. Aside from paper discussions, the students are expected to do a small practical project (e.g., validating, stress testing or extending the existing method)	
Assessment	The final grade of the course consists of the following components: Research proposal (weighting 65%) Presentation (weighting 25%) Class participation (weighting 10%) Final grade calculation: 0.65 * Research proposal + 0.25 * Presentation + 0.1 * Class participation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Research proposal: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Artificial intelligence	

CS4365	Applied Image Processing	5
Responsible Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Weinmann	
Instructor	P. Kellnhofer	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Basic knowledge of programming concepts, linear algebra, and calculus. Prior experience with Python and C++ is useful but it is not required.	
Course Contents	The course discusses image processing techniques and their application to visual data problems in computer graphics and computer vision including neural image processing techniques. The practical part of the course focuses on the efficient implementation of image processing algorithms on parallel computer hardware. The final project guides students to work with literature sources and state-of-the-art surveys in an area. This course can serve as an introduction to the field of visual computing, which spans the domains of Computer Graphics, Computer Vision, Computational Photography and more. During the course, you will receive an overview of image processing with a deeper focus on applications for visual data. This will include: image representations, color models, linear filters, image sampling, digital photography (including HDR and tone mapping), image retargeting, image abstraction and non-photorealism, multiscale image processing (pyramids, Fourier transform), geometrical image transformations (warping, morphing), stereoscopic imaging, and neural image processing (concept, texture synthesis, style transfer, super-resolution, denoising, editing, depth map estimation). The course will cover classical signal processing and advanced image techniques, as well as approaches utilizing neural networks and machine learning. During lab sessions, you will learn about efficient implementations of the relevant algorithms with a focus on parallel data processing. This will include use of C++ with OpenMP and the Python machine-learning library Pytorch. Introduction to these technologies will be provided before their use. Assignments are solved by using the information presented in the lectures and lab sessions. Formative feedback will be provided during the lab sessions.	
Study Goals	By the end of this course, you should be able to: - Explain the process in which a digital image is formed. - Efficiently conceive, apply, and implement algorithms for processing digital images in C++ or Pytorch. - Analyze algorithms for image processing based on their computational costs, qualitative performance, and application areas.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of image processing, common problems and their solutions. Various algorithms and their properties will be discussed. During the guided lab sessions (4 hours each), you will be introduced to guidelines for efficient implementation of selected algorithms, and you will receive formative feedback on your approach to practical assignments. The final project will focus on survey and assessment of state-of-the-art approaches to a specified image processing problem.	
Literature and Study Materials	Required literature: - Lecture slides with additional references. Recommended literature: - Szeliski, Richard. Computer vision: algorithms and applications. Springer Science & Business Media, 2010. - Chapters 3, 4 and 10. - Burger, Wilhelm, and Mark J. Burge. Digital image processing: an algorithmic introduction using Java. Springer, 2016. - Chapters 22-26. - Sundararajan, Duraisamy. Digital image processing: a signal processing and algorithmic approach. Springer, 2017. - Chapters 2-7. - Goodfellow, I., Bengio, Y. and Courville, A.. Deep Learning. MIT Press, 2016. - Chapters 6-9 and 14.	
Assessment	The final grade of the course consists of the following components: - 3 assignments during the quarter (weighting 55%) - a final project (weighting 45%) Final grade calculation: 0.55 * Assignments + 0.45 * Final project The course is passed if all components have at least a grade of 5.0 and the weighted average is a 6 (rounded on 0.5). Repair possibilities (under conditions described in TER Implementation Regulations Art 5, sub 5): - Assignments: An individual replacement assignment with a maximal grade 6.0 in the following quarter. - Final project: An individual replacement project with a maximal grade 6.0 in the following quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

CS4375	Artificial Intelligence Techniques	5
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. M.T.J. Spaan	
Instructor	L. Cavalcante Siebert	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	We expect students to have programming expertise at the Bachelor level of Computer Science, in particular knowledge is expected of algorithms (e.g. search algorithms), and probability theory. Programming skills are required for the practical assignments which can be done in Python.	
Course Contents	The course focuses on concepts and techniques from probabilistic Artificial Intelligence for building intelligent agents, and decision making and -support systems. The specific topics include Bayesian networks, inference, belief monitoring, (partially observable) Markov decision processes, reinforcement learning and multiagent systems.	
Study Goals	After completing the course, student will be able 1. to comprehensively summarize the covered topics within probabilistic artificial intelligence and discuss their main issues; 2. to explain the main concepts and techniques; 3. to relate practical problems to appropriate AI techniques and apply the formal models covered in the course to these problems; 4. to implement and evaluate algorithms for complex decision making.	
Education Method	Lectures, lab work (programming assignments).	
Literature and Study Materials	Stuart J. Russel and Peter Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Prentice-Hall. ISBN 9781292401133 Global edition. See website http://aima.cs.berkeley.edu/global-index.html for additional information that goes with the book.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	The practical assignments will be done in groups of 4 to 5 students. The ethical standards of working are expected of all students. Work can be divided over students, but all students are responsible for the overall quality and originality of the work. Students that do not do their share of the work have to be reported by the group to the supervisor so that action can be taken.	
Tags	Artificial intelligence	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IN4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Reuired: Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations Strongly advised: some experience with a programming language (e.g., C) is preferred.	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas are considered. The scaling behavior of these algorithms in case of increasing problem size and/or an increasing number of processors is analyzed. Attention is paid to those aspects of computer architectures that are important to understand the resulting performance, such as the memory hierarchy and the interconnection network. By analyzing a number of case studies (applications) with respect to their computing-intensive character, possible bottlenecks will be determined. Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to get high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and the necessary optimization for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found the Brightspace.	
Course Contents Continuation	High Performance Computing, parallel programming, parallel algorithm	
Study Goals	1. Is able to categorize high-performance computer systems 2. Is able to explain and classify parallel and distributed architectures, and programming models; 3. Is able to explain the concepts of data decomposition and parallel algorithms; 4. Is able to apply various parallelization methods to benefit from the potential parallel computer systems; 5. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 6. Capable to evaluate and analyze the performance of parallel programs.	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): implementing (small) parallel programs with C, MPI and Cuda.	
Literature and Study Materials	Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Lab work (weighting 50%) Quizzes, bonus of max 1 point can be obtained (only valid on the first examination occasion after the lectures) Final grade calculation: 0.5 * Written exam + 0.5 * lab work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Via Brightspace	

IN4191	Security and Cryptography	5
Responsible Instructor	Z. Erkin	
Instructor	J.A. Martinez Castaneda	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	UT-201500042 Privacy Enhancing Technologies (Q4)	
Expected prior knowledge	Basic understanding on the following is suggested. -Probability and statistics -Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Learning outcomes: The goal is to make students familiar with the basic concepts applied cryptography, including classical cryptography and modern secret key and public key cryptography. In particular, the students will acquire A sound understanding of the notion of security An understanding of the confidentiality, integrity and authenticity needs of the society Understand the role of cryptographic primitives including the differences between symmetric and asymmetric cryptography, the role of hash functions, digital signatures and PKI Understand the advanced topics in cryptography needed for the modern society with untrustworthy entities Among others things, the following topics are covered: -Classical systems -Information theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures, -*Secret Sharing. (if time permits) -*Zero Knowledge proofs (if time permits)	
Study Goals	It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security, as well as is familiar with present applications. Through assignments, students are expected to have the chance to work on the topics covered in the lectures.	
Education Method	Lectures, assignments and weekly exercises. Attention: This course requires full effort of 140 hours. Even more, if you lack the background (probability and modular arithmetic) Planned Workload: Lectures: 28 x 45minutes sessions, total 22 hours Practice session: 7 x 90 min. total 12 hours Assignments: 3 x 20 hour, total 60 hours Weekly study: 7 x 4 hours, total 28 hours Exam preparation: 20 hours Exam: 3 hours	
Literature and Study Materials	 -Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition -Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) Handouts of lectures	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%) Mandatory assignments (weighting 30%) Final grade calculation: 0.7 * Written exam + 0.3 * Mandatory assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mandatory assignments: re-submit deliverable	

	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	In case of in person examination at campus: The exam is closed book. A cheat sheet of size A4, hand written is allowed for the written exam. Name and student number has to be present on each side. In case of remote exam from home: open book: textbook, slides and self-made notes only. randomised/customised exam.
Permitted Materials during Tests	Only non-scientific calculators.

IN4252	Web Science & Engineering	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	J. Yang	
Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The expected entry level is (equivalent to) standard bachelor-level computer science.	
Course Contents	The main subject of the course is the Web, and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g. Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan / research paper draft in collaboration with others in a small group to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies, and verbally present and defend this plan / draft.	
Education Method	The education methods include: - Lectures, before which and after which students study material by themselves, to get an understanding of the relevant material; - Small assignments and hands-on exercises, to apply the understanding of relevant material; - One large assignment, with a number of feedback moments, to learn how to write a web science paper / research project plan. Lectures will not be in each week in the class period (1+2): in between lectures there is time reserved for studying before and after lectures, for small assignments and exercises, and for writing the large assignment paper. The writing of the large assignment paper / project plan happens throughout the class period (1+2) to enable frequent feedback.	
Literature and Study Materials	Will be provided in class, depending on the topics chosen for the assignments and final paper.	
Assessment	Assessments are based on two components: the "small assignments" (accompanying the lectures) which provide a total of 20% of the grade (these are a mix of literature and programming homework), and the "final paper" which provides 80% of the final grade. To pass, each of the components ("small assignments" and "final paper") need to be graded with a minimum of 5.0. The final grade must be at least a 5.8. If these criteria are not met, the "small assignments" and the "final paper" can be repaired provided that the to-be-repaired component was at least graded with a 4 in accordance with OER/TER article 17. This repair is not a resit, and is limited to a maximum of 6.0 as a result. Components that have been graded with less than a 4.0 cannot be repaired. If the course is still not passed after the repair or cannot be repaired, all components need to be redone if the course is retaken in a later year. Both the small assignments and the final paper are group works of groups of preferably 2 members.	
Special Information	Students are asked to register/enrol on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	The expected workload is 5ects and that is principally distributed uniformly over the two quarters.	

IN4307	Medical Visualization	5
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge of linear algebra, calculus and programming is needed. The project will be implemented in MeVisLab and Python. Learning MeVisLab will be part of the practice assignments.	
Course Contents	Theory and practice (Notice project extends to Q2) of medical visualization. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measuremen; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for medical data; advanced applications.	
Study Goals	By the end of the course, you should be able to LO1: Explain medical visualization algorithms and their applicability to medical problems. LO2: Discuss the advantages and disadvantage of medical visualization algorithms. LO3: Build a medical visualization system for a given problem: a. Discuss a suitable visualization for a given medical problem. b. Implement the most suitable solution. c. Judge the performance of the implemented solution.	
Education Method	The course will be based on a combination of lectures and practical assignments. A final project will be developed in Q2	
Literature and Study Materials	Course slides, instructions for project and assignments, and selected literature via Brightspace. Chapters from the following book can be used as alternate view on the topics but are not mandatory: Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Book available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam Q1 (weighting 40%) Final project Q2 (weighting 60%) The final project will be the design and implementation of a visualization system for a given medical problem. The final project will be carried out in teams of three. The deliverables for the final project will be a report (paper), the results (e.g., code) and a short video presenting the project (i.e. screencast). Final grade calculation: 0.4 * Written exam + 0.6 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Q2 In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final project: re-submit deliverable (the project resit is not automatic and must be requested within 2 weeks after publishing the grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Notes and written material. No computers.	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	
Co-Instructor	Prof.dr. E. Eisemann	

IN4344	Advanced Algorithms	5

Responsible Instructor	Prof.dr. M.M. de Weerdt
Instructor	Y. Murakami
Contact Hours / Week x/x/x/x	6/0/0/0
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	English
Expected prior knowledge	Basic courses in Algorithmics and Complexity Theory
Course Contents	The course is on solving (abstract models of) complex real-world problems, with a focus on solving intractable problems exactly. The course consists of two main parts: * modeling and solving using linear programming * exact algorithms using search trees, dynamic programming, and/or decision diagrams
Study Goals	By the end of this course, students will have knowledge of and experience with the following advanced algorithmic techniques: (Part 1:) - linear programming (LP) and LP modelling - duality and simplex algorithm - integer LP and integer modelling - LP relaxation and branch and bound (Part 2:) - complete and bounded search trees - tree decomposition - dynamic programming - preprocessing - decision diagrams Furthermore, they will be able to * recognize situations where they can use these techniques and adapt them to different applications. construct new algorithms that are similar to these techniques. construct proofs that are similar to a selected set of proofs from the material. analyze the run time of algorithms. implement an algorithm that is given in pseudo-code. experimentally evaluate the quality and the runtime of an algorithm on a set of instances.
Education Method	Interactive lectures (usually 2x2 per week), optional homework exercises, programming assignments (2 hours lab support per week) The expected workload is 30% studying the written material and the recorded lectures, and participating in the interactive sessions 30% making the homework exercises and preparation for the exams 40% working on the programming assignments
Literature and Study Materials	Part 1 of the course will be mainly based on chapters 1-8, some of 9-11 (most of 9-11 is assumed to be known), and 12-14 of the syllabus "Optimization" by Karen Aardal, Leo van Iersel and Remie Janssen, which can be ordered via https://www.webedu.nl/bestellen/tudelft/ Part 2 will use chapter 10 of the following textbook: J. Kleinberg and E. Tardos, Algorithm Design, Pearson Education, 2006. ISBN 0-321-37291-3 As well as some supplemental study material, which will be provided via BrightSpace.
Assessment	The final mark depends on the marks obtained for (a) written reports and evaluation of programming assignments, PA (weight 40%) (b) a written exam, EX (2 parts, weight 60%) Each programming assignment is graded on a scale from 0 to 10. The final mark for the programming assignment (PA) is the average of the mark obtained for the assignments. Programming exercises can be completed by 2 students working together. If the average grade for the four programming assignments (PA) is between 4.0 and 6.0 (excluding 6.0), a repair is possible by making one extra assignment to replace the lowest grade of the four assignments. This leads to a repaired PA score, maximized at 6. (E.g. if grades were 3.5, 4, 4, 4.5, the 3.5 can be replaced. If the grade for the replacement assignment is a 7.5, the repaired PA score is a 5. If grades were 5, 5.5, 5.5, 6, the 5 can be replaced. If the replacement is a 9, the repaired PA score is not 6.25 but 6.) The exam consists of two parts. Each part will be examined after the lectures about that part have been delivered and will be graded on a scale from 1 to 10. The final mark for the exam (EX) is the average of the marks for the parts. Each exam part contains one or more challenging assignments; during each part of the course two representative homework assignments will be made available to prepare for this. There is a resit for the exam where any of the two parts can be redone. The result for a part after the resits is determined by the maximum score obtained for the original exam and the resit. The final mark for the course is determined as follows: - if the PA and the EX mark are above 5, the final mark is the weighted average of these three marks: 60% EX, 40% PA - if at least one of PA, EX is less than or equal to 5, the final mark is the minimum of the results obtained for PA or EX. Partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Special Information	For content questions / of general interest, please use StackOverflow. For personal questions, please use the course email address: aa-cs-ewi@tudelft.nl
Tags	Algorithmics Artificial intelligence Mathematics

WM-ITAV-4010	Scientific Writing	2
Module Manager	L. Meester	
Instructor	Drs. W.J. Blokzijl	
Instructor	Drs. B.M.D. van der Laaken	
Instructor	Drs. P.C. Post	
Instructor	Drs. A.E. Kam	
Instructor	M.J.Y. Wackers	
Instructor	S. Baars	
Instructor	M. Bliekendaal	
Instructor	L.C. Schroten	
Instructor	A. Glasbergen-Plas	
Instructor	M. Looij	
Co-responsible for assignments	Drs. B.M.D. van der Laaken	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1 2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you learn to write a scientific article, either a research article based on your own research data or a literature review about a subject of your own choice. This is a necessary skill for anyone who wants to pursue an academic career after their graduation, but it can also be used immediately for all academic texts you will write during your Master program, such as your Master thesis. In seven weeks, we will go through all steps of the writing process, from formulating a good main question and finding relevant literature, to the actual writing, re-writing and final editing. You will exercise with finding, reading and managing relevant academic literature, writing in an academic style, building a comprehensive argumentation, reviewing fellow students' articles and using other students and the instructor's comments to improve your own work.	
Study Goals	The purpose of this course is to learn how to write a scientific text. To achieve this, at the end of this course you will: know what the main characteristics are of a scientific text be able to formulate a main question be able to find, critically read and manage scientific literature be able to use literature properly and avoid plagiarism be able to build up your argumentation be able to structure an article according to the conventions in your field of study be able to use scientific English style be able to use tables and figures to support and communicate your results be able to give feedback on somebody elses article be able to use feedback for improving your work.	
Education Method	Practical, in 6 sessions (attendance mandatory). Every week you have to read some background information about scientific writing and hand in a part of your text. Participants must attend all sessions - one missed session is allowed only - and hand in all assignments in time. Students who receive a pass for this course are rewarded with 2 ects. This equals 56 hours of study. A total of 12 hours is spent on attending (online) classes, in which you can ask questions, discuss the feedback you received on your work and discuss aspects of scientific writing with your fellow students; the remaining 44 hours is for self study, writing and revising. In seven weeks, from preparing lecture 2 up to handing in your final article in week 8, you will have to spend at least 6 hours of self study on this course, every week. It is important that you make sure you have this time in your personal schedule. At the beginning of the course you will mainly be reading up about scientific writing and the subject of your text. As the course proceeds, you will be spending more of your time on writing, giving feedback and revising your own text.	
Books	Theory about academic writing will be made available through Brightspace.	
Assessment	You write a scientific article of 2000-2400 words (excluding the list of references and the abstract). You have to hand in parts of your article every week. Your final grade is based on the final article. An evaluation form for the grading of the article is available throughout the course.	
Elective	Yes	
Category	MSc level	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialistievakken start kwartaal 2 2023

CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	- basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	The course intends to bring the student into the position to: - Explain the fundamental concepts and terminology of real-time systems - Construct task schedules using different scheduling policies under a given set of realistic system constraints - Analyze the timing behavior of a system for a given system model and scheduling policy - Discuss advantages and disadvantages of different scheduling policies for a given platform or system - Discuss the effect of hardware and software interferences on the timing behavior of a given system - Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system - Derive (reverse engineer) the system specification from a given implementation (in the lab) - Evaluate the scheduling overheads of a given implementation (in the lab) - Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	Written exam (grade) + lab work; the exam has a resit Repair option for the Individual Assignment = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet has become of critical importance to society. However, the large size of networks and abundance of protocols have made network management very complex. The novel concept of network programmability addresses this complexity and has resulted in a paradigm shift in how networks are (or can be) operated. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network programmability and which treats fundamental networking concepts like Quality of Service and network resilience.	
Study Goals	The learning objectives of this course are twofold: (1) The student should gain knowledge of the treated networking technologies. (2) The student should be able to apply and work with the programmable network technologies in a network emulator (Mininet).	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final assessment (and hence 100% of the grade) will be based on an exam that covers both the theory from the slides as well as the content from the reader. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CS4015	Behaviour Change Support Systems	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	none	
Course Contents	Behavior change support systems (BCSS) are computer-based systems that support individuals to form, alter or reinforce cognitions, attitudes or behaviors without using coercion or deception. They can serve individuals throughout the various stages of a change process, such as awareness development, contemplation, action strategy development, development of new behaviors, and maintaining these new behaviors. Virtual healthcare coaches, negotiation support systems, and applications that provide individuals with personalised financial guidance are three examples of these systems. To establish, modify or maintain change BCSS can deploy computerized persuasive strategies (e.g. reducing effort to establish target behavior, or argumentation reflection strategies, and also algorithmic nudge, including data-, expert-, and theory-driven), simulations (e.g. serious gaming, virtual reality), relational software agents (e.g. ePartners, virtual coaches), and personalization based on longitudinal user data. BCSSs are found in many domains, including education, sales, negotiation, management, and particularly in the health domain. The main topics are: Understanding the behaviour; strategies for behaviour change, action funnel, human decision making Identifying change support strategies; Behaviour change wheel, product vision, target outcome, target user, population diversity Conceptual design of BCSS; Behaviour plan, decision-making environment, story-editing technique Behaviour change and technology acceptance; Motivation, Trans-theoretical model, Theory of reasoned action, theory of planned behaviour, Technology acceptance model, Unified Theory of Acceptance and Use of Technology, technology adaption lifecycle, Fogg behaviour model, Self-Determination theory Persuasive technology; functional triad, levels of persuasion, Nudge Theory, principles of influence, triggers Algorithmic nudge/Persuasion and support Algorithms; concept, key characteristics, and feasibility Learning principles; Behaviouristic learning principles, cognitive learning principles, constructivism based learning principles, social cognitive (learning) theory Communication and framing; Behavioural economic principles, elaboration likelihood model (ELM), health belief model, protection motivation theory, persuasive communication Adherence and Gamification; Implementation Intentions, system 1 and system 2, treatment adherence, gamification Application domains eHealth and Cyber security; Self-monitoring, self-management, virtual health agents, virtual reality therapy, internet cognitive behavioural therapy (iCBT)	
Study Goals	This course provides students the opportunity to develop and demonstrate understanding, knowledge and competence in the field of BCSS. The learning outcomes for the course are that students will be able to: 1. Apply psychological theories, principles, and concepts in the design of BCSS systems. 2. Apply BCSS practices, principles and techniques. 3. Construct an implementation of an algorithmic nudge/boost/persuasion for a specific real-world application domain and objective 4. Design a BCSS system using BCSS principles, concepts, theories, and design methods.	
Education Method	The course uses a flipped-classroom approach, where students watch the pre-recorded lectures in their own time and complete online quizzes. In the on-campus lectures, students and teachers discuss small challenges related to the lecture topic. In the pre-recorded video material, theories, principles and methods are presented, discussed and illustrated with examples from the field. The video material is supported by online self-tests. In their own time, students work in small groups on coursework assignments to develop a product design for a BCSS with a computer simulation of supporting algorithmic nudge/boost/persuasion. In the on-campus session, student groups present the progress of their coursework and receive formative feedback. The presentations are done three times, following topic as the project progress over time: 1. The targeted behaviour 2. Behaviour plan 3. Conceptual design and the algorithmic nudge/boost/persuasion	
Literature and Study Materials	Available on brightspace	
Books	Wendel, S. (2020). Designing for Behavior Change: Applying Psychology and Behavioral Economics. O'Reilly Media, Inc,. or the older edition Wendel, S. (2013). Designing for Behavior Change: Applying Psychology and Behavioral Economics. " O'Reilly Media, Inc.	
Assessment	The final grade of the course consists of the following components: Computerised examination (or oral exam) (weighting 60%) If the expected number of students registering for the exam or resit is small, the teacher might decide to replace the computerized examination with an oral examination. Design BCSS Coursework Project (resulting in presentation slides and demonstration of computer simulation of a supporting algorithmic nudge/boost/persuasion, including question and answer round where individual group members are assessed on the coursework) (weighting 40%) Final grade calculation: 0.6 * Computerised examination (or oral exam) + 0.4 * Design BCSS Coursework Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Computerised examination (or oral exam): Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Design BCSS Coursework Project: Resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	3	
Co-Instructor	M.L. Tielman	

CS4135	Software Verification	5
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	Ir. K.F. Wullaert	
Contact Hours / Week x/x/x/x	0/2/0/0 + 0/4/0/0 practicum	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	How can we ensure that software cannot crash and is guaranteed to be correct? In this course we tackle this question by viewing programs and programming languages as mathematical objects. That way we can use logic to prove properties about programs and thereby guarantee that software is correct. To make reasoning about actual programs and programming languages feasible, we will not be doing these proofs by hand, but instead use a tool called a proof assistant to build proofs that can be checked by a computer. As we will show during this course, proof assistants turn the activity of doing proofs into programming. This course is a specialization course for programming languages and software engineering Prior knowledge: - Required: - Strongly advised: some familiarity with functional programming and propositional and predicate logic.	
Study Goals	Learning Objectives: - State and prove properties of functional programs in logic. - Specify the semantics of a programming language in logic. - State and prove the correctness of imperative programs. - Use a proof assistant to perform a mechanized proof.	
Education Method	This course is being taught in a "flipped classroom" style. It consists of 2 hours of studying (by reading or watching lecture videos, for greater flexibility and avoiding clashes with other courses - material will be provided) and two lab sessions of 2 hours each. During the lab sessions students will work on stating and proving theorems. Towards the end of the course students will carry out research projects that apply the ideas of the course.	
Literature and Study Materials	Supplementary material: Free online text book "Logic and Proof": https://leanprover.github.io/logic_and_proof/ Free online text book "The Hitchhikers Guide to Logical Verification": https://github.com/blanchette/logical_verification_2021/raw/main/hitchhikers_guide.pdf	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the individual project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4270	Conversational Agents	5
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Responsible Instructor	F. Broz	
Instructor	M.L. Tielman	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming skills (e.g. Python and Java) Probability theory and statistics	
Course Contents	Chatbots, embodied and conversational virtual agents, and social robots are becoming more and more popular. Many people are owning an Alexa, Cortana or Echo or are talking to their virtual assistant on their phone. Indeed, such technologies have the potential of making our lives easier and relieve people from the more repetitive tasks. For example, it is imaginable that such systems are being used for financial applications by helping customers with frequently asked questions but also to advise them on in the long term more impactful decisions such as their pension plans. Further applications can be imagined in the area of healthcare and education, some of which are already in existence today. In this course, attention will be given to different verbal and nonverbal behavioral characteristics, like speech, intonation, gaze and gestures that humans show when communicating with both other people and machines. This behavior is then related to different dialogue functions, including turn-taking, addressing others, and backchanneling, that give shape to the communication process. This course introduces conversational agent technology. We cover agent related technologies which can be grouped into: Dialog Management NLP speech synthesis social robotics	
Study Goals	After this course you have learned to: 1) Apply relevant linguistic and psychological theory to conversational agent systems 2) Design the conversational memory of your agent focusing on a perception/generational model 3) Analyse human-human conversational data to better design ML models for adding generation or perception capabilities to conversational agents 4) Develop the conversational memory in your agent 5) Evaluate the effects of affect and embodiment on human-agent interaction 6) Discuss your results and the limitations of your implementation	
Education Method	There are 2 lectures scheduled per week. Students work in groups of 4 on a group project. Lectures: 26 hours (13 × 2 hours lectures) Reading time: 39 hours Preparation basis tool use: 25 hours Coursework project, including writing report and prepare a video demonstration: 50 hours (10 × 5 hours)	
Literature and Study Materials	We use the book "The conversational interface " by Michael McTear, Zoraida Callejas, David Griol. This book is freely available through the TU Delft library. https://link-springer-com.tudelft.idm.oclc.org/book/10.1007%2F978-3-319-32967-3 Other relevant material will be provided on Brightspace.	
Assessment	The final grade of the course consists of the following components: Online Examination (weighting 30%) Group Assignment (weighting 70%). This assignment will result in a group report and a video demonstration. 3 Lab assignments (pass/fail) Final grade calculation: 0.3 * online examination + 0.7 * Group assignments + P/F * Lab assignments A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Online examination: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Group Assignment: re-submit deliverable Lab assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4400	Deep Reinforcement Learning	5
Responsible Instructor	Dr. J.W. Böhmer	
Instructor	Dr. M.T.J. Spaan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students must have passed CS4375 "Artificial Intelligence Techniques", or have acquired equivalent knowledge about: - basic probability theory, analysis and algebra - general machine learning methodology, e.g. regression - fully and partially observable Markov decision processes - tabular reinforcement learning methods, e.g. Q-learning - the exploration/exploitation trade-off, e.g. RMAX or UCB - multi-agent learning, e.g. centralized training and decentralized execution The course will require a lot of math and implementations of deep-learning algorithms. Students are encouraged to close any gaps in the above knowledge and to familiarize themselves with the Python/PyTorch deep-learning framework before the start of the course.	
Course Contents	This course will cover the breadth of modern model-free RL methods, discuss their limitations and introduce a variety of current research topics. In particular, we expect to cover the following: - deep learning methodology and architectures - stabilization of approximated value estimation - modern actor-critic methods - planning as inference - exploration with deep networks - offline reinforcement learning - deep multi-agent reinforcement learning - multi-task and meta learning	
Study Goals	After successful completion of this course, students - can list the strengths and limitations of modern deep RL approaches, - explain the underlying concepts of the discussed methods, and how they differ from each other, - derive the objectives and constraints of selected algorithms, - can implement selected algorithms/architectures, and - can analyze a new task to decide which algorithms/architectures to apply.	
Education Method	The course will be taught in lectures and the content will be solidified in homework submissions, which will be presented in tutorials.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Exercises (Pass/Fail) To be eligible for the exam, students must hand in homework exercises. Homework will not be individually graded, but at least 75% of the answers must be of sufficient quality (in terms of time commitment, not necessarily correctness) to be eligible to take the exam. Final grade calculation: P/F * Exercises + 1 * Witten exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Exercises: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4089	Data Visualization	5
Responsible Instructor	T. Höllt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/2/0/0 & lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Master course MKE/ST/DS	
Expected prior knowledge	IN2905-A/IN4152/CSE2215 Computer Graphics (recommended, not required). The practicals will be implemented in HTML/Javascript/D3 (InfoVis) and C++ (VolVis). We consider programming skills as a requirement but not necessarily in the mentioned languages and no advanced skills are needed. Relevant topics will be introduced and experience in other programming languages should make adaption feasible.	
Course Contents	Data visualization is the visual representation of data by computer generated images. The data sets can be results of numerical simulations or measurements (scientific visualization), or other data collections such as databases (information visualization). The goal is to improve insight, understanding and/or communication of data. Data visualizations use a combination of methods from a very diverse variety of disciplines: perception, computer graphics, human computer interaction, algorithmics, image processing, machine learning, numerical analysis, optimization, The course has two main parts; information and scientific visualization that will involve knowledge of diverse disciplines. As a computer science course, affinity to algorithmic thinking and programing skills will be needed. Topics covered: models of the visualization process; colour models and use of colour; information visualization; representation and processing of data; volume visualization; interactive visual data analysis; visualization of vector fields and flows. Guest lectures might be given on selected topics.	
Study Goals	The goal of the course is to get knowledge on the fundamentals that are part of data visualization. The main principles and techniques that are the basis of generating effective visual representations of data. Techniques and cases of data visualization are discussed. There are several applications for the techniques, such as medical, engineering, finances, economics, game analytics, and more. By the end of the course, you should be able to LO1: Discuss a large range of visualization techniques. LO2: Discuss a perception principle of visualization. LO3: Explain mathematical principles and algorithms of visualization techniques. LO4: Design suitable visualization systems for a given practical data analysis problem. LO5: Implement visualization systems for a given practical data analysis problem.	
Education Method	Lectures, practical assignments, self-study, and projects.	
Literature and Study Materials	Course slides, instructions for projects, and selected literature via Brightspace. Chapters from: Visualization Analysis and Design Author: Tamara Munzner CRC Press Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Books available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 30%). You are allowed to have the slides and material of the course during the exam. No computer or laptop is allowed. Information visualization project (weighting 35%) Volume visualization project (weighting 35%) The projects will be developed in groups of 3 and are evaluated based on the reasoning/justification of the techniques used based on the material given at the course, effectiveness of the results, technical contribution or implementation, quality of the documentation and presentation Final grade calculation: 0.3 * Written exam + 0.35 * Information visualization project + 0.35 * Volume visualization project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Information visualization project: re-submit deliverable Volume visualization project: re-submit deliverable The project resits are not automatic and must be requested within 2 weeks after publishing the grades. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

IN4150	Distributed Algorithms	6
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for network overlays, message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, broadcast, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	Understand the fundamental problems that distributed systems face Understand the most important distributed algorithms that solve these problems Reason about the execution of distributed algorithms Program distributed algorithms Select and summarize relevant literature on distributed algorithms	
Education Method	Lab work executed in groups of two students	
Literature and Study Materials	Lecture notes and lecture slides (available on Brightspace)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 75%). In case in-person, on-campus written exams are not possible because of unforeseen circumstances or measures, the written exam will be replaced by individual online oral exams. Paper summary (weighting 25%) Final grade calculation: 0.75 * Written exam + 0.25 * Paper summary A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper summary: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None except the list of algorithms	
Special Information	Email: in4150-ewi@tudelft.nl. Please send any emails related to the course to this address.	
Remarks	Lab work is 40 hrs.	

IN4302TU	Building Serious Games	5
Responsible Instructor	Dr.ir. A.R. Bidarra	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	For CS students: programming experience with some object-oriented language; experience with graphics, AI and/or some game engine(s) is a plus. For all students: though not compulsory, it may be convenient to have followed the course SEN9235 (Game Design Project), which is taught in the first quarter.	
Course Contents	Project-based interdisciplinary course, open to MSc students of all faculties. The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, training, health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose. Assignments for this course will be provided by real-world end-users (e.g. companies or the Science Centre Delft), to whom the group will be reporting throughout the term of the project. Project-based interdisciplinary course, open to MSc students of all faculties (of LDE universities). The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, professional- and health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose, solving a variety of possibly conflicting, technical challenges. Assignments for this course are provided by real-world commissioners (from startups to NGOs to research institutes), to whom the group will be reporting throughout the project. In previous years, various serious game prototypes originated in this course were actually developed and eventually deployed for their intended application; and many more were published and presented by students in international research venues (incl. IEEE CoG, FDG, ICIDS, GALA, ISAGA).	
Study Goals	At the end of the project, the student will demonstrate proficiency in the following aspects: o identifying and valuing the soft skills necessary to work effectively in interdisciplinary teams o responsibly interacting in a team, integrating its members' talents and expertise in a variety of team roles o adapting with flexibility to the dynamic requirements of a complex external assignment o translating feedback received into proactive personal development steps o formulating research questions, identifying research contributions, and describe them in a scientific publication Additionally, the CS student will demonstrate proficiency in the following specific aspects: o identifying, selecting and deploying the most adequate game technologies for the given serious game domain and constraints o deepening programming skills while building a complex and large software system in an agile context	
Education Method	Project: teams work intensively as a small game studio, according to a tight planning. Also a few plenary sessions and/or lectures.	
Assessment	The final grade of the course consists of the following components: Product grade (weighting 50%). Unique for the whole group, based on both the game itself and the required documentation. Process grade (individual) (weighting 45%). Including personal contribution, performance, attitude, and peer evaluation. Final presentation (weighting 5%) The commissioner will be involved both as advisor and as assessor. The final documentation includes writing a scientific paper and preparing it for actual submission to a conference on serious games and/or their application. Final grade calculation: 0.5 * Product grade + 0.45 * Process grade + 0.05 * Final presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final presentation: resit There is no repair possibility for the other components (product and process), since their assessment necessarily requires an extended development period. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. R. Marroquim	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Literature and Study Materials	We follow the book Performance Analysis of Complex Networks and Systems, by P. Van Mieghem, Cambridge University Press (2014). See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Written and closed book. A formularium is provided that can be consulted at the examination. Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. M. Blaauboer	
Responsible Instructor	Dr. L. di Carlo	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	30% homework assignments, 10% online quizzes, 60% final exam. A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course.	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialistievakken start kwartaal 3 2023

AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morpological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 6 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: ((0.85*EX + 0.15)+ AS)/ 2 or without the bonus point from the practicals: (EX+AS)/2 With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

CESE4030	Embedded Systems Laboratory	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. G. Lan	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This highly multi-disciplinary course comes with a lab project where teams of 4 students each will have to develop an embedded control unit for a tethered electrical model quad rotor aerial vehicle (the Quadrupel drone), in order to provide stabilization such that it can hover and (ideally!) fly, with only limited user control (one joystick). The control algorithm (which is given) must be mapped onto a home-brew PCB holding a modern RF SoC interfacing a sensor module and the motor controllers. The students will be exposed to simple physics, signal processing, sensors (gyros, accelerometers), actuators (motors, servos), and basic control principles. The embedded software running on the flight controller board (FCB) driving the drone must be written in the Rust programming language. The project work (including written report) covers the entire duration of the course period, and will take approximately 128 hours, of which 32 hours are spent at the lab facilities.	
Study Goals	Student is acquainted with real-time programming in an embedded context, along with a basic understanding of embedded systems, real-time communication, sensor data processing, actuator control, control theory, and simulation. Moreover, the student has had exposure to integrating the various multidisciplinary aspects at the system level.	
Education Method	Lectures (6*2hrs), lab work (8*4hrs), coding@home (8*12hrs), report (8hrs), so on average 2 days per week	
Prerequisites	Literacy in Rust (CESE4000 or equivalent, and CSE4015)	
Assessment	Lab. project (75%) + written report (25%), no exam, no resit. Failed student has to do an additional assignment/report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for CESE students- they get preference over other MSc students.	

CESE4050	Measuring and Simulating the Internet	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/2/0 (Lectures(2x2h), Pract)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Strongly advised: (Advanced) Networking course (e.g., CESE4045) and Programming skills.	
Course Contents	The Internet is a complex network without a fixed structure. Hence, measuring the Internet is crucial to acquire knowledge about the Internet infrastructure (topology), traffic, and performance (e.g., loss, delay, bandwidth, etc.). This course will discuss the design requirements and challenges in measuring and simulating the Internet, and the existing measurement methodologies (how/where/when to measure). Knowledge of how to conduct and evaluate Internet measurements enables the design and enhancement of a large set of applications, including: capacity planning and traffic engineering, network management and trouble-shooting, detecting network abuse and intrusions, etc.	
Study Goals	The goal of this course is to introduce the students to basic Internet measurement tools, as well as the state-of-the-art in Internet measurements research. The students will learn several Internet measurement techniques (e.g., active vs. passive measurements), and different software tools. Through a measurement assignment, the students will learn how to define/formulate a research problem, choose a specific approach, and complete a measurements-related research project.	
Education Method	Lectures + weekly instructions (discussion of papers and progress) + independent project work.	
Literature and Study Materials	Papers	
Assessment	Groups of students will be assigned a project that requires the students to put the theory on measuring and simulating the Internet into practice. The students have approximately 1 month to complete their assignment. The final assessment is based on the presentation (via report and/or demonstration) of the project assignment results and on the individual contribution and level of participation. Students within a group may thus receive different grades. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Maximum number of participants	Because this is a project-based course, we can only admit a limited number of students (typically around 30, but the actual number depends on the number of TAs involved).	

CESE4060	Wireless IoT and Local Area Networks	5
Responsible Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students are advised to follow the course Wireless Communications (ET4358) before taking this Wireless Networking course. An advantage is to have entry-level programming skills (Matlab, Python, C/C++). Nonetheless, students with little knowledge of programming will be helped.	
Course Contents	The following modules will be discussed during the lectures: Introduction (example topics): - What is wireless networking - Where to search for (academic) wireless network literature and resources Medium Access Control (example topics): - WiFi: hidden/exposed terminal problem, Carrier Sense Multiple Access - Bluetooth standard: in-depth look into the channel hopping, protocol specifications WiFi (example topics): - Review of IEEE 802.11 standards - Protocol format - ISM band regulation - Adaptive Modulation and Coding - WiFi Matlab class (assignment) IoT networking standards (example topics): - LoRa: protocol specifications, energy consumption, modulation format, network design Review of wireless tools (example topics): - Introduction to wireless packet sniffing and analysis using Wireshark (assignment) - Simple simulations of WiFi network with NS3 RFID networking (example topics): - Principles of backscatter - Protocol formats: EPC C1G2 - RFID hackathon (assignment) Cognitive radio (example topics): - Basics of spectrum management - White Space Databases - Theory of spectrum sensing	
Study Goals	At the end of the course students will be able to: (i) to understand how practical wireless systems work and get a deeper understanding of how the theoretical concepts of wireless communications apply to practice; (ii) employ their own analysis methodology to assess new wireless network systems (especially at the physical layer); (iii) understand rapid prototyping of new wireless systems (for instance, with software defined radio).	
Education Method	Lecture presentations, mini-project assignments, assigned paper reading and its critical analysis and presentation.	
Computer Use	Each student should have its own laptop (preferably with a Linux distribution, where Linux must not be installed on a virtual machine). We will be using Matlab, and/or NS3 and/or GNURadio and/or Wireshark for the assignments.	
Literature and Study Materials	WiFi Matlab WLAN toolbox: https://nl.mathworks.com/help/wlan/; Wireshark learn page: https://www.wireshark.org/#learnWS; tutorial on NS3 network simulator: https://www.nsnam.org/documentation/; specific chapters from books provided at the beginning of each lecture.	
Prerequisites	Background in programming (Matlab, Python, Bash)	
Assessment	Points from the mini-project assignments. A research paper analysis from conferences such as IEEE INFOCOM, ACM MobiCom, ACM SIGCOMM will be required to pass the course. Repair option: Individual Assignment = Re-submit deliverable Individual Report = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CS4090	Quantum Communication and Cryptography	5

Responsible Instructor	Dr. S.D.C. Wehner
Contact Hours / Week x/x/x/x	0/0/4/0
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information)
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.
Study Goals	The student will acquire: A good understanding of the fundamental concepts of quantum communication Insight into the differences between classical and quantum communication and cryptography Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session
Literature and Study Materials	Primary: Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.
Assessment	The final grade of the course consists of the following components: Homework (weighting 70%) Final Project (weighting 30%) Final grade calculation: 0.7 * Homework + 0.3 * Final Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Homework: re-submit deliverable Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication

CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5

Responsible Instructor	Ir. S.E. Verwer
Responsible Instructor	Dr. A. Panichella
Contact Hours / Week x/x/x/x	0/0/4/0
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Course Contents	Software is one of the most complex artifacts of mankind has ever created, but complexity is the enemy of correctness. Modern software testing and validation tools use a multitude of techniques geared toward correct computer code, most of these are base on artificial intelligence. In this course, we study these techniques in details, specifically we will understand and implement: Execution monitoring and taint analysis Branch distance computation Hill-climbing and genetic algorithms Concrete and symbolic (concolic) execution Active state machine learning Genetic programming The goal is to better understand and test software using artificial intelligence. Using the taught techniques you will be able to automatically: Discover which code is reachable Find (security) bugs in software Write tests that cover all reachable code Reverse engineer a code's functionality Patch code to remove bugs and failing tests
Study Goals	The student will: Understand modern AI techniques for software testing. Be able to implement several such techniques from scratch: - smart fuzzing (probing software with input to find crashes/bugs), - symbolic execution (using logic to construct inputs that trigger specific code branches), - fault localization (given that a program fails, find the line of code responsible for the failure), and - automated program repair (using a patch library and genetic programming to improve code) Be able to apply this technology to locate bugs in real-world software implementations.
Education Method	The main part of the course will consist of 3 lab assignments covering the theory (fuzzing&tainting, symbolic execution, automated program repair), and one lab assignment for the application to real software. The students will implement the taught techniques from scratch in the first 3 assignments, which will be scored with a pass/fail. All three assignments need to be passed to complete the course. The final lab will contain a recap from the first three assignments and an application of a state-of-the-art tool on real software. The final lab will be graded and be the final course grade. There will be instruction sessions where students can work on their assignment and ask the teachers for assistance.
Assessment	The final grade of the course consists of the following components: Three lab assignments (pass/fail) Final lab (weighting 100%) Final grade calculation: P/F * 3 lab assignments + 1 * Final lab A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: 3 lab assignments: re-submit deliverable Final lab: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Tags	Artificial intelligence Software

CS4160	Blockchain Engineering	5
Responsible Instructor	Dr.ir. J.A. Pouwelse	
Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	In this course you will learn all aspects of blockchain technology, including tamper-proof data structures, digital identities, transitive trust, fault tolerance, distributed consensus, smart contracts and applications. Ledgers and blockchains are an emerging technology with the potential to radically improve financial transactions, supply-chain flows, transactions in general, and distributed databases. The first three weeks of the course will provide a fast-paced introduction to Bitcoin, Ethereum, and TrustChain developed by TUDelft itself. The main component in this course is a team-based complex engineering project. This course is designed for computer scientists to understand blockchain technology and to produce significant hands-on experience. To provide a deep understanding of blockchain technology and understand why it is special you need to experience first-hand how it operates at a detailed technical level. Students design, implement, and test their own independent project in teams of 3-5 students. Students can choose from a pool of possible project ideas. This course requires you to like software engineering. Topics covered: -Blockchain basics and evolution Bitcoin 1st generation, smart contract generation, future 3rd generation (trust or trust in math) -identity and transitive trust Authentication and security primitives, tamper-proof identities, trust models, MITM attacks, Sybil attacks, and TrustChain by TUDelft -Consensus models Proof-of-work, permissioned, Proof-of-stake, Corda no-global-consensus, TUDelft bottom-up fast consensus model -Smart Contract pro/con encrypted data, Bitcoin scripts, Etherium execution model, Hyperledger + Docker issues, Corda Jar file approach, Tezos difficult to use, powerful technology, vision of the future: trusted verified execution -Markets and exchanges Disruption by open markets, winner-takes-all, and multi-sided market platforms, Uber, Airbnb, 22 years of eBay, Silk Road, honesty among drug dealers, the role of trust in markets, P2P exchange markets -Decentralized Autonomous Organization, novel method to collaborate and organise any economic activity -Web3: how the Internet can be re-decentralised using DAOs Within this 2023 edition "the Delft DAO" will be prominently featured. TUDelft achieved a world-first in DAO research. We devised a full end-to-end proof-of-principle of a DAO which is capable of 0) near unbounded scalability 1) controlling money 2) democratic decision making and 3) continuous sustained self-evolution. This course provides you with the knowledge to work with this advanced technology. After this course you will have a firm grasp on the current operational blockchain-based systems, realistic view of real-world applications that may be built on top of ledger technology. You will be able to reason and discuss the open challenges and questions that still need to be resolved. This course is a key course for distributed systems students.	
Study Goals	After this course students are able to design and engineer complex blockchain-based systems. Students are able to describe blockchain technology, the various consensus model, smart contracts, markets, and relation to existing database technology. Student are able to setup a new architecture for blockchain applications.	
Education Method	This course consists of four 2-hour lectures. Each lecture is followed by a 4-hour homework period in the same week focused on understanding the background material. In week 1 you will form teams and initiate work on your blockchain engineering project. A list of projects to select from will be provided at the start of this course.	
Literature and Study Materials	Online course textbook: Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction by Narayanan, Bonneau, Felten, Miller and Goldfeder.	
Prerequisites	It is highly recommended to follow this course (see remarks): Security and Cryptography (Q1) Distributed Algorithms (Q2)	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) You will write down your project results in a single-page report, markdown format. You will be graded on your open source efforts located on Github and single-page report. Your grade will be expressed on a scale of 0 to 10. Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This class has a limited capacity (50). If there is a larger number of enrollments than the capacity of the class, students will be assigned to their preferred blockchain engineering project based on their background, engineering experience level, and match to the course goals. Students who followed Security and Cryptography (Q1) and are also enrolled in Distributed Algorithms (Q2) will have priority for placement. Mathematics students are exempts from this, if they can show some minimal software development experience (e.g. Github profile). Finally, students with a Grade Point Average of 8.0 or higher are eligible for the challenging scientific projects, resulting in a research paper. These project receive intense guidance, but have no capacity limits.	

CS4195	Modeling and Data Analysis in Complex Networks	5
Responsible Instructor	H. Wang	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The assignment and final project require basic programming skill.	
Course Contents	Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	After this course, students could construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies, model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. Students could obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 20%) Final project (weighting 80%) Final grade calculation: 0.2 * Assignments + 0.8 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4210-A	Algorithms for Intelligent Decision Making	5
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. V. Robu	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: CS4375: Artificial Intelligence Techniques, or equivalent; and/or IN4344: Advanced Algorithms, or equivalent Required: basic course(s) in algorithm design and analysis, logic and probability; basic programming (in Python)	
Course Contents	Decision making is at the centre of artificial intelligence. This course gives you practical skills on a solid theoretical base. The course looks at solving mathematical models of NP-hard discrete optimisation problems. These kinds of problems lie at the heart of AI techniques such as planning, machine learning and mechanism design, and more generally combinatorial optimisation. You will learn about a range of modelling techniques from boolean satisfiability to constraint programming, and how advanced solvers for these models work. The course has plenty of real-world case studies as well as theoretical results. Apply the skills you learn in this course by taking CS4210-B: Intelligent Decision Making Project in quarter 4!	
Study Goals	By the end of this course, you will be able to identify features of real-world combinatorial decision problems, and be able to model and design systems for simplified instances of these problems using boolean satisfiability, mixed integer programming, and constraint programming over finite and real domains. You will be able to explain how SAT, CP and LCG solvers work in some detail, and how MIP solvers work at a high level. You will also be able to explain the basics of program synthesis and combinatorial game theory.	
Education Method	Lectures, homework exercises (optional, and not counted towards the course grade), and programming assignment(s) (mandatory, but not counted towards the course grade). The expected workload is: 30% lectures (including preparation for the exams) 30% homework exercises (optional) 40% programming assignments	
Literature and Study Materials	Provided on Brightspace	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Besides the summative assessment, formative coursework will be offered. This includes a mandatory assignment, performed in pairs. Students must submit a non-trivial attempt at the assignment in order to pass the course. The assignment receives formative feedback, but is not counted in computing the course grade. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Elective	Yes	
Tags	Algorithmics Artificial intelligence Group work Modelling Optimalisation Programming Projects Small groups	

CS4225	Educational Technologies	5
Responsible Instructor	Prof. M.M. Specht	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	* Theories of Human Information Processing and Learning * Learning Management Systems * Learning Analytics * Personalisation and Adaptive Educational Systems * Mobile and Seamless Learning Technologies * Artificial Intelligence in Education * Realtime Learning Technologies * Project Design * Project Implementation	
Study Goals	The course will enable you to classify, understand, design and implement the core functionalities and systems for supporting human learning processes. As well current practices implemented as also approaches for technology enhanced learning currently researched will be presented. You will learn how educational technologies provide human learning process support, implement guidance and recommendation, create personalised learning support, as also give real-time feedback and support reflection of learners. In the final project you will identify a problem, design a solution based on the presented approaches and implement your own educational technology solution.	
Education Method	Lectures, weekly assignments and quiz questions, final project	
Assessment	The final grade of the course consists of the following components: Weekly assignments (weighting 30%) Final project (weighting 70%) Final grade calculation: 0.3 * weekly assignments + 0.7 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4230	Machine Learning 2	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Instructor	Dr. D.M.J. Tax	
Instructor	T.J. Viering	
Instructor	Dr. F.A. Oliehoek	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Required: This course is the more advanced and research oriented follow-up to Machine Learning 1 (CS4220). CS4220 or an equivalent machine learning course is therefore expected as prior knowledge.	
Course Contents	The course will treat a number of machine learning settings and theories spanning the breadth of machine learning research in detail and on an advanced level. Topics that are covered include (subject to change): learning theory, learning curves, latent variable models, causal inference and discovery, adversarial learning, online learning and neuromorphic computing.	
Study Goals	After completing the course, you should be able to: 1. Identify which setting beyond (un)supervised learning a problem belongs to 2. Explain the essential problem that (adversarial/causal/online/PAC/neuromorphic/...) learning addresses and how it differs from the classical ML setting 3. Evaluate methods that are proposed in these learning scenarios, both theoretically and empirically 4. Mathematically prove basic properties of methods in these scenarios 5. Apply common methods in these learning scenarios on real-world datasets 6. Develop machine learning techniques adapted for a particular learning task	
Education Method	Each week: 1 lecture and 1 Q&A session where the weekly exercise set will be discussed.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) 4 combined assignments (pass/fail). There will be multiple small assignments (in weeks 2,4,6,8). Final grade calculation: P/F * assignments + 1 * Written exam A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: 4 combined assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4235	Socio-Cognitive Engineering	5
Responsible Instructor	Prof.dr. M.A. Neerincx	
Instructor	F. Broz	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic prior knowledge on human-computer interaction is helpful, but not required.	
Course Contents	Whether you are playing a game in virtual reality, driving a semi-autonomous car, educating yourself with a mobile app, or harmonizing your health and lifestyle with a robot assistant; nowadays intelligent networked information and communication technology is omnipresent. This course focuses on the design of human-aware intelligence into such environments, to support joint human-technology performances that bring about positive human experiences. The focus will be on social robots in general, with practical design and test assignments on robots that support people with dementia, http://rejam.tudelft.nl). In the Socio-Cognitive Engineering (SCE) course (MSc level), you will become acquainted with the application of a coherent set of methods for the design and evaluation of human--agent collaboration. Based on the SCE-method, we will elaborate on the state of the art of intelligent user interfaces (ePartners), such as virtual assistants, robotic teammates, eCoaches, and companion agents. The main topics of study are: - Design methods: Cognitive Engineering, Value Sensitive Design, Scenario-based Design, Claims Analysis, Design Rationale, Design Patterns. - Design for collective intelligence: Knowledge Representation, Ontology Engineering, Mental Models, Theory of Mind, ePartners, Adaptive Automation, Social Robots. - Design Evaluation: Prototyping, Test Methods, Measures, Questionnaires, Ethics. - Human Factors Theories and Models: Human Cognition & Learning, Memory, Emotion, Task Load, Human-Agent Teamwork, Behavior Change and Persuasive Technology.	
Study Goals	At the end of this course, students will be able to: 1. Explain the essential concepts of the design methods addressed in the course. 2. Explain the (dis)advantages of various design methods and their complementarity. 3. Apply the design methods addressed in the course in their research and design projects. 4. Explain what a design rationale is. 5. Construct a design rationale. 6. Create design specifications that are grounded in a design rationale. 7. Evaluate the strengths and weaknesses of a design rationale, e.g. using human-centered evaluations that test the design rationale. 8. Explain some of the state of the art human factors theories, models, and methods relevant to intelligent user interfaces, human-robot collaboration, and ePartner technology. 9. Write a structured report about a design-test cycle, with sufficient detail for a new group of researchers to continue the research. 10. Present work on a design project to an academic audience. 11. Work in a group on collaborative assignments.	
Education Method	LECTURES During the lectures, the teachers will present a range of theories, models, and methods relevant to socio-cognitive engineering. Students are required to read a number of scientific papers which are made available on Brightspace, along with the sheets/slides of the lectures. Together, the sheets/slides and the papers provide the students with the required theoretical knowledge to work on the practical project, and to learn about relevant design methods, human factors theories, conceptual solutions, and design principles. Most of the lectures include practical assignments and discussions stimulating the students to apply the contents of the lecture to their project (also see Project). PROJECT In the project, students work in groups to apply the knowledge acquired during the lectures. Students are required to plan, execute, present, and report on a complete design cycle (i.e. design, prototype, and evaluation) for a given design problem. This year (like the past years), the design problem is a social robot for older adults with dementia, and their social environment (https://rejam.tudelft.nl). The objective of the social robot is to improve humans physical, social, cognitive, and emotional well-being. The students will use the Wiki Socio-Cognitive Engineering (WiSCE) tool to specify the design rationale and its evaluation, step-by-step (see also https://xwiki.ewi.tudelft.nl/). Throughout the course, students will give presentations about their progress, on the design and evaluation of their prototype. Literature and study material consist of: - Papers from scientific journals on Brightspace. - Lecture notes on Brightspace	
Literature and Study Materials	Papers from scientific journals on Brightspace. Lecture notes on Brightspace.	
Assessment	The module assessment concerns the processing and application of the theory and methods; the construction of the design (rationale) and the evaluation; and the provision of the resulting concise and coherent report (including the lessons learned): The final grade of the course consists of the following components: Presentations (weighting 15%) Prototype (weighting 15%) Project report according to the prescribed format (weighting 60%) Individual reflection (weighting 10%) Final grade calculation: 0.15 * Presentations + 0.15 * Prototype + 0.6 * Project report + 0.1 * Individual reflection A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentations: resit Prototype: re-submit deliverable Project report according to the prescribed format: re-submit deliverable Individual reflection: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Exam Hours	There is no exam. The assessment is based on a paper, presentation and report. During the course, students will receive feedback on interim work. There is no resit after the end of the course.

CS4240	Deep Learning	5

Responsible Instructor	Dr. J.C. van Gemert
Instructor	Dr. D.M.J. Tax
Contact Hours / Week x/x/x/x	0/0/8/0
Education Period	3
Start Education	3
Exam Period	3 4
Course Language	English
Required for	Seminar Computer Vision by Deep Learning
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy). Experience in programming and installing programming languages/libraries is required. Without affinity/experience with python (and numpy) programming this course will become too time consuming as the project will become difficult.
Course Contents	In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ - https://udlbook.github.io/udlbook/ Research papers will be made available through Brightspace. Assignments are based on PyTorch: https://pytorch.org/
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Digital Exam (weighting 90%). Exam about the papers, lab assignments, and the theory (lectures). Project (Pass/Fail). In a small group of students you work on a deep learning paper reproducibility project. Final grade calculation: (pass project) * (0.1 * Presentation + 0.9 * Exam) A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

CS4345	Seminar Formal Methods for Learned Systems	5
Responsible Instructor	Dr. A. Lukina	
Instructor	Dr. M.T.J. Spaan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course will cover several topics at the intersection of formal methods and machine learning, for instance, abstraction refinement for analyzing deep neural networks, algorithms for reliable learning-based control, interpretable model design, and formal methods for verification and monitoring of learned systems. The course should be useful for students of both formal methods and machine learning, and lies at the intersection of these areas. Since the area is relatively new, the course material will be primarily based on research papers.	
Study Goals	After this course students will acquire formal methods knowledge and skills such as: - modeling complex systems, - formally defining critical properties, - applying verification tools to learned systems, - synthesizing controllers for learned systems, - providing interpretability for learned models.	
Education Method	The course will consist of one lecture and one discussion session per week. The students will be assigned projects in groups of two, where they will need to apply the material from the lectures, e.g., train a learned system, define a property and verify it using one of the open-source tools from state-of-the-art research papers. Discussion sessions will be used for questions about the projects.	
Assessment	The final grade of the course consists of the following components: Participation (weighting 15%) Presenting Papers (weighting 15%) Assignment #1 (weighting 15%) Assignment #2 (weighting 15%) Final Report (weighting 40%) Final grade calculation: 0.15 * Participation + 0.15 * Presenting Papers + 0.15 * Assignment #1 + 0.15 * Assignment #2 + 0.4 * Final report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presenting Papers: resit Assignment #1: re-submit deliverable Assignment #1: re-submit deliverable Final report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4370	Testing and Validation for AI-intensive Systems	5
Responsible Instructor	C.C.S. Liem	
Responsible Instructor	Dr. A. Panichella	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of machine learning. Knowledge of software testing is preferred but not required.	
Summary	How do you know and assess whether an AI-intensive system is working as intended? On the one hand, quality assurance for such systems can be considered to be a software testing challenge; in parallel, the machine learning components within such systems are normally validated according to machine learning-specific evaluation methodology. In this course, you will learn more about these two complementary takes to evaluation and quality assurance, while learning about (and getting hands-on experience in) state-of-the-art advances in combining these two takes.	
Course Contents	- Basics of testing (assertions, white-box vs. black-box testing) - Basics of machine learning evaluation (validation & validity) - Adversarial attacks - Testing techniques (e.g. fuzzing, metamorphic testing, combinatorial testing, differential testing) - Data labels (oracle) validation	
Study Goals	After completing this course, students are able to: 1. Apply state-of-the-art techniques to validate machine learning models and training routines 2. Explain, analyze and improve robustness considerations in AI-intensive systems 3. Design and implement techniques to generate adversarial attacks 4. Apply fuzzing, metamorphic and combinatorial testing techniques to machine learning models 5. Connect the oracle problem in software testing to questions of validity in datasets 6. Complement classical evaluation reporting of machine learning models with insights obtained through testing	
Education Method	2 hours lecture, 2 hours project	
Literature and Study Materials	The study material will be provided online by the lecturers	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4405	Analysis of Concurrent and Distributed Programs	5
Responsible Instructor	Dr. B. Özkan	
Responsible Instructor	Dr. S.S. Chakraborty	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	

Course Contents

Software systems are becoming highly concurrent and distributed to utilize modern multicore architectures and increasing speed and bandwidth in networks. Shared-memory concurrency in multicore programs and message-passing concurrency in distributed programs share many common abstractions and problems.

In the multicore era, all performance-critical software employs some form of concurrent programming; typically shared memory concurrency. In this setting,
programmers use a number of primitives to develop efficient and correct concurrent programs. To do so the programmers have to understand the behaviors of the primitives and reason about them. It is also important to match the programming paradigms and underlying architectures. For instance, traditionally programmers have assumed that a multithreaded program executed simply by interleaving the executions of its threadsa model known as sequential consistency (SC). This assumption is, however, invalidated both by mainstream multicore architectures, which often execute instructions out of order, and by compilers, whose optimizations affect the outcomes of concurrent programs. As a result, concurrent programs have more outcomes than SC allows.

In the distributed setting, the units of concurrency are independent processes that do not share memory but communicate by exchanging asynchronous messages. The execution of such a system involves two main sources of nondeterminism: concurrency and partial failures. As the processes run concurrently, the exchanged messages can be delivered and processed in many different orderings. The distributed set of processes is also prone to network of process failures. The trade-off between the systems availability in the existence of failures and the consistency between the processes gives rise to a spectrum of weak consistency notions. It is important to reason about concurrency, possible failures, and consistency guarantees to implement distributed programs correctly and understand their behavior.

This course aims to explore analysis techniques for concurrent and distributed programs.

Outline of Lectures:

Shared memory concurrency:
- Abstractions for shared memory concurrency
- Relaxed memory concurrency
- Correctness of concurrent programs

Distributed concurrency:
- Distributed system components, models and assumptions
- Fundamental abstractions for distributed systems

Study Goals

This course aims to give students a deep understanding of concurrency and distribution in modern systems and hands-on experience for analyzing these systems.
By the end of this course, you should be able to:

- Describe the fundamental concurrency models in multicore and distributed systems

- Explain the concurrency nondeterminism in the executions of concurrent and distributed programs

- Discover concurrency bugs in multicore and distributed programs by program analysis and testing techniques

- Evaluate the pros and cons of program analysis and testing techniques for specific multicore and distributed programs

Education Method

The course consists of the following education methods:

- Lectures for reviewing concurrency and distribution concepts
- Homeworks/assignments
- Developing a course project, writing a report, and presenting it (course project)

To finish the course, students (in teams) will have to:

- Study several papers which will be discussed during the lectures
- Deliver their assignments
- Deliver and present their implementation project

Assessment

The final grade of the course consists of the following components:
Research project implementation (weighting 40%)
Research project report (weighting 20%)
Research project presentation (weighting 20%)
Homework assignments (weighting 20%)

Final grade calculation: 0.4 * Research project implementation + 0.2 * Research project report + 0.2 * Research project presentation + 0.2 * Homework assignments
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

Resit/ Repair opportunities:
In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:
Research project report: re-submit deliverable
Research project presentation: resit
Homework assignments: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

CS4410	Category Theory for Programmers	5
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	Ir. K.F. Wullaert	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Categorical structures occur in programming languages on different levels: (1) within programming languages, providing design principles and guidance on how to write modular and correct-by-design programmes (as demonstrated in the practical programming language Haskell) and (2) in the design and study of programming languages, as a guiding meta-theory. In particular, category theory provides a mathematical justification for recursion schemes for inductive datatypes. This course aims to provide solid foundations on both (1) and (2). Lecture notes for the course are available at https://github.com/benediktahrens/CT4P. This course is suitable for people with an interest in functional programming, abstract mathematics, and logical reasoning. Prior knowledge: - Required: None - Strongly advised: None	
Study Goals	Learning Objectives: - Use categorical constructions (e.g., monads) in the design and structuring of computer programmes - Specify datatypes as initial algebras - Prove properties of computer programmes, guided by categorical intuition - Use categorical fusion laws to optimize functions - Use the theory of infinite data structures to define functions on infinite data	
Education Method	Learning in this course is achieved through lectures, problem sessions, and guided self-study.	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4415	Sustainable Software Engineering	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Sustainable Software Engineering is an overarching discipline that addresses the long-term consequences of designing, building, and releasing a software project. By definition, sustainability covers five main perspectives: environmental, social, individual, economic, technical. This course mainly focuses on the first, also known as Green Software Engineering. On top of that, we also cover fundamental aspects of social and individual sustainability of software projects. Software Engineering (SE) has long addressed sustainability by narrowing it down to economic and technical sustainability. However, our society is facing major sustainability challenges that can no longer be overlooked by software engineers and computer scientists. It is estimated that, by 2040, the ICT sector will contribute to 14% of the global carbon footprint. Hence, the environmental, social, and individual perspectives ought to be part of the equation when it comes to design, build, and release software systems. The problem is far from simple and we need expert computer scientists to bring sustainability into the core values of the next generation of tech-leading organisations.	
Study Goals	After attending this course, you will be able to: LO1. Explain the fundamental sustainability principles and their relevance to software engineering. LO2. Assess the energy consumption of software systems using measurement and testing techniques. LO3. Create actionable strategies to address sustainability issues within software organisations. LO4. Evaluate and compare different solutions for sustainable software systems based on their effectiveness in achieving sustainability goals and their potential impact on various stakeholders.	
Education Method	To meet these objectives, you will be involved in a broad set of learning activities: lectures, paper reading, software analysis, software development, essay writing and presentation. These heterogenous set of activities aim at building a strong set of hard skills for energy-efficient code development combined with a strong set of soft-skills and critical thinking. You will work on ideas that will help real-world software projects embrace a green software culture.	
Assessment	The final grade of the course consists of the following components: Software project (weighting 45%) Essay (weighting 45%) Presentation (weighting 10%) Final grade calculation: 0.45 * Software project+ 0.45 * Essay + 0.1 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Software project: re-submit deliverable Essay: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4430	Network Security	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best practices in computer and network security. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course's primary focus will be on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design, review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics, as well as conducting measurements on the effectiveness of attack and defense schemes.	
Study Goals	See course contents.	
Education Method	Lectures, Labs, and Project.	
Assessment	The final grade of the course consists of the following components: Assignments (Pass/Fail) Final project (Report + Presentation) (weighting 100%) Final grade calculation: P/F * Assignments + 1* Final project A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable, resit presentation Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l0 norm is shown to be NP-hard, and several classical approximate algorithms like l1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IN4152	3D Computer Graphics and Animation	5
Responsible Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. R. Marroquim	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Students that haven't followed any previous Computer Graphics courses (like CSE2215) will be able to participate, but might have to invest some more time to catch up in the first lectures.	
Course Contents	Have you ever wondered how Toy Story was made, why the game Last of Us 2 looks so beautiful, or have you ever wanted to create your own graphics application or game? Then you should consider following this course. If not, then you should still follow it... maybe, you will become interested! In this course, you will get a good idea of Computer Graphics in general. The topic is of very high relevance for the industry and the research community and has numerous applications in different domains, such as scientific visualization, video games, simulators, special effects, animated movies and many more. Here, you will learn about basic algorithms, as well as modern techniques. We will address several topics: the principles of image synthesis, object representations, geometric and hierarchical transformations, graphics cards and the graphics pipeline, realistic rendering (including global illumination and effects, such as reflections), expressive rendering, physics simulations, rendering control (including previsualization systems used by professionals in the movie industry), and perceptual rendering, which relies on properties of the human visual system to enhance the quality of the images. Besides course sessions on the theory of Computer Graphics, some of the algorithms will also be reproduced in practice, and deepened during the final project.	
Study Goals	The course teaches computer graphics techniques on an advanced level. After the course the student is able to classify the different modeling, shading, and display techniques. The student can reproduce the basic mathematical and algorithmic notions associated with these concepts, can comment on the weak and strong points of these techniques, and can apply the core concepts within a graphics program in practice.	
Education Method	lectures, instructions, research papers, lab work	
Literature and Study Materials	Research Papers in domain of selected topics, lecture sheets, online sources, optional books (see below)	
Books	Fundamentals of Computer Graphics by Shirley et al. - CRC Press Real-time Rendering by Tomas Akenine-Möller, Eric Haines, Naty Hoffman - Peters, Wellesley Real-Time Shadows by Elmar Eisemann, Michael Schwarz, Ulf Assarsson, Michael Wimmer - Taylor & Francis Computer Graphics. Principles and Practice by James D. Foley, Andries VanDam, Steven K. Feiner - Addison Wesley Advanced Material: Physically Based Rendering, From Theory to Implementation by Matt Pharr, Wenzel Jakob and Greg Humphreys - MIT Press	
Assessment	The final grade of the course consists of the following components (details of both elements will be presented during the lecture): Project (weighting 60%). The project grade is the result of a project and its presentation that is building upon the assignments that are handed out (roughly) weekly during the duration of this course. Paper (weighting 40%). The paper grade is the result of the presentation of a scientific paper and the development of an associated practical implementation. Final grade calculation: 0.6 * Project + 0.4 * Paper A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Paper: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4253ET	"Hacking Lab"-Applied Security Analysis	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Necessary background differs per student project, see first lecture or contact instructors for details	
Course Contents	The security of computer and telecommunication systems is becoming an increasing concern. In this course, we will review the current state of the art on security research and gain practical experience in assessing the security and vulnerabilities of communication systems. Engineers are typically taught to focus on performance, correctness, scalability, and maintainability when building communication and information processing systems. However, an additional set of design principles are required to achieve security. In this course, we discuss security principles, common pitfalls and vulnerabilities. The weekly lectures provide an introduction into security research, with a focus on real-world security, privacy-enhancing technology and common security pitfalls. Each student participates in a "Hack Project", with a group of one to four students. Students can select between a wide range of available Hack Project outlines within the first week. The goal may be to evaluate the security of a real-world IT system, developing a proof-of-concept exposing a vulnerability or focussed on preserving privacy in a post-Snowden world. Students may propose their own Hack Project based on their background knowledge and skills. Such Hack Projects need to be approved and shaped together with the instructor. Example of possible outlined hardware-oriented projects are: development of a wifi tracker, programing an FPGA system to break passwords, assess the security of RFID cards, or to transparently intercept Ethernet traffic. Concrete software projects are: hacking Bitcoin, improving the TOR anonymity protocol and create Android-based tools for human rights activists in Iran, Egypt and Russia, reprogramming neural networks attacks. Each Hack Project is documented with a written report. This can be in the form of a 6-8 page IEEE-style scientific article or a traditional more lengthy report. All results, experiences and findings are presented to the entire class in the last week of the course. Hack Projects also report their progress several times during the course, after the weekly lectures.	
Study Goals	After this course, the student will have a thorough knowledge of security in real-world systems, and will be able to explore the literature on this topic independently. The student will be aware of the poor state of security in real-world computer systems. The student can explain the common pitfalls, why these known failures still occur and reasons behind the poor state of security in general.	
Education Method	Lectures, student presentations, written final report and active participation. Attendance and active participation during lectures is mandatory. This sadly means telelecturing is not possible.	
Literature and Study Materials	Customize literature lists and study materials are provided per project topic	
Assessment	The final grade of the course consists of the following components: Written Hack Project report (weighting 60%) Final presentation of result (weighting 10%) Presentation of ongoing project progress (weighting 20%) Participation in discussions, overall quality of the practical work and class attendance (weighting 10%) Attendance to lectures is mandatory. Final grade calculation: 0.6 * Written Hack Project report + 0.1 * Final presentation of result + 0.2 * Presentation of ongoing project progress + 0.1 * Participation in discussions, overall quality of the practical work and class attendance A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Written Hack Project report: re-submit deliverable Final presentation of result: resit Presentation of ongoing project progress: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
maximum aantal deelnemers	If there is an unexpected high demand for this course, then enrollment will be based on past performance in relevant courses.	
Co-Instructor	Dr. S. Picek	

IN4315	Software Architecture	5
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software engineering	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four, in which the actual architectures of existing open source systems are analyzed in considerable detail. These systems are taken from github, and student teams are challenged to actually contribute to the systems under analysis in the course.	
Study Goals	Bring students into the position that they can (1) explain the key architectural concepts and methods for modeling software architectures; (2) apply viewpoints and perspectives to model software architectures; (3) discuss the benefits of architecting and the role of the software architect; (4) evaluate and validate software architectures; (5) explain and discuss the concepts of component-based and plugin architectures, service-oriented architectures, and software product lines; (6) explain and recognize technical debt and have an understanding of possible refactorings.	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books; Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; and Coplien and Bjørnvig, "Lean Architecture", Wiley, 2010. Additional reading material will be announced in the lectures.	
Assessment	Grades will be based on lab assignment including essay writing, coding, (video) presentation, peer reviewing, participation. The final grade of the course consists of the following components: Essays (weighting 60%) Coding (weighting 20%) Presentation (weighting 20%) Final grade calculation: 0.6 * Essays + 0.2 * Coding + 0.2 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Essays: re-submit deliverable Coding: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Course web site: https://se.ewi.tudelft.nl/delftswa/index.html	
Co-Instructor	M. Finavaro Aniche	

IN4325	Information Retrieval	5
Responsible Instructor	J. Yang	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language. Knowledge of Web information systems and software engineering can be helpful.	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. The World Wide Web is now the primary source of information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, analysed, and exploited so that each user can quickly find information that is both relevant and comprehensive for their needs. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation of Web and Enterprise search engines. Covered topics include: = Information Retrieval Models; = Indexing Techniques; = Web Search; = Information Seeking Paradigms; = Evaluation of information retrieval systems;	
Study Goals	At the completion of this course, students will be able to: = Describe the different information retrieval models, and compare their strenghts and weaknesses. [Learning Objective 1] = Describe and implement different indexing techniques. [Learning Objective 2] = Describe and analyze querying techniques with respect to their most suited application domains. [Learning Objective 3] = Analyse the effectiveness of an information retrieval system through proper use of evaluation metrics. [Learning Objective 4] = Design and implement (Web) Information Retrieval systems, possibly using advanced social and semantic search functionalities. Support and defend the relevance and correctness the choices with regards to the adopted information retrieval model, indexing technique, and querying technique. [Learning Objective 5]	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 80 hours for the group project and 15 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books - all resources are available on Brightspace.	
Books	Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. Introduction to Information Retrieval. Cambridge University Press, New York, NY, USA. Stefan Büttcher, Charles Clarke, and Gordon V. Cormack. 2010. Information Retrieval: Implementing and Evaluating Search Engines. The MIT Press.	
Assessment	The final grade of the course consists of the following components: Individual assignment (weighting 10%) Weekly individual assignment (weighting 20%) Group project (weighting 70%) Final grade calculation: 0.1 * Individual assignment + 0.2 * Weekly individual assignment + 0.7 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual assignment: re-submit deliverable Weekly individual assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialistievakken start kwartaal 4 2023

CESE4120	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics". We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming.	
Course Contents	The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. After taking this course students will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non--trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	Written reports + project presentation + oral exam Overall, the final grade is determined by: 1) Two intermediate reports (10% of grade each, 2 pages each) 2) Final report (10 % of grade, 4 pages) 3) Final project demonstration (70% of grade) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. There is no resit for this course. repair option:A new but related project will be given for the Report (W), Oral(0) and Demonstration (IP). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

CS4035	Cyber Data Analytics	5
Responsible Instructor	Ir. S.E. Verwer	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course provides theoretical and practical background for applying data analytics in the field of cyber security. Cyber data analytics is a huge field with a great diversity of techniques and applications. The course is centered on a selection of seven such techniques: learning from imbalanced data; behavioral profiling and anomaly detection; sequential data mining; data stream mining; learning from software data; adversarial machine learning; and privacy-aware data mining; Anomaly detection is one of the main topics in cyber security. Specific difficulties that the student will learn to handle are the huge amounts of data and the large number of false positives. Behavioral profiling applies to both people and software processes. Different techniques will be taught to construct profiles from software logs. While building such profiles, care should be taken to not infringe upon the privacy of individuals the data is collected from. Finally, attackers will modify their behavior in order to avoid being detected, a cyber data analytics engineer tries to make their models/profiles robust against such modifications.	
Study Goals	The student will be able to: Apply machine learning to real data Understand and modify machine learning algorithms Learn models from time series Detect anomalies in multidimensional time-series Use distributed processing to speed up machine learning Learn models from data streams with limited memory Learn sequential models Use machine learning for fingerprinting and profiling Preserve the privacy of data owners while learning models Learn robust models that can detect evasive attackers Use machine learning to detect fraud, attacks, and botnets	
Education Method	There will be two lectures for each of the seven topics, and 3 large lab assignments on fraud detection, anomaly detection, and behavioral profiling, and 1 smaller lab on adversarial robustness. There is no exam. Teams of two students will work on these assignments which contain both individual and collaborative components. Deadlines are strict as peer-review will be used to both learn of other possible solutions, provide feedback, and get initial estimates on the obtained grade.	
Assessment	The final grade of the course consists of the following components: Lab assignment fraud detection (weighting 30%) Lab assignment anomaly detection (weighting 30%) Lab assignment behavioral profiling (weighting 30%) Lab assignment adversarial robustness (weighting 10%) Final grade calculation: 0.3 * Lab assignment + 0.3 * Lab assignment + 0.3 * Lab assignment + 0.1 * Lab assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignment fraud detection: Re-submit deliverable Lab assignment anomaly detection: Re-submit deliverable Lab assignment behavioral profiling: Re-submit deliverable Lab assignment adversarial robustness: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4125	Seminar Research Methodology for Data Science	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Instructor	Dr. K.A. Hildebrandt	
Instructor	J. Urbano Merino	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	basic knowledge in mathematics (linear algebra, calculus, probability and statistics)	
Course Contents	The course focuses on research methods for data science. It looks at underlying principles and concepts for data collection, analysis and data processing, as well as the use of tools to do this. The main topics of study are: Conceptualizing research questions and experimental design Frequentist and Bayesian data analysis Generalized linear models for statistical analysis Multilevel modelling for hierarchical and longitudinal data analysis Measuring and sampling, validity and reliability Linear and nonlinear dimensional reduction Principles of statistical testing In the course, students will be using software tools such as R, and Matlab	
Study Goals	The main aims of this module for the student is to achieve an understanding of research methods for data science and obtain practical experience with data analysis and data processing methods. This module provides students with the opportunity to develop and demonstrate their understanding, knowledge, and competence. The learning outcomes for the module are that students will be able to: 1. Appreciate and comprehend strategies for collecting and processing data to answer research questions based on data 2. Understand and reproduce key principles underlying statistical data and data processing analysis 3. Learn to identify and avoid typical biases, paradoxes and misunderstandings in data-driven research 4. Apply and select appropriate data modelling techniques to analyse data and data processing	
Education Method	Lectures/Assignments Some lectures have a flipped classroom set-up, where it is expected that students prepare themselves by watching a series of videos. In the lecture, students work together on a set of classroom challenges under the guidance of the lecturer. In some others, students work on preliminary solutions to some exercises that will be discussed during the lecture. Expected Workload Lectures: 26 hours (13 × 2 hours lectures) Reading time: 39 hours Preparation basis tool use: 25 hours (5 × 5 hours for each tool) Coursework project, including writing reports and preparing for presentation: 50 hours (10 × 5 hours) Total = 140 hours	
Literature and Study Materials	Will be provided online	
Assessment	The final grade of the course consists of the following components: Analysis of experimental research data (weighting 40%) Exploration of real-world data set (weighting 30%) Linear and nonlinear dimensional reduction (weighting 30%) Students work in small groups on the 3 assignments. For the assignment, the student group submits reports and gives a presentation, including a question and answer round where individual group members are assessed on the coursework. Final grade calculation: 0.4 * Analysis of experimental research data + 0.3 * Exploration of real-world data set + 0.3 * Linear and nonlinear dimensional reduction A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Analysis of experimental research data: re-submit deliverable Exploration of real-world data set: re-submit deliverable Linear and nonlinear dimensional reduction: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	NA	

CS4145	Crowd Computing	5
Responsible Instructor	U.K. Gadiraju	
Instructor	A. Bozzon	
Instructor	J. Yang	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic knowledge of artificial intelligence and/or human computer interaction is advised. Proficiency in at least one programming language.	
Course Contents	Crowd Computing is an emerging field that sits at the intersection of computer science and data science. Crowd computing studies how large groups of people can solve complex tasks that are currently beyond the capabilities of artificial intelligence algorithms, and that cannot be solved by a single person alone. It involves the algorithmic engagement and coordination of people by means of Web-enabled platforms. These complex tasks are mainly focused on the creation, enrichment, and interpretation of data, making crowd computing a building block of data science. Examples of such tasks include the coordinated creation of data about real world events when electronic sensors are not available; the annotation of existing data sets to create ground truth data for the training of machine learning algorithms; and the analysis and interpretation of Web data to spot identify inappropriate content (e.g.,hate speech, or fake news). Crowd computing is an essential tool for any data-driven company: from Facebook to Microsoft, from Google to IBM, from Spotify to Pandora, all major companies employ crowd computing to fulfil their data needs, both by involving employees, and by reaching out to anonymous crowds through online marketplaces like Amazon Mechanical Turk or Appen. Crowd computing methods therefore play an important role in the design, development and evaluation of a variety of products, services, and systems in a variety of domains. The objective of the Crowd Computing course is to introduce the scientific and technical underpinnings of crowd computing, and to investigate how it can be used for computer science applications (e.g., information retrieval, machine learning, next-generation interfaces, and data mining) and for real world applications (e.g., cultural heritage preservation, online knowledge creation, smart cities, etc.) The course is designed around one key challenge, the creation and consumption of (high quality) data, and will be organized around three themes: 1) Establishing data needs; 2) Fulfilling data needs with crowd computing; and 3) Evaluating the quality of the retrieved data with respect to the original data need. Covered topics include: 1) Establishing Data Needs: - Requirement Elicitation - Requirement Analysis - User Modelling Properties 2) Fulfilling Data Needs with Crowd Computation: - Systems for/with collective intelligence (e.g., recommendation, semiautonomous systems, citizen science, crowdsourcing, and human computation systems) - Multi-modal Interaction (e.g., conversational systems) - Human Computation (e.g., worker modelling, task modelling, incentives, task assignment, recruitment) - Games with a purpose - Algorithms for Crowd Computing - Computational Methods for User Modelling - Interfaces for Crowd Computing Systems 3) Evaluating Retrieved Data: - Expert Evaluation - User Evaluation - Explanation of the output of Crowd Computing Systems 4) Study of Application Domains When applicable, the course will also feature invited lectures from selected academics and professionals in the field. Since instructors of this course are also directing the Design@Scale Delft AI lab, students of this course will have the opportunity to engage with cutting-edge research projects relevant to this lab. This Crowd Computing course is an elective for students following the Data Science and Technology Track and the Software Technology Track. It adds to the master education offer by addressing topics that are complementary to courses like IN4325 Information Retrieval, IN4252 Web Science & Engineering, CS4065 Multimedia Search and Recommendation, and IN4010 Artificial Intelligence Techniques.	
Study Goals	After this course, students will be able to: - Identify the requirements for a Crowd Computing system [LO1] - Design and develop Crowd Computing systems. Support and defend the relevance and correctness of his/her choices [LO2] - Describe and compare several Crowd Computing techniques. [LO3] - Describe and compare design decisions in the context of Crowd Computing interaction paradigms [LO4] - Determine which Crowd Computing technique(s) is most appropriate for being used in a certain problem domain [LO5] - Apply the appropriate Crowd Computing technique to an application domain and evaluate the obtained results. [LO6] - Analyse the performance of a Crowd Computing system by applying the proper evaluation measures. [LO7]	
Education Method	This course consists of 16 2-hour lectures. Each week, a 30-minute assignment tests the knowledge acquired on the discussed topics. Starting from Week 1, students form groups and work on a project, to be presented in week 9. Students are expected to work 6 hours per week (each) on the project assignment. Expected workload is 32 hours for attending lectures, 24 hours of reading study material and preparing lectures, 55 hours for weekly assignments and group assignment, 24 hours for preparing final survey, and 5 hours for exam and plenary presentations (total 140 hours).	
Literature and Study Materials	Books: - Human Computation. Author(s): Edith Law and Luis von Ahn. Synthesis Lectures on Artificial Intelligence and Machine	

	Learning, June 2011, Vol. 5, No. 3. http://www.morganclaypool.com/doi/abs/10.2200/S00371ED1V01Y201107AIM013 - A. Marcus and A. Parameswaran. Crowdsourced Data Management: Industry and Academic Perspectives. Foundations and TrendsR in Databases, vol. 6, no. 1-2, pp. 1161, 2013. DOI: 10.1561/1900000044. https://people.eecs.berkeley.edu/~adityagp/papers/crowd-book.pdf - An Introduction to Hybrid Human-Machine Information Systems. Demartini, G., Difallah, D.E., Gadiraju, U. and Catasta, M., 2017. Foundations and Trends in Web Science, 7(1), pp.1-87. https://edu.nl/np4th Slides: available on Brightspace Articles: available on Brightspace Recommended reading: - Interaction Design: Beyond Human-Computer Interaction (4th Ed, 2015). Authors: Jenny Preece, Helen Sharp, Yvonne Rogers
Assessment	The final grade of the course consists of the following components: Weekly Individual assignment (weighting 15%) Group Assignment (weighting 55%). The group assignment is performed collectively, but graded individually. Final Individual Assignment (weighting 30%) Final grade calculation: 0.15 * Weekly Individual assignment + 0.55 * Group Assignment + 0.3 * Final Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly Individual assignment: re-submit deliverable Group Assignment: re-submit deliverable Final Individual Assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Tags	Algorithmics Artificial intelligence Design Programming Research Methods Software Technology

CS4205	Evolutionary Algorithms	5
Responsible Instructor	P.A.N. Bosman	
Instructor	Dr.ir. P.A. Bouter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Must have the ability to program, predominantly in Python (but also Java and C/C++ are very useful) so that you can create your own (extensions of) evolutionary algorithms. Further, a basic understanding of key aspects in linear algebra (e.g., matrix multiplications, inversions, decompositions), probability theory and statistics (e.g., normal distribution, statistical hypothesis testing), and complexity theory (e.g., big-O notation), will be very helpful.	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, and real-world applications. The course is planned to have 10 lectures, multiple small practical assignemnts and 1 larger practical assignment. The first practical assignments pertains to experimenting with already implemented EAs on predefined problems. The second practical assignment offers more freedom, allowing you, in a group, to build your own EA (this may vary depending on student numbers and other circumstances).	
Study Goals	Upon successful completion of this course, students will be able to: 1) Explain and apply key concepts underlying the main streams in Evolutionary Algorithm (EA) research, in particular genetic algorithms, evolution strategies, genetic programming, evolutionary multi-objective optimization, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2) Explain and apply key ingredients underlying the rationale of when these algorithms work and when they do not work and several ways to overcome this. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3) Name and explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4) Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5) Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	10 Lectures Several smaller, individual, lab projects One larger lab project in groups of several students	
Literature and Study Materials	Papers and slides that will be made available.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 60%) Lab practical work (weighting 40%) Final grade calculation: 0.6 * written exam + 0.4 * lab practical work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab practical work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Tags	Algorithmics Artificial intelligence Optimalisation	

CS4210-B	Intelligent Decision Making Project	5
Responsible Instructor	Dr. M.T.J. Spaan	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Dr. V. Robu	
Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. J.W. Böhmer	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Instructor	Dr. A. Lukina	
Contact Hours / Week x/x/x/x	0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Theoretical knowledge regarding algorithms for decision making in Artificial Intelligence, obtained for instance by passing one of the following courses: - CS4210-A Algorithms for Intelligent Decision Making - CS4400 Deep Reinforcement Learning - CS4375 Artificial Intelligence Techniques - IN4344 Advanced Algorithms.	
Course Contents	Intelligent decision making is a key skill of computational agents. Research on this topic focuses on building models and algorithms that enable AI systems to take appropriate decisions. Building upon theoretical knowledge gained in other courses, students collaborate in small groups on a distinct research project per group, for instance on decision-making problems in transport, logistics or smart energy grids. Purely algorithmic challenges will also be provided. The research projects provide a good opportunity to learn about topics suitable for Masters projects.	
Study Goals	After completing the Intelligent Decision Making Project course, the student is able to: 1. Apply algorithms for decision making to problem domains, and can compare and evaluate them. 2. Design and implement an extension of a decision-making algorithm. 3. Identify and discuss relevant topics in the research field of algorithms for intelligent decision making. 4. Describe and apply the appropriate research methodology. 5. Communicate his/her findings effectively.	
Education Method	A research project in a small group.	
Literature and Study Materials	Mainly survey papers and book chapters. Details are provided via Brightspace.	
Assessment	The final grade of the course consists of the following components: Quality of work of the research project (weighting 40%) A scientific report of the research project (including peer review of a report) (weighting 20%) Performance during the project (weighting 30%) Oral presentation of the research project (weighting 10%) Final grade calculation: 0.4 * Quality of work of the research project + 0.2 * A scientific report of the research project + 0.3 * Performance during the project + 0.1 * Oral presentation of the research project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Quality of work of the research project: re-submit deliverable A scientific report of the research project: re-submit deliverable Oral presentation of the research project: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Only a limited number of students can participate in this course. In order to be admitted, please submit a short motivation letter (max 200 words) via Brightspace. Attending the first lecture is compulsory.	
Tags	Artificial intelligence	
maximum aantal deelnemers	40	

CS4245	Seminar Computer Vision by Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	N. Tömen	
Instructor	Dr. X. Zhang	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	MSc thesis in the Computer Vision lab	
Expected prior knowledge	Deep Learning (CS4240)	
Course Contents	The recent boom in computer vision and automatic image understanding represents an inflection point in human productivity, permeating wide aspects of the economy and society. Examples of visual tasks which are repetitive or require expert knowledge include medical diagnosis, industrial inspection, autonomous vehicles, etc. When machines can meaningfully assist or even completely take-over such tasks it will change the world as we know it. The breakthrough in the 2012 ImageNet automatic image recognition competition shows all previously existing methods decisively defeated by a deep neural network. Deep learning replaces feature engineering methods and is able to successfully learn image features from huge annotated datasets. This course is on automatically understanding visual content such as images and videos by deep learning. The range of topics (although, not limited to): Fundamentals in Vision, object detection, per-pixel labelings, video recognition, image similarity learning, efficiency, self-supervision, 3D computer vision, adverserial attacks, explainability, generative models, etc. The course will have lectures, a seminar and a lab practical: - The lectures will be on established topics based on the current literature. - The seminar will have students read, critique, and present relevant computer vision research papers. You will have to read 2 papers per week, for 7 weeks (14 papers). - The lab will have students apply and design their own (small) computer vision project. The course build on top of the Deep Learning course (CS4240) and follows a similar setup.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the deep learning techniques reviewed in the course for computer vision applications such as image classification, object detection, per-pixel labelings, video recognition, image similarity learning. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [L04]. Quickly identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (we focus on Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) Computer Vision problem. [LO8]. Write clearly and concisely about your code, method, results, and analyis.	
Education Method	Lectures Lab project: design and execute your own Computer Vision project. Seminar: paper reading, critiquing, and presenting.	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Exam (weighting 60%). Exam about all material. Project (weighting 30%). in a small group of students you work on a computer vision project. Final grade calculation: 0.1 * Presentation + 0.3 * project + 0.6 * Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4280	Language-Based Software Security	5
Responsible Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	This course has no formal prerequisites. However, for the homework assignments you will have to implement several program analysis techniques using the Scala programming language. If you have not used Scala before, you are thus expected to learn the basics of the language through self-study.	
Course Contents	Security vulnerabilities often arise due to programming errors in the source code of an application. Recent programming errors with severe security implications include Heartbleed (buffer over-read), Shellshock (code injection), and goto-fail (ill-formated code). Rather than hunt for individual vulnerabilities in programs, a more structural approach to improve security is to improve the programming language. This is the goal of language-based security: to rule out whole classes of potential security vulnerabilities in one go. This course studies various security properties and program analysis techniques for enforcing these properties at the level of the programming language to improve software security. In particular, we will study the following properties: - Memory safety: prevent buffer overflows and overreads - Type safety: prevent undefined behaviour - Information flow control: prevent data leaks and code injection attacks We will study techniques to address these problems at the language level through dynamic analysis, static analysis, and language design. To facilitate a precise study and comparison, we will define the above techniques formally in class. To facilitate student experimentation and exploration of trade-offs, students will implement the above techniques in homework assignments.	
Study Goals	After taking this course, students should be able to: 1. Describe the nature and causes of security vulnerabilities in software systems, and give concrete examples of how these security vulnerabilities can be exploited. 2. Explain the properties that can be enforced at the level of the programming language to rule out security vulnerabilities, such as memory safety, type safety, and non-interference. 3. Formally define the semantics of a simple programming language. 4. Formally define dynamic and static analysis techniques for enforcing these security properties. 5. Implement these techniques for a small programming language. 6. Discuss and evaluate the importance of soundness and precision of a given program analysis. 7. Contrast programming languages based on the set of countermeasures they provide, and give an appropriate recommendation for a specific application. 8. Analyse and apply results from scientific literature in the area of language based security.	
Education Method	The course work consists of the following activities: 1 or 2 instruction sessions per week. Weekly homework assignments consisting of theoretical questions, programming assignments, and reading assignments	
Assessment	The final grade of the course consists of the following components: Weekly homework assignments (weighting 40%). The weekly homework assignments will test your ability to design and implement (variants of) the techniques discussed in the lectures (study goals 3-5). Written or oral exam (weighting 60%). The final written or oral exam will test your theoretical understanding of the security vulnerabilities and their countermeasures discussed in class (study goals 1-2) and your ability to discuss and contrast the different aspects of these techniques (study goals 6-8). Final grade calculation: 0.4 * Weekly homework assignments + 0.6 * Written or oral exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written or oral exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4290	Seminar on Distributed Machine Learning Systems	5
Responsible Instructor	Dr. Y. Chen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Machine learning systems are often conventionally designed for centralized processing in that they first collect data from distributed sources and then execute algorithms on a single server. Due to the limited scalability of processing large amount of data and the long latency delay, there is a strong demand for a paradigm shift to distributed or decentralized ML systems which execute ML algorithms on multiple and in some cases even geographically dispersed nodes. The aim of this seminar course is to let students learn how to design and build distributed ML systems via paper reading, presentation, discussion, and project prototyping. We provide a broad overview on the design of the state-of-the-art distributed ML systems, with a strong focus on the scalability, resource efficiency, data requirements, and robustness of the solutions. We will present an array of methodologies and techniques that can efficiently scale ML analysis to a large number of distributed nodes against all operation conditions, e.g., system failures and malicious attacks. The specific course topics are listed below. The course materials will be based on a mixture of classic and recently published papers. For each topic, the basic concepts and technology landscape will be first provided and then two state-of-the art of papers will be presented and discussed by students. We offer a testbed of a distributed (deep) ML system in which students are encouraged to apply different techniques to jointly improve its scalability and resilience. Course topics include Overview of distributed machine learning systems Performance and scalability of state-of-the-art systems Acceleration of ML workloads Slim distributed ML systems on small data Robust deep learning systems Federated machine learning systems	
Study Goals	Students are able to argue and reason about distributed ML from a systems perspective. Students understand the behavior and tradeoffs of distributed ML in terms of performance and scalability. Students can estimate the importance of data inputs via different techniques, i.e., core set and decomposition methods, for distributed ML systems. Students understand data poison attacks and design defense strategy for distributed ML systems. Students can analyze the state-of-the art federated machine learning systems and design the failure-resilient communication protocols. Students are able to design and implement methods and techniques for making distributed ML systems more efficient.	
Education Method	Lectures: 7 weeks X 2h Papers: one paper presentation, two paper reviews, and paper discussion. Practical: apply system and algorithmic optimization techniques learned in the lecture to improve the performance of distributed machine learning systems, e.g., image recognition on CIFAR 10. The testbed environment, learning algorithms, and dataset will be given. Deliverables include git commit of functioning code and a report summarizing the contribution	
Assessment	The final grade of the course consists of the following components: Paper presentation by group (10%): each group of 2 to 3 students needs to choose from a given set a papers to present (15 minutes) and lead the discussion (10 minutes). Paper reviews by individual (30%): each student needs to write three reviews of papers assigned from a given set and those two papers have to be different from the paper for the presented. Each review will account for 10% of the grade. Questionnaires by individual (0%): each student needs to hand in a list of questions at the beginning of the lectures that have paper presentation from other students. This is not graded. Individual project (60%): The project is collaborative among the entire class and competitive as a whole group. The objective is to continuously improve the performance of the given distributed ML system. The students need to hand in a final project report in style of a short scientific paper, stating their individual contribution to the overall system performance. Final grade calculation: 0.1 * Paper presentation group + 0.3 * Paper presentation individual + 0.6 * Individual project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper presentation by group: resit Paper reviews by individual: resit Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4295	Release Engineering for Machine Learning Applications	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Course Contents

The world of Software Engineering has been revolutionized in the last decade. Instead of releasing software updates yearly, companies can now release multiple times per week, sometimes even per day, to their customers. This allows quick reactions to market demands, software failures, and it is crucial for increasing the business value of software. These improvements have been mostly enabled by advances in release engineering and, in this course, we will learn about the techniques and technologies that build the foundation for modern release engineering.

This course is addressing two main target audiences: 1) software engineers that would like to learn how to systematically release their applications and make them scalable in an orchestrated environment and 2) machine learning experts that would like to learn how to make their proof-of-concept applications production ready.

Together, we will go on a journey that starts at continuous integration and then moves on to continuous delivery, continuous deployment, and continuous experimentation. We will discuss the theory and the current research on various related subjects like containerization, testing, or monitoring and will put the learned theory into practice. As a running example, we will build a pipeline for a machine learning application, which -compared to traditional release engineering- poses additional challenges, like data versioning or model deployment.

Important: Make sure to register for a group in Brightspace AT THE LATEST during the first lecture.

Study Goals

After following this course, you will be able to...

Explain the concepts of DevOps and MLOps

Automate the development and operation process
... Automate packaging, versioning, and releasing in modern build environments like GitLab or GitHub
... Release artifacts in public registries like the GitHub Maven Registry.
... Provision servers with Vagrant and Ansible
... Automate the model training pipeline using DVC

Use modern containerization technology to build a scalable micro-service architecture
... Use Docker to isolate a complex, web-based application in a container
... Use Kubernetes to orchestrate a non-trivial, distributed application
... Create an Istio service mesh in Kubernetes that enables custom routing strategies
... Use Prometheus and Grafana to monitor the state of a Kubernetes-based application

Deploy and maintain machine learning models in production
... Embed a typical ML model in an HTTP micro-service
... Apply version-control techniques to data and models in a machine learning project
... Implement quality-control techniques in a machine learning application
... Design a testing strategy that covers data, model, infrastructure, and monitoring aspects

Document the state and extensions of release engineering pipeline
... Find related work and inspiration for release engineering tasks
... Identify extension opportunities in existing release pipelines

Education Method

Following this course will require you to...

Follow interactive lectures and pre-recorded material for self-study
Execute a programming project in small teams
Transfer lecture contents into a project through practical assignments
Assess the work of yourself and others through rubric-supported peer-feedback
Actively participate in Q&A sessions
Document a release engineering pipeline in a report
Present your project results and answer questions about the lecture contents.

Assessment

--- Formative Assessment ---

You can check your individual learning progress through weekly rubrics
... You will receive feedback from other groups as a neutral outsider
... You can check your own progress via self-assessment
Joint Q&A sessions will discuss common points of the peer feedback and concrete questions

--- Individual Knock-Out Elements ---

The course enforces several criteria to protect the active members of a team. Students not reaching the following criteria will be removed from the course:

Upholding a professional attitude when interacting with peers and staff.
Actively participating in the team process and contributing to all team deadlines and deliverables
Meaningful and visible project contribution throughout each of the seven project weeks (W2-W4, W6-W9)
... 1+ commit to one of the team repositories
... Create 1+ PR and approve 1+ PR that is not your own
... The changes must contain a meaningful contribution to code, config, or report
Constructive participation in all peer reviews for assignments (one for another team and one for your own)
Ability to explain lecture contents and defend all aspects of the team project in the oral examination
Availability of team results for grading (repositories, report, artifacts)

We offer repair options for the following criteria:

Not having meaningful and visible contributions in one week can be repaired in the following week. Failing to satisfy the criteria in more than one week will lead to removal.
Failing to submit a peer review in time can be repaired in the following week. Missing a review a second time will lead to removal
The inability to explain or justify the project in the oral examination can be repaired in a second oral examination.

	--- Summative Assessment --- Students attending the course will be graded on the following grading components. ALL graded components are rounded to .1 and must at least be "almost passed" (>5.0). Their weighted average will be rounded to .5 to calculate the final team grade. Project ... Versioning & Releases (10%) ... Containers & Orchestration (10%) ... Kubernetes & Monitoring (10%) ... Provisioning (10%) ... Istio Service Mesh (10%) ... ML Configuration Management (10%) ... ML Testing (10%) Report (30%) Presentation & Oral Examination (Pass/Fail) In case of large performance differences in a group, we may choose to assess the team members individually, based on the individual contributions to the deliverables and the interaction traces that we find in the repository. --- Please Note --- All assignment deadlines throughout the course are fixed, we do not accept late submissions. Late submissions in systems that do not enforce due dates will be treated as not submitted (e.g., Brightspace, Buddycheck). Failing to submit a deliverable cannot be repaired. Insufficient deliverables (>=4.0 and <6.0) can be repaired, but the grade of the deliverable is then capped at 6.0. There is NO resit opportunity for this course. Partial grades are not carried over to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Enrolment / Application	In addition to your regular enrollment of the course, you must also register in Brightspace, AT THE LATEST during the first lecture, so you can be assigned to a group. Late registrations after the group creation cannot be considered anymore.
Co-Instructor	Prof.dr. A.E. Zaidman

CS4350	Machine Learning for Graph Data	5
Responsible Instructor	M. Khosla	
Responsible Instructor	E. Isufi	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics Machine Learning: data division in train-validation-test, overfitting, kernels, regularization; Basics Deep Learning: back-propagation, stochastic gradient descent (SGD), multi-layer perceptron. Basics in programming: Python or equivalents, minimal experience with Pytorch / TensorFlow or equivalents is helpful but not strongly expected.	
Course Contents	Graph data are present in a myriad of modern computer sciences systems and applications. Examples include data generated over social, brain, financial, power, water and sensor networks. Because these data have a complicated structure they require different tools conventionally used in machine and deep learning to develop end to end solutions. These solutions falls generally under the umbrella of graph-based machine learning and can be used to perform recommendations, detect anomalies in the brain, predict financial crisis, estimate the sate of a power or water network, and coordinate group of autonomous moving sensor, to name a few. This course deals with the foundations and principles of machine and deep learning for network data. Topics include: unsupervised and semi-supervised learning on graphs; graph representation learning; graph signal processing; graph convolutions; graph neural networks; spatiotemporal learning on graphs; scalable algorithms; explainability.	
Study Goals	By the end of this course, you should be able to: Lo1: Recognise key challenges in learning from graph-structured data. You are able to recognise challenges in real applications involving graph-based data, explain them with a machine learning language, and discuss strategies to solve them. You are able to identify the issues of lack of interpretability in graph-based machine learning models and describe some possible solutions. You are also able to explain a few techniques to improve scalability of existing techniques for large graphs. Lo2: Explain and compare representation learning techniques on graphs. You are able to explain, and compare different graph-based machine learning techniques, identifying advantages and limitations in each of them, and discuss the latter with respect to a specific application. Lo3: Implement and analyse machine and deep learning techniques for graph data. You are able to apply different graph-based machine learning techniques including graph kernels, graph neural networks, graph-time learning solutions to real data. You are also able to analyse the different trade-off of these solutions, compare them with each other and inspects their behaviour. LO4: Design machine learning solutions for real world graph data tasks. You are able to create a novel solution for a practical graph-based challenge, discusses different graph-machine learning strategies for solving the task, test one or more of them, and contrast the performance of the developed approach with alternative baselines.	
Education Method	The course consists of: - 12 lectures in which the main concepts are explained and discussed; - 3 hands-on homework labs, in which practical assignments need to be solved individually; - At least two Q&A and feedback sessions about the final project;	
Literature and Study Materials	The instructors will provide online accessible material for the specific lectures. Also the slides used during the lectures are made available via Brightspace. Practical final project examples will be made available for download via Brightspace. During the course several pointer to on-line video lectures will be given.	
Assessment	The final grade of the course consists of the following components: Lab assignments (Pass/Fail). The labs assignments must be solved individually and consists of only pass/fail grade. Students can solve their assignment at home or during the lab sessions. During the labs instructors and TAs (if available) will provide feedback on the questions regarding the assignment. Students need to pass all the three lab assignments before the final project submission to pass the course. Final project (weighting 100%). The final project will be solved in groups and a group grade 1-10 will be given. The project grade will be evaluated on both the written final document as well as the question-answering session during the final interview. Final grade calculation: P/F * Lab assignments + 1 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignments: re-submit deliverable Final project: resit Q&A Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4360	Natural Language Processing	5

Responsible Instructor	J. Yang
Instructor	Dr. C. Lofi
Instructor	Dr. A. Anand
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	Proficiency in at least one programming language. Knowledge of basic algebra. Knowledge of Web Information Systems and Machine Learning can be helpful.
Course Contents	Natural Language Processing (NLP) concerns the interaction between computers and human language, in particular how to build systems to process and analyze large amounts of human language data. It has become an important technology in many parts of computer science, facilitating key applications of large-scale text understanding, automatic question answering, knowledge base construction, information retrieval, text or speech-driven dialogue agents, translation, text summarization, etc. This course aims to introduce the principles and techniques for Natural Language Processing. Through the lectures, assignments, and the group project, students will gain an understanding of classic and modern techniques for language understanding and generation and learn the skills to design, develop, and evaluate their own Natural Language Processing systems and applications. Covered topics include: = Lexical, Syntactic and semantic analysis = Language models = Word, and subword embedding techniques = Attention mechanism and Transformers = Transfer Learning = Knowledge-driven language processing = NLP applications (e.g., QA, summarization, machine translation) = Responsible NLP (explainability, fairness) = Evaluation of NLP systems
Study Goals	At the completion of this course, students will be able to: [LO1] Recognize typical NLP tasks and components of an NLP system [LO2] Describe lexical analysis techniques such as language modeling and smoothing [LO3] Describe syntactic and semantic analysis techniques [LO4] Describe and compare different token embedding techniques, common neural architectures (including attention mechanisms) for text processing and their implementation in neural language models (e.g., GPT-3 and BERT). [LO5] Describe information extraction techniques and its application to NLP tasks [LO6] Apply NLP techniques to information retrieval, question answering, text summarization [LO7] Explain the implications of data-driven models in terms of transparency, accountability, and fairness [LO8] Design, develop, and evaluate NLP systems, possibly using neural methods with semantic enrichment and user adaptation functionalities. Support and defend the relevance and correctness of the design choices.
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 65 hours for the group project and 30 hours for the literature survey.
Literature and Study Materials	Scientific papers, course slides, course books, authentic blog posts - all resources are available on Brightspace.
Books	Dan Jurafsky and James H. Martin. 2014. Speech and language processing. Pearson. Yoav Goldberg. 2015. A Primer on Neural Network Models for Natural Language Processing. Steven Bird, Ewan Klein, and Edward Loper. 2009. Natural language processing with Python: analyzing text with the natural language toolkit. " O'Reilly Media, Inc.".
Assessment	The final grade of the course consists of the following components: Weekly individual assignment (weighting 20%) Group project (weighting 80%). The group assignment is performed collectively, but graded individually. Final grade calculation: 0.2 * Weekly individual assignment + 0.8 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly individual assignment: re-submit deliverable Group project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor anaylsis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: 0.5 * Oral discussion 1 + 0.5 * Oral discussion 2. Both partial grades must be sufficient, i.e., .a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case the take-home assignments are insufficient, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4030	Error Correcting Codes	4
Responsible Instructor	Dr.ir. J.H. Weber	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic linear algebra and probability theory, as being treated in B.Sc. programmes like Electrical Engineering, Computer Science, and Mathematics	
Course Contents	Introduction into error-correcting codes; mathematical basics; block codes fundamentals; cyclic codes; co-operating codes; soft-decision decoding; convolutional codes; iterative decoding (turbo codes, LDPC codes); applications.	
Study Goals	The general overall objectives of this course are that the students can - reproduce the error control concepts as presented in the lecture notes; - explain the meaning and usefulness of these concepts; - apply the concepts to transmission and storage systems, including basic calculations, at the level of the exercises in the lecture notes and previous exams. More specifically, the students are able to - discuss the basic trade-offs between efficiency, reliability, and complexity in designing coding schemes for transmission and storage systems; - apply finite fields theory in order to construct (cyclic) codes; - encode data into code sequences that are resistant to a certain number of random and/or burst errors; - decode received sequences using (iterative) hard or soft decision techniques.	
Education Method	Lectures; expected workload is 20 hours attending lectures, 60 hours preparing for the lectures, studying the lecture notes, and making suggested exercises, and 32 hours for preparing and making the exam.	
Literature and Study Materials	Lecture notes "Error-Correcting Codes" by J.H. Weber, available on Brightspace. Also previous exams as well as demos are available via Brightspace.	
Assessment	The final grade will be fully determined by a scheduled written exam, which will be held at the end of Q4. The resit will take place as an oral exam in Q5, on appointment with the lecturer. The exam format is closed-book in any case. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	Actual course information available on Brightspace.	

IN4255	Geometric Data Processing	5
Responsible Instructor	Dr. K.A. Hildebrandt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic knowledge in mathematics (linear algebra, calculus): CSE1205, CSE1200 or comparable courses. Students who haven't followed any of these courses can follow the course, but should be willing to invest more time.	
Course Contents	Geometric data processing is concerned with the representation, analysis, manipulation, and optimization of digital shapes. Thanks to the advances in 3D acquisition and manufacturing technologies (like 3D-Scanning and 3D-printing), the usage of geometric data is continuously increasing and an efficient processing of digital shapes plays an important role for a variety of applications in areas such as computer graphics, computer-aided design and engineering, medical imaging and surgery planning, architecture, and entertainment. In this course, we will study concepts and algorithms for creating, analyzing, editing and optimizing digital geometric shapes.	
Study Goals	After successfully completing this course, the student is able to: - describe the fundamental techniques used for representing, analyzing, processing and modeling digital 3D-shapes treated in the course and to explain the mathematical and algorithmic concepts associated with them - apply the learned mathematical concepts to solve basic geometric problems arising in geometric modeling applications - design algorithms that can solve simple geometric processing tasks and evaluate the drawbacks, benefits and limitations of the proposed algorithms - implement the designed algorithms in a geometric processing software framework	
Education Method	The course combines lectures, tutorials, practical project work, and homework assignments.	
Literature and Study Materials	References to textbooks and recent research and survey papers are given in the lectures.	
Assessment	The final grade of the course consists of the following components: Practical projects (weighting 60%) Theoretical assignment (weighting 40%) Final grade calculation: 0.6 * Practical projects + 0.4 * Theoretical assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Practical projects: re-submit deliverable Theoretical assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4331	Web-scale Data Management	5
Responsible Instructor	Dr. A. Katsifodimos	
Instructor	Dr. C. Lofi	
Instructor	Dr. R. Hai	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor level courses in database management systems and operating systems. A prior course in distributed systems or middleware would be helpful but is not required. Programming skills are important for the final assignment.	
Course Contents	This course addresses the challenges of Data Management at Web-scale. Especially, it covers the need for large-scale distributed data storage systems. The lecture therefore introduces step-by-step increasingly complex distributed storage systems, leading up to modern implementations of different NoSQL data storage systems. The challenges arising from such systems are presented and discussed, especially focusing on the CAP theorem and the resulting trade-offs with respect to data models, transactional power, query expressivity, and replication consistency. These discussions lead to different variants of NoSQL database systems, like Key-Value Stores, Document Stores, Wide-Columnar stores, and Graph Databases. The advantages, disadvantages, and general properties of these systems are discusses in more detail. There is special focus on distributed transactions and consistency guarantees of different data management systems and methods.	
Study Goals	At the end of this course the student can - assess the nature of a given storage problem, and can select a suitable technology for solving it - understands the different data models encountered in Web Data Management, and their impact on modelling and querying - understands the issues arising from distributing and replicating data, especially with respect to the CAP theorem -understands the trade-offs which can be chosen within the design space of the CAP theorem -categorize and explain modern NoSQL databases within the framework of the previously mentioned trade-offs	
Education Method	Lectures and assignments	
Literature and Study Materials	Course slides and Lecture Videos	
Books	Literature mentioned in the lecture, mostly research papers.	
Assessment	There is a group assignment (80%) and an individual exam (20%). In order to pass the class, a student has to have a grade of at least 5.0 for the group assignment as well as the individual exam, with a weighted average of 5.8 overall. The exam can be resit in the following academic quarter. If the student achieves a 4.0 - 6.0 in the group assignment, a repair opportunity can also be given (e.g., extra work, corrections on the group assignment, etc.).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Track Data Science & Technology 2023

Introduction 1

In the Data Science and Technology track of the Computer Science MSc programme, you will learn how to engineer and develop systems capable of processing and interpreting massive data sets to extract important information. Fundamental and practical issues of the analysis of data will be addressed, including e.g. security of data and software, visualization of information, decision making from data, and high performance computing algorithms.

The richness and importance of the information conveyed by data has led to a rapid increase in the influence of data on individuals and society. Data of various kinds, such as the enormous data collections on the Internet, has become omnipresent in virtually all aspects of our society. Digital data has become the key to innovations in both social and scientific domains ranging from energy, economy, health and climate to bioinformatics and web science.

THE CURRICULUM
1. An Individual Exam Programma (IEP) in this track consists of
a. a common core,
b. courses offered by the faculty EEMCS,
c. a seminar offered by the programme CS or a Literature survey (IN4306)
d. free electives,
e. a thesis project (IN5000 Final project) worth 45 credits and
f. if required, homologation.

The IEP must be drawn up in agreement with the thesis coordinator of the research group in which the student wishes to carry out his or her thesis project. The thesis coordinator is a member of the scientific staff of that research group.

The seminar of the research group in which the thesis is performed or the Literature Study (IN4306) is part of said IEP.

Free elective courses
- the number of credits spent on free electives in said IEP is no higher than 25 credits,
- the number of credits spent on homologation in said IEP is no higher than 15 credits,
- at least 40 credits of the courses in the IEP (notwithstanding the thesis project) should be computer Science courses. A list of these courses is published annually in the digital study guide.

Free electives - language course list:
Up to 3 credits may be spent on language courses. These may only be chosen if required. Placement tests showing the necessity to take one or more of these courses must be taken and submitted to the master coordinator.

TPM018A English Grammar for the University 2 EC
TPM303A Intermediate Writing in English for the University 2 EC
TPM304A Advanced Writing in English for the University 2 EC
TPM305A Writing a Masters Thesis in English 2 EC
WM1115TU Dutch Elementary 1 3 EC
WM1116TU Dutch Elementary 2 3 EC
WM1117TU Dutch intermediate 1 3 EC
WM1135TU Advanced English for the University 3 EC

The free elective space may also be used for an extra project:
TUD4040 Joint Interdisciplinary Project (JIP) 15 EC
IFEEMCS520200 Interdisciplinary Advanced Artificial Intelligence Project 15 EC

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Common Core DST 2023	
Introduction 1	Common Core MSc CS - DST (at least 20 EC): Choose 4 out of 7. Students follow the masters programme of the year in which they have started their programme. E.g. students who started their masters programme in 2021-2022 follow the Teaching and Examination Regulations (TER) of 2021-2022..

CS4035	Cyber Data Analytics	5
Responsible Instructor	Ir. S.E. Verwer	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course provides theoretical and practical background for applying data analytics in the field of cyber security. Cyber data analytics is a huge field with a great diversity of techniques and applications. The course is centered on a selection of seven such techniques: learning from imbalanced data; behavioral profiling and anomaly detection; sequential data mining; data stream mining; learning from software data; adversarial machine learning; and privacy-aware data mining; Anomaly detection is one of the main topics in cyber security. Specific difficulties that the student will learn to handle are the huge amounts of data and the large number of false positives. Behavioral profiling applies to both people and software processes. Different techniques will be taught to construct profiles from software logs. While building such profiles, care should be taken to not infringe upon the privacy of individuals the data is collected from. Finally, attackers will modify their behavior in order to avoid being detected, a cyber data analytics engineer tries to make their models/profiles robust against such modifications.	
Study Goals	The student will be able to: Apply machine learning to real data Understand and modify machine learning algorithms Learn models from time series Detect anomalies in multidimensional time-series Use distributed processing to speed up machine learning Learn models from data streams with limited memory Learn sequential models Use machine learning for fingerprinting and profiling Preserve the privacy of data owners while learning models Learn robust models that can detect evasive attackers Use machine learning to detect fraud, attacks, and botnets	
Education Method	There will be two lectures for each of the seven topics, and 3 large lab assignments on fraud detection, anomaly detection, and behavioral profiling, and 1 smaller lab on adversarial robustness. There is no exam. Teams of two students will work on these assignments which contain both individual and collaborative components. Deadlines are strict as peer-review will be used to both learn of other possible solutions, provide feedback, and get initial estimates on the obtained grade.	
Assessment	The final grade of the course consists of the following components: Lab assignment fraud detection (weighting 30%) Lab assignment anomaly detection (weighting 30%) Lab assignment behavioral profiling (weighting 30%) Lab assignment adversarial robustness (weighting 10%) Final grade calculation: 0.3 * Lab assignment + 0.3 * Lab assignment + 0.3 * Lab assignment + 0.1 * Lab assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignment fraud detection: Re-submit deliverable Lab assignment anomaly detection: Re-submit deliverable Lab assignment behavioral profiling: Re-submit deliverable Lab assignment adversarial robustness: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4375	Artificial Intelligence Techniques	5
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. M.T.J. Spaan	
Instructor	L. Cavalcante Siebert	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	We expect students to have programming expertise at the Bachelor level of Computer Science, in particular knowledge is expected of algorithms (e.g. search algorithms), and probability theory. Programming skills are required for the practical assignments which can be done in Python.	
Course Contents	The course focuses on concepts and techniques from probabilistic Artificial Intelligence for building intelligent agents, and decision making and -support systems. The specific topics include Bayesian networks, inference, belief monitoring, (partially observable) Markov decision processes, reinforcement learning and multiagent systems.	
Study Goals	After completing the course, student will be able 1. to comprehensively summarize the covered topics within probabilistic artificial intelligence and discuss their main issues; 2. to explain the main concepts and techniques; 3. to relate practical problems to appropriate AI techniques and apply the formal models covered in the course to these problems; 4. to implement and evaluate algorithms for complex decision making.	
Education Method	Lectures, lab work (programming assignments).	
Literature and Study Materials	Stuart J. Russel and Peter Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Prentice-Hall. ISBN 9781292401133 Global edition. See website http://aima.cs.berkeley.edu/global-index.html for additional information that goes with the book.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	The practical assignments will be done in groups of 4 to 5 students. The ethical standards of working are expected of all students. Work can be divided over students, but all students are responsible for the overall quality and originality of the work. Students that do not do their share of the work have to be reported by the group to the supervisor so that action can be taken.	
Tags	Artificial intelligence	

IN4089	Data Visualization	5
Responsible Instructor	T. Höllt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/2/0/0 & lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Master course MKE/ST/DS	
Expected prior knowledge	IN2905-A/IN4152/CSE2215 Computer Graphics (recommended, not required). The practicals will be implemented in HTML/Javascript/D3 (InfoVis) and C++ (VolVis). We consider programming skills as a requirement but not necessarily in the mentioned languages and no advanced skills are needed. Relevant topics will be introduced and experience in other programming languages should make adaption feasible.	
Course Contents	Data visualization is the visual representation of data by computer generated images. The data sets can be results of numerical simulations or measurements (scientific visualization), or other data collections such as databases (information visualization). The goal is to improve insight, understanding and/or communication of data. Data visualizations use a combination of methods from a very diverse variety of disciplines: perception, computer graphics, human computer interaction, algorithmics, image processing, machine learning, numerical analysis, optimization, The course has two main parts; information and scientific visualization that will involve knowledge of diverse disciplines. As a computer science course, affinity to algorithmic thinking and programing skills will be needed. Topics covered: models of the visualization process; colour models and use of colour; information visualization; representation and processing of data; volume visualization; interactive visual data analysis; visualization of vector fields and flows. Guest lectures might be given on selected topics.	
Study Goals	The goal of the course is to get knowledge on the fundamentals that are part of data visualization. The main principles and techniques that are the basis of generating effective visual representations of data. Techniques and cases of data visualization are discussed. There are several applications for the techniques, such as medical, engineering, finances, economics, game analytics, and more. By the end of the course, you should be able to LO1: Discuss a large range of visualization techniques. LO2: Discuss a perception principle of visualization. LO3: Explain mathematical principles and algorithms of visualization techniques. LO4: Design suitable visualization systems for a given practical data analysis problem. LO5: Implement visualization systems for a given practical data analysis problem.	
Education Method	Lectures, practical assignments, self-study, and projects.	
Literature and Study Materials	Course slides, instructions for projects, and selected literature via Brightspace. Chapters from: Visualization Analysis and Design Author: Tamara Munzner CRC Press Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Books available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 30%). You are allowed to have the slides and material of the course during the exam. No computer or laptop is allowed. Information visualization project (weighting 35%) Volume visualization project (weighting 35%) The projects will be developed in groups of 3 and are evaluated based on the reasoning/justification of the techniques used based on the material given at the course, effectiveness of the results, technical contribution or implementation, quality of the documentation and presentation Final grade calculation: 0.3 * Written exam + 0.35 * Information visualization project + 0.35 * Volume visualization project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Information visualization project: re-submit deliverable Volume visualization project: re-submit deliverable The project resits are not automatic and must be requested within 2 weeks after publishing the grades. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

IN4252	Web Science & Engineering	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	J. Yang	
Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The expected entry level is (equivalent to) standard bachelor-level computer science.	
Course Contents	The main subject of the course is the Web, and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g. Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan / research paper draft in collaboration with others in a small group to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies, and verbally present and defend this plan / draft.	
Education Method	The education methods include: - Lectures, before which and after which students study material by themselves, to get an understanding of the relevant material; - Small assignments and hands-on exercises, to apply the understanding of relevant material; - One large assignment, with a number of feedback moments, to learn how to write a web science paper / research project plan. Lectures will not be in each week in the class period (1+2): in between lectures there is time reserved for studying before and after lectures, for small assignments and exercises, and for writing the large assignment paper. The writing of the large assignment paper / project plan happens throughout the class period (1+2) to enable frequent feedback.	
Literature and Study Materials	Will be provided in class, depending on the topics chosen for the assignments and final paper.	
Assessment	Assessments are based on two components: the "small assignments" (accompanying the lectures) which provide a total of 20% of the grade (these are a mix of literature and programming homework), and the "final paper" which provides 80% of the final grade. To pass, each of the components ("small assignments" and "final paper") need to be graded with a minimum of 5.0. The final grade must be at least a 5.8. If these criteria are not met, the "small assignments" and the "final paper" can be repaired provided that the to-be-repaired component was at least graded with a 4 in accordance with OER/TER article 17. This repair is not a resit, and is limited to a maximum of 6.0 as a result. Components that have been graded with less than a 4.0 cannot be repaired. If the course is still not passed after the repair or cannot be repaired, all components need to be redone if the course is retaken in a later year. Both the small assignments and the final paper are group works of groups of preferably 2 members.	
Special Information	Students are asked to register/enrol on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	The expected workload is 5ects and that is principally distributed uniformly over the two quarters.	

IN4315	Software Architecture	5
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software engineering	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four, in which the actual architectures of existing open source systems are analyzed in considerable detail. These systems are taken from github, and student teams are challenged to actually contribute to the systems under analysis in the course.	
Study Goals	Bring students into the position that they can (1) explain the key architectural concepts and methods for modeling software architectures; (2) apply viewpoints and perspectives to model software architectures; (3) discuss the benefits of architecting and the role of the software architect; (4) evaluate and validate software architectures; (5) explain and discuss the concepts of component-based and plugin architectures, service-oriented architectures, and software product lines; (6) explain and recognize technical debt and have an understanding of possible refactorings.	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books; Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; and Coplien and Bjørnvig, "Lean Architecture", Wiley, 2010. Additional reading material will be announced in the lectures.	
Assessment	Grades will be based on lab assignment including essay writing, coding, (video) presentation, peer reviewing, participation. The final grade of the course consists of the following components: Essays (weighting 60%) Coding (weighting 20%) Presentation (weighting 20%) Final grade calculation: 0.6 * Essays + 0.2 * Coding + 0.2 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Essays: re-submit deliverable Coding: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Course web site: https://se.ewi.tudelft.nl/delftswa/index.html	
Co-Instructor	M. Finavaro Aniche	

IN4344	Advanced Algorithms	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic courses in Algorithmics and Complexity Theory	
Course Contents	The course is on solving (abstract models of) complex real-world problems, with a focus on solving intractable problems exactly. The course consists of two main parts: * modeling and solving using linear programming * exact algorithms using search trees, dynamic programming, and/or decision diagrams	
Study Goals	By the end of this course, students will have knowledge of and experience with the following advanced algorithmic techniques: (Part 1:) - linear programming (LP) and LP modelling - duality and simplex algorithm - integer LP and integer modelling - LP relaxation and branch and bound (Part 2:) - complete and bounded search trees - tree decomposition - dynamic programming - preprocessing - decision diagrams Furthermore, they will be able to * recognize situations where they can use these techniques and adapt them to different applications. construct new algorithms that are similar to these techniques. construct proofs that are similar to a selected set of proofs from the material. analyze the run time of algorithms. implement an algorithm that is given in pseudo-code. experimentally evaluate the quality and the runtime of an algorithm on a set of instances.	
Education Method	Interactive lectures (usually 2x2 per week), optional homework exercises, programming assignments (2 hours lab support per week) The expected workload is 30% studying the written material and the recorded lectures, and participating in the interactive sessions 30% making the homework exercises and preparation for the exams 40% working on the programming assignments	
Literature and Study Materials	Part 1 of the course will be mainly based on chapters 1-8, some of 9-11 (most of 9-11 is assumed to be known), and 12-14 of the syllabus "Optimization" by Karen Aardal, Leo van Iersel and Remie Janssen, which can be ordered via https://www.webedu.nl/bestellen/tudelft/ Part 2 will use chapter 10 of the following textbook: J. Kleinberg and E. Tardos, Algorithm Design, Pearson Education, 2006. ISBN 0-321-37291-3 As well as some supplemental study material, which will be provided via BrightSpace.	
Assessment	The final mark depends on the marks obtained for (a) written reports and evaluation of programming assignments, PA (weight 40%) (b) a written exam, EX (2 parts, weight 60%) Each programming assignment is graded on a scale from 0 to 10. The final mark for the programming assignment (PA) is the average of the mark obtained for the assignments. Programming exercises can be completed by 2 students working together. If the average grade for the four programming assignments (PA) is between 4.0 and 6.0 (excluding 6.0), a repair is possible by making one extra assignment to replace the lowest grade of the four assignments. This leads to a repaired PA score, maximized at 6. (E.g. if grades were 3.5, 4, 4, 4.5, the 3.5 can be replaced. If the grade for the replacement assignment is a 7.5, the repaired PA score is a 5. If grades were 5, 5.5, 5.5, 6, the 5 can be replaced. If the replacement is a 9, the repaired PA score is not 6.25 but 6.) The exam consists of two parts. Each part will be examined after the lectures about that part have been delivered and will be graded on a scale from 1 to 10. The final mark for the exam (EX) is the average of the marks for the parts. Each exam part contains one or more challenging assignments; during each part of the course two representative homework assignments will be made available to prepare for this. There is a resit for the exam where any of the two parts can be redone. The result for a part after the resits is determined by the maximum score obtained for the original exam and the resit. The final mark for the course is determined as follows: - if the PA and the EX mark are above 5, the final mark is the weighted average of these three marks: 60% EX, 40% PA - if at least one of PA, EX is less than or equal to 5, the final mark is the minimum of the results obtained for PA or EX. Partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	For content questions / of general interest, please use StackOverflow. For personal questions, please use the course email address: aa-cs-ewi@tudelft.nl	
Tags	Algorithmics Artificial intelligence Mathematics	

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CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ------------------------- Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. -- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) o Definition of ad-hoc networks o Comparison with infrastructure based systems o Typical applications o Advantages and challenges o Radio technologies for ad-hoc networks o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks o Propagation models o Topology models based on graph theory o Degree and hopcount o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) o Introduction to MAC protocols o Issues and design goals o Classification o Directional, muli-channel MAC protocols o Energy efficiency in MAC protocols o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) o Flooding o Node discovery, neighbour discovery o Route establishment o Topology maintenance, localisation o Proactive, reactive and hybrid routing o Typical protocols o Energy efficiency in routing o Broadcast and multicast o Effects of mobility on connectivity and capacity o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks o Wireless sensor networks (Chapter 12 of the textbook and papers) o Cooperation (Reference papers) o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consisting of slides presented at the lectures (Slides are only teaching aid and they are not substitute for	

	textbooks, research papers, etc). 3. Some recent journal papers 4. Optional Reference Books 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms) 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar , C Puttamadappa , and T. G Basavaraju, Auerbach Publications, 2008. This book is avaliable online in the library. 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is avaliable online in the library. 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer. 5. OPNET/ns-2 web pages, tutorials and video lectures
Books	Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.
Assessment	1. There will be totally three written/oral tests/examinations for this course (including the final exam). 2. The students will carry out a project in a group and submit a short report. 3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis. Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class. Breakup: 2 Tests + Final Exam = 55% Project 40% Self assessment + Reflection 3% Activities on Feedback Fruits or any Online platform 2% Resit: written/oral exam [+ carry over from projects]. == Changes due to COVID19 in 2021 is below; in case there is a similar situation, we use the following methods. == There are three chapters Chapter 1: Network modelling Chapter 2: MAC protocol Chapter 3: Routing + Lab: Simple experiments using laptops (individual) + simulations (group-wise) Marks Breakup -------------- Homework/Assignments/Group works Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks Part B1: Individual experiment + report Part B2: Group-wise simulations + report + demo Part B1 + Part B2 - 50 Marks Part C: Self-assessment + peer activities -- 5 marks Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit. == (More information will be given in the first class) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Permitted Materials during Tests	Different conditions for different test/exams. Conditions will be informed 1 week before the exams/test.

CS4070	Multivariate Data Analysis	5
Responsible Instructor	Dr. J. Söhl	
Responsible Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	

Expected prior knowledge

Required:

* Introduction Probability Theory and Statistics: see for instance

A Modern Introduction to Probability and Statistics
Understanding Why and How
Series: Springer Texts in Statistics
Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E.
2005, XVI, 488 p. 120 illus., Hardcover
ISBN: 1-85233-896-2

* Basic calculus

* Linear Algebra: matrix multiplication, the inverse of a matrix, the transpose of a matrix, least square solution.
see:

David C. Lay: Linear Algebra and Its Applications
ISBN-10: 0321385179 ISBN-13: 9780321385178
©2012 Pearson)

Course Contents

PART I:
Stochastic models will be developed on the basis of probability theory. Probability theory describes the behavior of certain phenomena in terms of how likely it is that certain values will occur. Central features of the models will be discussed are random variables, probability density functions, and the expected value operator. In describing random processes and signals, the correlation function and conditional probabilities play a central role.

It addresses the following subjects:
1. Random variables. Matlab exercise on estimation of PDF, expected value and variance.
2. Refresher correlation. Calculating with correlation functions.
3. Random processes, correlation function, stationarity, wide sense stationarity, estimation of correlation function (Matlab exercise).
4. Random signal processing, power spectral density function, white noise.
5. AR processes, linear prediction: theory and computer exercise.
6. Markov chains.

PART II:
A course in advanced statistics about linear models, Bayesian inference, classification problems, Gaussian processes and Markov Chain Monte Carlo.

Study Goals

PART I:
1. Probability Theory
- Conditional) probabilities, the law of total probability, and Bayes rule.
- Solve probability problems that require the use of axioms of probability.

2. Definition and Description of Random Variables and Processes
PDF, PMF, CDF, Covariance, Correlation- Determine if a given PDF, PMF, CDF, variance, (auto/cross-)correlation(-function), (auto/cross-)covariance(-function), power spectral density complies with (theoretical and analytical) requirements.
- Convert the description of a probabilistic problem into a probabilistic model using PDF, PMF, or CDF.

3. PDF/PMF and Expected Value
Calculate the various forms of expected value of (combinations of) random variables and random processes
- For a given (amplitude continuous/discrete and time continuous/discrete) probability model calculate the following probabilistic (marginal, joint and conditional) characterizations: PDF, PMF, CDF, probability of an event, expected value, variance, covariance, correlation, correlation coefficient, auto/crosscorrelation function, auto/crosscovariance function, (cross) power spectral density.
- Calculate the PDF, PMF, expected value and variance of a derived random variable.

4. Properties of Random Processes
- Independence, orthogonality, uncorrelated, whiteness, IID- Determine if random variables/processes have the following properties: independent, orthogonal, uncorrelated, white, Poisson, Gaussian, Bernoulli, Markov, IID, stationary, WSS, ergodic.
- Calculate the expected value, variance, auto/crosscorrelation(function), auto/crosscovariance(function), power spectral density of a linear combination of random variables and of a linearly filtered (WSS, amplitude discrete/continuous, time discrete/continuous) random process.

5. Large NumbersCentral limit theorem, law of large numbers
- Solve problems that require the use of the central limit theorem in an engineering context
- Explain the law of the large numbers in an engineering context.

6. Statistical Estimators
- Estimated mean, variance, and correlation function
- Given a set of outcomes, sample functions or realizations, calculate estimators for expected value, variance, and (auto-)correlation function.

7. Application to Engineering Problems and Simulations
- Select and translate a simple electrical engineering or computer science problem into mathematical probability model. The emphasis is on problems in signal and image processing, telecommunication, and media and knowledge technology. The class of probability models encompasses the following random variables/processes: Bernoulli, exponential, binomial, Poisson, Gaussian, uniform.
- Justify and reflect on the approach taken in calculating or simulating (MatLab) the following probabilistic properties: PDF,

	PMF, expected value, variance, autocorrelation function, autocovariance function. PART II: By the end of the course, you should be able to: - compute properties of frequentist and Bayesian estimators in the linear model - derive posterior distributions and their properties - apply hyperparameter optimization in Bayesian models - apply computational methods for posterior approximation - derive properties of frequentist and Bayesian classification methods - derive properties of MCMC methods for sampling from the posterior distributions - derive properties of Gaussian process regression
Education Method	PART I: Lectures, working groups (problem solving), laboratory work (a Matlab exercise) Workload is around 15 hours for attending lectures, 5 hours of reading study material and preparing lectures, 15 hours for the lab course, 20 hours for preparing the exam, 3 hours for the exam, and 8 hours for a final report (66 hours in total). PART II: Classes and weekly exercises.
Books	PART I: R.D. Yates and D.J. Goodman, "Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers", ISBN 0-471-17837-3, John Wiley and Sons, New York, 2005, Second Edition. PART II: Simon Rogers and Mark Girolami "A first coruse in machine learning, 2nd edition" Chapman & Hall It is planned to cover chapters 1--4, 8 and 9 from this book. The precise material will be determined during the course.
Assessment	The final grade of the course consists of the following components: Part I (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Part II (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Final grade calculation: 0.5 * Part I + 0.5 * Part II A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Part I written exam: Resit (written or oral exam, depending on the number of students) Part II written exam: Resit (written or oral exam, depending on the number of students) In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Part I lab assignments: Re-submit deliverable Part II lab assignments: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	PART I: Exam of 3 hours. PART II: Exam of 3 hours.
Permitted Materials during Tests	PART I: Self made notes on a two-sided written A4 sheet. Calculator. PART II: none
Remarks	PART II: This course is particularly interesting for students that are interested in statistical exploratory and quantitative techniques to analyse multivariate data.

CS4200-A	Compiler Construction	5
Responsible Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	- programming (required) - software engineering (recommended) - functional programming (recommended) - formal languages and automata (recommended)	
Course Contents	Compilers translate the source code of programs in a high-level programming language into executable (virtual) machine code. In this course you will study the architecture of compilers, and important concepts and techniques underlying the components of that architecture. You will study these techniques by applying them in practice to implement your own compiler for a small language. The course covers the following topics: - Syntax and parsing - Static semantics and type checking - Virtual machines, assemby code, byte code - Interpretation vs. compilation - Code generation - Memory management, garbage collection - Data-flow analysis - Optimization	
Study Goals	At the end of this course, you will be able to: - Explain the architecture of a compiler and the role of each component of that architecture - Explain the algorithms and techniques for the implementation of compiler components and apply these techniques to examples - Implement a code generator that translates source language abstract syntax trees to target language instructions - Implement transformations on abstract syntax terms to simplify programs - Implement simple type checkers - Implement simple data-flow analyzers - Integrate these components into a working compiler	
Education Method	Attending the course involves: - Attending weekly lectures - Attending weekly lab session (which may start with a group tutorial) - Reading lecture material and papers - Making homework assignments (building a compiler)	
Literature and Study Materials	Required reading: - We will use selected chapters from the book "Essentials of Compilation" by Jeremy Siek, available online: https://github.com/IUCompilerCourse/Essentials-of-Compilation - Lecture slides and selected papers from the literature Optional reading: - We will use Scala to implement our compilers. The following book is optional reading which may be helpful in getting up to speed with programming in Scala: https://www.artima.com/shop/programming_in_scala_5ed (Additional) reading materials will be made available on Brightspace. We will use WebLab (https://weblab.tudelft.nl/cs4200/) for the submission of homework assignments.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Homework assignments (weighting 50%) Final grade calculation: 0.5 * Written exam + 0.5 * Homework assignments	

	A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Permitted Materials during Tests	No materials are permitted during the exam.
Co-Instructor	Dr. S.S. Chakraborty

CS4340	**Seminar Probabilistic Programming**	**5**
Responsible Instructor	S. Dumani	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Probabilistic reasoning is one of the core aspects of artificial intelligence. This course will look into probabilistic programming, the most powerful paradigm of probabilistic reasoning, which specifies probabilistic models as computer programs. Because of the representational power of programming languages, any computable process can be modelled as a probabilistic program. The course covers the topics of probabilistic modelling and inference, but the main focus will be on developing a working probabilistic programming language. The topics include: Probabilistic modelling techniques, Importance sampling, Particle filters, Metropolis-Hastings algorithm, Hamiltonian Monte Carlo and automated differentiation, Variational inference, Discrete and logic probabilistic programming languages, Probabilistic circuits, Compiled/Amortised inference, Deep generative models	
Study Goals	By the end of the course, you will be able to explain the standard probabilistic inference procedure, solve real-world problems using probabilistic programs, identify the strengths and weaknesses in recent probabilistic programming literature, and design a research proposal to address an open problem in the field	
Education Method	This is a seminar course: we will meet twice a week to discuss a research paper. The paper should be read before the lecture and the participants need to submit a short summary (max 0.5 - 1 page). The students are expected to study every discussed paper (but not to the same amount of detail), prepare a presentation and lead a discussion for a few papers, and actively participate in discussions. Aside from paper discussions, the students are expected to do a small practical project (e.g., validating, stress testing or extending the existing method)	
Assessment	The final grade of the course consists of the following components: Research proposal (weighting 65%) Presentation (weighting 25%) Class participation (weighting 10%) Final grade calculation: 0.65 * Research proposal + 0.25 * Presentation + 0.1 * Class participation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Research proposal: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Artificial intelligence	

CS4365	Applied Image Processing	5
Responsible Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Weinmann	
Instructor	P. Kellnhofer	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Basic knowledge of programming concepts, linear algebra, and calculus. Prior experience with Python and C++ is useful but it is not required.	
Course Contents	The course discusses image processing techniques and their application to visual data problems in computer graphics and computer vision including neural image processing techniques. The practical part of the course focuses on the efficient implementation of image processing algorithms on parallel computer hardware. The final project guides students to work with literature sources and state-of-the-art surveys in an area. This course can serve as an introduction to the field of visual computing, which spans the domains of Computer Graphics, Computer Vision, Computational Photography and more. During the course, you will receive an overview of image processing with a deeper focus on applications for visual data. This will include: image representations, color models, linear filters, image sampling, digital photography (including HDR and tone mapping), image retargeting, image abstraction and non-photorealism, multiscale image processing (pyramids, Fourier transform), geometrical image transformations (warping, morphing), stereoscopic imaging, and neural image processing (concept, texture synthesis, style transfer, super-resolution, denoising, editing, depth map estimation). The course will cover classical signal processing and advanced image techniques, as well as approaches utilizing neural networks and machine learning. During lab sessions, you will learn about efficient implementations of the relevant algorithms with a focus on parallel data processing. This will include use of C++ with OpenMP and the Python machine-learning library Pytorch. Introduction to these technologies will be provided before their use. Assignments are solved by using the information presented in the lectures and lab sessions. Formative feedback will be provided during the lab sessions.	
Study Goals	By the end of this course, you should be able to: - Explain the process in which a digital image is formed. - Efficiently conceive, apply, and implement algorithms for processing digital images in C++ or Pytorch. - Analyze algorithms for image processing based on their computational costs, qualitative performance, and application areas.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of image processing, common problems and their solutions. Various algorithms and their properties will be discussed. During the guided lab sessions (4 hours each), you will be introduced to guidelines for efficient implementation of selected algorithms, and you will receive formative feedback on your approach to practical assignments. The final project will focus on survey and assessment of state-of-the-art approaches to a specified image processing problem.	
Literature and Study Materials	Required literature: - Lecture slides with additional references. Recommended literature: - Szeliski, Richard. Computer vision: algorithms and applications. Springer Science & Business Media, 2010. - Chapters 3, 4 and 10. - Burger, Wilhelm, and Mark J. Burge. Digital image processing: an algorithmic introduction using Java. Springer, 2016. - Chapters 22-26. - Sundararajan, Duraisamy. Digital image processing: a signal processing and algorithmic approach. Springer, 2017. - Chapters 2-7. - Goodfellow, I., Bengio, Y. and Courville, A.. Deep Learning. MIT Press, 2016. - Chapters 6-9 and 14.	
Assessment	The final grade of the course consists of the following components: - 3 assignments during the quarter (weighting 55%) - a final project (weighting 45%) Final grade calculation: 0.55 * Assignments + 0.45 * Final project The course is passed if all components have at least a grade of 5.0 and the weighted average is a 6 (rounded on 0.5). Repair possibilities (under conditions described in TER Implementation Regulations Art 5, sub 5): - Assignments: An individual replacement assignment with a maximal grade 6.0 in the following quarter. - Final project: An individual replacement project with a maximal grade 6.0 in the following quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

CS4375	Artificial Intelligence Techniques	5
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. M.T.J. Spaan	
Instructor	L. Cavalcante Siebert	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	We expect students to have programming expertise at the Bachelor level of Computer Science, in particular knowledge is expected of algorithms (e.g. search algorithms), and probability theory. Programming skills are required for the practical assignments which can be done in Python.	
Course Contents	The course focuses on concepts and techniques from probabilistic Artificial Intelligence for building intelligent agents, and decision making and -support systems. The specific topics include Bayesian networks, inference, belief monitoring, (partially observable) Markov decision processes, reinforcement learning and multiagent systems.	
Study Goals	After completing the course, student will be able 1. to comprehensively summarize the covered topics within probabilistic artificial intelligence and discuss their main issues; 2. to explain the main concepts and techniques; 3. to relate practical problems to appropriate AI techniques and apply the formal models covered in the course to these problems; 4. to implement and evaluate algorithms for complex decision making.	
Education Method	Lectures, lab work (programming assignments).	
Literature and Study Materials	Stuart J. Russel and Peter Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Prentice-Hall. ISBN 9781292401133 Global edition. See website http://aima.cs.berkeley.edu/global-index.html for additional information that goes with the book.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	The practical assignments will be done in groups of 4 to 5 students. The ethical standards of working are expected of all students. Work can be divided over students, but all students are responsible for the overall quality and originality of the work. Students that do not do their share of the work have to be reported by the group to the supervisor so that action can be taken.	
Tags	Artificial intelligence	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IN4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Reuired: Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations Strongly advised: some experience with a programming language (e.g., C) is preferred.	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas are considered. The scaling behavior of these algorithms in case of increasing problem size and/or an increasing number of processors is analyzed. Attention is paid to those aspects of computer architectures that are important to understand the resulting performance, such as the memory hierarchy and the interconnection network. By analyzing a number of case studies (applications) with respect to their computing-intensive character, possible bottlenecks will be determined. Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to get high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and the necessary optimization for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found the Brightspace.	
Course Contents Continuation	High Performance Computing, parallel programming, parallel algorithm	
Study Goals	1. Is able to categorize high-performance computer systems 2. Is able to explain and classify parallel and distributed architectures, and programming models; 3. Is able to explain the concepts of data decomposition and parallel algorithms; 4. Is able to apply various parallelization methods to benefit from the potential parallel computer systems; 5. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 6. Capable to evaluate and analyze the performance of parallel programs.	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): implementing (small) parallel programs with C, MPI and Cuda.	
Literature and Study Materials	Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Lab work (weighting 50%) Quizzes, bonus of max 1 point can be obtained (only valid on the first examination occasion after the lectures) Final grade calculation: 0.5 * Written exam + 0.5 * lab work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Via Brightspace	

IN4191	Security and Cryptography	5
Responsible Instructor	Z. Erkin	
Instructor	J.A. Martinez Castaneda	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	UT-201500042 Privacy Enhancing Technologies (Q4)	
Expected prior knowledge	Basic understanding on the following is suggested. -Probability and statistics -Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Learning outcomes: The goal is to make students familiar with the basic concepts applied cryptography, including classical cryptography and modern secret key and public key cryptography. In particular, the students will acquire A sound understanding of the notion of security An understanding of the confidentiality, integrity and authenticity needs of the society Understand the role of cryptographic primitives including the differences between symmetric and asymmetric cryptography, the role of hash functions, digital signatures and PKI Understand the advanced topics in cryptography needed for the modern society with untrustworthy entities Among others things, the following topics are covered: -Classical systems -Information theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures, -*Secret Sharing. (if time permits) -*Zero Knowledge proofs (if time permits)	
Study Goals	It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security, as well as is familiar with present applications. Through assignments, students are expected to have the chance to work on the topics covered in the lectures.	
Education Method	Lectures, assignments and weekly exercises. Attention: This course requires full effort of 140 hours. Even more, if you lack the background (probability and modular arithmetic) Planned Workload: Lectures: 28 x 45minutes sessions, total 22 hours Practice session: 7 x 90 min. total 12 hours Assignments: 3 x 20 hour, total 60 hours Weekly study: 7 x 4 hours, total 28 hours Exam preparation: 20 hours Exam: 3 hours	
Literature and Study Materials	-Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition -Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) Handouts of lectures	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%) Mandatory assignments (weighting 30%) Final grade calculation: 0.7 * Written exam + 0.3 * Mandatory assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mandatory assignments: re-submit deliverable	

	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	In case of in person examination at campus: The exam is closed book. A cheat sheet of size A4, hand written is allowed for the written exam. Name and student number has to be present on each side. In case of remote exam from home: open book: textbook, slides and self-made notes only. randomised/customised exam.
Permitted Materials during Tests	Only non-scientific calculators.

IN4252	Web Science & Engineering	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	J. Yang	
Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The expected entry level is (equivalent to) standard bachelor-level computer science.	
Course Contents	The main subject of the course is the Web, and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g. Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan / research paper draft in collaboration with others in a small group to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies, and verbally present and defend this plan / draft.	
Education Method	The education methods include: - Lectures, before which and after which students study material by themselves, to get an understanding of the relevant material; - Small assignments and hands-on exercises, to apply the understanding of relevant material; - One large assignment, with a number of feedback moments, to learn how to write a web science paper / research project plan. Lectures will not be in each week in the class period (1+2): in between lectures there is time reserved for studying before and after lectures, for small assignments and exercises, and for writing the large assignment paper. The writing of the large assignment paper / project plan happens throughout the class period (1+2) to enable frequent feedback.	
Literature and Study Materials	Will be provided in class, depending on the topics chosen for the assignments and final paper.	
Assessment	Assessments are based on two components: the "small assignments" (accompanying the lectures) which provide a total of 20% of the grade (these are a mix of literature and programming homework), and the "final paper" which provides 80% of the final grade. To pass, each of the components ("small assignments" and "final paper") need to be graded with a minimum of 5.0. The final grade must be at least a 5.8. If these criteria are not met, the "small assignments" and the "final paper" can be repaired provided that the to-be-repaired component was at least graded with a 4 in accordance with OER/TER article 17. This repair is not a resit, and is limited to a maximum of 6.0 as a result. Components that have been graded with less than a 4.0 cannot be repaired. If the course is still not passed after the repair or cannot be repaired, all components need to be redone if the course is retaken in a later year. Both the small assignments and the final paper are group works of groups of preferably 2 members.	
Special Information	Students are asked to register/enrol on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	The expected workload is 5ects and that is principally distributed uniformly over the two quarters.	

IN4307	Medical Visualization	5
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge of linear algebra, calculus and programming is needed. The project will be implemented in MeVisLab and Python. Learning MeVisLab will be part of the practice assignments.	
Course Contents	Theory and practice (Notice project extends to Q2) of medical visualization. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measuremen; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for medical data; advanced applications.	
Study Goals	By the end of the course, you should be able to LO1: Explain medical visualization algorithms and their applicability to medical problems. LO2: Discuss the advantages and disadvantage of medical visualization algorithms. LO3: Build a medical visualization system for a given problem: a. Discuss a suitable visualization for a given medical problem. b. Implement the most suitable solution. c. Judge the performance of the implemented solution.	
Education Method	The course will be based on a combination of lectures and practical assignments. A final project will be developed in Q2	
Literature and Study Materials	Course slides, instructions for project and assignments, and selected literature via Brightspace. Chapters from the following book can be used as alternate view on the topics but are not mandatory: Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Book available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam Q1 (weighting 40%) Final project Q2 (weighting 60%) The final project will be the design and implementation of a visualization system for a given medical problem. The final project will be carried out in teams of three. The deliverables for the final project will be a report (paper), the results (e.g., code) and a short video presenting the project (i.e. screencast). Final grade calculation: 0.4 * Written exam + 0.6 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Q2 In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final project: re-submit deliverable (the project resit is not automatic and must be requested within 2 weeks after publishing the grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Notes and written material. No computers.	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	
Co-Instructor	Prof.dr. E. Eisemann	

IN4344	Advanced Algorithms	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic courses in Algorithmics and Complexity Theory	
Course Contents	The course is on solving (abstract models of) complex real-world problems, with a focus on solving intractable problems exactly. The course consists of two main parts: * modeling and solving using linear programming * exact algorithms using search trees, dynamic programming, and/or decision diagrams	
Study Goals	By the end of this course, students will have knowledge of and experience with the following advanced algorithmic techniques: (Part 1:) - linear programming (LP) and LP modelling - duality and simplex algorithm - integer LP and integer modelling - LP relaxation and branch and bound (Part 2:) - complete and bounded search trees - tree decomposition - dynamic programming - preprocessing - decision diagrams Furthermore, they will be able to * recognize situations where they can use these techniques and adapt them to different applications. construct new algorithms that are similar to these techniques. construct proofs that are similar to a selected set of proofs from the material. analyze the run time of algorithms. implement an algorithm that is given in pseudo-code. experimentally evaluate the quality and the runtime of an algorithm on a set of instances.	
Education Method	Interactive lectures (usually 2x2 per week), optional homework exercises, programming assignments (2 hours lab support per week) The expected workload is 30% studying the written material and the recorded lectures, and participating in the interactive sessions 30% making the homework exercises and preparation for the exams 40% working on the programming assignments	
Literature and Study Materials	Part 1 of the course will be mainly based on chapters 1-8, some of 9-11 (most of 9-11 is assumed to be known), and 12-14 of the syllabus "Optimization" by Karen Aardal, Leo van Iersel and Remie Janssen, which can be ordered via https://www.webedu.nl/bestellen/tudelft/ Part 2 will use chapter 10 of the following textbook: J. Kleinberg and E. Tardos, Algorithm Design, Pearson Education, 2006. ISBN 0-321-37291-3 As well as some supplemental study material, which will be provided via BrightSpace.	
Assessment	The final mark depends on the marks obtained for (a) written reports and evaluation of programming assignments, PA (weight 40%) (b) a written exam, EX (2 parts, weight 60%) Each programming assignment is graded on a scale from 0 to 10. The final mark for the programming assignment (PA) is the average of the mark obtained for the assignments. Programming exercises can be completed by 2 students working together. If the average grade for the four programming assignments (PA) is between 4.0 and 6.0 (excluding 6.0), a repair is possible by making one extra assignment to replace the lowest grade of the four assignments. This leads to a repaired PA score, maximized at 6. (E.g. if grades were 3.5, 4, 4, 4.5, the 3.5 can be replaced. If the grade for the replacement assignment is a 7.5, the repaired PA score is a 5. If grades were 5, 5.5, 5.5, 6, the 5 can be replaced. If the replacement is a 9, the repaired PA score is not 6.25 but 6.) The exam consists of two parts. Each part will be examined after the lectures about that part have been delivered and will be graded on a scale from 1 to 10. The final mark for the exam (EX) is the average of the marks for the parts. Each exam part contains one or more challenging assignments; during each part of the course two representative homework assignments will be made available to prepare for this. There is a resit for the exam where any of the two parts can be redone. The result for a part after the resits is determined by the maximum score obtained for the original exam and the resit. The final mark for the course is determined as follows: - if the PA and the EX mark are above 5, the final mark is the weighted average of these three marks: 60% EX, 40% PA - if at least one of PA, EX is less than or equal to 5, the final mark is the minimum of the results obtained for PA or EX. Partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	For content questions / of general interest, please use StackOverflow. For personal questions, please use the course email address: aa-cs-ewi@tudelft.nl	
Tags	Algorithmics Artificial intelligence Mathematics	

WM-ITAV-4010	Scientific Writing	2
Module Manager	L. Meester	
Instructor	Drs. W.J. Blokzijl	
Instructor	Drs. B.M.D. van der Laaken	
Instructor	Drs. P.C. Post	
Instructor	Drs. A.E. Kam	
Instructor	M.J.Y. Wackers	
Instructor	S. Baars	
Instructor	M. Bliekendaal	
Instructor	L.C. Schroten	
Instructor	A. Glasbergen-Plas	
Instructor	M. Looij	
Co-responsible for assignments	Drs. B.M.D. van der Laaken	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1 2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you learn to write a scientific article, either a research article based on your own research data or a literature review about a subject of your own choice. This is a necessary skill for anyone who wants to pursue an academic career after their graduation, but it can also be used immediately for all academic texts you will write during your Master program, such as your Master thesis. In seven weeks, we will go through all steps of the writing process, from formulating a good main question and finding relevant literature, to the actual writing, re-writing and final editing. You will exercise with finding, reading and managing relevant academic literature, writing in an academic style, building a comprehensive argumentation, reviewing fellow students' articles and using other students and the instructor's comments to improve your own work.	
Study Goals	The purpose of this course is to learn how to write a scientific text. To achieve this, at the end of this course you will: know what the main characteristics are of a scientific text be able to formulate a main question be able to find, critically read and manage scientific literature be able to use literature properly and avoid plagiarism be able to build up your argumentation be able to structure an article according to the conventions in your field of study be able to use scientific English style be able to use tables and figures to support and communicate your results be able to give feedback on somebody elses article be able to use feedback for improving your work.	
Education Method	Practical, in 6 sessions (attendance mandatory). Every week you have to read some background information about scientific writing and hand in a part of your text. Participants must attend all sessions - one missed session is allowed only - and hand in all assignments in time. Students who receive a pass for this course are rewarded with 2 ects. This equals 56 hours of study. A total of 12 hours is spent on attending (online) classes, in which you can ask questions, discuss the feedback you received on your work and discuss aspects of scientific writing with your fellow students; the remaining 44 hours is for self study, writing and revising. In seven weeks, from preparing lecture 2 up to handing in your final article in week 8, you will have to spend at least 6 hours of self study on this course, every week. It is important that you make sure you have this time in your personal schedule. At the beginning of the course you will mainly be reading up about scientific writing and the subject of your text. As the course proceeds, you will be spending more of your time on writing, giving feedback and revising your own text.	
Books	Theory about academic writing will be made available through Brightspace.	
Assessment	You write a scientific article of 2000-2400 words (excluding the list of references and the abstract). You have to hand in parts of your article every week. Your final grade is based on the final article. An evaluation form for the grading of the article is available throughout the course.	
Elective	Yes	
Category	MSc level	

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CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	- basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	The course intends to bring the student into the position to: - Explain the fundamental concepts and terminology of real-time systems - Construct task schedules using different scheduling policies under a given set of realistic system constraints - Analyze the timing behavior of a system for a given system model and scheduling policy - Discuss advantages and disadvantages of different scheduling policies for a given platform or system - Discuss the effect of hardware and software interferences on the timing behavior of a given system - Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system - Derive (reverse engineer) the system specification from a given implementation (in the lab) - Evaluate the scheduling overheads of a given implementation (in the lab) - Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	Written exam (grade) + lab work; the exam has a resit Repair option for the Individual Assignment = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet has become of critical importance to society. However, the large size of networks and abundance of protocols have made network management very complex. The novel concept of network programmability addresses this complexity and has resulted in a paradigm shift in how networks are (or can be) operated. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network programmability and which treats fundamental networking concepts like Quality of Service and network resilience.	
Study Goals	The learning objectives of this course are twofold: (1) The student should gain knowledge of the treated networking technologies. (2) The student should be able to apply and work with the programmable network technologies in a network emulator (Mininet).	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final assessment (and hence 100% of the grade) will be based on an exam that covers both the theory from the slides as well as the content from the reader. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CS4015	Behaviour Change Support Systems	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	none	
Course Contents	Behavior change support systems (BCSS) are computer-based systems that support individuals to form, alter or reinforce cognitions, attitudes or behaviors without using coercion or deception. They can serve individuals throughout the various stages of a change process, such as awareness development, contemplation, action strategy development, development of new behaviors, and maintaining these new behaviors. Virtual healthcare coaches, negotiation support systems, and applications that provide individuals with personalised financial guidance are three examples of these systems. To establish, modify or maintain change BCSS can deploy computerized persuasive strategies (e.g. reducing effort to establish target behavior, or argumentation reflection strategies, and also algorithmic nudge, including data-, expert-, and theory-driven), simulations (e.g. serious gaming, virtual reality), relational software agents (e.g. ePartners, virtual coaches), and personalization based on longitudinal user data. BCSSs are found in many domains, including education, sales, negotiation, management, and particularly in the health domain. The main topics are: Understanding the behaviour; strategies for behaviour change, action funnel, human decision making Identifying change support strategies; Behaviour change wheel, product vision, target outcome, target user, population diversity Conceptual design of BCSS; Behaviour plan, decision-making environment, story-editing technique Behaviour change and technology acceptance; Motivation, Trans-theoretical model, Theory of reasoned action, theory of planned behaviour, Technology acceptance model, Unified Theory of Acceptance and Use of Technology, technology adaption lifecycle, Fogg behaviour model, Self-Determination theory Persuasive technology; functional triad, levels of persuasion, Nudge Theory, principles of influence, triggers Algorithmic nudge/Persuasion and support Algorithms; concept, key characteristics, and feasibility Learning principles; Behaviouristic learning principles, cognitive learning principles, constructivism based learning principles, social cognitive (learning) theory Communication and framing; Behavioural economic principles, elaboration likelihood model (ELM), health belief model, protection motivation theory, persuasive communication Adherence and Gamification; Implementation Intentions, system 1 and system 2, treatment adherence, gamification Application domains eHealth and Cyber security; Self-monitoring, self-management, virtual health agents, virtual reality therapy, internet cognitive behavioural therapy (iCBT)	
Study Goals	This course provides students the opportunity to develop and demonstrate understanding, knowledge and competence in the field of BCSS. The learning outcomes for the course are that students will be able to: 1. Apply psychological theories, principles, and concepts in the design of BCSS systems. 2. Apply BCSS practices, principles and techniques. 3. Construct an implementation of an algorithmic nudge/boost/persuasion for a specific real-world application domain and objective 4. Design a BCSS system using BCSS principles, concepts, theories, and design methods.	
Education Method	The course uses a flipped-classroom approach, where students watch the pre-recorded lectures in their own time and complete online quizzes. In the on-campus lectures, students and teachers discuss small challenges related to the lecture topic. In the pre-recorded video material, theories, principles and methods are presented, discussed and illustrated with examples from the field. The video material is supported by online self-tests. In their own time, students work in small groups on coursework assignments to develop a product design for a BCSS with a computer simulation of supporting algorithmic nudge/boost/persuasion. In the on-campus session, student groups present the progress of their coursework and receive formative feedback. The presentations are done three times, following topic as the project progress over time: 1. The targeted behaviour 2. Behaviour plan 3. Conceptual design and the algorithmic nudge/boost/persuasion	
Literature and Study Materials	Available on brightspace	
Books	Wendel, S. (2020). Designing for Behavior Change: Applying Psychology and Behavioral Economics. O'Reilly Media, Inc,. or the older edition Wendel, S. (2013). Designing for Behavior Change: Applying Psychology and Behavioral Economics. " O'Reilly Media, Inc.	
Assessment	The final grade of the course consists of the following components: Computerised examination (or oral exam) (weighting 60%) If the expected number of students registering for the exam or resit is small, the teacher might decide to replace the computerized examination with an oral examination. Design BCSS Coursework Project (resulting in presentation slides and demonstration of computer simulation of a supporting algorithmic nudge/boost/persuasion, including question and answer round where individual group members are assessed on the coursework) (weighting 40%) Final grade calculation: 0.6 * Computerised examination (or oral exam) + 0.4 * Design BCSS Coursework Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Computerised examination (or oral exam): Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Design BCSS Coursework Project: Resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	3	
Co-Instructor	M.L. Tielman	

CS4135	Software Verification	5
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	Ir. K.F. Wullaert	
Contact Hours / Week x/x/x/x	0/2/0/0 + 0/4/0/0 practicum	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	How can we ensure that software cannot crash and is guaranteed to be correct? In this course we tackle this question by viewing programs and programming languages as mathematical objects. That way we can use logic to prove properties about programs and thereby guarantee that software is correct. To make reasoning about actual programs and programming languages feasible, we will not be doing these proofs by hand, but instead use a tool called a proof assistant to build proofs that can be checked by a computer. As we will show during this course, proof assistants turn the activity of doing proofs into programming. This course is a specialization course for programming languages and software engineering Prior knowledge: - Required: - Strongly advised: some familiarity with functional programming and propositional and predicate logic.	
Study Goals	Learning Objectives: - State and prove properties of functional programs in logic. - Specify the semantics of a programming language in logic. - State and prove the correctness of imperative programs. - Use a proof assistant to perform a mechanized proof.	
Education Method	This course is being taught in a "flipped classroom" style. It consists of 2 hours of studying (by reading or watching lecture videos, for greater flexibility and avoiding clashes with other courses - material will be provided) and two lab sessions of 2 hours each. During the lab sessions students will work on stating and proving theorems. Towards the end of the course students will carry out research projects that apply the ideas of the course.	
Literature and Study Materials	Supplementary material: Free online text book "Logic and Proof": https://leanprover.github.io/logic_and_proof/ Free online text book "The Hitchhikers Guide to Logical Verification": https://github.com/blanchette/logical_verification_2021/raw/main/hitchhikers_guide.pdf	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the individual project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4270	Conversational Agents	5
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Responsible Instructor	F. Broz	
Instructor	M.L. Tielman	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming skills (e.g. Python and Java) Probability theory and statistics	
Course Contents	Chatbots, embodied and conversational virtual agents, and social robots are becoming more and more popular. Many people are owning an Alexa, Cortana or Echo or are talking to their virtual assistant on their phone. Indeed, such technologies have the potential of making our lives easier and relieve people from the more repetitive tasks. For example, it is imaginable that such systems are being used for financial applications by helping customers with frequently asked questions but also to advise them on in the long term more impactful decisions such as their pension plans. Further applications can be imagined in the area of healthcare and education, some of which are already in existence today. In this course, attention will be given to different verbal and nonverbal behavioral characteristics, like speech, intonation, gaze and gestures that humans show when communicating with both other people and machines. This behavior is then related to different dialogue functions, including turn-taking, addressing others, and backchanneling, that give shape to the communication process. This course introduces conversational agent technology. We cover agent related technologies which can be grouped into: Dialog Management NLP speech synthesis social robotics	
Study Goals	After this course you have learned to: 1) Apply relevant linguistic and psychological theory to conversational agent systems 2) Design the conversational memory of your agent focusing on a perception/generational model 3) Analyse human-human conversational data to better design ML models for adding generation or perception capabilities to conversational agents 4) Develop the conversational memory in your agent 5) Evaluate the effects of affect and embodiment on human-agent interaction 6) Discuss your results and the limitations of your implementation	
Education Method	There are 2 lectures scheduled per week. Students work in groups of 4 on a group project. Lectures: 26 hours (13 × 2 hours lectures) Reading time: 39 hours Preparation basis tool use: 25 hours Coursework project, including writing report and prepare a video demonstration: 50 hours (10 × 5 hours)	
Literature and Study Materials	We use the book "The conversational interface " by Michael McTear, Zoraida Callejas, David Griol. This book is freely available through the TU Delft library. https://link-springer-com.tudelft.idm.oclc.org/book/10.1007%2F978-3-319-32967-3 Other relevant material will be provided on Brightspace.	
Assessment	The final grade of the course consists of the following components: Online Examination (weighting 30%) Group Assignment (weighting 70%). This assignment will result in a group report and a video demonstration. 3 Lab assignments (pass/fail) Final grade calculation: 0.3 * online examination + 0.7 * Group assignments + P/F * Lab assignments A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Online examination: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Group Assignment: re-submit deliverable Lab assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4400	Deep Reinforcement Learning	5

Responsible Instructor
Dr. J.W. Böhmer

Instructor
Dr. M.T.J. Spaan

Contact Hours / Week x/x/x/x
0/4/0/0

Education Period
2

Start Education
2

Exam Period
2
3

Course Language
English

Expected prior knowledge
Students must have passed CS4375 "Artificial Intelligence Techniques", or have acquired equivalent knowledge about:
- basic probability theory, analysis and algebra
- general machine learning methodology, e.g. regression
- fully and partially observable Markov decision processes
- tabular reinforcement learning methods, e.g. Q-learning
- the exploration/exploitation trade-off, e.g. RMAX or UCB
- multi-agent learning, e.g. centralized training and decentralized execution

The course will require a lot of math and implementations of deep-learning algorithms. Students are encouraged to close any gaps in the above knowledge and to familiarize themselves with the Python/PyTorch deep-learning framework before the start of the course.

Course Contents
This course will cover the breadth of modern model-free RL methods, discuss their limitations and introduce a variety of current research topics. In particular, we expect to cover the following:
- deep learning methodology and architectures
- stabilization of approximated value estimation
- modern actor-critic methods
- planning as inference
- exploration with deep networks
- offline reinforcement learning
- deep multi-agent reinforcement learning
- multi-task and meta learning

Study Goals
After successful completion of this course, students
- can list the strengths and limitations of modern deep RL approaches,
- explain the underlying concepts of the discussed methods, and how they differ from each other,
- derive the objectives and constraints of selected algorithms,
- can implement selected algorithms/architectures, and
- can analyze a new task to decide which algorithms/architectures to apply.

Education Method
The course will be taught in lectures and the content will be solidified in homework submissions, which will be presented in tutorials.

Assessment
The final grade of the course consists of the following components:
Written exam (weighting 100%).
Exercises (Pass/Fail) To be eligible for the exam, students must hand in homework exercises. Homework will not be individually graded, but at least 75% of the answers must be of sufficient quality (in terms of time commitment, not necessarily correctness) to be eligible to take the exam.

Final grade calculation: P/F * Exercises + 1 * Witten exam
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

Resit/ Repair opportunities:
Written exam: resit
In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:
Exercises: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

IN4089	Data Visualization	5
Responsible Instructor	T. Höllt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/2/0/0 & lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Master course MKE/ST/DS	
Expected prior knowledge	IN2905-A/IN4152/CSE2215 Computer Graphics (recommended, not required). The practicals will be implemented in HTML/Javascript/D3 (InfoVis) and C++ (VolVis). We consider programming skills as a requirement but not necessarily in the mentioned languages and no advanced skills are needed. Relevant topics will be introduced and experience in other programming languages should make adaption feasible.	
Course Contents	Data visualization is the visual representation of data by computer generated images. The data sets can be results of numerical simulations or measurements (scientific visualization), or other data collections such as databases (information visualization). The goal is to improve insight, understanding and/or communication of data. Data visualizations use a combination of methods from a very diverse variety of disciplines: perception, computer graphics, human computer interaction, algorithmics, image processing, machine learning, numerical analysis, optimization, The course has two main parts; information and scientific visualization that will involve knowledge of diverse disciplines. As a computer science course, affinity to algorithmic thinking and programing skills will be needed. Topics covered: models of the visualization process; colour models and use of colour; information visualization; representation and processing of data; volume visualization; interactive visual data analysis; visualization of vector fields and flows. Guest lectures might be given on selected topics.	
Study Goals	The goal of the course is to get knowledge on the fundamentals that are part of data visualization. The main principles and techniques that are the basis of generating effective visual representations of data. Techniques and cases of data visualization are discussed. There are several applications for the techniques, such as medical, engineering, finances, economics, game analytics, and more. By the end of the course, you should be able to LO1: Discuss a large range of visualization techniques. LO2: Discuss a perception principle of visualization. LO3: Explain mathematical principles and algorithms of visualization techniques. LO4: Design suitable visualization systems for a given practical data analysis problem. LO5: Implement visualization systems for a given practical data analysis problem.	
Education Method	Lectures, practical assignments, self-study, and projects.	
Literature and Study Materials	Course slides, instructions for projects, and selected literature via Brightspace. Chapters from: Visualization Analysis and Design Author: Tamara Munzner CRC Press Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Books available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 30%). You are allowed to have the slides and material of the course during the exam. No computer or laptop is allowed. Information visualization project (weighting 35%) Volume visualization project (weighting 35%) The projects will be developed in groups of 3 and are evaluated based on the reasoning/justification of the techniques used based on the material given at the course, effectiveness of the results, technical contribution or implementation, quality of the documentation and presentation Final grade calculation: 0.3 * Written exam + 0.35 * Information visualization project + 0.35 * Volume visualization project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Information visualization project: re-submit deliverable Volume visualization project: re-submit deliverable The project resits are not automatic and must be requested within 2 weeks after publishing the grades. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

IN4150	Distributed Algorithms	6
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for network overlays, message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, broadcast, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	Understand the fundamental problems that distributed systems face Understand the most important distributed algorithms that solve these problems Reason about the execution of distributed algorithms Program distributed algorithms Select and summarize relevant literature on distributed algorithms	
Education Method	Lab work executed in groups of two students	
Literature and Study Materials	Lecture notes and lecture slides (available on Brightspace)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 75%). In case in-person, on-campus written exams are not possible because of unforeseen circumstances or measures, the written exam will be replaced by individual online oral exams. Paper summary (weighting 25%) Final grade calculation: 0.75 * Written exam + 0.25 * Paper summary A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper summary: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None except the list of algorithms	
Special Information	Email: in4150-ewi@tudelft.nl. Please send any emails related to the course to this address.	
Remarks	Lab work is 40 hrs.	

IN4302TU	Building Serious Games	5
Responsible Instructor	Dr.ir. A.R. Bidarra	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	For CS students: programming experience with some object-oriented language; experience with graphics, AI and/or some game engine(s) is a plus. For all students: though not compulsory, it may be convenient to have followed the course SEN9235 (Game Design Project), which is taught in the first quarter.	
Course Contents	Project-based interdisciplinary course, open to MSc students of all faculties. The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, training, health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose. Assignments for this course will be provided by real-world end-users (e.g. companies or the Science Centre Delft), to whom the group will be reporting throughout the term of the project. Project-based interdisciplinary course, open to MSc students of all faculties (of LDE universities). The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, professional- and health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose, solving a variety of possibly conflicting, technical challenges. Assignments for this course are provided by real-world commissioners (from startups to NGOs to research institutes), to whom the group will be reporting throughout the project. In previous years, various serious game prototypes originated in this course were actually developed and eventually deployed for their intended application; and many more were published and presented by students in international research venues (incl. IEEE CoG, FDG, ICIDS, GALA, ISAGA).	
Study Goals	At the end of the project, the student will demonstrate proficiency in the following aspects: o identifying and valuing the soft skills necessary to work effectively in interdisciplinary teams o responsibly interacting in a team, integrating its members' talents and expertise in a variety of team roles o adapting with flexibility to the dynamic requirements of a complex external assignment o translating feedback received into proactive personal development steps o formulating research questions, identifying research contributions, and describe them in a scientific publication Additionally, the CS student will demonstrate proficiency in the following specific aspects: o identifying, selecting and deploying the most adequate game technologies for the given serious game domain and constraints o deepening programming skills while building a complex and large software system in an agile context	
Education Method	Project: teams work intensively as a small game studio, according to a tight planning. Also a few plenary sessions and/or lectures.	
Assessment	The final grade of the course consists of the following components: Product grade (weighting 50%). Unique for the whole group, based on both the game itself and the required documentation. Process grade (individual) (weighting 45%). Including personal contribution, performance, attitude, and peer evaluation. Final presentation (weighting 5%) The commissioner will be involved both as advisor and as assessor. The final documentation includes writing a scientific paper and preparing it for actual submission to a conference on serious games and/or their application. Final grade calculation: 0.5 * Product grade + 0.45 * Process grade + 0.05 * Final presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final presentation: resit There is no repair possibility for the other components (product and process), since their assessment necessarily requires an extended development period. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. R. Marroquim	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Literature and Study Materials	We follow the book Performance Analysis of Complex Networks and Systems, by P. Van Mieghem, Cambridge University Press (2014). See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Written and closed book. A formularium is provided that can be consulted at the examination. Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. M. Blaauboer	
Responsible Instructor	Dr. L. di Carlo	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	30% homework assignments, 10% online quizzes, 60% final exam. A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course.	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialistievakken start kwartaal 3 2023

AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morpological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 6 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: ((0.85*EX + 0.15)+ AS)/ 2 or without the bonus point from the practicals: (EX+AS)/2 With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

CESE4030	Embedded Systems Laboratory	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. G. Lan	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This highly multi-disciplinary course comes with a lab project where teams of 4 students each will have to develop an embedded control unit for a tethered electrical model quad rotor aerial vehicle (the Quadrupel drone), in order to provide stabilization such that it can hover and (ideally!) fly, with only limited user control (one joystick). The control algorithm (which is given) must be mapped onto a home-brew PCB holding a modern RF SoC interfacing a sensor module and the motor controllers. The students will be exposed to simple physics, signal processing, sensors (gyros, accelerometers), actuators (motors, servos), and basic control principles. The embedded software running on the flight controller board (FCB) driving the drone must be written in the Rust programming language. The project work (including written report) covers the entire duration of the course period, and will take approximately 128 hours, of which 32 hours are spent at the lab facilities.	
Study Goals	Student is acquainted with real-time programming in an embedded context, along with a basic understanding of embedded systems, real-time communication, sensor data processing, actuator control, control theory, and simulation. Moreover, the student has had exposure to integrating the various multidisciplinary aspects at the system level.	
Education Method	Lectures (6*2hrs), lab work (8*4hrs), coding@home (8*12hrs), report (8hrs), so on average 2 days per week	
Prerequisites	Literacy in Rust (CESE4000 or equivalent, and CSE4015)	
Assessment	Lab. project (75%) + written report (25%), no exam, no resit. Failed student has to do an additional assignment/report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for CESE students- they get preference over other MSc students.	

CESE4050	Measuring and Simulating the Internet	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/2/0 (Lectures(2x2h), Pract)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Strongly advised: (Advanced) Networking course (e.g., CESE4045) and Programming skills.	
Course Contents	The Internet is a complex network without a fixed structure. Hence, measuring the Internet is crucial to acquire knowledge about the Internet infrastructure (topology), traffic, and performance (e.g., loss, delay, bandwidth, etc.). This course will discuss the design requirements and challenges in measuring and simulating the Internet, and the existing measurement methodologies (how/where/when to measure). Knowledge of how to conduct and evaluate Internet measurements enables the design and enhancement of a large set of applications, including: capacity planning and traffic engineering, network management and trouble-shooting, detecting network abuse and intrusions, etc.	
Study Goals	The goal of this course is to introduce the students to basic Internet measurement tools, as well as the state-of-the-art in Internet measurements research. The students will learn several Internet measurement techniques (e.g., active vs. passive measurements), and different software tools. Through a measurement assignment, the students will learn how to define/formulate a research problem, choose a specific approach, and complete a measurements-related research project.	
Education Method	Lectures + weekly instructions (discussion of papers and progress) + independent project work.	
Literature and Study Materials	Papers	
Assessment	Groups of students will be assigned a project that requires the students to put the theory on measuring and simulating the Internet into practice. The students have approximately 1 month to complete their assignment. The final assessment is based on the presentation (via report and/or demonstration) of the project assignment results and on the individual contribution and level of participation. Students within a group may thus receive different grades. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Maximum number of participants	Because this is a project-based course, we can only admit a limited number of students (typically around 30, but the actual number depends on the number of TAs involved).	

CESE4060	Wireless IoT and Local Area Networks	5
Responsible Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students are advised to follow the course Wireless Communications (ET4358) before taking this Wireless Networking course. An advantage is to have entry-level programming skills (Matlab, Python, C/C++). Nonetheless, students with little knowledge of programming will be helped.	
Course Contents	The following modules will be discussed during the lectures: Introduction (example topics): - What is wireless networking - Where to search for (academic) wireless network literature and resources Medium Access Control (example topics): - WiFi: hidden/exposed terminal problem, Carrier Sense Multiple Access - Bluetooth standard: in-depth look into the channel hopping, protocol specifications WiFi (example topics): - Review of IEEE 802.11 standards - Protocol format - ISM band regulation - Adaptive Modulation and Coding - WiFi Matlab class (assignment) IoT networking standards (example topics): - LoRa: protocol specifications, energy consumption, modulation format, network design Review of wireless tools (example topics): - Introduction to wireless packet sniffing and analysis using Wireshark (assignment) - Simple simulations of WiFi network with NS3 RFID networking (example topics): - Principles of backscatter - Protocol formats: EPC C1G2 - RFID hackathon (assignment) Cognitive radio (example topics): - Basics of spectrum management - White Space Databases - Theory of spectrum sensing	
Study Goals	At the end of the course students will be able to: (i) to understand how practical wireless systems work and get a deeper understanding of how the theoretical concepts of wireless communications apply to practice; (ii) employ their own analysis methodology to assess new wireless network systems (especially at the physical layer); (iii) understand rapid prototyping of new wireless systems (for instance, with software defined radio).	
Education Method	Lecture presentations, mini-project assignments, assigned paper reading and its critical analysis and presentation.	
Computer Use	Each student should have its own laptop (preferably with a Linux distribution, where Linux must not be installed on a virtual machine). We will be using Matlab, and/or NS3 and/or GNURadio and/or Wireshark for the assignments.	
Literature and Study Materials	WiFi Matlab WLAN toolbox: https://nl.mathworks.com/help/wlan/; Wireshark learn page: https://www.wireshark.org/#learnWS; tutorial on NS3 network simulator: https://www.nsnam.org/documentation/; specific chapters from books provided at the beginning of each lecture.	
Prerequisites	Background in programming (Matlab, Python, Bash)	
Assessment	Points from the mini-project assignments. A research paper analysis from conferences such as IEEE INFOCOM, ACM MobiCom, ACM SIGCOMM will be required to pass the course. Repair option: Individual Assignment = Re-submit deliverable Individual Report = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: A good understanding of the fundamental concepts of quantum communication Insight into the differences between classical and quantum communication and cryptography Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)	
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session	
Literature and Study Materials	Primary: Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: Homework (weighting 70%) Final Project (weighting 30%) Final grade calculation: 0.7 * Homework + 0.3 * Final Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Homework: re-submit deliverable Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5

Responsible Instructor	Ir. S.E. Verwer
Responsible Instructor	Dr. A. Panichella
Contact Hours / Week x/x/x/x	0/0/4/0
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Course Contents	Software is one of the most complex artifacts of mankind has ever created, but complexity is the enemy of correctness. Modern software testing and validation tools use a multitude of techniques geared toward correct computer code, most of these are base on artificial intelligence. In this course, we study these techniques in details, specifically we will understand and implement: Execution monitoring and taint analysis Branch distance computation Hill-climbing and genetic algorithms Concrete and symbolic (concolic) execution Active state machine learning Genetic programming The goal is to better understand and test software using artificial intelligence. Using the taught techniques you will be able to automatically: Discover which code is reachable Find (security) bugs in software Write tests that cover all reachable code Reverse engineer a code's functionality Patch code to remove bugs and failing tests
Study Goals	The student will: Understand modern AI techniques for software testing. Be able to implement several such techniques from scratch: - smart fuzzing (probing software with input to find crashes/bugs), - symbolic execution (using logic to construct inputs that trigger specific code branches), - fault localization (given that a program fails, find the line of code responsible for the failure), and - automated program repair (using a patch library and genetic programming to improve code) Be able to apply this technology to locate bugs in real-world software implementations.
Education Method	The main part of the course will consist of 3 lab assignments covering the theory (fuzzing&tainting, symbolic execution, automated program repair), and one lab assignment for the application to real software. The students will implement the taught techniques from scratch in the first 3 assignments, which will be scored with a pass/fail. All three assignments need to be passed to complete the course. The final lab will contain a recap from the first three assignments and an application of a state-of-the-art tool on real software. The final lab will be graded and be the final course grade. There will be instruction sessions where students can work on their assignment and ask the teachers for assistance.
Assessment	The final grade of the course consists of the following components: Three lab assignments (pass/fail) Final lab (weighting 100%) Final grade calculation: P/F * 3 lab assignments + 1 * Final lab A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: 3 lab assignments: re-submit deliverable Final lab: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Tags	Artificial intelligence Software

CS4160	Blockchain Engineering	5
Responsible Instructor	Dr.ir. J.A. Pouwelse	
Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	In this course you will learn all aspects of blockchain technology, including tamper-proof data structures, digital identities, transitive trust, fault tolerance, distributed consensus, smart contracts and applications. Ledgers and blockchains are an emerging technology with the potential to radically improve financial transactions, supply-chain flows, transactions in general, and distributed databases. The first three weeks of the course will provide a fast-paced introduction to Bitcoin, Ethereum, and TrustChain developed by TUDelft itself. The main component in this course is a team-based complex engineering project. This course is designed for computer scientists to understand blockchain technology and to produce significant hands-on experience. To provide a deep understanding of blockchain technology and understand why it is special you need to experience first-hand how it operates at a detailed technical level. Students design, implement, and test their own independent project in teams of 3-5 students. Students can choose from a pool of possible project ideas. This course requires you to like software engineering. Topics covered: -Blockchain basics and evolution Bitcoin 1st generation, smart contract generation, future 3rd generation (trust or trust in math) -identity and transitive trust Authentication and security primitives, tamper-proof identities, trust models, MITM attacks, Sybil attacks, and TrustChain by TUDelft -Consensus models Proof-of-work, permissioned, Proof-of-stake, Corda no-global-consensus, TUDelft bottom-up fast consensus model -Smart Contract pro/con encrypted data, Bitcoin scripts, Etherium execution model, Hyperledger + Docker issues, Corda Jar file approach, Tezos difficult to use, powerful technology, vision of the future: trusted verified execution -Markets and exchanges Disruption by open markets, winner-takes-all, and multi-sided market platforms, Uber, Airbnb, 22 years of eBay, Silk Road, honesty among drug dealers, the role of trust in markets, P2P exchange markets -Decentralized Autonomous Organization, novel method to collaborate and organise any economic activity -Web3: how the Internet can be re-decentralised using DAOs Within this 2023 edition "the Delft DAO" will be prominently featured. TUDelft achieved a world-first in DAO research. We devised a full end-to-end proof-of-principle of a DAO which is capable of 0) near unbounded scalability 1) controlling money 2) democratic decision making and 3) continuous sustained self-evolution. This course provides you with the knowledge to work with this advanced technology. After this course you will have a firm grasp on the current operational blockchain-based systems, realistic view of real-world applications that may be built on top of ledger technology. You will be able to reason and discuss the open challenges and questions that still need to be resolved. This course is a key course for distributed systems students.	
Study Goals	After this course students are able to design and engineer complex blockchain-based systems. Students are able to describe blockchain technology, the various consensus model, smart contracts, markets, and relation to existing database technology. Student are able to setup a new architecture for blockchain applications.	
Education Method	This course consists of four 2-hour lectures. Each lecture is followed by a 4-hour homework period in the same week focused on understanding the background material. In week 1 you will form teams and initiate work on your blockchain engineering project. A list of projects to select from will be provided at the start of this course.	
Literature and Study Materials	Online course textbook: Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction by Narayanan, Bonneau, Felten, Miller and Goldfeder.	
Prerequisites	It is highly recommended to follow this course (see remarks): Security and Cryptography (Q1) Distributed Algorithms (Q2)	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) You will write down your project results in a single-page report, markdown format. You will be graded on your open source efforts located on Github and single-page report. Your grade will be expressed on a scale of 0 to 10. Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This class has a limited capacity (50). If there is a larger number of enrollments than the capacity of the class, students will be assigned to their preferred blockchain engineering project based on their background, engineering experience level, and match to the course goals. Students who followed Security and Cryptography (Q1) and are also enrolled in Distributed Algorithms (Q2) will have priority for placement. Mathematics students are exempts from this, if they can show some minimal software development experience (e.g. Github profile). Finally, students with a Grade Point Average of 8.0 or higher are eligible for the challenging scientific projects, resulting in a research paper. These project receive intense guidance, but have no capacity limits.	

CS4195	Modeling and Data Analysis in Complex Networks	5
Responsible Instructor	H. Wang	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The assignment and final project require basic programming skill.	
Course Contents	Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	After this course, students could construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies, model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. Students could obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 20%) Final project (weighting 80%) Final grade calculation: 0.2 * Assignments + 0.8 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4210-A	Algorithms for Intelligent Decision Making	5
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. V. Robu	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: CS4375: Artificial Intelligence Techniques, or equivalent; and/or IN4344: Advanced Algorithms, or equivalent Required: basic course(s) in algorithm design and analysis, logic and probability; basic programming (in Python)	
Course Contents	Decision making is at the centre of artificial intelligence. This course gives you practical skills on a solid theoretical base. The course looks at solving mathematical models of NP-hard discrete optimisation problems. These kinds of problems lie at the heart of AI techniques such as planning, machine learning and mechanism design, and more generally combinatorial optimisation. You will learn about a range of modelling techniques from boolean satisfiability to constraint programming, and how advanced solvers for these models work. The course has plenty of real-world case studies as well as theoretical results. Apply the skills you learn in this course by taking CS4210-B: Intelligent Decision Making Project in quarter 4!	
Study Goals	By the end of this course, you will be able to identify features of real-world combinatorial decision problems, and be able to model and design systems for simplified instances of these problems using boolean satisfiability, mixed integer programming, and constraint programming over finite and real domains. You will be able to explain how SAT, CP and LCG solvers work in some detail, and how MIP solvers work at a high level. You will also be able to explain the basics of program synthesis and combinatorial game theory.	
Education Method	Lectures, homework exercises (optional, and not counted towards the course grade), and programming assignment(s) (mandatory, but not counted towards the course grade). The expected workload is: 30% lectures (including preparation for the exams) 30% homework exercises (optional) 40% programming assignments	
Literature and Study Materials	Provided on Brightspace	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Besides the summative assessment, formative coursework will be offered. This includes a mandatory assignment, performed in pairs. Students must submit a non-trivial attempt at the assignment in order to pass the course. The assignment receives formative feedback, but is not counted in computing the course grade. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Elective	Yes	
Tags	Algorithmics Artificial intelligence Group work Modelling Optimalisation Programming Projects Small groups	

CS4225	Educational Technologies	5
Responsible Instructor	Prof. M.M. Specht	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	* Theories of Human Information Processing and Learning * Learning Management Systems * Learning Analytics * Personalisation and Adaptive Educational Systems * Mobile and Seamless Learning Technologies * Artificial Intelligence in Education * Realtime Learning Technologies * Project Design * Project Implementation	
Study Goals	The course will enable you to classify, understand, design and implement the core functionalities and systems for supporting human learning processes. As well current practices implemented as also approaches for technology enhanced learning currently researched will be presented. You will learn how educational technologies provide human learning process support, implement guidance and recommendation, create personalised learning support, as also give real-time feedback and support reflection of learners. In the final project you will identify a problem, design a solution based on the presented approaches and implement your own educational technology solution.	
Education Method	Lectures, weekly assignments and quiz questions, final project	
Assessment	The final grade of the course consists of the following components: Weekly assignments (weighting 30%) Final project (weighting 70%) Final grade calculation: 0.3 * weekly assignments + 0.7 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4230	Machine Learning 2	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Instructor	Dr. D.M.J. Tax	
Instructor	T.J. Viering	
Instructor	Dr. F.A. Oliehoek	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Required: This course is the more advanced and research oriented follow-up to Machine Learning 1 (CS4220). CS4220 or an equivalent machine learning course is therefore expected as prior knowledge.	
Course Contents	The course will treat a number of machine learning settings and theories spanning the breadth of machine learning research in detail and on an advanced level. Topics that are covered include (subject to change): learning theory, learning curves, latent variable models, causal inference and discovery, adversarial learning, online learning and neuromorphic computing.	
Study Goals	After completing the course, you should be able to: 1. Identify which setting beyond (un)supervised learning a problem belongs to 2. Explain the essential problem that (adversarial/causal/online/PAC/neuromorphic/...) learning addresses and how it differs from the classical ML setting 3. Evaluate methods that are proposed in these learning scenarios, both theoretically and empirically 4. Mathematically prove basic properties of methods in these scenarios 5. Apply common methods in these learning scenarios on real-world datasets 6. Develop machine learning techniques adapted for a particular learning task	
Education Method	Each week: 1 lecture and 1 Q&A session where the weekly exercise set will be discussed.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) 4 combined assignments (pass/fail). There will be multiple small assignments (in weeks 2,4,6,8). Final grade calculation: P/F * assignments + 1 * Written exam A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: 4 combined assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4235	Socio-Cognitive Engineering	5
Responsible Instructor	Prof.dr. M.A. Neerincx	
Instructor	F. Broz	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic prior knowledge on human-computer interaction is helpful, but not required.	
Course Contents	Whether you are playing a game in virtual reality, driving a semi-autonomous car, educating yourself with a mobile app, or harmonizing your health and lifestyle with a robot assistant; nowadays intelligent networked information and communication technology is omnipresent. This course focuses on the design of human-aware intelligence into such environments, to support joint human-technology performances that bring about positive human experiences. The focus will be on social robots in general, with practical design and test assignments on robots that support people with dementia, http://rejam.tudelft.nl). In the Socio-Cognitive Engineering (SCE) course (MSc level), you will become acquainted with the application of a coherent set of methods for the design and evaluation of human--agent collaboration. Based on the SCE-method, we will elaborate on the state of the art of intelligent user interfaces (ePartners), such as virtual assistants, robotic teammates, eCoaches, and companion agents. The main topics of study are: - Design methods: Cognitive Engineering, Value Sensitive Design, Scenario-based Design, Claims Analysis, Design Rationale, Design Patterns. - Design for collective intelligence: Knowledge Representation, Ontology Engineering, Mental Models, Theory of Mind, ePartners, Adaptive Automation, Social Robots. - Design Evaluation: Prototyping, Test Methods, Measures, Questionnaires, Ethics. - Human Factors Theories and Models: Human Cognition & Learning, Memory, Emotion, Task Load, Human-Agent Teamwork, Behavior Change and Persuasive Technology.	
Study Goals	At the end of this course, students will be able to: 1. Explain the essential concepts of the design methods addressed in the course. 2. Explain the (dis)advantages of various design methods and their complementarity. 3. Apply the design methods addressed in the course in their research and design projects. 4. Explain what a design rationale is. 5. Construct a design rationale. 6. Create design specifications that are grounded in a design rationale. 7. Evaluate the strengths and weaknesses of a design rationale, e.g. using human-centered evaluations that test the design rationale. 8. Explain some of the state of the art human factors theories, models, and methods relevant to intelligent user interfaces, human-robot collaboration, and ePartner technology. 9. Write a structured report about a design-test cycle, with sufficient detail for a new group of researchers to continue the research. 10. Present work on a design project to an academic audience. 11. Work in a group on collaborative assignments.	
Education Method	LECTURES During the lectures, the teachers will present a range of theories, models, and methods relevant to socio-cognitive engineering. Students are required to read a number of scientific papers which are made available on Brightspace, along with the sheets/slides of the lectures. Together, the sheets/slides and the papers provide the students with the required theoretical knowledge to work on the practical project, and to learn about relevant design methods, human factors theories, conceptual solutions, and design principles. Most of the lectures include practical assignments and discussions stimulating the students to apply the contents of the lecture to their project (also see Project). PROJECT In the project, students work in groups to apply the knowledge acquired during the lectures. Students are required to plan, execute, present, and report on a complete design cycle (i.e. design, prototype, and evaluation) for a given design problem. This year (like the past years), the design problem is a social robot for older adults with dementia, and their social environment (https://rejam.tudelft.nl). The objective of the social robot is to improve humans physical, social, cognitive, and emotional well-being. The students will use the Wiki Socio-Cognitive Engineering (WiSCE) tool to specify the design rationale and its evaluation, step-by-step (see also https://xwiki.ewi.tudelft.nl/). Throughout the course, students will give presentations about their progress, on the design and evaluation of their prototype. Literature and study material consist of: - Papers from scientific journals on Brightspace. - Lecture notes on Brightspace	
Literature and Study Materials	Papers from scientific journals on Brightspace. Lecture notes on Brightspace.	
Assessment	The module assessment concerns the processing and application of the theory and methods; the construction of the design (rationale) and the evaluation; and the provision of the resulting concise and coherent report (including the lessons learned): The final grade of the course consists of the following components: Presentations (weighting 15%) Prototype (weighting 15%) Project report according to the prescribed format (weighting 60%) Individual reflection (weighting 10%) Final grade calculation: 0.15 * Presentations + 0.15 * Prototype + 0.6 * Project report + 0.1 * Individual reflection A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentations: resit Prototype: re-submit deliverable Project report according to the prescribed format: re-submit deliverable Individual reflection: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Exam Hours	There is no exam. The assessment is based on a paper, presentation and report. During the course, students will receive feedback on interim work. There is no resit after the end of the course.

CS4240	Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Seminar Computer Vision by Deep Learning	
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy). Experience in programming and installing programming languages/libraries is required. Without affinity/experience with python (and numpy) programming this course will become too time consuming as the project will become difficult.	
Course Contents	In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.	
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.	
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ - https://udlbook.github.io/udlbook/ Research papers will be made available through Brightspace. Assignments are based on PyTorch: https://pytorch.org/	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Digital Exam (weighting 90%). Exam about the papers, lab assignments, and the theory (lectures). Project (Pass/Fail). In a small group of students you work on a deep learning paper reproducibility project. Final grade calculation: (pass project) * (0.1 * Presentation + 0.9 * Exam) A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4345	Seminar Formal Methods for Learned Systems	5
Responsible Instructor	Dr. A. Lukina	
Instructor	Dr. M.T.J. Spaan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course will cover several topics at the intersection of formal methods and machine learning, for instance, abstraction refinement for analyzing deep neural networks, algorithms for reliable learning-based control, interpretable model design, and formal methods for verification and monitoring of learned systems. The course should be useful for students of both formal methods and machine learning, and lies at the intersection of these areas. Since the area is relatively new, the course material will be primarily based on research papers.	
Study Goals	After this course students will acquire formal methods knowledge and skills such as: - modeling complex systems, - formally defining critical properties, - applying verification tools to learned systems, - synthesizing controllers for learned systems, - providing interpretability for learned models.	
Education Method	The course will consist of one lecture and one discussion session per week. The students will be assigned projects in groups of two, where they will need to apply the material from the lectures, e.g., train a learned system, define a property and verify it using one of the open-source tools from state-of-the-art research papers. Discussion sessions will be used for questions about the projects.	
Assessment	The final grade of the course consists of the following components: Participation (weighting 15%) Presenting Papers (weighting 15%) Assignment #1 (weighting 15%) Assignment #2 (weighting 15%) Final Report (weighting 40%) Final grade calculation: 0.15 * Participation + 0.15 * Presenting Papers + 0.15 * Assignment #1 + 0.15 * Assignment #2 + 0.4 * Final report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presenting Papers: resit Assignment #1: re-submit deliverable Assignment #1: re-submit deliverable Final report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4370	Testing and Validation for AI-intensive Systems	5
Responsible Instructor	C.C.S. Liem	
Responsible Instructor	Dr. A. Panichella	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of machine learning. Knowledge of software testing is preferred but not required.	
Summary	How do you know and assess whether an AI-intensive system is working as intended? On the one hand, quality assurance for such systems can be considered to be a software testing challenge; in parallel, the machine learning components within such systems are normally validated according to machine learning-specific evaluation methodology. In this course, you will learn more about these two complementary takes to evaluation and quality assurance, while learning about (and getting hands-on experience in) state-of-the-art advances in combining these two takes.	
Course Contents	- Basics of testing (assertions, white-box vs. black-box testing) - Basics of machine learning evaluation (validation & validity) - Adversarial attacks - Testing techniques (e.g. fuzzing, metamorphic testing, combinatorial testing, differential testing) - Data labels (oracle) validation	
Study Goals	After completing this course, students are able to: 1. Apply state-of-the-art techniques to validate machine learning models and training routines 2. Explain, analyze and improve robustness considerations in AI-intensive systems 3. Design and implement techniques to generate adversarial attacks 4. Apply fuzzing, metamorphic and combinatorial testing techniques to machine learning models 5. Connect the oracle problem in software testing to questions of validity in datasets 6. Complement classical evaluation reporting of machine learning models with insights obtained through testing	
Education Method	2 hours lecture, 2 hours project	
Literature and Study Materials	The study material will be provided online by the lecturers	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4405	Analysis of Concurrent and Distributed Programs	5

Responsible Instructor	Dr. B. Özkan
Responsible Instructor	Dr. S.S. Chakraborty
Contact Hours / Week x/x/x/x	0/0/4/0
Education Period	3
Start Education	3
Exam Period	3
Course Language	English

Course Contents

Software systems are becoming highly concurrent and distributed to utilize modern multicore architectures and increasing speed and bandwidth in networks. Shared-memory concurrency in multicore programs and message-passing concurrency in distributed programs share many common abstractions and problems.

In the multicore era, all performance-critical software employs some form of concurrent programming; typically shared memory concurrency. In this setting,
programmers use a number of primitives to develop efficient and correct concurrent programs. To do so the programmers have to understand the behaviors of the primitives and reason about them. It is also important to match the programming paradigms and underlying architectures. For instance, traditionally programmers have assumed that a multithreaded program executed simply by interleaving the executions of its threadsa model known as sequential consistency (SC). This assumption is, however, invalidated both by mainstream multicore architectures, which often execute instructions out of order, and by compilers, whose optimizations affect the outcomes of concurrent programs. As a result, concurrent programs have more outcomes than SC allows.

In the distributed setting, the units of concurrency are independent processes that do not share memory but communicate by exchanging asynchronous messages. The execution of such a system involves two main sources of nondeterminism: concurrency and partial failures. As the processes run concurrently, the exchanged messages can be delivered and processed in many different orderings. The distributed set of processes is also prone to network of process failures. The trade-off between the systems availability in the existence of failures and the consistency between the processes gives rise to a spectrum of weak consistency notions. It is important to reason about concurrency, possible failures, and consistency guarantees to implement distributed programs correctly and understand their behavior.

This course aims to explore analysis techniques for concurrent and distributed programs.

Outline of Lectures:

Shared memory concurrency:
- Abstractions for shared memory concurrency
- Relaxed memory concurrency
- Correctness of concurrent programs

Distributed concurrency:
- Distributed system components, models and assumptions
- Fundamental abstractions for distributed systems

Study Goals

This course aims to give students a deep understanding of concurrency and distribution in modern systems and hands-on experience for analyzing these systems.
By the end of this course, you should be able to:

- Describe the fundamental concurrency models in multicore and distributed systems

- Explain the concurrency nondeterminism in the executions of concurrent and distributed programs

- Discover concurrency bugs in multicore and distributed programs by program analysis and testing techniques

- Evaluate the pros and cons of program analysis and testing techniques for specific multicore and distributed programs

Education Method

The course consists of the following education methods:

- Lectures for reviewing concurrency and distribution concepts
- Homeworks/assignments
- Developing a course project, writing a report, and presenting it (course project)

To finish the course, students (in teams) will have to:

- Study several papers which will be discussed during the lectures
- Deliver their assignments
- Deliver and present their implementation project

Assessment

The final grade of the course consists of the following components:
Research project implementation (weighting 40%)
Research project report (weighting 20%)
Research project presentation (weighting 20%)
Homework assignments (weighting 20%)

Final grade calculation: 0.4 * Research project implementation + 0.2 * Research project report + 0.2 * Research project presentation + 0.2 * Homework assignments
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

Resit/ Repair opportunities:
In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:
Research project report: re-submit deliverable
Research project presentation: resit
Homework assignments: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

CS4410	Category Theory for Programmers	5
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	Ir. K.F. Wullaert	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Categorical structures occur in programming languages on different levels: (1) within programming languages, providing design principles and guidance on how to write modular and correct-by-design programmes (as demonstrated in the practical programming language Haskell) and (2) in the design and study of programming languages, as a guiding meta-theory. In particular, category theory provides a mathematical justification for recursion schemes for inductive datatypes. This course aims to provide solid foundations on both (1) and (2). Lecture notes for the course are available at https://github.com/benediktahrens/CT4P. This course is suitable for people with an interest in functional programming, abstract mathematics, and logical reasoning. Prior knowledge: - Required: None - Strongly advised: None	
Study Goals	Learning Objectives: - Use categorical constructions (e.g., monads) in the design and structuring of computer programmes - Specify datatypes as initial algebras - Prove properties of computer programmes, guided by categorical intuition - Use categorical fusion laws to optimize functions - Use the theory of infinite data structures to define functions on infinite data	
Education Method	Learning in this course is achieved through lectures, problem sessions, and guided self-study.	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4415	Sustainable Software Engineering	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Sustainable Software Engineering is an overarching discipline that addresses the long-term consequences of designing, building, and releasing a software project. By definition, sustainability covers five main perspectives: environmental, social, individual, economic, technical. This course mainly focuses on the first, also known as Green Software Engineering. On top of that, we also cover fundamental aspects of social and individual sustainability of software projects. Software Engineering (SE) has long addressed sustainability by narrowing it down to economic and technical sustainability. However, our society is facing major sustainability challenges that can no longer be overlooked by software engineers and computer scientists. It is estimated that, by 2040, the ICT sector will contribute to 14% of the global carbon footprint. Hence, the environmental, social, and individual perspectives ought to be part of the equation when it comes to design, build, and release software systems. The problem is far from simple and we need expert computer scientists to bring sustainability into the core values of the next generation of tech-leading organisations.	
Study Goals	After attending this course, you will be able to: LO1. Explain the fundamental sustainability principles and their relevance to software engineering. LO2. Assess the energy consumption of software systems using measurement and testing techniques. LO3. Create actionable strategies to address sustainability issues within software organisations. LO4. Evaluate and compare different solutions for sustainable software systems based on their effectiveness in achieving sustainability goals and their potential impact on various stakeholders.	
Education Method	To meet these objectives, you will be involved in a broad set of learning activities: lectures, paper reading, software analysis, software development, essay writing and presentation. These heterogenous set of activities aim at building a strong set of hard skills for energy-efficient code development combined with a strong set of soft-skills and critical thinking. You will work on ideas that will help real-world software projects embrace a green software culture.	
Assessment	The final grade of the course consists of the following components: Software project (weighting 45%) Essay (weighting 45%) Presentation (weighting 10%) Final grade calculation: 0.45 * Software project+ 0.45 * Essay + 0.1 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Software project: re-submit deliverable Essay: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4430	Network Security	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best practices in computer and network security. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course's primary focus will be on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design, review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics, as well as conducting measurements on the effectiveness of attack and defense schemes.	
Study Goals	See course contents.	
Education Method	Lectures, Labs, and Project.	
Assessment	The final grade of the course consists of the following components: Assignments (Pass/Fail) Final project (Report + Presentation) (weighting 100%) Final grade calculation: P/F * Assignments + 1* Final project A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable, resit presentation Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l0 norm is shown to be NP-hard, and several classical approximate algorithms like l1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IN4152	3D Computer Graphics and Animation	5
Responsible Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. R. Marroquim	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Students that haven't followed any previous Computer Graphics courses (like CSE2215) will be able to participate, but might have to invest some more time to catch up in the first lectures.	
Course Contents	Have you ever wondered how Toy Story was made, why the game Last of Us 2 looks so beautiful, or have you ever wanted to create your own graphics application or game? Then you should consider following this course. If not, then you should still follow it... maybe, you will become interested! In this course, you will get a good idea of Computer Graphics in general. The topic is of very high relevance for the industry and the research community and has numerous applications in different domains, such as scientific visualization, video games, simulators, special effects, animated movies and many more. Here, you will learn about basic algorithms, as well as modern techniques. We will address several topics: the principles of image synthesis, object representations, geometric and hierarchical transformations, graphics cards and the graphics pipeline, realistic rendering (including global illumination and effects, such as reflections), expressive rendering, physics simulations, rendering control (including previsualization systems used by professionals in the movie industry), and perceptual rendering, which relies on properties of the human visual system to enhance the quality of the images. Besides course sessions on the theory of Computer Graphics, some of the algorithms will also be reproduced in practice, and deepened during the final project.	
Study Goals	The course teaches computer graphics techniques on an advanced level. After the course the student is able to classify the different modeling, shading, and display techniques. The student can reproduce the basic mathematical and algorithmic notions associated with these concepts, can comment on the weak and strong points of these techniques, and can apply the core concepts within a graphics program in practice.	
Education Method	lectures, instructions, research papers, lab work	
Literature and Study Materials	Research Papers in domain of selected topics, lecture sheets, online sources, optional books (see below)	
Books	Fundamentals of Computer Graphics by Shirley et al. - CRC Press Real-time Rendering by Tomas Akenine-Möller, Eric Haines, Naty Hoffman - Peters, Wellesley Real-Time Shadows by Elmar Eisemann, Michael Schwarz, Ulf Assarsson, Michael Wimmer - Taylor & Francis Computer Graphics. Principles and Practice by James D. Foley, Andries VanDam, Steven K. Feiner - Addison Wesley Advanced Material: Physically Based Rendering, From Theory to Implementation by Matt Pharr, Wenzel Jakob and Greg Humphreys - MIT Press	
Assessment	The final grade of the course consists of the following components (details of both elements will be presented during the lecture): Project (weighting 60%). The project grade is the result of a project and its presentation that is building upon the assignments that are handed out (roughly) weekly during the duration of this course. Paper (weighting 40%). The paper grade is the result of the presentation of a scientific paper and the development of an associated practical implementation. Final grade calculation: 0.6 * Project + 0.4 * Paper A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Paper: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4253ET	"Hacking Lab"-Applied Security Analysis	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Necessary background differs per student project, see first lecture or contact instructors for details	
Course Contents	The security of computer and telecommunication systems is becoming an increasing concern. In this course, we will review the current state of the art on security research and gain practical experience in assessing the security and vulnerabilities of communication systems. Engineers are typically taught to focus on performance, correctness, scalability, and maintainability when building communication and information processing systems. However, an additional set of design principles are required to achieve security. In this course, we discuss security principles, common pitfalls and vulnerabilities. The weekly lectures provide an introduction into security research, with a focus on real-world security, privacy-enhancing technology and common security pitfalls. Each student participates in a "Hack Project", with a group of one to four students. Students can select between a wide range of available Hack Project outlines within the first week. The goal may be to evaluate the security of a real-world IT system, developing a proof-of-concept exposing a vulnerability or focussed on preserving privacy in a post-Snowden world. Students may propose their own Hack Project based on their background knowledge and skills. Such Hack Projects need to be approved and shaped together with the instructor. Example of possible outlined hardware-oriented projects are: development of a wifi tracker, programing an FPGA system to break passwords, assess the security of RFID cards, or to transparently intercept Ethernet traffic. Concrete software projects are: hacking Bitcoin, improving the TOR anonymity protocol and create Android-based tools for human rights activists in Iran, Egypt and Russia, reprogramming neural networks attacks. Each Hack Project is documented with a written report. This can be in the form of a 6-8 page IEEE-style scientific article or a traditional more lengthy report. All results, experiences and findings are presented to the entire class in the last week of the course. Hack Projects also report their progress several times during the course, after the weekly lectures.	
Study Goals	After this course, the student will have a thorough knowledge of security in real-world systems, and will be able to explore the literature on this topic independently. The student will be aware of the poor state of security in real-world computer systems. The student can explain the common pitfalls, why these known failures still occur and reasons behind the poor state of security in general.	
Education Method	Lectures, student presentations, written final report and active participation. Attendance and active participation during lectures is mandatory. This sadly means telelecturing is not possible.	
Literature and Study Materials	Customize literature lists and study materials are provided per project topic	
Assessment	The final grade of the course consists of the following components: Written Hack Project report (weighting 60%) Final presentation of result (weighting 10%) Presentation of ongoing project progress (weighting 20%) Participation in discussions, overall quality of the practical work and class attendance (weighting 10%) Attendance to lectures is mandatory. Final grade calculation: 0.6 * Written Hack Project report + 0.1 * Final presentation of result + 0.2 * Presentation of ongoing project progress + 0.1 * Participation in discussions, overall quality of the practical work and class attendance A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Written Hack Project report: re-submit deliverable Final presentation of result: resit Presentation of ongoing project progress: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
maximum aantal deelnemers	If there is an unexpected high demand for this course, then enrollment will be based on past performance in relevant courses.	
Co-Instructor	Dr. S. Picek	

IN4315	Software Architecture	5
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software engineering	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four, in which the actual architectures of existing open source systems are analyzed in considerable detail. These systems are taken from github, and student teams are challenged to actually contribute to the systems under analysis in the course.	
Study Goals	Bring students into the position that they can (1) explain the key architectural concepts and methods for modeling software architectures; (2) apply viewpoints and perspectives to model software architectures; (3) discuss the benefits of architecting and the role of the software architect; (4) evaluate and validate software architectures; (5) explain and discuss the concepts of component-based and plugin architectures, service-oriented architectures, and software product lines; (6) explain and recognize technical debt and have an understanding of possible refactorings.	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books; Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; and Coplien and Bjørnvig, "Lean Architecture", Wiley, 2010. Additional reading material will be announced in the lectures.	
Assessment	Grades will be based on lab assignment including essay writing, coding, (video) presentation, peer reviewing, participation. The final grade of the course consists of the following components: Essays (weighting 60%) Coding (weighting 20%) Presentation (weighting 20%) Final grade calculation: 0.6 * Essays + 0.2 * Coding + 0.2 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Essays: re-submit deliverable Coding: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Course web site: https://se.ewi.tudelft.nl/delftswa/index.html	
Co-Instructor	M. Finavaro Aniche	

IN4325	Information Retrieval	5
Responsible Instructor	J. Yang	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language. Knowledge of Web information systems and software engineering can be helpful.	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. The World Wide Web is now the primary source of information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, analysed, and exploited so that each user can quickly find information that is both relevant and comprehensive for their needs. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation of Web and Enterprise search engines. Covered topics include: = Information Retrieval Models; = Indexing Techniques; = Web Search; = Information Seeking Paradigms; = Evaluation of information retrieval systems;	
Study Goals	At the completion of this course, students will be able to: = Describe the different information retrieval models, and compare their strenghts and weaknesses. [Learning Objective 1] = Describe and implement different indexing techniques. [Learning Objective 2] = Describe and analyze querying techniques with respect to their most suited application domains. [Learning Objective 3] = Analyse the effectiveness of an information retrieval system through proper use of evaluation metrics. [Learning Objective 4] = Design and implement (Web) Information Retrieval systems, possibly using advanced social and semantic search functionalities. Support and defend the relevance and correctness the choices with regards to the adopted information retrieval model, indexing technique, and querying technique. [Learning Objective 5]	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 80 hours for the group project and 15 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books - all resources are available on Brightspace.	
Books	Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. Introduction to Information Retrieval. Cambridge University Press, New York, NY, USA. Stefan Büttcher, Charles Clarke, and Gordon V. Cormack. 2010. Information Retrieval: Implementing and Evaluating Search Engines. The MIT Press.	
Assessment	The final grade of the course consists of the following components: Individual assignment (weighting 10%) Weekly individual assignment (weighting 20%) Group project (weighting 70%) Final grade calculation: 0.1 * Individual assignment + 0.2 * Weekly individual assignment + 0.7 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual assignment: re-submit deliverable Weekly individual assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialistievakken start kwartaal 4 2023

CESE4120	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics". We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming.	
Course Contents	The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. After taking this course students will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non--trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	Written reports + project presentation + oral exam Overall, the final grade is determined by: 1) Two intermediate reports (10% of grade each, 2 pages each) 2) Final report (10 % of grade, 4 pages) 3) Final project demonstration (70% of grade) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. There is no resit for this course. repair option:A new but related project will be given for the Report (W), Oral(0) and Demonstration (IP). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

CS4035	Cyber Data Analytics	5
Responsible Instructor	Ir. S.E. Verwer	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course provides theoretical and practical background for applying data analytics in the field of cyber security. Cyber data analytics is a huge field with a great diversity of techniques and applications. The course is centered on a selection of seven such techniques: learning from imbalanced data; behavioral profiling and anomaly detection; sequential data mining; data stream mining; learning from software data; adversarial machine learning; and privacy-aware data mining; Anomaly detection is one of the main topics in cyber security. Specific difficulties that the student will learn to handle are the huge amounts of data and the large number of false positives. Behavioral profiling applies to both people and software processes. Different techniques will be taught to construct profiles from software logs. While building such profiles, care should be taken to not infringe upon the privacy of individuals the data is collected from. Finally, attackers will modify their behavior in order to avoid being detected, a cyber data analytics engineer tries to make their models/profiles robust against such modifications.	
Study Goals	The student will be able to: Apply machine learning to real data Understand and modify machine learning algorithms Learn models from time series Detect anomalies in multidimensional time-series Use distributed processing to speed up machine learning Learn models from data streams with limited memory Learn sequential models Use machine learning for fingerprinting and profiling Preserve the privacy of data owners while learning models Learn robust models that can detect evasive attackers Use machine learning to detect fraud, attacks, and botnets	
Education Method	There will be two lectures for each of the seven topics, and 3 large lab assignments on fraud detection, anomaly detection, and behavioral profiling, and 1 smaller lab on adversarial robustness. There is no exam. Teams of two students will work on these assignments which contain both individual and collaborative components. Deadlines are strict as peer-review will be used to both learn of other possible solutions, provide feedback, and get initial estimates on the obtained grade.	
Assessment	The final grade of the course consists of the following components: Lab assignment fraud detection (weighting 30%) Lab assignment anomaly detection (weighting 30%) Lab assignment behavioral profiling (weighting 30%) Lab assignment adversarial robustness (weighting 10%) Final grade calculation: 0.3 * Lab assignment + 0.3 * Lab assignment + 0.3 * Lab assignment + 0.1 * Lab assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignment fraud detection: Re-submit deliverable Lab assignment anomaly detection: Re-submit deliverable Lab assignment behavioral profiling: Re-submit deliverable Lab assignment adversarial robustness: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4125	Seminar Research Methodology for Data Science	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Instructor	Dr. K.A. Hildebrandt	
Instructor	J. Urbano Merino	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	basic knowledge in mathematics (linear algebra, calculus, probability and statistics)	
Course Contents	The course focuses on research methods for data science. It looks at underlying principles and concepts for data collection, analysis and data processing, as well as the use of tools to do this. The main topics of study are: Conceptualizing research questions and experimental design Frequentist and Bayesian data analysis Generalized linear models for statistical analysis Multilevel modelling for hierarchical and longitudinal data analysis Measuring and sampling, validity and reliability Linear and nonlinear dimensional reduction Principles of statistical testing In the course, students will be using software tools such as R, and Matlab	
Study Goals	The main aims of this module for the student is to achieve an understanding of research methods for data science and obtain practical experience with data analysis and data processing methods. This module provides students with the opportunity to develop and demonstrate their understanding, knowledge, and competence. The learning outcomes for the module are that students will be able to: 1. Appreciate and comprehend strategies for collecting and processing data to answer research questions based on data 2. Understand and reproduce key principles underlying statistical data and data processing analysis 3. Learn to identify and avoid typical biases, paradoxes and misunderstandings in data-driven research 4. Apply and select appropriate data modelling techniques to analyse data and data processing	
Education Method	Lectures/Assignments Some lectures have a flipped classroom set-up, where it is expected that students prepare themselves by watching a series of videos. In the lecture, students work together on a set of classroom challenges under the guidance of the lecturer. In some others, students work on preliminary solutions to some exercises that will be discussed during the lecture. Expected Workload Lectures: 26 hours (13 × 2 hours lectures) Reading time: 39 hours Preparation basis tool use: 25 hours (5 × 5 hours for each tool) Coursework project, including writing reports and preparing for presentation: 50 hours (10 × 5 hours) Total = 140 hours	
Literature and Study Materials	Will be provided online	
Assessment	The final grade of the course consists of the following components: Analysis of experimental research data (weighting 40%) Exploration of real-world data set (weighting 30%) Linear and nonlinear dimensional reduction (weighting 30%) Students work in small groups on the 3 assignments. For the assignment, the student group submits reports and gives a presentation, including a question and answer round where individual group members are assessed on the coursework. Final grade calculation: 0.4 * Analysis of experimental research data + 0.3 * Exploration of real-world data set + 0.3 * Linear and nonlinear dimensional reduction A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Analysis of experimental research data: re-submit deliverable Exploration of real-world data set: re-submit deliverable Linear and nonlinear dimensional reduction: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	NA	

CS4145	Crowd Computing	5
Responsible Instructor	U.K. Gadiraju	
Instructor	A. Bozzon	
Instructor	J. Yang	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic knowledge of artificial intelligence and/or human computer interaction is advised. Proficiency in at least one programming language.	
Course Contents	Crowd Computing is an emerging field that sits at the intersection of computer science and data science. Crowd computing studies how large groups of people can solve complex tasks that are currently beyond the capabilities of artificial intelligence algorithms, and that cannot be solved by a single person alone. It involves the algorithmic engagement and coordination of people by means of Web-enabled platforms. These complex tasks are mainly focused on the creation, enrichment, and interpretation of data, making crowd computing a building block of data science. Examples of such tasks include the coordinated creation of data about real world events when electronic sensors are not available; the annotation of existing data sets to create ground truth data for the training of machine learning algorithms; and the analysis and interpretation of Web data to spot identify inappropriate content (e.g.,hate speech, or fake news). Crowd computing is an essential tool for any data-driven company: from Facebook to Microsoft, from Google to IBM, from Spotify to Pandora, all major companies employ crowd computing to fulfil their data needs, both by involving employees, and by reaching out to anonymous crowds through online marketplaces like Amazon Mechanical Turk or Appen. Crowd computing methods therefore play an important role in the design, development and evaluation of a variety of products, services, and systems in a variety of domains. The objective of the Crowd Computing course is to introduce the scientific and technical underpinnings of crowd computing, and to investigate how it can be used for computer science applications (e.g., information retrieval, machine learning, next-generation interfaces, and data mining) and for real world applications (e.g., cultural heritage preservation, online knowledge creation, smart cities, etc.) The course is designed around one key challenge, the creation and consumption of (high quality) data, and will be organized around three themes: 1) Establishing data needs; 2) Fulfilling data needs with crowd computing; and 3) Evaluating the quality of the retrieved data with respect to the original data need. Covered topics include: 1) Establishing Data Needs: - Requirement Elicitation - Requirement Analysis - User Modelling Properties 2) Fulfilling Data Needs with Crowd Computation: - Systems for/with collective intelligence (e.g., recommendation, semiautonomous systems, citizen science, crowdsourcing, and human computation systems) - Multi-modal Interaction (e.g., conversational systems) - Human Computation (e.g., worker modelling, task modelling, incentives, task assignment, recruitment) - Games with a purpose - Algorithms for Crowd Computing - Computational Methods for User Modelling - Interfaces for Crowd Computing Systems 3) Evaluating Retrieved Data: - Expert Evaluation - User Evaluation - Explanation of the output of Crowd Computing Systems 4) Study of Application Domains When applicable, the course will also feature invited lectures from selected academics and professionals in the field. Since instructors of this course are also directing the Design@Scale Delft AI lab, students of this course will have the opportunity to engage with cutting-edge research projects relevant to this lab. This Crowd Computing course is an elective for students following the Data Science and Technology Track and the Software Technology Track. It adds to the master education offer by addressing topics that are complementary to courses like IN4325 Information Retrieval, IN4252 Web Science & Engineering, CS4065 Multimedia Search and Recommendation, and IN4010 Artificial Intelligence Techniques.	
Study Goals	After this course, students will be able to: - Identify the requirements for a Crowd Computing system [LO1] - Design and develop Crowd Computing systems. Support and defend the relevance and correctness of his/her choices [LO2] - Describe and compare several Crowd Computing techniques. [LO3] - Describe and compare design decisions in the context of Crowd Computing interaction paradigms [LO4] - Determine which Crowd Computing technique(s) is most appropriate for being used in a certain problem domain [LO5] - Apply the appropriate Crowd Computing technique to an application domain and evaluate the obtained results. [LO6] - Analyse the performance of a Crowd Computing system by applying the proper evaluation measures. [LO7]	
Education Method	This course consists of 16 2-hour lectures. Each week, a 30-minute assignment tests the knowledge acquired on the discussed topics. Starting from Week 1, students form groups and work on a project, to be presented in week 9. Students are expected to work 6 hours per week (each) on the project assignment. Expected workload is 32 hours for attending lectures, 24 hours of reading study material and preparing lectures, 55 hours for weekly assignments and group assignment, 24 hours for preparing final survey, and 5 hours for exam and plenary presentations (total 140 hours).	
Literature and Study Materials	Books: - Human Computation. Author(s): Edith Law and Luis von Ahn. Synthesis Lectures on Artificial Intelligence and Machine	

	Learning, June 2011, Vol. 5, No. 3. http://www.morganclaypool.com/doi/abs/10.2200/S00371ED1V01Y201107AIM013 - A. Marcus and A. Parameswaran. Crowdsourced Data Management: Industry and Academic Perspectives. Foundations and TrendsR in Databases, vol. 6, no. 1-2, pp. 1161, 2013. DOI: 10.1561/1900000044. https://people.eecs.berkeley.edu/~adityagp/papers/crowd-book.pdf - An Introduction to Hybrid Human-Machine Information Systems. Demartini, G., Difallah, D.E., Gadiraju, U. and Catasta, M., 2017. Foundations and Trends in Web Science, 7(1), pp.1-87. https://edu.nl/np4th Slides: available on Brightspace Articles: available on Brightspace Recommended reading: - Interaction Design: Beyond Human-Computer Interaction (4th Ed, 2015). Authors: Jenny Preece, Helen Sharp, Yvonne Rogers
Assessment	The final grade of the course consists of the following components: Weekly Individual assignment (weighting 15%) Group Assignment (weighting 55%). The group assignment is performed collectively, but graded individually. Final Individual Assignment (weighting 30%) Final grade calculation: 0.15 * Weekly Individual assignment + 0.55 * Group Assignment + 0.3 * Final Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly Individual assignment: re-submit deliverable Group Assignment: re-submit deliverable Final Individual Assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Tags	Algorithmics Artificial intelligence Design Programming Research Methods Software Technology

CS4205	Evolutionary Algorithms	5
Responsible Instructor	P.A.N. Bosman	
Instructor	Dr.ir. P.A. Bouter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Must have the ability to program, predominantly in Python (but also Java and C/C++ are very useful) so that you can create your own (extensions of) evolutionary algorithms. Further, a basic understanding of key aspects in linear algebra (e.g., matrix multiplications, inversions, decompositions), probability theory and statistics (e.g., normal distribution, statistical hypothesis testing), and complexity theory (e.g., big-O notation), will be very helpful.	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, and real-world applications. The course is planned to have 10 lectures, multiple small practical assignemnts and 1 larger practical assignment. The first practical assignments pertains to experimenting with already implemented EAs on predefined problems. The second practical assignment offers more freedom, allowing you, in a group, to build your own EA (this may vary depending on student numbers and other circumstances).	
Study Goals	Upon successful completion of this course, students will be able to: 1) Explain and apply key concepts underlying the main streams in Evolutionary Algorithm (EA) research, in particular genetic algorithms, evolution strategies, genetic programming, evolutionary multi-objective optimization, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2) Explain and apply key ingredients underlying the rationale of when these algorithms work and when they do not work and several ways to overcome this. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3) Name and explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4) Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5) Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	10 Lectures Several smaller, individual, lab projects One larger lab project in groups of several students	
Literature and Study Materials	Papers and slides that will be made available.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 60%) Lab practical work (weighting 40%) Final grade calculation: 0.6 * written exam + 0.4 * lab practical work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab practical work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Tags	Algorithmics Artificial intelligence Optimalisation	

CS4210-B	Intelligent Decision Making Project	5
Responsible Instructor	Dr. M.T.J. Spaan	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Dr. V. Robu	
Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. J.W. Böhmer	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Instructor	Dr. A. Lukina	
Contact Hours / Week x/x/x/x	0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Theoretical knowledge regarding algorithms for decision making in Artificial Intelligence, obtained for instance by passing one of the following courses: - CS4210-A Algorithms for Intelligent Decision Making - CS4400 Deep Reinforcement Learning - CS4375 Artificial Intelligence Techniques - IN4344 Advanced Algorithms.	
Course Contents	Intelligent decision making is a key skill of computational agents. Research on this topic focuses on building models and algorithms that enable AI systems to take appropriate decisions. Building upon theoretical knowledge gained in other courses, students collaborate in small groups on a distinct research project per group, for instance on decision-making problems in transport, logistics or smart energy grids. Purely algorithmic challenges will also be provided. The research projects provide a good opportunity to learn about topics suitable for Masters projects.	
Study Goals	After completing the Intelligent Decision Making Project course, the student is able to: 1. Apply algorithms for decision making to problem domains, and can compare and evaluate them. 2. Design and implement an extension of a decision-making algorithm. 3. Identify and discuss relevant topics in the research field of algorithms for intelligent decision making. 4. Describe and apply the appropriate research methodology. 5. Communicate his/her findings effectively.	
Education Method	A research project in a small group.	
Literature and Study Materials	Mainly survey papers and book chapters. Details are provided via Brightspace.	
Assessment	The final grade of the course consists of the following components: Quality of work of the research project (weighting 40%) A scientific report of the research project (including peer review of a report) (weighting 20%) Performance during the project (weighting 30%) Oral presentation of the research project (weighting 10%) Final grade calculation: 0.4 * Quality of work of the research project + 0.2 * A scientific report of the research project + 0.3 * Performance during the project + 0.1 * Oral presentation of the research project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Quality of work of the research project: re-submit deliverable A scientific report of the research project: re-submit deliverable Oral presentation of the research project: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Only a limited number of students can participate in this course. In order to be admitted, please submit a short motivation letter (max 200 words) via Brightspace. Attending the first lecture is compulsory.	
Tags	Artificial intelligence	
maximum aantal deelnemers	40	

CS4245	Seminar Computer Vision by Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	N. Tömen	
Instructor	Dr. X. Zhang	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	MSc thesis in the Computer Vision lab	
Expected prior knowledge	Deep Learning (CS4240)	
Course Contents	The recent boom in computer vision and automatic image understanding represents an inflection point in human productivity, permeating wide aspects of the economy and society. Examples of visual tasks which are repetitive or require expert knowledge include medical diagnosis, industrial inspection, autonomous vehicles, etc. When machines can meaningfully assist or even completely take-over such tasks it will change the world as we know it. The breakthrough in the 2012 ImageNet automatic image recognition competition shows all previously existing methods decisively defeated by a deep neural network. Deep learning replaces feature engineering methods and is able to successfully learn image features from huge annotated datasets. This course is on automatically understanding visual content such as images and videos by deep learning. The range of topics (although, not limited to): Fundamentals in Vision, object detection, per-pixel labelings, video recognition, image similarity learning, efficiency, self-supervision, 3D computer vision, adverserial attacks, explainability, generative models, etc. The course will have lectures, a seminar and a lab practical: - The lectures will be on established topics based on the current literature. - The seminar will have students read, critique, and present relevant computer vision research papers. You will have to read 2 papers per week, for 7 weeks (14 papers). - The lab will have students apply and design their own (small) computer vision project. The course build on top of the Deep Learning course (CS4240) and follows a similar setup.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the deep learning techniques reviewed in the course for computer vision applications such as image classification, object detection, per-pixel labelings, video recognition, image similarity learning. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [L04]. Quickly identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (we focus on Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) Computer Vision problem. [LO8]. Write clearly and concisely about your code, method, results, and analyis.	
Education Method	Lectures Lab project: design and execute your own Computer Vision project. Seminar: paper reading, critiquing, and presenting.	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Exam (weighting 60%). Exam about all material. Project (weighting 30%). in a small group of students you work on a computer vision project. Final grade calculation: 0.1 * Presentation + 0.3 * project + 0.6 * Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4280	Language-Based Software Security	5
Responsible Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	This course has no formal prerequisites. However, for the homework assignments you will have to implement several program analysis techniques using the Scala programming language. If you have not used Scala before, you are thus expected to learn the basics of the language through self-study.	
Course Contents	Security vulnerabilities often arise due to programming errors in the source code of an application. Recent programming errors with severe security implications include Heartbleed (buffer over-read), Shellshock (code injection), and goto-fail (ill-formated code). Rather than hunt for individual vulnerabilities in programs, a more structural approach to improve security is to improve the programming language. This is the goal of language-based security: to rule out whole classes of potential security vulnerabilities in one go. This course studies various security properties and program analysis techniques for enforcing these properties at the level of the programming language to improve software security. In particular, we will study the following properties: - Memory safety: prevent buffer overflows and overreads - Type safety: prevent undefined behaviour - Information flow control: prevent data leaks and code injection attacks We will study techniques to address these problems at the language level through dynamic analysis, static analysis, and language design. To facilitate a precise study and comparison, we will define the above techniques formally in class. To facilitate student experimentation and exploration of trade-offs, students will implement the above techniques in homework assignments.	
Study Goals	After taking this course, students should be able to: 1. Describe the nature and causes of security vulnerabilities in software systems, and give concrete examples of how these security vulnerabilities can be exploited. 2. Explain the properties that can be enforced at the level of the programming language to rule out security vulnerabilities, such as memory safety, type safety, and non-interference. 3. Formally define the semantics of a simple programming language. 4. Formally define dynamic and static analysis techniques for enforcing these security properties. 5. Implement these techniques for a small programming language. 6. Discuss and evaluate the importance of soundness and precision of a given program analysis. 7. Contrast programming languages based on the set of countermeasures they provide, and give an appropriate recommendation for a specific application. 8. Analyse and apply results from scientific literature in the area of language based security.	
Education Method	The course work consists of the following activities: 1 or 2 instruction sessions per week. Weekly homework assignments consisting of theoretical questions, programming assignments, and reading assignments	
Assessment	The final grade of the course consists of the following components: Weekly homework assignments (weighting 40%). The weekly homework assignments will test your ability to design and implement (variants of) the techniques discussed in the lectures (study goals 3-5). Written or oral exam (weighting 60%). The final written or oral exam will test your theoretical understanding of the security vulnerabilities and their countermeasures discussed in class (study goals 1-2) and your ability to discuss and contrast the different aspects of these techniques (study goals 6-8). Final grade calculation: 0.4 * Weekly homework assignments + 0.6 * Written or oral exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written or oral exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4290	Seminar on Distributed Machine Learning Systems	5
Responsible Instructor	Dr. Y. Chen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Machine learning systems are often conventionally designed for centralized processing in that they first collect data from distributed sources and then execute algorithms on a single server. Due to the limited scalability of processing large amount of data and the long latency delay, there is a strong demand for a paradigm shift to distributed or decentralized ML systems which execute ML algorithms on multiple and in some cases even geographically dispersed nodes. The aim of this seminar course is to let students learn how to design and build distributed ML systems via paper reading, presentation, discussion, and project prototyping. We provide a broad overview on the design of the state-of-the-art distributed ML systems, with a strong focus on the scalability, resource efficiency, data requirements, and robustness of the solutions. We will present an array of methodologies and techniques that can efficiently scale ML analysis to a large number of distributed nodes against all operation conditions, e.g., system failures and malicious attacks. The specific course topics are listed below. The course materials will be based on a mixture of classic and recently published papers. For each topic, the basic concepts and technology landscape will be first provided and then two state-of-the art of papers will be presented and discussed by students. We offer a testbed of a distributed (deep) ML system in which students are encouraged to apply different techniques to jointly improve its scalability and resilience. Course topics include Overview of distributed machine learning systems Performance and scalability of state-of-the-art systems Acceleration of ML workloads Slim distributed ML systems on small data Robust deep learning systems Federated machine learning systems	
Study Goals	Students are able to argue and reason about distributed ML from a systems perspective. Students understand the behavior and tradeoffs of distributed ML in terms of performance and scalability. Students can estimate the importance of data inputs via different techniques, i.e., core set and decomposition methods, for distributed ML systems. Students understand data poison attacks and design defense strategy for distributed ML systems. Students can analyze the state-of-the art federated machine learning systems and design the failure-resilient communication protocols. Students are able to design and implement methods and techniques for making distributed ML systems more efficient.	
Education Method	Lectures: 7 weeks X 2h Papers: one paper presentation, two paper reviews, and paper discussion. Practical: apply system and algorithmic optimization techniques learned in the lecture to improve the performance of distributed machine learning systems, e.g., image recognition on CIFAR 10. The testbed environment, learning algorithms, and dataset will be given. Deliverables include git commit of functioning code and a report summarizing the contribution	
Assessment	The final grade of the course consists of the following components: Paper presentation by group (10%): each group of 2 to 3 students needs to choose from a given set a papers to present (15 minutes) and lead the discussion (10 minutes). Paper reviews by individual (30%): each student needs to write three reviews of papers assigned from a given set and those two papers have to be different from the paper for the presented. Each review will account for 10% of the grade. Questionnaires by individual (0%): each student needs to hand in a list of questions at the beginning of the lectures that have paper presentation from other students. This is not graded. Individual project (60%): The project is collaborative among the entire class and competitive as a whole group. The objective is to continuously improve the performance of the given distributed ML system. The students need to hand in a final project report in style of a short scientific paper, stating their individual contribution to the overall system performance. Final grade calculation: 0.1 * Paper presentation group + 0.3 * Paper presentation individual + 0.6 * Individual project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper presentation by group: resit Paper reviews by individual: resit Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4295	Release Engineering for Machine Learning Applications	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Course Contents

The world of Software Engineering has been revolutionized in the last decade. Instead of releasing software updates yearly, companies can now release multiple times per week, sometimes even per day, to their customers. This allows quick reactions to market demands, software failures, and it is crucial for increasing the business value of software. These improvements have been mostly enabled by advances in release engineering and, in this course, we will learn about the techniques and technologies that build the foundation for modern release engineering.

This course is addressing two main target audiences: 1) software engineers that would like to learn how to systematically release their applications and make them scalable in an orchestrated environment and 2) machine learning experts that would like to learn how to make their proof-of-concept applications production ready.

Together, we will go on a journey that starts at continuous integration and then moves on to continuous delivery, continuous deployment, and continuous experimentation. We will discuss the theory and the current research on various related subjects like containerization, testing, or monitoring and will put the learned theory into practice. As a running example, we will build a pipeline for a machine learning application, which -compared to traditional release engineering- poses additional challenges, like data versioning or model deployment.

Important: Make sure to register for a group in Brightspace AT THE LATEST during the first lecture.

Study Goals

After following this course, you will be able to...

Explain the concepts of DevOps and MLOps

Automate the development and operation process
... Automate packaging, versioning, and releasing in modern build environments like GitLab or GitHub
... Release artifacts in public registries like the GitHub Maven Registry.
... Provision servers with Vagrant and Ansible
... Automate the model training pipeline using DVC

Use modern containerization technology to build a scalable micro-service architecture
... Use Docker to isolate a complex, web-based application in a container
... Use Kubernetes to orchestrate a non-trivial, distributed application
... Create an Istio service mesh in Kubernetes that enables custom routing strategies
... Use Prometheus and Grafana to monitor the state of a Kubernetes-based application

Deploy and maintain machine learning models in production
... Embed a typical ML model in an HTTP micro-service
... Apply version-control techniques to data and models in a machine learning project
... Implement quality-control techniques in a machine learning application
... Design a testing strategy that covers data, model, infrastructure, and monitoring aspects

Document the state and extensions of release engineering pipeline
... Find related work and inspiration for release engineering tasks
... Identify extension opportunities in existing release pipelines

Education Method

Following this course will require you to...

Follow interactive lectures and pre-recorded material for self-study
Execute a programming project in small teams
Transfer lecture contents into a project through practical assignments
Assess the work of yourself and others through rubric-supported peer-feedback
Actively participate in Q&A sessions
Document a release engineering pipeline in a report
Present your project results and answer questions about the lecture contents.

Assessment

--- Formative Assessment ---

You can check your individual learning progress through weekly rubrics
... You will receive feedback from other groups as a neutral outsider
... You can check your own progress via self-assessment
Joint Q&A sessions will discuss common points of the peer feedback and concrete questions

--- Individual Knock-Out Elements ---

The course enforces several criteria to protect the active members of a team. Students not reaching the following criteria will be removed from the course:

Upholding a professional attitude when interacting with peers and staff.
Actively participating in the team process and contributing to all team deadlines and deliverables
Meaningful and visible project contribution throughout each of the seven project weeks (W2-W4, W6-W9)
... 1+ commit to one of the team repositories
... Create 1+ PR and approve 1+ PR that is not your own
... The changes must contain a meaningful contribution to code, config, or report
Constructive participation in all peer reviews for assignments (one for another team and one for your own)
Ability to explain lecture contents and defend all aspects of the team project in the oral examination
Availability of team results for grading (repositories, report, artifacts)

We offer repair options for the following criteria:

Not having meaningful and visible contributions in one week can be repaired in the following week. Failing to satisfy the criteria in more than one week will lead to removal.
Failing to submit a peer review in time can be repaired in the following week. Missing a review a second time will lead to removal
The inability to explain or justify the project in the oral examination can be repaired in a second oral examination.

	--- Summative Assessment --- Students attending the course will be graded on the following grading components. ALL graded components are rounded to .1 and must at least be "almost passed" (>5.0). Their weighted average will be rounded to .5 to calculate the final team grade. Project ... Versioning & Releases (10%) ... Containers & Orchestration (10%) ... Kubernetes & Monitoring (10%) ... Provisioning (10%) ... Istio Service Mesh (10%) ... ML Configuration Management (10%) ... ML Testing (10%) Report (30%) Presentation & Oral Examination (Pass/Fail) In case of large performance differences in a group, we may choose to assess the team members individually, based on the individual contributions to the deliverables and the interaction traces that we find in the repository. --- Please Note --- All assignment deadlines throughout the course are fixed, we do not accept late submissions. Late submissions in systems that do not enforce due dates will be treated as not submitted (e.g., Brightspace, Buddycheck). Failing to submit a deliverable cannot be repaired. Insufficient deliverables (>=4.0 and <6.0) can be repaired, but the grade of the deliverable is then capped at 6.0. There is NO resit opportunity for this course. Partial grades are not carried over to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Enrolment / Application	In addition to your regular enrollment of the course, you must also register in Brightspace, AT THE LATEST during the first lecture, so you can be assigned to a group. Late registrations after the group creation cannot be considered anymore.
Co-Instructor	Prof.dr. A.E. Zaidman

CS4350	Machine Learning for Graph Data	5
Responsible Instructor	M. Khosla	
Responsible Instructor	E. Isufi	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics Machine Learning: data division in train-validation-test, overfitting, kernels, regularization; Basics Deep Learning: back-propagation, stochastic gradient descent (SGD), multi-layer perceptron. Basics in programming: Python or equivalents, minimal experience with Pytorch / TensorFlow or equivalents is helpful but not strongly expected.	
Course Contents	Graph data are present in a myriad of modern computer sciences systems and applications. Examples include data generated over social, brain, financial, power, water and sensor networks. Because these data have a complicated structure they require different tools conventionally used in machine and deep learning to develop end to end solutions. These solutions falls generally under the umbrella of graph-based machine learning and can be used to perform recommendations, detect anomalies in the brain, predict financial crisis, estimate the sate of a power or water network, and coordinate group of autonomous moving sensor, to name a few. This course deals with the foundations and principles of machine and deep learning for network data. Topics include: unsupervised and semi-supervised learning on graphs; graph representation learning; graph signal processing; graph convolutions; graph neural networks; spatiotemporal learning on graphs; scalable algorithms; explainability.	
Study Goals	By the end of this course, you should be able to: Lo1: Recognise key challenges in learning from graph-structured data. You are able to recognise challenges in real applications involving graph-based data, explain them with a machine learning language, and discuss strategies to solve them. You are able to identify the issues of lack of interpretability in graph-based machine learning models and describe some possible solutions. You are also able to explain a few techniques to improve scalability of existing techniques for large graphs. Lo2: Explain and compare representation learning techniques on graphs. You are able to explain, and compare different graph-based machine learning techniques, identifying advantages and limitations in each of them, and discuss the latter with respect to a specific application. Lo3: Implement and analyse machine and deep learning techniques for graph data. You are able to apply different graph-based machine learning techniques including graph kernels, graph neural networks, graph-time learning solutions to real data. You are also able to analyse the different trade-off of these solutions, compare them with each other and inspects their behaviour. LO4: Design machine learning solutions for real world graph data tasks. You are able to create a novel solution for a practical graph-based challenge, discusses different graph-machine learning strategies for solving the task, test one or more of them, and contrast the performance of the developed approach with alternative baselines.	
Education Method	The course consists of: - 12 lectures in which the main concepts are explained and discussed; - 3 hands-on homework labs, in which practical assignments need to be solved individually; - At least two Q&A and feedback sessions about the final project;	
Literature and Study Materials	The instructors will provide online accessible material for the specific lectures. Also the slides used during the lectures are made available via Brightspace. Practical final project examples will be made available for download via Brightspace. During the course several pointer to on-line video lectures will be given.	
Assessment	The final grade of the course consists of the following components: Lab assignments (Pass/Fail). The labs assignments must be solved individually and consists of only pass/fail grade. Students can solve their assignment at home or during the lab sessions. During the labs instructors and TAs (if available) will provide feedback on the questions regarding the assignment. Students need to pass all the three lab assignments before the final project submission to pass the course. Final project (weighting 100%). The final project will be solved in groups and a group grade 1-10 will be given. The project grade will be evaluated on both the written final document as well as the question-answering session during the final interview. Final grade calculation: P/F * Lab assignments + 1 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignments: re-submit deliverable Final project: resit Q&A Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4360	Natural Language Processing	5
Responsible Instructor	J. Yang	
Instructor	Dr. C. Lofi	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Proficiency in at least one programming language. Knowledge of basic algebra. Knowledge of Web Information Systems and Machine Learning can be helpful.	
Course Contents	Natural Language Processing (NLP) concerns the interaction between computers and human language, in particular how to build systems to process and analyze large amounts of human language data. It has become an important technology in many parts of computer science, facilitating key applications of large-scale text understanding, automatic question answering, knowledge base construction, information retrieval, text or speech-driven dialogue agents, translation, text summarization, etc. This course aims to introduce the principles and techniques for Natural Language Processing. Through the lectures, assignments, and the group project, students will gain an understanding of classic and modern techniques for language understanding and generation and learn the skills to design, develop, and evaluate their own Natural Language Processing systems and applications. Covered topics include: = Lexical, Syntactic and semantic analysis = Language models = Word, and subword embedding techniques = Attention mechanism and Transformers = Transfer Learning = Knowledge-driven language processing = NLP applications (e.g., QA, summarization, machine translation) = Responsible NLP (explainability, fairness) = Evaluation of NLP systems	
Study Goals	At the completion of this course, students will be able to: [LO1] Recognize typical NLP tasks and components of an NLP system [LO2] Describe lexical analysis techniques such as language modeling and smoothing [LO3] Describe syntactic and semantic analysis techniques [LO4] Describe and compare different token embedding techniques, common neural architectures (including attention mechanisms) for text processing and their implementation in neural language models (e.g., GPT-3 and BERT). [LO5] Describe information extraction techniques and its application to NLP tasks [LO6] Apply NLP techniques to information retrieval, question answering, text summarization [LO7] Explain the implications of data-driven models in terms of transparency, accountability, and fairness [LO8] Design, develop, and evaluate NLP systems, possibly using neural methods with semantic enrichment and user adaptation functionalities. Support and defend the relevance and correctness of the design choices.	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 65 hours for the group project and 30 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books, authentic blog posts - all resources are available on Brightspace.	
Books	Dan Jurafsky and James H. Martin. 2014. Speech and language processing. Pearson. Yoav Goldberg. 2015. A Primer on Neural Network Models for Natural Language Processing. Steven Bird, Ewan Klein, and Edward Loper. 2009. Natural language processing with Python: analyzing text with the natural language toolkit. " O'Reilly Media, Inc.".	
Assessment	The final grade of the course consists of the following components: Weekly individual assignment (weighting 20%) Group project (weighting 80%). The group assignment is performed collectively, but graded individually. Final grade calculation: 0.2 * Weekly individual assignment + 0.8 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly individual assignment: re-submit deliverable Group project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor anaylsis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: 0.5 * Oral discussion 1 + 0.5 * Oral discussion 2. Both partial grades must be sufficient, i.e., .a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case the take-home assignments are insufficient, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4030	Error Correcting Codes	4
Responsible Instructor	Dr.ir. J.H. Weber	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic linear algebra and probability theory, as being treated in B.Sc. programmes like Electrical Engineering, Computer Science, and Mathematics	
Course Contents	Introduction into error-correcting codes; mathematical basics; block codes fundamentals; cyclic codes; co-operating codes; soft-decision decoding; convolutional codes; iterative decoding (turbo codes, LDPC codes); applications.	
Study Goals	The general overall objectives of this course are that the students can - reproduce the error control concepts as presented in the lecture notes; - explain the meaning and usefulness of these concepts; - apply the concepts to transmission and storage systems, including basic calculations, at the level of the exercises in the lecture notes and previous exams. More specifically, the students are able to - discuss the basic trade-offs between efficiency, reliability, and complexity in designing coding schemes for transmission and storage systems; - apply finite fields theory in order to construct (cyclic) codes; - encode data into code sequences that are resistant to a certain number of random and/or burst errors; - decode received sequences using (iterative) hard or soft decision techniques.	
Education Method	Lectures; expected workload is 20 hours attending lectures, 60 hours preparing for the lectures, studying the lecture notes, and making suggested exercises, and 32 hours for preparing and making the exam.	
Literature and Study Materials	Lecture notes "Error-Correcting Codes" by J.H. Weber, available on Brightspace. Also previous exams as well as demos are available via Brightspace.	
Assessment	The final grade will be fully determined by a scheduled written exam, which will be held at the end of Q4. The resit will take place as an oral exam in Q5, on appointment with the lecturer. The exam format is closed-book in any case. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	Actual course information available on Brightspace.	

IN4255	Geometric Data Processing	5
Responsible Instructor	Dr. K.A. Hildebrandt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic knowledge in mathematics (linear algebra, calculus): CSE1205, CSE1200 or comparable courses. Students who haven't followed any of these courses can follow the course, but should be willing to invest more time.	
Course Contents	Geometric data processing is concerned with the representation, analysis, manipulation, and optimization of digital shapes. Thanks to the advances in 3D acquisition and manufacturing technologies (like 3D-Scanning and 3D-printing), the usage of geometric data is continuously increasing and an efficient processing of digital shapes plays an important role for a variety of applications in areas such as computer graphics, computer-aided design and engineering, medical imaging and surgery planning, architecture, and entertainment. In this course, we will study concepts and algorithms for creating, analyzing, editing and optimizing digital geometric shapes.	
Study Goals	After successfully completing this course, the student is able to: - describe the fundamental techniques used for representing, analyzing, processing and modeling digital 3D-shapes treated in the course and to explain the mathematical and algorithmic concepts associated with them - apply the learned mathematical concepts to solve basic geometric problems arising in geometric modeling applications - design algorithms that can solve simple geometric processing tasks and evaluate the drawbacks, benefits and limitations of the proposed algorithms - implement the designed algorithms in a geometric processing software framework	
Education Method	The course combines lectures, tutorials, practical project work, and homework assignments.	
Literature and Study Materials	References to textbooks and recent research and survey papers are given in the lectures.	
Assessment	The final grade of the course consists of the following components: Practical projects (weighting 60%) Theoretical assignment (weighting 40%) Final grade calculation: 0.6 * Practical projects + 0.4 * Theoretical assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Practical projects: re-submit deliverable Theoretical assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4331	Web-scale Data Management	5
Responsible Instructor	Dr. A. Katsifodimos	
Instructor	Dr. C. Lofi	
Instructor	Dr. R. Hai	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor level courses in database management systems and operating systems. A prior course in distributed systems or middleware would be helpful but is not required. Programming skills are important for the final assignment.	
Course Contents	This course addresses the challenges of Data Management at Web-scale. Especially, it covers the need for large-scale distributed data storage systems. The lecture therefore introduces step-by-step increasingly complex distributed storage systems, leading up to modern implementations of different NoSQL data storage systems. The challenges arising from such systems are presented and discussed, especially focusing on the CAP theorem and the resulting trade-offs with respect to data models, transactional power, query expressivity, and replication consistency. These discussions lead to different variants of NoSQL database systems, like Key-Value Stores, Document Stores, Wide-Columnar stores, and Graph Databases. The advantages, disadvantages, and general properties of these systems are discusses in more detail. There is special focus on distributed transactions and consistency guarantees of different data management systems and methods.	
Study Goals	At the end of this course the student can - assess the nature of a given storage problem, and can select a suitable technology for solving it - understands the different data models encountered in Web Data Management, and their impact on modelling and querying - understands the issues arising from distributing and replicating data, especially with respect to the CAP theorem -understands the trade-offs which can be chosen within the design space of the CAP theorem -categorize and explain modern NoSQL databases within the framework of the previously mentioned trade-offs	
Education Method	Lectures and assignments	
Literature and Study Materials	Course slides and Lecture Videos	
Books	Literature mentioned in the lecture, mostly research papers.	
Assessment	There is a group assignment (80%) and an individual exam (20%). In order to pass the class, a student has to have a grade of at least 5.0 for the group assignment as well as the individual exam, with a weighted average of 5.8 overall. The exam can be resit in the following academic quarter. If the student achieves a 4.0 - 6.0 in the group assignment, a repair opportunity can also be given (e.g., extra work, corrections on the group assignment, etc.).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Track Artificial Intelligence Technology 2023

Introduction 1

The Artificial Intelligence Technology MSc Track will cover the algorithmic foundations of AI systems, addressing topics in machine learning and intelligent algorithms, but also foundational topics in system and software engineering and data management. A wide selection of specializations allows you to cater and steer the study programme according to personal preferences, and allows to focus on chosen core technologies or application areas.

Artificial Intelligence (AI) is currently at the center of a plethora of disruptive societal and scientific innovations and is predicted to be one of the key enabler technologies for the coming decade. Applications range from intelligent personal assistants, to self-driving cars, smart infrastructures and cities to smart industries, but also cover health, education, and legal applications. These developments are driven by artificial intelligence systems - software systems which combine data with intelligent algorithms embedded in a larger software-ecosystem of the application domain. In the Artificial Intelligence Technology (AIT) track of the Computer Science MSc programme, you will learn to analyze and evaluate intelligent algorithms as well as how to design and engineer AI software systems.

THE CURRICULUM
1. An Individual Exam Programma (IEP) in this track consists of
a. a common core,
b. courses offered by the faculty EEMCS,
c. a seminar offered by the programme CS or a Literature survey (IN4306)
d. free electives,
e. a thesis project (IN5000 Final project) worth 45 credits and
f. if required, homologation.

The IEP must be drawn up in agreement with the thesis coordinator of the research group in which the student wishes to carry out his or her thesis project. The thesis coordinator is a member of the scientific staff of that research group.

The seminar of the research group in which the thesis is performed or the Literature Study (IN4306) is part of said IEP.

Free elective courses
- the number of credits spent on free electives in said IEP is no higher than 25 credits,
- the number of credits spent on homologation in said IEP is no higher than 15 credits,
- at least 40 credits of the courses in the IEP (notwithstanding the thesis project) should be computer Science courses. A list of these courses is published annually in the digital study guide.

Free electives - language course list:
Up to 3 credits may be spent on language courses. These may only be chosen if required. Placement tests showing the necessity to take one or more of these courses must be taken and submitted to the master coordinator.

TPM018A English Grammar for the University 2 EC
TPM303A Intermediate Writing in English for the University 2 EC
TPM304A Advanced Writing in English for the University 2 EC
TPM305A Writing a Masters Thesis in English 2 EC
WM1115TU Dutch Elementary 1 3 EC
WM1116TU Dutch Elementary 2 3 EC
WM1117TU Dutch intermediate 1 3 EC
WM1135TU Advanced English for the University 3 EC

The free elective space may also be used for an extra project:
TUD4040 Joint Interdisciplinary Project (JIP) 15 EC
IFEEMCS520200 Interdisciplinary Advanced Artificial Intelligence Project 15 EC

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Common Core AIT 2023	
Introduction 1	Common Core MSc CS - DST (at least 20 EC): Choose 4 out of 8. Students follow the masters programme of the year in which they have started their programme. E.g. students who started their masters programme in 2021-2022 follow the Teaching and Examination Regulations (TER) of 2021-2022.

CS4205	Evolutionary Algorithms	5
Responsible Instructor	P.A.N. Bosman	
Instructor	Dr.ir. P.A. Bouter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Must have the ability to program, predominantly in Python (but also Java and C/C++ are very useful) so that you can create your own (extensions of) evolutionary algorithms. Further, a basic understanding of key aspects in linear algebra (e.g., matrix multiplications, inversions, decompositions), probability theory and statistics (e.g., normal distribution, statistical hypothesis testing), and complexity theory (e.g., big-O notation), will be very helpful.	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, and real-world applications. The course is planned to have 10 lectures, multiple small practical assignemnts and 1 larger practical assignment. The first practical assignments pertains to experimenting with already implemented EAs on predefined problems. The second practical assignment offers more freedom, allowing you, in a group, to build your own EA (this may vary depending on student numbers and other circumstances).	
Study Goals	Upon successful completion of this course, students will be able to: 1) Explain and apply key concepts underlying the main streams in Evolutionary Algorithm (EA) research, in particular genetic algorithms, evolution strategies, genetic programming, evolutionary multi-objective optimization, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2) Explain and apply key ingredients underlying the rationale of when these algorithms work and when they do not work and several ways to overcome this. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3) Name and explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4) Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5) Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	10 Lectures Several smaller, individual, lab projects One larger lab project in groups of several students	
Literature and Study Materials	Papers and slides that will be made available.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 60%) Lab practical work (weighting 40%) Final grade calculation: 0.6 * written exam + 0.4 * lab practical work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab practical work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Tags	Algorithmics Artificial intelligence Optimalisation	

CS4210-A	Algorithms for Intelligent Decision Making	5
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. V. Robu	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: CS4375: Artificial Intelligence Techniques, or equivalent; and/or IN4344: Advanced Algorithms, or equivalent Required: basic course(s) in algorithm design and analysis, logic and probability; basic programming (in Python)	
Course Contents	Decision making is at the centre of artificial intelligence. This course gives you practical skills on a solid theoretical base. The course looks at solving mathematical models of NP-hard discrete optimisation problems. These kinds of problems lie at the heart of AI techniques such as planning, machine learning and mechanism design, and more generally combinatorial optimisation. You will learn about a range of modelling techniques from boolean satisfiability to constraint programming, and how advanced solvers for these models work. The course has plenty of real-world case studies as well as theoretical results. Apply the skills you learn in this course by taking CS4210-B: Intelligent Decision Making Project in quarter 4!	
Study Goals	By the end of this course, you will be able to identify features of real-world combinatorial decision problems, and be able to model and design systems for simplified instances of these problems using boolean satisfiability, mixed integer programming, and constraint programming over finite and real domains. You will be able to explain how SAT, CP and LCG solvers work in some detail, and how MIP solvers work at a high level. You will also be able to explain the basics of program synthesis and combinatorial game theory.	
Education Method	Lectures, homework exercises (optional, and not counted towards the course grade), and programming assignment(s) (mandatory, but not counted towards the course grade). The expected workload is: 30% lectures (including preparation for the exams) 30% homework exercises (optional) 40% programming assignments	
Literature and Study Materials	Provided on Brightspace	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Besides the summative assessment, formative coursework will be offered. This includes a mandatory assignment, performed in pairs. Students must submit a non-trivial attempt at the assignment in order to pass the course. The assignment receives formative feedback, but is not counted in computing the course grade. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Elective	Yes	
Tags	Algorithmics Artificial intelligence Group work Modelling Optimalisation Programming Projects Small groups	

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4240	Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Seminar Computer Vision by Deep Learning	
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy). Experience in programming and installing programming languages/libraries is required. Without affinity/experience with python (and numpy) programming this course will become too time consuming as the project will become difficult.	
Course Contents	In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.	
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.	
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ - https://udlbook.github.io/udlbook/ Research papers will be made available through Brightspace. Assignments are based on PyTorch: https://pytorch.org/	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Digital Exam (weighting 90%). Exam about the papers, lab assignments, and the theory (lectures). Project (Pass/Fail). In a small group of students you work on a deep learning paper reproducibility project. Final grade calculation: (pass project) * (0.1 * Presentation + 0.9 * Exam) A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4270	Conversational Agents	5
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Responsible Instructor	F. Broz	
Instructor	M.L. Tielman	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming skills (e.g. Python and Java) Probability theory and statistics	
Course Contents	Chatbots, embodied and conversational virtual agents, and social robots are becoming more and more popular. Many people are owning an Alexa, Cortana or Echo or are talking to their virtual assistant on their phone. Indeed, such technologies have the potential of making our lives easier and relieve people from the more repetitive tasks. For example, it is imaginable that such systems are being used for financial applications by helping customers with frequently asked questions but also to advise them on in the long term more impactful decisions such as their pension plans. Further applications can be imagined in the area of healthcare and education, some of which are already in existence today. In this course, attention will be given to different verbal and nonverbal behavioral characteristics, like speech, intonation, gaze and gestures that humans show when communicating with both other people and machines. This behavior is then related to different dialogue functions, including turn-taking, addressing others, and backchanneling, that give shape to the communication process. This course introduces conversational agent technology. We cover agent related technologies which can be grouped into: Dialog Management NLP speech synthesis social robotics	
Study Goals	After this course you have learned to: 1) Apply relevant linguistic and psychological theory to conversational agent systems 2) Design the conversational memory of your agent focusing on a perception/generational model 3) Analyse human-human conversational data to better design ML models for adding generation or perception capabilities to conversational agents 4) Develop the conversational memory in your agent 5) Evaluate the effects of affect and embodiment on human-agent interaction 6) Discuss your results and the limitations of your implementation	
Education Method	There are 2 lectures scheduled per week. Students work in groups of 4 on a group project. Lectures: 26 hours (13 × 2 hours lectures) Reading time: 39 hours Preparation basis tool use: 25 hours Coursework project, including writing report and prepare a video demonstration: 50 hours (10 × 5 hours)	
Literature and Study Materials	We use the book "The conversational interface " by Michael McTear, Zoraida Callejas, David Griol. This book is freely available through the TU Delft library. https://link-springer-com.tudelft.idm.oclc.org/book/10.1007%2F978-3-319-32967-3 Other relevant material will be provided on Brightspace.	
Assessment	The final grade of the course consists of the following components: Online Examination (weighting 30%) Group Assignment (weighting 70%). This assignment will result in a group report and a video demonstration. 3 Lab assignments (pass/fail) Final grade calculation: 0.3 * online examination + 0.7 * Group assignments + P/F * Lab assignments A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Online examination: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Group Assignment: re-submit deliverable Lab assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4375	Artificial Intelligence Techniques	5
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. M.T.J. Spaan	
Instructor	L. Cavalcante Siebert	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	We expect students to have programming expertise at the Bachelor level of Computer Science, in particular knowledge is expected of algorithms (e.g. search algorithms), and probability theory. Programming skills are required for the practical assignments which can be done in Python.	
Course Contents	The course focuses on concepts and techniques from probabilistic Artificial Intelligence for building intelligent agents, and decision making and -support systems. The specific topics include Bayesian networks, inference, belief monitoring, (partially observable) Markov decision processes, reinforcement learning and multiagent systems.	
Study Goals	After completing the course, student will be able 1. to comprehensively summarize the covered topics within probabilistic artificial intelligence and discuss their main issues; 2. to explain the main concepts and techniques; 3. to relate practical problems to appropriate AI techniques and apply the formal models covered in the course to these problems; 4. to implement and evaluate algorithms for complex decision making.	
Education Method	Lectures, lab work (programming assignments).	
Literature and Study Materials	Stuart J. Russel and Peter Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Prentice-Hall. ISBN 9781292401133 Global edition. See website http://aima.cs.berkeley.edu/global-index.html for additional information that goes with the book.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	The practical assignments will be done in groups of 4 to 5 students. The ethical standards of working are expected of all students. Work can be divided over students, but all students are responsible for the overall quality and originality of the work. Students that do not do their share of the work have to be reported by the group to the supervisor so that action can be taken.	
Tags	Artificial intelligence	

IN4315	Software Architecture	5
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software engineering	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four, in which the actual architectures of existing open source systems are analyzed in considerable detail. These systems are taken from github, and student teams are challenged to actually contribute to the systems under analysis in the course.	
Study Goals	Bring students into the position that they can (1) explain the key architectural concepts and methods for modeling software architectures; (2) apply viewpoints and perspectives to model software architectures; (3) discuss the benefits of architecting and the role of the software architect; (4) evaluate and validate software architectures; (5) explain and discuss the concepts of component-based and plugin architectures, service-oriented architectures, and software product lines; (6) explain and recognize technical debt and have an understanding of possible refactorings.	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books; Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; and Coplien and Bjørnvig, "Lean Architecture", Wiley, 2010. Additional reading material will be announced in the lectures.	
Assessment	Grades will be based on lab assignment including essay writing, coding, (video) presentation, peer reviewing, participation. The final grade of the course consists of the following components: Essays (weighting 60%) Coding (weighting 20%) Presentation (weighting 20%) Final grade calculation: 0.6 * Essays + 0.2 * Coding + 0.2 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Essays: re-submit deliverable Coding: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Course web site: https://se.ewi.tudelft.nl/delftswa/index.html	
Co-Instructor	M. Finavaro Aniche	

IN4325	Information Retrieval	5
Responsible Instructor	J. Yang	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language. Knowledge of Web information systems and software engineering can be helpful.	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. The World Wide Web is now the primary source of information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, analysed, and exploited so that each user can quickly find information that is both relevant and comprehensive for their needs. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation of Web and Enterprise search engines. Covered topics include: = Information Retrieval Models; = Indexing Techniques; = Web Search; = Information Seeking Paradigms; = Evaluation of information retrieval systems;	
Study Goals	At the completion of this course, students will be able to: = Describe the different information retrieval models, and compare their strenghts and weaknesses. [Learning Objective 1] = Describe and implement different indexing techniques. [Learning Objective 2] = Describe and analyze querying techniques with respect to their most suited application domains. [Learning Objective 3] = Analyse the effectiveness of an information retrieval system through proper use of evaluation metrics. [Learning Objective 4] = Design and implement (Web) Information Retrieval systems, possibly using advanced social and semantic search functionalities. Support and defend the relevance and correctness the choices with regards to the adopted information retrieval model, indexing technique, and querying technique. [Learning Objective 5]	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 80 hours for the group project and 15 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books - all resources are available on Brightspace.	
Books	Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. Introduction to Information Retrieval. Cambridge University Press, New York, NY, USA. Stefan Büttcher, Charles Clarke, and Gordon V. Cormack. 2010. Information Retrieval: Implementing and Evaluating Search Engines. The MIT Press.	
Assessment	The final grade of the course consists of the following components: Individual assignment (weighting 10%) Weekly individual assignment (weighting 20%) Group project (weighting 70%) Final grade calculation: 0.1 * Individual assignment + 0.2 * Weekly individual assignment + 0.7 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual assignment: re-submit deliverable Weekly individual assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialistievakken start kwartaal 1 2023

CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ------------------------ Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. -- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) o Definition of ad-hoc networks o Comparison with infrastructure based systems o Typical applications o Advantages and challenges o Radio technologies for ad-hoc networks o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks o Propagation models o Topology models based on graph theory o Degree and hopcount o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) o Introduction to MAC protocols o Issues and design goals o Classification o Directional, muli-channel MAC protocols o Energy efficiency in MAC protocols o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) o Flooding o Node discovery, neighbour discovery o Route establishment o Topology maintenance, localisation o Proactive, reactive and hybrid routing o Typical protocols o Energy efficiency in routing o Broadcast and multicast o Effects of mobility on connectivity and capacity o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks o Wireless sensor networks (Chapter 12 of the textbook and papers) o Cooperation (Reference papers) o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consisting of slides presented at the lectures (Slides are only teaching aid and they are not substitute for	

	textbooks, research papers, etc). 3. Some recent journal papers 4. Optional Reference Books 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms) 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar , C Puttamadappa , and T. G Basavaraju, Auerbach Publications, 2008. This book is avaliable online in the library. 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is avaliable online in the library. 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer. 5. OPNET/ns-2 web pages, tutorials and video lectures
Books	Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.
Assessment	1. There will be totally three written/oral tests/examinations for this course (including the final exam). 2. The students will carry out a project in a group and submit a short report. 3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis. Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class. Breakup: 2 Tests + Final Exam = 55% Project 40% Self assessment + Reflection 3% Activities on Feedback Fruits or any Online platform 2% Resit: written/oral exam [+ carry over from projects]. == Changes due to COVID19 in 2021 is below; in case there is a similar situation, we use the following methods. == There are three chapters Chapter 1: Network modelling Chapter 2: MAC protocol Chapter 3: Routing + Lab: Simple experiments using laptops (individual) + simulations (group-wise) Marks Breakup -------------- Homework/Assignments/Group works Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks Part B1: Individual experiment + report Part B2: Group-wise simulations + report + demo Part B1 + Part B2 - 50 Marks Part C: Self-assessment + peer activities -- 5 marks Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit. == (More information will be given in the first class) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Permitted Materials during Tests	Different conditions for different test/exams. Conditions will be informed 1 week before the exams/test.

CS4070	Multivariate Data Analysis	5
Responsible Instructor	Dr. J. Söhl	
Responsible Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	

Expected prior knowledge

Required:

* Introduction Probability Theory and Statistics: see for instance

A Modern Introduction to Probability and Statistics
Understanding Why and How
Series: Springer Texts in Statistics
Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E.
2005, XVI, 488 p. 120 illus., Hardcover
ISBN: 1-85233-896-2

* Basic calculus

* Linear Algebra: matrix multiplication, the inverse of a matrix, the transpose of a matrix, least square solution.
see:

David C. Lay: Linear Algebra and Its Applications
ISBN-10: 0321385179 ISBN-13: 9780321385178
©2012 Pearson)

Course Contents

PART I:
Stochastic models will be developed on the basis of probability theory. Probability theory describes the behavior of certain phenomena in terms of how likely it is that certain values will occur. Central features of the models will be discussed are random variables, probability density functions, and the expected value operator. In describing random processes and signals, the correlation function and conditional probabilities play a central role.

It addresses the following subjects:
1. Random variables. Matlab exercise on estimation of PDF, expected value and variance.
2. Refresher correlation. Calculating with correlation functions.
3. Random processes, correlation function, stationarity, wide sense stationarity, estimation of correlation function (Matlab exercise).
4. Random signal processing, power spectral density function, white noise.
5. AR processes, linear prediction: theory and computer exercise.
6. Markov chains.

PART II:
A course in advanced statistics about linear models, Bayesian inference, classification problems, Gaussian processes and Markov Chain Monte Carlo.

Study Goals

PART I:
1. Probability Theory
- Conditional) probabilities, the law of total probability, and Bayes rule.
- Solve probability problems that require the use of axioms of probability.

2. Definition and Description of Random Variables and Processes
PDF, PMF, CDF, Covariance, Correlation- Determine if a given PDF, PMF, CDF, variance, (auto/cross-)correlation(-function), (auto/cross-)covariance(-function), power spectral density complies with (theoretical and analytical) requirements.
- Convert the description of a probabilistic problem into a probabilistic model using PDF, PMF, or CDF.

3. PDF/PMF and Expected Value
Calculate the various forms of expected value of (combinations of) random variables and random processes
- For a given (amplitude continuous/discrete and time continuous/discrete) probability model calculate the following probabilistic (marginal, joint and conditional) characterizations: PDF, PMF, CDF, probability of an event, expected value, variance, covariance, correlation, correlation coefficient, auto/crosscorrelation function, auto/crosscovariance function, (cross) power spectral density.
- Calculate the PDF, PMF, expected value and variance of a derived random variable.

4. Properties of Random Processes
- Independence, orthogonality, uncorrelated, whiteness, IID- Determine if random variables/processes have the following properties: independent, orthogonal, uncorrelated, white, Poisson, Gaussian, Bernoulli, Markov, IID, stationary, WSS, ergodic.
- Calculate the expected value, variance, auto/crosscorrelation(function), auto/crosscovariance(function), power spectral density of a linear combination of random variables and of a linearly filtered (WSS, amplitude discrete/continuous, time discrete/continuous) random process.

5. Large NumbersCentral limit theorem, law of large numbers
- Solve problems that require the use of the central limit theorem in an engineering context
- Explain the law of the large numbers in an engineering context.

6. Statistical Estimators
- Estimated mean, variance, and correlation function
- Given a set of outcomes, sample functions or realizations, calculate estimators for expected value, variance, and (auto-)correlation function.

7. Application to Engineering Problems and Simulations
- Select and translate a simple electrical engineering or computer science problem into mathematical probability model. The emphasis is on problems in signal and image processing, telecommunication, and media and knowledge technology. The class of probability models encompasses the following random variables/processes: Bernoulli, exponential, binomial, Poisson, Gaussian, uniform.
- Justify and reflect on the approach taken in calculating or simulating (MatLab) the following probabilistic properties: PDF,

	PMF, expected value, variance, autocorrelation function, autocovariance function. PART II: By the end of the course, you should be able to: - compute properties of frequentist and Bayesian estimators in the linear model - derive posterior distributions and their properties - apply hyperparameter optimization in Bayesian models - apply computational methods for posterior approximation - derive properties of frequentist and Bayesian classification methods - derive properties of MCMC methods for sampling from the posterior distributions - derive properties of Gaussian process regression
Education Method	PART I: Lectures, working groups (problem solving), laboratory work (a Matlab exercise) Workload is around 15 hours for attending lectures, 5 hours of reading study material and preparing lectures, 15 hours for the lab course, 20 hours for preparing the exam, 3 hours for the exam, and 8 hours for a final report (66 hours in total). PART II: Classes and weekly exercises.
Books	PART I: R.D. Yates and D.J. Goodman, "Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers", ISBN 0-471-17837-3, John Wiley and Sons, New York, 2005, Second Edition. PART II: Simon Rogers and Mark Girolami "A first coruse in machine learning, 2nd edition" Chapman & Hall It is planned to cover chapters 1--4, 8 and 9 from this book. The precise material will be determined during the course.
Assessment	The final grade of the course consists of the following components: Part I (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Part II (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Final grade calculation: 0.5 * Part I + 0.5 * Part II A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Part I written exam: Resit (written or oral exam, depending on the number of students) Part II written exam: Resit (written or oral exam, depending on the number of students) In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Part I lab assignments: Re-submit deliverable Part II lab assignments: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	PART I: Exam of 3 hours. PART II: Exam of 3 hours.
Permitted Materials during Tests	PART I: Self made notes on a two-sided written A4 sheet. Calculator. PART II: none
Remarks	PART II: This course is particularly interesting for students that are interested in statistical exploratory and quantitative techniques to analyse multivariate data.

CS4200-A	Compiler Construction	5

Responsible Instructor	C.B. Poulsen
Contact Hours / Week x/x/x/x	6/0/0/0
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	English
Expected prior knowledge	- programming (required) - software engineering (recommended) - functional programming (recommended) - formal languages and automata (recommended)
Course Contents	Compilers translate the source code of programs in a high-level programming language into executable (virtual) machine code. In this course you will study the architecture of compilers, and important concepts and techniques underlying the components of that architecture. You will study these techniques by applying them in practice to implement your own compiler for a small language. The course covers the following topics: - Syntax and parsing - Static semantics and type checking - Virtual machines, assemby code, byte code - Interpretation vs. compilation - Code generation - Memory management, garbage collection - Data-flow analysis - Optimization
Study Goals	At the end of this course, you will be able to: - Explain the architecture of a compiler and the role of each component of that architecture - Explain the algorithms and techniques for the implementation of compiler components and apply these techniques to examples - Implement a code generator that translates source language abstract syntax trees to target language instructions - Implement transformations on abstract syntax terms to simplify programs - Implement simple type checkers - Implement simple data-flow analyzers - Integrate these components into a working compiler
Education Method	Attending the course involves: - Attending weekly lectures - Attending weekly lab session (which may start with a group tutorial) - Reading lecture material and papers - Making homework assignments (building a compiler)
Literature and Study Materials	Required reading: - We will use selected chapters from the book "Essentials of Compilation" by Jeremy Siek, available online: https://github.com/IUCompilerCourse/Essentials-of-Compilation - Lecture slides and selected papers from the literature Optional reading: - We will use Scala to implement our compilers. The following book is optional reading which may be helpful in getting up to speed with programming in Scala: https://www.artima.com/shop/programming_in_scala_5ed (Additional) reading materials will be made available on Brightspace. We will use WebLab (https://weblab.tudelft.nl/cs4200/) for the submission of homework assignments.
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Homework assignments (weighting 50%) Final grade calculation: 0.5 * Written exam + 0.5 * Homework assignments

	A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Permitted Materials during Tests	No materials are permitted during the exam.
Co-Instructor	Dr. S.S. Chakraborty

CS4340	**Seminar Probabilistic Programming**	**5**
Responsible Instructor	S. Dumani	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Probabilistic reasoning is one of the core aspects of artificial intelligence. This course will look into probabilistic programming, the most powerful paradigm of probabilistic reasoning, which specifies probabilistic models as computer programs. Because of the representational power of programming languages, any computable process can be modelled as a probabilistic program. The course covers the topics of probabilistic modelling and inference, but the main focus will be on developing a working probabilistic programming language. The topics include: Probabilistic modelling techniques, Importance sampling, Particle filters, Metropolis-Hastings algorithm, Hamiltonian Monte Carlo and automated differentiation, Variational inference, Discrete and logic probabilistic programming languages, Probabilistic circuits, Compiled/Amortised inference, Deep generative models	
Study Goals	By the end of the course, you will be able to explain the standard probabilistic inference procedure, solve real-world problems using probabilistic programs, identify the strengths and weaknesses in recent probabilistic programming literature, and design a research proposal to address an open problem in the field	
Education Method	This is a seminar course: we will meet twice a week to discuss a research paper. The paper should be read before the lecture and the participants need to submit a short summary (max 0.5 - 1 page). The students are expected to study every discussed paper (but not to the same amount of detail), prepare a presentation and lead a discussion for a few papers, and actively participate in discussions. Aside from paper discussions, the students are expected to do a small practical project (e.g., validating, stress testing or extending the existing method)	
Assessment	The final grade of the course consists of the following components: Research proposal (weighting 65%) Presentation (weighting 25%) Class participation (weighting 10%) Final grade calculation: 0.65 * Research proposal + 0.25 * Presentation + 0.1 * Class participation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Research proposal: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Artificial intelligence	

CS4365	Applied Image Processing	5
Responsible Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Weinmann	
Instructor	P. Kellnhofer	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Basic knowledge of programming concepts, linear algebra, and calculus. Prior experience with Python and C++ is useful but it is not required.	
Course Contents	The course discusses image processing techniques and their application to visual data problems in computer graphics and computer vision including neural image processing techniques. The practical part of the course focuses on the efficient implementation of image processing algorithms on parallel computer hardware. The final project guides students to work with literature sources and state-of-the-art surveys in an area. This course can serve as an introduction to the field of visual computing, which spans the domains of Computer Graphics, Computer Vision, Computational Photography and more. During the course, you will receive an overview of image processing with a deeper focus on applications for visual data. This will include: image representations, color models, linear filters, image sampling, digital photography (including HDR and tone mapping), image retargeting, image abstraction and non-photorealism, multiscale image processing (pyramids, Fourier transform), geometrical image transformations (warping, morphing), stereoscopic imaging, and neural image processing (concept, texture synthesis, style transfer, super-resolution, denoising, editing, depth map estimation). The course will cover classical signal processing and advanced image techniques, as well as approaches utilizing neural networks and machine learning. During lab sessions, you will learn about efficient implementations of the relevant algorithms with a focus on parallel data processing. This will include use of C++ with OpenMP and the Python machine-learning library Pytorch. Introduction to these technologies will be provided before their use. Assignments are solved by using the information presented in the lectures and lab sessions. Formative feedback will be provided during the lab sessions.	
Study Goals	By the end of this course, you should be able to: - Explain the process in which a digital image is formed. - Efficiently conceive, apply, and implement algorithms for processing digital images in C++ or Pytorch. - Analyze algorithms for image processing based on their computational costs, qualitative performance, and application areas.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of image processing, common problems and their solutions. Various algorithms and their properties will be discussed. During the guided lab sessions (4 hours each), you will be introduced to guidelines for efficient implementation of selected algorithms, and you will receive formative feedback on your approach to practical assignments. The final project will focus on survey and assessment of state-of-the-art approaches to a specified image processing problem.	
Literature and Study Materials	Required literature: - Lecture slides with additional references. Recommended literature: - Szeliski, Richard. Computer vision: algorithms and applications. Springer Science & Business Media, 2010. - Chapters 3, 4 and 10. - Burger, Wilhelm, and Mark J. Burge. Digital image processing: an algorithmic introduction using Java. Springer, 2016. - Chapters 22-26. - Sundararajan, Duraisamy. Digital image processing: a signal processing and algorithmic approach. Springer, 2017. - Chapters 2-7. - Goodfellow, I., Bengio, Y. and Courville, A.. Deep Learning. MIT Press, 2016. - Chapters 6-9 and 14.	
Assessment	The final grade of the course consists of the following components: - 3 assignments during the quarter (weighting 55%) - a final project (weighting 45%) Final grade calculation: 0.55 * Assignments + 0.45 * Final project The course is passed if all components have at least a grade of 5.0 and the weighted average is a 6 (rounded on 0.5). Repair possibilities (under conditions described in TER Implementation Regulations Art 5, sub 5): - Assignments: An individual replacement assignment with a maximal grade 6.0 in the following quarter. - Final project: An individual replacement project with a maximal grade 6.0 in the following quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

CS4375	Artificial Intelligence Techniques	5
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. M.T.J. Spaan	
Instructor	L. Cavalcante Siebert	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	We expect students to have programming expertise at the Bachelor level of Computer Science, in particular knowledge is expected of algorithms (e.g. search algorithms), and probability theory. Programming skills are required for the practical assignments which can be done in Python.	
Course Contents	The course focuses on concepts and techniques from probabilistic Artificial Intelligence for building intelligent agents, and decision making and -support systems. The specific topics include Bayesian networks, inference, belief monitoring, (partially observable) Markov decision processes, reinforcement learning and multiagent systems.	
Study Goals	After completing the course, student will be able 1. to comprehensively summarize the covered topics within probabilistic artificial intelligence and discuss their main issues; 2. to explain the main concepts and techniques; 3. to relate practical problems to appropriate AI techniques and apply the formal models covered in the course to these problems; 4. to implement and evaluate algorithms for complex decision making.	
Education Method	Lectures, lab work (programming assignments).	
Literature and Study Materials	Stuart J. Russel and Peter Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Prentice-Hall. ISBN 9781292401133 Global edition. See website http://aima.cs.berkeley.edu/global-index.html for additional information that goes with the book.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	The practical assignments will be done in groups of 4 to 5 students. The ethical standards of working are expected of all students. Work can be divided over students, but all students are responsible for the overall quality and originality of the work. Students that do not do their share of the work have to be reported by the group to the supervisor so that action can be taken.	
Tags	Artificial intelligence	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IN4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Reuired: Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations Strongly advised: some experience with a programming language (e.g., C) is preferred.	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas are considered. The scaling behavior of these algorithms in case of increasing problem size and/or an increasing number of processors is analyzed. Attention is paid to those aspects of computer architectures that are important to understand the resulting performance, such as the memory hierarchy and the interconnection network. By analyzing a number of case studies (applications) with respect to their computing-intensive character, possible bottlenecks will be determined. Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to get high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and the necessary optimization for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found the Brightspace.	
Course Contents Continuation	High Performance Computing, parallel programming, parallel algorithm	
Study Goals	1. Is able to categorize high-performance computer systems 2. Is able to explain and classify parallel and distributed architectures, and programming models; 3. Is able to explain the concepts of data decomposition and parallel algorithms; 4. Is able to apply various parallelization methods to benefit from the potential parallel computer systems; 5. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 6. Capable to evaluate and analyze the performance of parallel programs.	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): implementing (small) parallel programs with C, MPI and Cuda.	
Literature and Study Materials	Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Lab work (weighting 50%) Quizzes, bonus of max 1 point can be obtained (only valid on the first examination occasion after the lectures) Final grade calculation: 0.5 * Written exam + 0.5 * lab work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Via Brightspace	

IN4191	Security and Cryptography	5

Responsible Instructor	Z. Erkin
Instructor	J.A. Martinez Castaneda
Contact Hours / Week x/x/x/x	6/0/0/0
Education Period	1
Start Education	1
Exam Period	1 2
Course Language	English
Required for	UT-201500042 Privacy Enhancing Technologies (Q4)
Expected prior knowledge	Basic understanding on the following is suggested. -Probability and statistics -Programming skills
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Learning outcomes: The goal is to make students familiar with the basic concepts applied cryptography, including classical cryptography and modern secret key and public key cryptography. In particular, the students will acquire A sound understanding of the notion of security An understanding of the confidentiality, integrity and authenticity needs of the society Understand the role of cryptographic primitives including the differences between symmetric and asymmetric cryptography, the role of hash functions, digital signatures and PKI Understand the advanced topics in cryptography needed for the modern society with untrustworthy entities Among others things, the following topics are covered: -Classical systems -Information theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures, -*Secret Sharing. (if time permits) -*Zero Knowledge proofs (if time permits)
Study Goals	It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security, as well as is familiar with present applications. Through assignments, students are expected to have the chance to work on the topics covered in the lectures.
Education Method	Lectures, assignments and weekly exercises. Attention: This course requires full effort of 140 hours. Even more, if you lack the background (probability and modular arithmetic) Planned Workload: Lectures: 28 x 45minutes sessions, total 22 hours Practice session: 7 x 90 min. total 12 hours Assignments: 3 x 20 hour, total 60 hours Weekly study: 7 x 4 hours, total 28 hours Exam preparation: 20 hours Exam: 3 hours
Literature and Study Materials	-Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition -Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) Handouts of lectures
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%) Mandatory assignments (weighting 30%) Final grade calculation: 0.7 * Written exam + 0.3 * Mandatory assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mandatory assignments: re-submit deliverable

	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	In case of in person examination at campus: The exam is closed book. A cheat sheet of size A4, hand written is allowed for the written exam. Name and student number has to be present on each side. In case of remote exam from home: open book: textbook, slides and self-made notes only. randomised/customised exam.
Permitted Materials during Tests	Only non-scientific calculators.

IN4252	Web Science & Engineering	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	J. Yang	
Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The expected entry level is (equivalent to) standard bachelor-level computer science.	
Course Contents	The main subject of the course is the Web, and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g. Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan / research paper draft in collaboration with others in a small group to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies, and verbally present and defend this plan / draft.	
Education Method	The education methods include: - Lectures, before which and after which students study material by themselves, to get an understanding of the relevant material; - Small assignments and hands-on exercises, to apply the understanding of relevant material; - One large assignment, with a number of feedback moments, to learn how to write a web science paper / research project plan. Lectures will not be in each week in the class period (1+2): in between lectures there is time reserved for studying before and after lectures, for small assignments and exercises, and for writing the large assignment paper. The writing of the large assignment paper / project plan happens throughout the class period (1+2) to enable frequent feedback.	
Literature and Study Materials	Will be provided in class, depending on the topics chosen for the assignments and final paper.	
Assessment	Assessments are based on two components: the "small assignments" (accompanying the lectures) which provide a total of 20% of the grade (these are a mix of literature and programming homework), and the "final paper" which provides 80% of the final grade. To pass, each of the components ("small assignments" and "final paper") need to be graded with a minimum of 5.0. The final grade must be at least a 5.8. If these criteria are not met, the "small assignments" and the "final paper" can be repaired provided that the to-be-repaired component was at least graded with a 4 in accordance with OER/TER article 17. This repair is not a resit, and is limited to a maximum of 6.0 as a result. Components that have been graded with less than a 4.0 cannot be repaired. If the course is still not passed after the repair or cannot be repaired, all components need to be redone if the course is retaken in a later year. Both the small assignments and the final paper are group works of groups of preferably 2 members.	
Special Information	Students are asked to register/enrol on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	The expected workload is 5ects and that is principally distributed uniformly over the two quarters.	

IN4307	Medical Visualization	5
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge of linear algebra, calculus and programming is needed. The project will be implemented in MeVisLab and Python. Learning MeVisLab will be part of the practice assignments.	
Course Contents	Theory and practice (Notice project extends to Q2) of medical visualization. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measuremen; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for medical data; advanced applications.	
Study Goals	By the end of the course, you should be able to LO1: Explain medical visualization algorithms and their applicability to medical problems. LO2: Discuss the advantages and disadvantage of medical visualization algorithms. LO3: Build a medical visualization system for a given problem: a. Discuss a suitable visualization for a given medical problem. b. Implement the most suitable solution. c. Judge the performance of the implemented solution.	
Education Method	The course will be based on a combination of lectures and practical assignments. A final project will be developed in Q2	
Literature and Study Materials	Course slides, instructions for project and assignments, and selected literature via Brightspace. Chapters from the following book can be used as alternate view on the topics but are not mandatory: Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Book available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam Q1 (weighting 40%) Final project Q2 (weighting 60%) The final project will be the design and implementation of a visualization system for a given medical problem. The final project will be carried out in teams of three. The deliverables for the final project will be a report (paper), the results (e.g., code) and a short video presenting the project (i.e. screencast). Final grade calculation: 0.4 * Written exam + 0.6 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Q2 In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final project: re-submit deliverable (the project resit is not automatic and must be requested within 2 weeks after publishing the grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Notes and written material. No computers.	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	
Co-Instructor	Prof.dr. E. Eisemann	

IN4344	Advanced Algorithms	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic courses in Algorithmics and Complexity Theory	
Course Contents	The course is on solving (abstract models of) complex real-world problems, with a focus on solving intractable problems exactly. The course consists of two main parts: * modeling and solving using linear programming * exact algorithms using search trees, dynamic programming, and/or decision diagrams	
Study Goals	By the end of this course, students will have knowledge of and experience with the following advanced algorithmic techniques: (Part 1:) - linear programming (LP) and LP modelling - duality and simplex algorithm - integer LP and integer modelling - LP relaxation and branch and bound (Part 2:) - complete and bounded search trees - tree decomposition - dynamic programming - preprocessing - decision diagrams Furthermore, they will be able to * recognize situations where they can use these techniques and adapt them to different applications. construct new algorithms that are similar to these techniques. construct proofs that are similar to a selected set of proofs from the material. analyze the run time of algorithms. implement an algorithm that is given in pseudo-code. experimentally evaluate the quality and the runtime of an algorithm on a set of instances.	
Education Method	Interactive lectures (usually 2x2 per week), optional homework exercises, programming assignments (2 hours lab support per week) The expected workload is 30% studying the written material and the recorded lectures, and participating in the interactive sessions 30% making the homework exercises and preparation for the exams 40% working on the programming assignments	
Literature and Study Materials	Part 1 of the course will be mainly based on chapters 1-8, some of 9-11 (most of 9-11 is assumed to be known), and 12-14 of the syllabus "Optimization" by Karen Aardal, Leo van Iersel and Remie Janssen, which can be ordered via https://www.webedu.nl/bestellen/tudelft/ Part 2 will use chapter 10 of the following textbook: J. Kleinberg and E. Tardos, Algorithm Design, Pearson Education, 2006. ISBN 0-321-37291-3 As well as some supplemental study material, which will be provided via BrightSpace.	
Assessment	The final mark depends on the marks obtained for (a) written reports and evaluation of programming assignments, PA (weight 40%) (b) a written exam, EX (2 parts, weight 60%) Each programming assignment is graded on a scale from 0 to 10. The final mark for the programming assignment (PA) is the average of the mark obtained for the assignments. Programming exercises can be completed by 2 students working together. If the average grade for the four programming assignments (PA) is between 4.0 and 6.0 (excluding 6.0), a repair is possible by making one extra assignment to replace the lowest grade of the four assignments. This leads to a repaired PA score, maximized at 6. (E.g. if grades were 3.5, 4, 4, 4.5, the 3.5 can be replaced. If the grade for the replacement assignment is a 7.5, the repaired PA score is a 5. If grades were 5, 5.5, 5.5, 6, the 5 can be replaced. If the replacement is a 9, the repaired PA score is not 6.25 but 6.) The exam consists of two parts. Each part will be examined after the lectures about that part have been delivered and will be graded on a scale from 1 to 10. The final mark for the exam (EX) is the average of the marks for the parts. Each exam part contains one or more challenging assignments; during each part of the course two representative homework assignments will be made available to prepare for this. There is a resit for the exam where any of the two parts can be redone. The result for a part after the resits is determined by the maximum score obtained for the original exam and the resit. The final mark for the course is determined as follows: - if the PA and the EX mark are above 5, the final mark is the weighted average of these three marks: 60% EX, 40% PA - if at least one of PA, EX is less than or equal to 5, the final mark is the minimum of the results obtained for PA or EX. Partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	For content questions / of general interest, please use StackOverflow. For personal questions, please use the course email address: aa-cs-ewi@tudelft.nl	
Tags	Algorithmics Artificial intelligence Mathematics	

WM-ITAV-4010	Scientific Writing	2
Module Manager	L. Meester	
Instructor	Drs. W.J. Blokzijl	
Instructor	Drs. B.M.D. van der Laaken	
Instructor	Drs. P.C. Post	
Instructor	Drs. A.E. Kam	
Instructor	M.J.Y. Wackers	
Instructor	S. Baars	
Instructor	M. Bliekendaal	
Instructor	L.C. Schroten	
Instructor	A. Glasbergen-Plas	
Instructor	M. Looij	
Co-responsible for assignments	Drs. B.M.D. van der Laaken	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1 2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you learn to write a scientific article, either a research article based on your own research data or a literature review about a subject of your own choice. This is a necessary skill for anyone who wants to pursue an academic career after their graduation, but it can also be used immediately for all academic texts you will write during your Master program, such as your Master thesis. In seven weeks, we will go through all steps of the writing process, from formulating a good main question and finding relevant literature, to the actual writing, re-writing and final editing. You will exercise with finding, reading and managing relevant academic literature, writing in an academic style, building a comprehensive argumentation, reviewing fellow students' articles and using other students and the instructor's comments to improve your own work.	
Study Goals	The purpose of this course is to learn how to write a scientific text. To achieve this, at the end of this course you will: know what the main characteristics are of a scientific text be able to formulate a main question be able to find, critically read and manage scientific literature be able to use literature properly and avoid plagiarism be able to build up your argumentation be able to structure an article according to the conventions in your field of study be able to use scientific English style be able to use tables and figures to support and communicate your results be able to give feedback on somebody elses article be able to use feedback for improving your work.	
Education Method	Practical, in 6 sessions (attendance mandatory). Every week you have to read some background information about scientific writing and hand in a part of your text. Participants must attend all sessions - one missed session is allowed only - and hand in all assignments in time. Students who receive a pass for this course are rewarded with 2 ects. This equals 56 hours of study. A total of 12 hours is spent on attending (online) classes, in which you can ask questions, discuss the feedback you received on your work and discuss aspects of scientific writing with your fellow students; the remaining 44 hours is for self study, writing and revising. In seven weeks, from preparing lecture 2 up to handing in your final article in week 8, you will have to spend at least 6 hours of self study on this course, every week. It is important that you make sure you have this time in your personal schedule. At the beginning of the course you will mainly be reading up about scientific writing and the subject of your text. As the course proceeds, you will be spending more of your time on writing, giving feedback and revising your own text.	
Books	Theory about academic writing will be made available through Brightspace.	
Assessment	You write a scientific article of 2000-2400 words (excluding the list of references and the abstract). You have to hand in parts of your article every week. Your final grade is based on the final article. An evaluation form for the grading of the article is available throughout the course.	
Elective	Yes	
Category	MSc level	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialistievakken start kwartaal 2 2023

CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	- basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	The course intends to bring the student into the position to: - Explain the fundamental concepts and terminology of real-time systems - Construct task schedules using different scheduling policies under a given set of realistic system constraints - Analyze the timing behavior of a system for a given system model and scheduling policy - Discuss advantages and disadvantages of different scheduling policies for a given platform or system - Discuss the effect of hardware and software interferences on the timing behavior of a given system - Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system - Derive (reverse engineer) the system specification from a given implementation (in the lab) - Evaluate the scheduling overheads of a given implementation (in the lab) - Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	Written exam (grade) + lab work; the exam has a resit Repair option for the Individual Assignment = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet has become of critical importance to society. However, the large size of networks and abundance of protocols have made network management very complex. The novel concept of network programmability addresses this complexity and has resulted in a paradigm shift in how networks are (or can be) operated. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network programmability and which treats fundamental networking concepts like Quality of Service and network resilience.	
Study Goals	The learning objectives of this course are twofold: (1) The student should gain knowledge of the treated networking technologies. (2) The student should be able to apply and work with the programmable network technologies in a network emulator (Mininet).	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final assessment (and hence 100% of the grade) will be based on an exam that covers both the theory from the slides as well as the content from the reader. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CS4015	Behaviour Change Support Systems	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	none	
Course Contents	Behavior change support systems (BCSS) are computer-based systems that support individuals to form, alter or reinforce cognitions, attitudes or behaviors without using coercion or deception. They can serve individuals throughout the various stages of a change process, such as awareness development, contemplation, action strategy development, development of new behaviors, and maintaining these new behaviors. Virtual healthcare coaches, negotiation support systems, and applications that provide individuals with personalised financial guidance are three examples of these systems. To establish, modify or maintain change BCSS can deploy computerized persuasive strategies (e.g. reducing effort to establish target behavior, or argumentation reflection strategies, and also algorithmic nudge, including data-, expert-, and theory-driven), simulations (e.g. serious gaming, virtual reality), relational software agents (e.g. ePartners, virtual coaches), and personalization based on longitudinal user data. BCSSs are found in many domains, including education, sales, negotiation, management, and particularly in the health domain. The main topics are: Understanding the behaviour; strategies for behaviour change, action funnel, human decision making Identifying change support strategies; Behaviour change wheel, product vision, target outcome, target user, population diversity Conceptual design of BCSS; Behaviour plan, decision-making environment, story-editing technique Behaviour change and technology acceptance; Motivation, Trans-theoretical model, Theory of reasoned action, theory of planned behaviour, Technology acceptance model, Unified Theory of Acceptance and Use of Technology, technology adaption lifecycle, Fogg behaviour model, Self-Determination theory Persuasive technology; functional triad, levels of persuasion, Nudge Theory, principles of influence, triggers Algorithmic nudge/Persuasion and support Algorithms; concept, key characteristics, and feasibility Learning principles; Behaviouristic learning principles, cognitive learning principles, constructivism based learning principles, social cognitive (learning) theory Communication and framing; Behavioural economic principles, elaboration likelihood model (ELM), health belief model, protection motivation theory, persuasive communication Adherence and Gamification; Implementation Intentions, system 1 and system 2, treatment adherence, gamification Application domains eHealth and Cyber security; Self-monitoring, self-management, virtual health agents, virtual reality therapy, internet cognitive behavioural therapy (iCBT)	
Study Goals	This course provides students the opportunity to develop and demonstrate understanding, knowledge and competence in the field of BCSS. The learning outcomes for the course are that students will be able to: 1. Apply psychological theories, principles, and concepts in the design of BCSS systems. 2. Apply BCSS practices, principles and techniques. 3. Construct an implementation of an algorithmic nudge/boost/persuasion for a specific real-world application domain and objective 4. Design a BCSS system using BCSS principles, concepts, theories, and design methods.	
Education Method	The course uses a flipped-classroom approach, where students watch the pre-recorded lectures in their own time and complete online quizzes. In the on-campus lectures, students and teachers discuss small challenges related to the lecture topic. In the pre-recorded video material, theories, principles and methods are presented, discussed and illustrated with examples from the field. The video material is supported by online self-tests. In their own time, students work in small groups on coursework assignments to develop a product design for a BCSS with a computer simulation of supporting algorithmic nudge/boost/persuasion. In the on-campus session, student groups present the progress of their coursework and receive formative feedback. The presentations are done three times, following topic as the project progress over time: 1. The targeted behaviour 2. Behaviour plan 3. Conceptual design and the algorithmic nudge/boost/persuasion	
Literature and Study Materials	Available on brightspace	
Books	Wendel, S. (2020). Designing for Behavior Change: Applying Psychology and Behavioral Economics. O'Reilly Media, Inc,. or the older edition Wendel, S. (2013). Designing for Behavior Change: Applying Psychology and Behavioral Economics. " O'Reilly Media, Inc.	
Assessment	The final grade of the course consists of the following components: Computerised examination (or oral exam) (weighting 60%) If the expected number of students registering for the exam or resit is small, the teacher might decide to replace the computerized examination with an oral examination. Design BCSS Coursework Project (resulting in presentation slides and demonstration of computer simulation of a supporting algorithmic nudge/boost/persuasion, including question and answer round where individual group members are assessed on the coursework) (weighting 40%) Final grade calculation: 0.6 * Computerised examination (or oral exam) + 0.4 * Design BCSS Coursework Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Computerised examination (or oral exam): Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Design BCSS Coursework Project: Resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	3	
Co-Instructor	M.L. Tielman	

CS4135	Software Verification	5
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	Ir. K.F. Wullaert	
Contact Hours / Week x/x/x/x	0/2/0/0 + 0/4/0/0 practicum	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	How can we ensure that software cannot crash and is guaranteed to be correct? In this course we tackle this question by viewing programs and programming languages as mathematical objects. That way we can use logic to prove properties about programs and thereby guarantee that software is correct. To make reasoning about actual programs and programming languages feasible, we will not be doing these proofs by hand, but instead use a tool called a proof assistant to build proofs that can be checked by a computer. As we will show during this course, proof assistants turn the activity of doing proofs into programming. This course is a specialization course for programming languages and software engineering Prior knowledge: - Required: - Strongly advised: some familiarity with functional programming and propositional and predicate logic.	
Study Goals	Learning Objectives: - State and prove properties of functional programs in logic. - Specify the semantics of a programming language in logic. - State and prove the correctness of imperative programs. - Use a proof assistant to perform a mechanized proof.	
Education Method	This course is being taught in a "flipped classroom" style. It consists of 2 hours of studying (by reading or watching lecture videos, for greater flexibility and avoiding clashes with other courses - material will be provided) and two lab sessions of 2 hours each. During the lab sessions students will work on stating and proving theorems. Towards the end of the course students will carry out research projects that apply the ideas of the course.	
Literature and Study Materials	Supplementary material: Free online text book "Logic and Proof": https://leanprover.github.io/logic_and_proof/ Free online text book "The Hitchhikers Guide to Logical Verification": https://github.com/blanchette/logical_verification_2021/raw/main/hitchhikers_guide.pdf	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the individual project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4270	Conversational Agents	5

Responsible Instructor Dr. C.R.M.M. Oertel Genannt Bierbach

Responsible Instructor F. Broz

Instructor M.L. Tielman

Contact Hours / Week x/x/x/x 0/4/0/0

Education Period 2

Start Education 2

Exam Period 2
3

Course Language English

Expected prior knowledge
Basic programming skills (e.g. Python and Java)
Probability theory and statistics

Course Contents

Chatbots, embodied and conversational virtual agents, and social robots are becoming more and more popular. Many people are owning an Alexa, Cortana or Echo or are talking to their virtual assistant on their phone. Indeed, such technologies have the potential of making our lives easier and relieve people from the more repetitive tasks. For example, it is imaginable that such systems are being used for financial applications by helping customers with frequently asked questions but also to advise them on in the long term more impactful decisions such as their pension plans. Further applications can be imagined in the area of healthcare and education, some of which are already in existence today.

In this course, attention will be given to different verbal and nonverbal behavioral characteristics, like speech, intonation, gaze and gestures that humans show when communicating with both other people and machines. This behavior is then related to different dialogue functions, including turn-taking, addressing others, and backchanneling, that give shape to the communication process.

This course introduces conversational agent technology.
We cover agent related technologies which can be grouped into:
Dialog Management
NLP
speech synthesis
social robotics

Study Goals

After this course you have learned to:
1) Apply relevant linguistic and psychological theory to conversational agent systems
2) Design the conversational memory of your agent focusing on a perception/generational model
3) Analyse human-human conversational data to better design ML models for adding generation or perception capabilities to conversational agents
4) Develop the conversational memory in your agent
5) Evaluate the effects of affect and embodiment on human-agent interaction
6) Discuss your results and the limitations of your implementation

Education Method

There are 2 lectures scheduled per week. Students work in groups of 4 on a group project.
Lectures: 26 hours (13 × 2 hours lectures)
Reading time: 39 hours
Preparation basis tool use: 25 hours
Coursework project, including writing report and prepare a video demonstration: 50 hours (10 × 5 hours)

Literature and Study Materials

We use the book "The conversational interface " by Michael McTear, Zoraida Callejas, David Griol. This book is freely available through the TU Delft library.
https://link-springer-com.tudelft.idm.oclc.org/book/10.1007%2F978-3-319-32967-3

Other relevant material will be provided on Brightspace.

Assessment

The final grade of the course consists of the following components:
Online Examination (weighting 30%)
Group Assignment (weighting 70%). This assignment will result in a group report and a video demonstration.
3 Lab assignments (pass/fail)

Final grade calculation: 0.3 * online examination + 0.7 * Group assignments + P/F * Lab assignments
A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8.

Resit/ Repair opportunities:
Online examination: resit
In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:
Group Assignment: re-submit deliverable
Lab assignments: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

CS4400	Deep Reinforcement Learning	5
Responsible Instructor	Dr. J.W. Böhmer	
Instructor	Dr. M.T.J. Spaan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students must have passed CS4375 "Artificial Intelligence Techniques", or have acquired equivalent knowledge about: - basic probability theory, analysis and algebra - general machine learning methodology, e.g. regression - fully and partially observable Markov decision processes - tabular reinforcement learning methods, e.g. Q-learning - the exploration/exploitation trade-off, e.g. RMAX or UCB - multi-agent learning, e.g. centralized training and decentralized execution The course will require a lot of math and implementations of deep-learning algorithms. Students are encouraged to close any gaps in the above knowledge and to familiarize themselves with the Python/PyTorch deep-learning framework before the start of the course.	
Course Contents	This course will cover the breadth of modern model-free RL methods, discuss their limitations and introduce a variety of current research topics. In particular, we expect to cover the following: - deep learning methodology and architectures - stabilization of approximated value estimation - modern actor-critic methods - planning as inference - exploration with deep networks - offline reinforcement learning - deep multi-agent reinforcement learning - multi-task and meta learning	
Study Goals	After successful completion of this course, students - can list the strengths and limitations of modern deep RL approaches, - explain the underlying concepts of the discussed methods, and how they differ from each other, - derive the objectives and constraints of selected algorithms, - can implement selected algorithms/architectures, and - can analyze a new task to decide which algorithms/architectures to apply.	
Education Method	The course will be taught in lectures and the content will be solidified in homework submissions, which will be presented in tutorials.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Exercises (Pass/Fail) To be eligible for the exam, students must hand in homework exercises. Homework will not be individually graded, but at least 75% of the answers must be of sufficient quality (in terms of time commitment, not necessarily correctness) to be eligible to take the exam. Final grade calculation: P/F * Exercises + 1 * Witten exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Exercises: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4089	Data Visualization	5
Responsible Instructor	T. Höllt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/2/0/0 & lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Master course MKE/ST/DS	
Expected prior knowledge	IN2905-A/IN4152/CSE2215 Computer Graphics (recommended, not required). The practicals will be implemented in HTML/Javascript/D3 (InfoVis) and C++ (VolVis). We consider programming skills as a requirement but not necessarily in the mentioned languages and no advanced skills are needed. Relevant topics will be introduced and experience in other programming languages should make adaption feasible.	
Course Contents	Data visualization is the visual representation of data by computer generated images. The data sets can be results of numerical simulations or measurements (scientific visualization), or other data collections such as databases (information visualization). The goal is to improve insight, understanding and/or communication of data. Data visualizations use a combination of methods from a very diverse variety of disciplines: perception, computer graphics, human computer interaction, algorithmics, image processing, machine learning, numerical analysis, optimization, The course has two main parts; information and scientific visualization that will involve knowledge of diverse disciplines. As a computer science course, affinity to algorithmic thinking and programing skills will be needed. Topics covered: models of the visualization process; colour models and use of colour; information visualization; representation and processing of data; volume visualization; interactive visual data analysis; visualization of vector fields and flows. Guest lectures might be given on selected topics.	
Study Goals	The goal of the course is to get knowledge on the fundamentals that are part of data visualization. The main principles and techniques that are the basis of generating effective visual representations of data. Techniques and cases of data visualization are discussed. There are several applications for the techniques, such as medical, engineering, finances, economics, game analytics, and more. By the end of the course, you should be able to LO1: Discuss a large range of visualization techniques. LO2: Discuss a perception principle of visualization. LO3: Explain mathematical principles and algorithms of visualization techniques. LO4: Design suitable visualization systems for a given practical data analysis problem. LO5: Implement visualization systems for a given practical data analysis problem.	
Education Method	Lectures, practical assignments, self-study, and projects.	
Literature and Study Materials	Course slides, instructions for projects, and selected literature via Brightspace. Chapters from: Visualization Analysis and Design Author: Tamara Munzner CRC Press Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Books available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 30%). You are allowed to have the slides and material of the course during the exam. No computer or laptop is allowed. Information visualization project (weighting 35%) Volume visualization project (weighting 35%) The projects will be developed in groups of 3 and are evaluated based on the reasoning/justification of the techniques used based on the material given at the course, effectiveness of the results, technical contribution or implementation, quality of the documentation and presentation Final grade calculation: 0.3 * Written exam + 0.35 * Information visualization project + 0.35 * Volume visualization project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Information visualization project: re-submit deliverable Volume visualization project: re-submit deliverable The project resits are not automatic and must be requested within 2 weeks after publishing the grades. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

IN4150	Distributed Algorithms	6
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for network overlays, message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, broadcast, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	Understand the fundamental problems that distributed systems face Understand the most important distributed algorithms that solve these problems Reason about the execution of distributed algorithms Program distributed algorithms Select and summarize relevant literature on distributed algorithms	
Education Method	Lab work executed in groups of two students	
Literature and Study Materials	Lecture notes and lecture slides (available on Brightspace)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 75%). In case in-person, on-campus written exams are not possible because of unforeseen circumstances or measures, the written exam will be replaced by individual online oral exams. Paper summary (weighting 25%) Final grade calculation: 0.75 * Written exam + 0.25 * Paper summary A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper summary: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None except the list of algorithms	
Special Information	Email: in4150-ewi@tudelft.nl. Please send any emails related to the course to this address.	
Remarks	Lab work is 40 hrs.	

IN4302TU	Building Serious Games	5
Responsible Instructor	Dr.ir. A.R. Bidarra	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	For CS students: programming experience with some object-oriented language; experience with graphics, AI and/or some game engine(s) is a plus. For all students: though not compulsory, it may be convenient to have followed the course SEN9235 (Game Design Project), which is taught in the first quarter.	
Course Contents	Project-based interdisciplinary course, open to MSc students of all faculties. The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, training, health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose. Assignments for this course will be provided by real-world end-users (e.g. companies or the Science Centre Delft), to whom the group will be reporting throughout the term of the project. Project-based interdisciplinary course, open to MSc students of all faculties (of LDE universities). The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, professional- and health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose, solving a variety of possibly conflicting, technical challenges. Assignments for this course are provided by real-world commissioners (from startups to NGOs to research institutes), to whom the group will be reporting throughout the project. In previous years, various serious game prototypes originated in this course were actually developed and eventually deployed for their intended application; and many more were published and presented by students in international research venues (incl. IEEE CoG, FDG, ICIDS, GALA, ISAGA).	
Study Goals	At the end of the project, the student will demonstrate proficiency in the following aspects: o identifying and valuing the soft skills necessary to work effectively in interdisciplinary teams o responsibly interacting in a team, integrating its members' talents and expertise in a variety of team roles o adapting with flexibility to the dynamic requirements of a complex external assignment o translating feedback received into proactive personal development steps o formulating research questions, identifying research contributions, and describe them in a scientific publication Additionally, the CS student will demonstrate proficiency in the following specific aspects: o identifying, selecting and deploying the most adequate game technologies for the given serious game domain and constraints o deepening programming skills while building a complex and large software system in an agile context	
Education Method	Project: teams work intensively as a small game studio, according to a tight planning. Also a few plenary sessions and/or lectures.	
Assessment	The final grade of the course consists of the following components: Product grade (weighting 50%). Unique for the whole group, based on both the game itself and the required documentation. Process grade (individual) (weighting 45%). Including personal contribution, performance, attitude, and peer evaluation. Final presentation (weighting 5%) The commissioner will be involved both as advisor and as assessor. The final documentation includes writing a scientific paper and preparing it for actual submission to a conference on serious games and/or their application. Final grade calculation: 0.5 * Product grade + 0.45 * Process grade + 0.05 * Final presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final presentation: resit There is no repair possibility for the other components (product and process), since their assessment necessarily requires an extended development period. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. R. Marroquim	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Literature and Study Materials	We follow the book Performance Analysis of Complex Networks and Systems, by P. Van Mieghem, Cambridge University Press (2014). See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Written and closed book. A formularium is provided that can be consulted at the examination. Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. M. Blaauboer	
Responsible Instructor	Dr. L. di Carlo	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	30% homework assignments, 10% online quizzes, 60% final exam. A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course.	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialistievakken start kwartaal 3 2023

AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morpological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 6 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: ((0.85*EX + 0.15)+ AS)/ 2 or without the bonus point from the practicals: (EX+AS)/2 With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

CESE4030	Embedded Systems Laboratory	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. G. Lan	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This highly multi-disciplinary course comes with a lab project where teams of 4 students each will have to develop an embedded control unit for a tethered electrical model quad rotor aerial vehicle (the Quadrupel drone), in order to provide stabilization such that it can hover and (ideally!) fly, with only limited user control (one joystick). The control algorithm (which is given) must be mapped onto a home-brew PCB holding a modern RF SoC interfacing a sensor module and the motor controllers. The students will be exposed to simple physics, signal processing, sensors (gyros, accelerometers), actuators (motors, servos), and basic control principles. The embedded software running on the flight controller board (FCB) driving the drone must be written in the Rust programming language. The project work (including written report) covers the entire duration of the course period, and will take approximately 128 hours, of which 32 hours are spent at the lab facilities.	
Study Goals	Student is acquainted with real-time programming in an embedded context, along with a basic understanding of embedded systems, real-time communication, sensor data processing, actuator control, control theory, and simulation. Moreover, the student has had exposure to integrating the various multidisciplinary aspects at the system level.	
Education Method	Lectures (6*2hrs), lab work (8*4hrs), coding@home (8*12hrs), report (8hrs), so on average 2 days per week	
Prerequisites	Literacy in Rust (CESE4000 or equivalent, and CSE4015)	
Assessment	Lab. project (75%) + written report (25%), no exam, no resit. Failed student has to do an additional assignment/report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for CESE students- they get preference over other MSc students.	

CESE4050	Measuring and Simulating the Internet	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/2/0 (Lectures(2x2h), Pract)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Strongly advised: (Advanced) Networking course (e.g., CESE4045) and Programming skills.	
Course Contents	The Internet is a complex network without a fixed structure. Hence, measuring the Internet is crucial to acquire knowledge about the Internet infrastructure (topology), traffic, and performance (e.g., loss, delay, bandwidth, etc.). This course will discuss the design requirements and challenges in measuring and simulating the Internet, and the existing measurement methodologies (how/where/when to measure). Knowledge of how to conduct and evaluate Internet measurements enables the design and enhancement of a large set of applications, including: capacity planning and traffic engineering, network management and trouble-shooting, detecting network abuse and intrusions, etc.	
Study Goals	The goal of this course is to introduce the students to basic Internet measurement tools, as well as the state-of-the-art in Internet measurements research. The students will learn several Internet measurement techniques (e.g., active vs. passive measurements), and different software tools. Through a measurement assignment, the students will learn how to define/formulate a research problem, choose a specific approach, and complete a measurements-related research project.	
Education Method	Lectures + weekly instructions (discussion of papers and progress) + independent project work.	
Literature and Study Materials	Papers	
Assessment	Groups of students will be assigned a project that requires the students to put the theory on measuring and simulating the Internet into practice. The students have approximately 1 month to complete their assignment. The final assessment is based on the presentation (via report and/or demonstration) of the project assignment results and on the individual contribution and level of participation. Students within a group may thus receive different grades. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Maximum number of participants	Because this is a project-based course, we can only admit a limited number of students (typically around 30, but the actual number depends on the number of TAs involved).	

CESE4060	Wireless IoT and Local Area Networks	5
Responsible Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students are advised to follow the course Wireless Communications (ET4358) before taking this Wireless Networking course. An advantage is to have entry-level programming skills (Matlab, Python, C/C++). Nonetheless, students with little knowledge of programming will be helped.	
Course Contents	The following modules will be discussed during the lectures: Introduction (example topics): - What is wireless networking - Where to search for (academic) wireless network literature and resources Medium Access Control (example topics): - WiFi: hidden/exposed terminal problem, Carrier Sense Multiple Access - Bluetooth standard: in-depth look into the channel hopping, protocol specifications WiFi (example topics): - Review of IEEE 802.11 standards - Protocol format - ISM band regulation - Adaptive Modulation and Coding - WiFi Matlab class (assignment) IoT networking standards (example topics): - LoRa: protocol specifications, energy consumption, modulation format, network design Review of wireless tools (example topics): - Introduction to wireless packet sniffing and analysis using Wireshark (assignment) - Simple simulations of WiFi network with NS3 RFID networking (example topics): - Principles of backscatter - Protocol formats: EPC C1G2 - RFID hackathon (assignment) Cognitive radio (example topics): - Basics of spectrum management - White Space Databases - Theory of spectrum sensing	
Study Goals	At the end of the course students will be able to: (i) to understand how practical wireless systems work and get a deeper understanding of how the theoretical concepts of wireless communications apply to practice; (ii) employ their own analysis methodology to assess new wireless network systems (especially at the physical layer); (iii) understand rapid prototyping of new wireless systems (for instance, with software defined radio).	
Education Method	Lecture presentations, mini-project assignments, assigned paper reading and its critical analysis and presentation.	
Computer Use	Each student should have its own laptop (preferably with a Linux distribution, where Linux must not be installed on a virtual machine). We will be using Matlab, and/or NS3 and/or GNURadio and/or Wireshark for the assignments.	
Literature and Study Materials	WiFi Matlab WLAN toolbox: https://nl.mathworks.com/help/wlan/; Wireshark learn page: https://www.wireshark.org/#learnWS; tutorial on NS3 network simulator: https://www.nsnam.org/documentation/; specific chapters from books provided at the beginning of each lecture.	
Prerequisites	Background in programming (Matlab, Python, Bash)	
Assessment	Points from the mini-project assignments. A research paper analysis from conferences such as IEEE INFOCOM, ACM MobiCom, ACM SIGCOMM will be required to pass the course. Repair option: Individual Assignment = Re-submit deliverable Individual Report = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: A good understanding of the fundamental concepts of quantum communication Insight into the differences between classical and quantum communication and cryptography Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)	
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session	
Literature and Study Materials	Primary: Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: Homework (weighting 70%) Final Project (weighting 30%) Final grade calculation: 0.7 * Homework + 0.3 * Final Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Homework: re-submit deliverable Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5

Responsible Instructor	Ir. S.E. Verwer
Responsible Instructor	Dr. A. Panichella
Contact Hours / Week x/x/x/x	0/0/4/0
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Course Contents	Software is one of the most complex artifacts of mankind has ever created, but complexity is the enemy of correctness. Modern software testing and validation tools use a multitude of techniques geared toward correct computer code, most of these are base on artificial intelligence. In this course, we study these techniques in details, specifically we will understand and implement: Execution monitoring and taint analysis Branch distance computation Hill-climbing and genetic algorithms Concrete and symbolic (concolic) execution Active state machine learning Genetic programming The goal is to better understand and test software using artificial intelligence. Using the taught techniques you will be able to automatically: Discover which code is reachable Find (security) bugs in software Write tests that cover all reachable code Reverse engineer a code's functionality Patch code to remove bugs and failing tests
Study Goals	The student will: Understand modern AI techniques for software testing. Be able to implement several such techniques from scratch: - smart fuzzing (probing software with input to find crashes/bugs), - symbolic execution (using logic to construct inputs that trigger specific code branches), - fault localization (given that a program fails, find the line of code responsible for the failure), and - automated program repair (using a patch library and genetic programming to improve code) Be able to apply this technology to locate bugs in real-world software implementations.
Education Method	The main part of the course will consist of 3 lab assignments covering the theory (fuzzing&tainting, symbolic execution, automated program repair), and one lab assignment for the application to real software. The students will implement the taught techniques from scratch in the first 3 assignments, which will be scored with a pass/fail. All three assignments need to be passed to complete the course. The final lab will contain a recap from the first three assignments and an application of a state-of-the-art tool on real software. The final lab will be graded and be the final course grade. There will be instruction sessions where students can work on their assignment and ask the teachers for assistance.
Assessment	The final grade of the course consists of the following components: Three lab assignments (pass/fail) Final lab (weighting 100%) Final grade calculation: P/F * 3 lab assignments + 1 * Final lab A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: 3 lab assignments: re-submit deliverable Final lab: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Tags	Artificial intelligence Software

CS4160	Blockchain Engineering	5
Responsible Instructor	Dr.ir. J.A. Pouwelse	
Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	In this course you will learn all aspects of blockchain technology, including tamper-proof data structures, digital identities, transitive trust, fault tolerance, distributed consensus, smart contracts and applications. Ledgers and blockchains are an emerging technology with the potential to radically improve financial transactions, supply-chain flows, transactions in general, and distributed databases. The first three weeks of the course will provide a fast-paced introduction to Bitcoin, Ethereum, and TrustChain developed by TUDelft itself. The main component in this course is a team-based complex engineering project. This course is designed for computer scientists to understand blockchain technology and to produce significant hands-on experience. To provide a deep understanding of blockchain technology and understand why it is special you need to experience first-hand how it operates at a detailed technical level. Students design, implement, and test their own independent project in teams of 3-5 students. Students can choose from a pool of possible project ideas. This course requires you to like software engineering. Topics covered: -Blockchain basics and evolution Bitcoin 1st generation, smart contract generation, future 3rd generation (trust or trust in math) -identity and transitive trust Authentication and security primitives, tamper-proof identities, trust models, MITM attacks, Sybil attacks, and TrustChain by TUDelft -Consensus models Proof-of-work, permissioned, Proof-of-stake, Corda no-global-consensus, TUDelft bottom-up fast consensus model -Smart Contract pro/con encrypted data, Bitcoin scripts, Etherium execution model, Hyperledger + Docker issues, Corda Jar file approach, Tezos difficult to use, powerful technology, vision of the future: trusted verified execution -Markets and exchanges Disruption by open markets, winner-takes-all, and multi-sided market platforms, Uber, Airbnb, 22 years of eBay, Silk Road, honesty among drug dealers, the role of trust in markets, P2P exchange markets -Decentralized Autonomous Organization, novel method to collaborate and organise any economic activity -Web3: how the Internet can be re-decentralised using DAOs Within this 2023 edition "the Delft DAO" will be prominently featured. TUDelft achieved a world-first in DAO research. We devised a full end-to-end proof-of-principle of a DAO which is capable of 0) near unbounded scalability 1) controlling money 2) democratic decision making and 3) continuous sustained self-evolution. This course provides you with the knowledge to work with this advanced technology. After this course you will have a firm grasp on the current operational blockchain-based systems, realistic view of real-world applications that may be built on top of ledger technology. You will be able to reason and discuss the open challenges and questions that still need to be resolved. This course is a key course for distributed systems students.	
Study Goals	After this course students are able to design and engineer complex blockchain-based systems. Students are able to describe blockchain technology, the various consensus model, smart contracts, markets, and relation to existing database technology. Student are able to setup a new architecture for blockchain applications.	
Education Method	This course consists of four 2-hour lectures. Each lecture is followed by a 4-hour homework period in the same week focused on understanding the background material. In week 1 you will form teams and initiate work on your blockchain engineering project. A list of projects to select from will be provided at the start of this course.	
Literature and Study Materials	Online course textbook: Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction by Narayanan, Bonneau, Felten, Miller and Goldfeder.	
Prerequisites	It is highly recommended to follow this course (see remarks): Security and Cryptography (Q1) Distributed Algorithms (Q2)	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) You will write down your project results in a single-page report, markdown format. You will be graded on your open source efforts located on Github and single-page report. Your grade will be expressed on a scale of 0 to 10. Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This class has a limited capacity (50). If there is a larger number of enrollments than the capacity of the class, students will be assigned to their preferred blockchain engineering project based on their background, engineering experience level, and match to the course goals. Students who followed Security and Cryptography (Q1) and are also enrolled in Distributed Algorithms (Q2) will have priority for placement. Mathematics students are exempts from this, if they can show some minimal software development experience (e.g. Github profile). Finally, students with a Grade Point Average of 8.0 or higher are eligible for the challenging scientific projects, resulting in a research paper. These project receive intense guidance, but have no capacity limits.	

CS4195	Modeling and Data Analysis in Complex Networks	5
Responsible Instructor	H. Wang	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The assignment and final project require basic programming skill.	
Course Contents	Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	After this course, students could construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies, model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. Students could obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 20%) Final project (weighting 80%) Final grade calculation: 0.2 * Assignments + 0.8 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4210-A	Algorithms for Intelligent Decision Making	5
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. V. Robu	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: CS4375: Artificial Intelligence Techniques, or equivalent; and/or IN4344: Advanced Algorithms, or equivalent Required: basic course(s) in algorithm design and analysis, logic and probability; basic programming (in Python)	
Course Contents	Decision making is at the centre of artificial intelligence. This course gives you practical skills on a solid theoretical base. The course looks at solving mathematical models of NP-hard discrete optimisation problems. These kinds of problems lie at the heart of AI techniques such as planning, machine learning and mechanism design, and more generally combinatorial optimisation. You will learn about a range of modelling techniques from boolean satisfiability to constraint programming, and how advanced solvers for these models work. The course has plenty of real-world case studies as well as theoretical results. Apply the skills you learn in this course by taking CS4210-B: Intelligent Decision Making Project in quarter 4!	
Study Goals	By the end of this course, you will be able to identify features of real-world combinatorial decision problems, and be able to model and design systems for simplified instances of these problems using boolean satisfiability, mixed integer programming, and constraint programming over finite and real domains. You will be able to explain how SAT, CP and LCG solvers work in some detail, and how MIP solvers work at a high level. You will also be able to explain the basics of program synthesis and combinatorial game theory.	
Education Method	Lectures, homework exercises (optional, and not counted towards the course grade), and programming assignment(s) (mandatory, but not counted towards the course grade). The expected workload is: 30% lectures (including preparation for the exams) 30% homework exercises (optional) 40% programming assignments	
Literature and Study Materials	Provided on Brightspace	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Besides the summative assessment, formative coursework will be offered. This includes a mandatory assignment, performed in pairs. Students must submit a non-trivial attempt at the assignment in order to pass the course. The assignment receives formative feedback, but is not counted in computing the course grade. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Elective	Yes	
Tags	Algorithmics Artificial intelligence Group work Modelling Optimalisation Programming Projects Small groups	

CS4225	Educational Technologies	5
Responsible Instructor	Prof. M.M. Specht	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	* Theories of Human Information Processing and Learning * Learning Management Systems * Learning Analytics * Personalisation and Adaptive Educational Systems * Mobile and Seamless Learning Technologies * Artificial Intelligence in Education * Realtime Learning Technologies * Project Design * Project Implementation	
Study Goals	The course will enable you to classify, understand, design and implement the core functionalities and systems for supporting human learning processes. As well current practices implemented as also approaches for technology enhanced learning currently researched will be presented. You will learn how educational technologies provide human learning process support, implement guidance and recommendation, create personalised learning support, as also give real-time feedback and support reflection of learners. In the final project you will identify a problem, design a solution based on the presented approaches and implement your own educational technology solution.	
Education Method	Lectures, weekly assignments and quiz questions, final project	
Assessment	The final grade of the course consists of the following components: Weekly assignments (weighting 30%) Final project (weighting 70%) Final grade calculation: 0.3 * weekly assignments + 0.7 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4230	Machine Learning 2	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Instructor	Dr. D.M.J. Tax	
Instructor	T.J. Viering	
Instructor	Dr. F.A. Oliehoek	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Required: This course is the more advanced and research oriented follow-up to Machine Learning 1 (CS4220). CS4220 or an equivalent machine learning course is therefore expected as prior knowledge.	
Course Contents	The course will treat a number of machine learning settings and theories spanning the breadth of machine learning research in detail and on an advanced level. Topics that are covered include (subject to change): learning theory, learning curves, latent variable models, causal inference and discovery, adversarial learning, online learning and neuromorphic computing.	
Study Goals	After completing the course, you should be able to: 1. Identify which setting beyond (un)supervised learning a problem belongs to 2. Explain the essential problem that (adversarial/causal/online/PAC/neuromorphic/...) learning addresses and how it differs from the classical ML setting 3. Evaluate methods that are proposed in these learning scenarios, both theoretically and empirically 4. Mathematically prove basic properties of methods in these scenarios 5. Apply common methods in these learning scenarios on real-world datasets 6. Develop machine learning techniques adapted for a particular learning task	
Education Method	Each week: 1 lecture and 1 Q&A session where the weekly exercise set will be discussed.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) 4 combined assignments (pass/fail). There will be multiple small assignments (in weeks 2,4,6,8). Final grade calculation: P/F * assignments + 1 * Written exam A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: 4 combined assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4235	Socio-Cognitive Engineering	5
Responsible Instructor	Prof.dr. M.A. Neerincx	
Instructor	F. Broz	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic prior knowledge on human-computer interaction is helpful, but not required.	
Course Contents	Whether you are playing a game in virtual reality, driving a semi-autonomous car, educating yourself with a mobile app, or harmonizing your health and lifestyle with a robot assistant; nowadays intelligent networked information and communication technology is omnipresent. This course focuses on the design of human-aware intelligence into such environments, to support joint human-technology performances that bring about positive human experiences. The focus will be on social robots in general, with practical design and test assignments on robots that support people with dementia, http://rejam.tudelft.nl). In the Socio-Cognitive Engineering (SCE) course (MSc level), you will become acquainted with the application of a coherent set of methods for the design and evaluation of human--agent collaboration. Based on the SCE-method, we will elaborate on the state of the art of intelligent user interfaces (ePartners), such as virtual assistants, robotic teammates, eCoaches, and companion agents. The main topics of study are: - Design methods: Cognitive Engineering, Value Sensitive Design, Scenario-based Design, Claims Analysis, Design Rationale, Design Patterns. - Design for collective intelligence: Knowledge Representation, Ontology Engineering, Mental Models, Theory of Mind, ePartners, Adaptive Automation, Social Robots. - Design Evaluation: Prototyping, Test Methods, Measures, Questionnaires, Ethics. - Human Factors Theories and Models: Human Cognition & Learning, Memory, Emotion, Task Load, Human-Agent Teamwork, Behavior Change and Persuasive Technology.	
Study Goals	At the end of this course, students will be able to: 1. Explain the essential concepts of the design methods addressed in the course. 2. Explain the (dis)advantages of various design methods and their complementarity. 3. Apply the design methods addressed in the course in their research and design projects. 4. Explain what a design rationale is. 5. Construct a design rationale. 6. Create design specifications that are grounded in a design rationale. 7. Evaluate the strengths and weaknesses of a design rationale, e.g. using human-centered evaluations that test the design rationale. 8. Explain some of the state of the art human factors theories, models, and methods relevant to intelligent user interfaces, human-robot collaboration, and ePartner technology. 9. Write a structured report about a design-test cycle, with sufficient detail for a new group of researchers to continue the research. 10. Present work on a design project to an academic audience. 11. Work in a group on collaborative assignments.	
Education Method	LECTURES During the lectures, the teachers will present a range of theories, models, and methods relevant to socio-cognitive engineering. Students are required to read a number of scientific papers which are made available on Brightspace, along with the sheets/slides of the lectures. Together, the sheets/slides and the papers provide the students with the required theoretical knowledge to work on the practical project, and to learn about relevant design methods, human factors theories, conceptual solutions, and design principles. Most of the lectures include practical assignments and discussions stimulating the students to apply the contents of the lecture to their project (also see Project). PROJECT In the project, students work in groups to apply the knowledge acquired during the lectures. Students are required to plan, execute, present, and report on a complete design cycle (i.e. design, prototype, and evaluation) for a given design problem. This year (like the past years), the design problem is a social robot for older adults with dementia, and their social environment (https://rejam.tudelft.nl). The objective of the social robot is to improve humans physical, social, cognitive, and emotional well-being. The students will use the Wiki Socio-Cognitive Engineering (WiSCE) tool to specify the design rationale and its evaluation, step-by-step (see also https://xwiki.ewi.tudelft.nl/). Throughout the course, students will give presentations about their progress, on the design and evaluation of their prototype. Literature and study material consist of: - Papers from scientific journals on Brightspace. - Lecture notes on Brightspace	
Literature and Study Materials	Papers from scientific journals on Brightspace. Lecture notes on Brightspace.	
Assessment	The module assessment concerns the processing and application of the theory and methods; the construction of the design (rationale) and the evaluation; and the provision of the resulting concise and coherent report (including the lessons learned): The final grade of the course consists of the following components: Presentations (weighting 15%) Prototype (weighting 15%) Project report according to the prescribed format (weighting 60%) Individual reflection (weighting 10%) Final grade calculation: 0.15 * Presentations + 0.15 * Prototype + 0.6 * Project report + 0.1 * Individual reflection A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentations: resit Prototype: re-submit deliverable Project report according to the prescribed format: re-submit deliverable Individual reflection: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Exam Hours	There is no exam. The assessment is based on a paper, presentation and report. During the course, students will receive feedback on interim work. There is no resit after the end of the course.

CS4240	Deep Learning	5

Responsible Instructor	Dr. J.C. van Gemert
Instructor	Dr. D.M.J. Tax
Contact Hours / Week x/x/x/x	0/0/8/0
Education Period	3
Start Education	3
Exam Period	3 4
Course Language	English
Required for	Seminar Computer Vision by Deep Learning
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy). Experience in programming and installing programming languages/libraries is required. Without affinity/experience with python (and numpy) programming this course will become too time consuming as the project will become difficult.
Course Contents	In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ - https://udlbook.github.io/udlbook/ Research papers will be made available through Brightspace. Assignments are based on PyTorch: https://pytorch.org/
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Digital Exam (weighting 90%). Exam about the papers, lab assignments, and the theory (lectures). Project (Pass/Fail). In a small group of students you work on a deep learning paper reproducibility project. Final grade calculation: (pass project) * (0.1 * Presentation + 0.9 * Exam) A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

CS4345	Seminar Formal Methods for Learned Systems	5
Responsible Instructor	Dr. A. Lukina	
Instructor	Dr. M.T.J. Spaan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course will cover several topics at the intersection of formal methods and machine learning, for instance, abstraction refinement for analyzing deep neural networks, algorithms for reliable learning-based control, interpretable model design, and formal methods for verification and monitoring of learned systems. The course should be useful for students of both formal methods and machine learning, and lies at the intersection of these areas. Since the area is relatively new, the course material will be primarily based on research papers.	
Study Goals	After this course students will acquire formal methods knowledge and skills such as: - modeling complex systems, - formally defining critical properties, - applying verification tools to learned systems, - synthesizing controllers for learned systems, - providing interpretability for learned models.	
Education Method	The course will consist of one lecture and one discussion session per week. The students will be assigned projects in groups of two, where they will need to apply the material from the lectures, e.g., train a learned system, define a property and verify it using one of the open-source tools from state-of-the-art research papers. Discussion sessions will be used for questions about the projects.	
Assessment	The final grade of the course consists of the following components: Participation (weighting 15%) Presenting Papers (weighting 15%) Assignment #1 (weighting 15%) Assignment #2 (weighting 15%) Final Report (weighting 40%) Final grade calculation: 0.15 * Participation + 0.15 * Presenting Papers + 0.15 * Assignment #1 + 0.15 * Assignment #2 + 0.4 * Final report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presenting Papers: resit Assignment #1: re-submit deliverable Assignment #1: re-submit deliverable Final report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4370	Testing and Validation for AI-intensive Systems	5
Responsible Instructor	C.C.S. Liem	
Responsible Instructor	Dr. A. Panichella	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of machine learning. Knowledge of software testing is preferred but not required.	
Summary	How do you know and assess whether an AI-intensive system is working as intended? On the one hand, quality assurance for such systems can be considered to be a software testing challenge; in parallel, the machine learning components within such systems are normally validated according to machine learning-specific evaluation methodology. In this course, you will learn more about these two complementary takes to evaluation and quality assurance, while learning about (and getting hands-on experience in) state-of-the-art advances in combining these two takes.	
Course Contents	- Basics of testing (assertions, white-box vs. black-box testing) - Basics of machine learning evaluation (validation & validity) - Adversarial attacks - Testing techniques (e.g. fuzzing, metamorphic testing, combinatorial testing, differential testing) - Data labels (oracle) validation	
Study Goals	After completing this course, students are able to: 1. Apply state-of-the-art techniques to validate machine learning models and training routines 2. Explain, analyze and improve robustness considerations in AI-intensive systems 3. Design and implement techniques to generate adversarial attacks 4. Apply fuzzing, metamorphic and combinatorial testing techniques to machine learning models 5. Connect the oracle problem in software testing to questions of validity in datasets 6. Complement classical evaluation reporting of machine learning models with insights obtained through testing	
Education Method	2 hours lecture, 2 hours project	
Literature and Study Materials	The study material will be provided online by the lecturers	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4405	Analysis of Concurrent and Distributed Programs	5

Responsible Instructor	Dr. B. Özkan
Responsible Instructor	Dr. S.S. Chakraborty
Contact Hours / Week x/x/x/x	0/0/4/0
Education Period	3
Start Education	3
Exam Period	3
Course Language	English

Course Contents

Software systems are becoming highly concurrent and distributed to utilize modern multicore architectures and increasing speed and bandwidth in networks. Shared-memory concurrency in multicore programs and message-passing concurrency in distributed programs share many common abstractions and problems.

In the multicore era, all performance-critical software employs some form of concurrent programming; typically shared memory concurrency. In this setting,
programmers use a number of primitives to develop efficient and correct concurrent programs. To do so the programmers have to understand the behaviors of the primitives and reason about them. It is also important to match the programming paradigms and underlying architectures. For instance, traditionally programmers have assumed that a multithreaded program executed simply by interleaving the executions of its threadsa model known as sequential consistency (SC). This assumption is, however, invalidated both by mainstream multicore architectures, which often execute instructions out of order, and by compilers, whose optimizations affect the outcomes of concurrent programs. As a result, concurrent programs have more outcomes than SC allows.

In the distributed setting, the units of concurrency are independent processes that do not share memory but communicate by exchanging asynchronous messages. The execution of such a system involves two main sources of nondeterminism: concurrency and partial failures. As the processes run concurrently, the exchanged messages can be delivered and processed in many different orderings. The distributed set of processes is also prone to network of process failures. The trade-off between the systems availability in the existence of failures and the consistency between the processes gives rise to a spectrum of weak consistency notions. It is important to reason about concurrency, possible failures, and consistency guarantees to implement distributed programs correctly and understand their behavior.

This course aims to explore analysis techniques for concurrent and distributed programs.

Outline of Lectures:

Shared memory concurrency:
- Abstractions for shared memory concurrency
- Relaxed memory concurrency
- Correctness of concurrent programs

Distributed concurrency:
- Distributed system components, models and assumptions
- Fundamental abstractions for distributed systems

Study Goals

This course aims to give students a deep understanding of concurrency and distribution in modern systems and hands-on experience for analyzing these systems.
By the end of this course, you should be able to:

- Describe the fundamental concurrency models in multicore and distributed systems

- Explain the concurrency nondeterminism in the executions of concurrent and distributed programs

- Discover concurrency bugs in multicore and distributed programs by program analysis and testing techniques

- Evaluate the pros and cons of program analysis and testing techniques for specific multicore and distributed programs

Education Method

The course consists of the following education methods:

- Lectures for reviewing concurrency and distribution concepts
- Homeworks/assignments
- Developing a course project, writing a report, and presenting it (course project)

To finish the course, students (in teams) will have to:

- Study several papers which will be discussed during the lectures
- Deliver their assignments
- Deliver and present their implementation project

Assessment

The final grade of the course consists of the following components:
Research project implementation (weighting 40%)
Research project report (weighting 20%)
Research project presentation (weighting 20%)
Homework assignments (weighting 20%)

Final grade calculation: 0.4 * Research project implementation + 0.2 * Research project report + 0.2 * Research project presentation + 0.2 * Homework assignments
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

Resit/ Repair opportunities:
In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:
Research project report: re-submit deliverable
Research project presentation: resit
Homework assignments: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

CS4410	Category Theory for Programmers	5
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	Ir. K.F. Wullaert	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Categorical structures occur in programming languages on different levels: (1) within programming languages, providing design principles and guidance on how to write modular and correct-by-design programmes (as demonstrated in the practical programming language Haskell) and (2) in the design and study of programming languages, as a guiding meta-theory. In particular, category theory provides a mathematical justification for recursion schemes for inductive datatypes. This course aims to provide solid foundations on both (1) and (2). Lecture notes for the course are available at https://github.com/benediktahrens/CT4P. This course is suitable for people with an interest in functional programming, abstract mathematics, and logical reasoning. Prior knowledge: - Required: None - Strongly advised: None	
Study Goals	Learning Objectives: - Use categorical constructions (e.g., monads) in the design and structuring of computer programmes - Specify datatypes as initial algebras - Prove properties of computer programmes, guided by categorical intuition - Use categorical fusion laws to optimize functions - Use the theory of infinite data structures to define functions on infinite data	
Education Method	Learning in this course is achieved through lectures, problem sessions, and guided self-study.	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4415	Sustainable Software Engineering	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Sustainable Software Engineering is an overarching discipline that addresses the long-term consequences of designing, building, and releasing a software project. By definition, sustainability covers five main perspectives: environmental, social, individual, economic, technical. This course mainly focuses on the first, also known as Green Software Engineering. On top of that, we also cover fundamental aspects of social and individual sustainability of software projects. Software Engineering (SE) has long addressed sustainability by narrowing it down to economic and technical sustainability. However, our society is facing major sustainability challenges that can no longer be overlooked by software engineers and computer scientists. It is estimated that, by 2040, the ICT sector will contribute to 14% of the global carbon footprint. Hence, the environmental, social, and individual perspectives ought to be part of the equation when it comes to design, build, and release software systems. The problem is far from simple and we need expert computer scientists to bring sustainability into the core values of the next generation of tech-leading organisations.	
Study Goals	After attending this course, you will be able to: LO1. Explain the fundamental sustainability principles and their relevance to software engineering. LO2. Assess the energy consumption of software systems using measurement and testing techniques. LO3. Create actionable strategies to address sustainability issues within software organisations. LO4. Evaluate and compare different solutions for sustainable software systems based on their effectiveness in achieving sustainability goals and their potential impact on various stakeholders.	
Education Method	To meet these objectives, you will be involved in a broad set of learning activities: lectures, paper reading, software analysis, software development, essay writing and presentation. These heterogenous set of activities aim at building a strong set of hard skills for energy-efficient code development combined with a strong set of soft-skills and critical thinking. You will work on ideas that will help real-world software projects embrace a green software culture.	
Assessment	The final grade of the course consists of the following components: Software project (weighting 45%) Essay (weighting 45%) Presentation (weighting 10%) Final grade calculation: 0.45 * Software project+ 0.45 * Essay + 0.1 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Software project: re-submit deliverable Essay: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4430	Network Security	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best practices in computer and network security. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course's primary focus will be on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design, review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics, as well as conducting measurements on the effectiveness of attack and defense schemes.	
Study Goals	See course contents.	
Education Method	Lectures, Labs, and Project.	
Assessment	The final grade of the course consists of the following components: Assignments (Pass/Fail) Final project (Report + Presentation) (weighting 100%) Final grade calculation: P/F * Assignments + 1* Final project A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable, resit presentation Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l0 norm is shown to be NP-hard, and several classical approximate algorithms like l1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (10% each) based on homework questions and coding assignments (15%). The course is closed by a project presentation (20%) and a report (35%). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IN4152	3D Computer Graphics and Animation	5
Responsible Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. R. Marroquim	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Students that haven't followed any previous Computer Graphics courses (like CSE2215) will be able to participate, but might have to invest some more time to catch up in the first lectures.	
Course Contents	Have you ever wondered how Toy Story was made, why the game Last of Us 2 looks so beautiful, or have you ever wanted to create your own graphics application or game? Then you should consider following this course. If not, then you should still follow it... maybe, you will become interested! In this course, you will get a good idea of Computer Graphics in general. The topic is of very high relevance for the industry and the research community and has numerous applications in different domains, such as scientific visualization, video games, simulators, special effects, animated movies and many more. Here, you will learn about basic algorithms, as well as modern techniques. We will address several topics: the principles of image synthesis, object representations, geometric and hierarchical transformations, graphics cards and the graphics pipeline, realistic rendering (including global illumination and effects, such as reflections), expressive rendering, physics simulations, rendering control (including previsualization systems used by professionals in the movie industry), and perceptual rendering, which relies on properties of the human visual system to enhance the quality of the images. Besides course sessions on the theory of Computer Graphics, some of the algorithms will also be reproduced in practice, and deepened during the final project.	
Study Goals	The course teaches computer graphics techniques on an advanced level. After the course the student is able to classify the different modeling, shading, and display techniques. The student can reproduce the basic mathematical and algorithmic notions associated with these concepts, can comment on the weak and strong points of these techniques, and can apply the core concepts within a graphics program in practice.	
Education Method	lectures, instructions, research papers, lab work	
Literature and Study Materials	Research Papers in domain of selected topics, lecture sheets, online sources, optional books (see below)	
Books	Fundamentals of Computer Graphics by Shirley et al. - CRC Press Real-time Rendering by Tomas Akenine-Möller, Eric Haines, Naty Hoffman - Peters, Wellesley Real-Time Shadows by Elmar Eisemann, Michael Schwarz, Ulf Assarsson, Michael Wimmer - Taylor & Francis Computer Graphics. Principles and Practice by James D. Foley, Andries VanDam, Steven K. Feiner - Addison Wesley Advanced Material: Physically Based Rendering, From Theory to Implementation by Matt Pharr, Wenzel Jakob and Greg Humphreys - MIT Press	
Assessment	The final grade of the course consists of the following components (details of both elements will be presented during the lecture): Project (weighting 60%). The project grade is the result of a project and its presentation that is building upon the assignments that are handed out (roughly) weekly during the duration of this course. Paper (weighting 40%). The paper grade is the result of the presentation of a scientific paper and the development of an associated practical implementation. Final grade calculation: 0.6 * Project + 0.4 * Paper A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Paper: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4253ET	"Hacking Lab"-Applied Security Analysis	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Necessary background differs per student project, see first lecture or contact instructors for details	
Course Contents	The security of computer and telecommunication systems is becoming an increasing concern. In this course, we will review the current state of the art on security research and gain practical experience in assessing the security and vulnerabilities of communication systems. Engineers are typically taught to focus on performance, correctness, scalability, and maintainability when building communication and information processing systems. However, an additional set of design principles are required to achieve security. In this course, we discuss security principles, common pitfalls and vulnerabilities. The weekly lectures provide an introduction into security research, with a focus on real-world security, privacy-enhancing technology and common security pitfalls. Each student participates in a "Hack Project", with a group of one to four students. Students can select between a wide range of available Hack Project outlines within the first week. The goal may be to evaluate the security of a real-world IT system, developing a proof-of-concept exposing a vulnerability or focussed on preserving privacy in a post-Snowden world. Students may propose their own Hack Project based on their background knowledge and skills. Such Hack Projects need to be approved and shaped together with the instructor. Example of possible outlined hardware-oriented projects are: development of a wifi tracker, programing an FPGA system to break passwords, assess the security of RFID cards, or to transparently intercept Ethernet traffic. Concrete software projects are: hacking Bitcoin, improving the TOR anonymity protocol and create Android-based tools for human rights activists in Iran, Egypt and Russia, reprogramming neural networks attacks. Each Hack Project is documented with a written report. This can be in the form of a 6-8 page IEEE-style scientific article or a traditional more lengthy report. All results, experiences and findings are presented to the entire class in the last week of the course. Hack Projects also report their progress several times during the course, after the weekly lectures.	
Study Goals	After this course, the student will have a thorough knowledge of security in real-world systems, and will be able to explore the literature on this topic independently. The student will be aware of the poor state of security in real-world computer systems. The student can explain the common pitfalls, why these known failures still occur and reasons behind the poor state of security in general.	
Education Method	Lectures, student presentations, written final report and active participation. Attendance and active participation during lectures is mandatory. This sadly means telelecturing is not possible.	
Literature and Study Materials	Customize literature lists and study materials are provided per project topic	
Assessment	The final grade of the course consists of the following components: Written Hack Project report (weighting 60%) Final presentation of result (weighting 10%) Presentation of ongoing project progress (weighting 20%) Participation in discussions, overall quality of the practical work and class attendance (weighting 10%) Attendance to lectures is mandatory. Final grade calculation: 0.6 * Written Hack Project report + 0.1 * Final presentation of result + 0.2 * Presentation of ongoing project progress + 0.1 * Participation in discussions, overall quality of the practical work and class attendance A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Written Hack Project report: re-submit deliverable Final presentation of result: resit Presentation of ongoing project progress: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
maximum aantal deelnemers	If there is an unexpected high demand for this course, then enrollment will be based on past performance in relevant courses.	
Co-Instructor	Dr. S. Picek	

IN4315	Software Architecture	5
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software engineering	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four, in which the actual architectures of existing open source systems are analyzed in considerable detail. These systems are taken from github, and student teams are challenged to actually contribute to the systems under analysis in the course.	
Study Goals	Bring students into the position that they can (1) explain the key architectural concepts and methods for modeling software architectures; (2) apply viewpoints and perspectives to model software architectures; (3) discuss the benefits of architecting and the role of the software architect; (4) evaluate and validate software architectures; (5) explain and discuss the concepts of component-based and plugin architectures, service-oriented architectures, and software product lines; (6) explain and recognize technical debt and have an understanding of possible refactorings.	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books; Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; and Coplien and Bjørnvig, "Lean Architecture", Wiley, 2010. Additional reading material will be announced in the lectures.	
Assessment	Grades will be based on lab assignment including essay writing, coding, (video) presentation, peer reviewing, participation. The final grade of the course consists of the following components: Essays (weighting 60%) Coding (weighting 20%) Presentation (weighting 20%) Final grade calculation: 0.6 * Essays + 0.2 * Coding + 0.2 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Essays: re-submit deliverable Coding: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Course web site: https://se.ewi.tudelft.nl/delftswa/index.html	
Co-Instructor	M. Finavaro Aniche	

IN4325	Information Retrieval	5
Responsible Instructor	J. Yang	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language. Knowledge of Web information systems and software engineering can be helpful.	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. The World Wide Web is now the primary source of information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, analysed, and exploited so that each user can quickly find information that is both relevant and comprehensive for their needs. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation of Web and Enterprise search engines. Covered topics include: = Information Retrieval Models; = Indexing Techniques; = Web Search; = Information Seeking Paradigms; = Evaluation of information retrieval systems;	
Study Goals	At the completion of this course, students will be able to: = Describe the different information retrieval models, and compare their strenghts and weaknesses. [Learning Objective 1] = Describe and implement different indexing techniques. [Learning Objective 2] = Describe and analyze querying techniques with respect to their most suited application domains. [Learning Objective 3] = Analyse the effectiveness of an information retrieval system through proper use of evaluation metrics. [Learning Objective 4] = Design and implement (Web) Information Retrieval systems, possibly using advanced social and semantic search functionalities. Support and defend the relevance and correctness the choices with regards to the adopted information retrieval model, indexing technique, and querying technique. [Learning Objective 5]	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 80 hours for the group project and 15 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books - all resources are available on Brightspace.	
Books	Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. Introduction to Information Retrieval. Cambridge University Press, New York, NY, USA. Stefan Büttcher, Charles Clarke, and Gordon V. Cormack. 2010. Information Retrieval: Implementing and Evaluating Search Engines. The MIT Press.	
Assessment	The final grade of the course consists of the following components: Individual assignment (weighting 10%) Weekly individual assignment (weighting 20%) Group project (weighting 70%) Final grade calculation: 0.1 * Individual assignment + 0.2 * Weekly individual assignment + 0.7 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual assignment: re-submit deliverable Weekly individual assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialistievakken start kwartaal 4 2023

CESE4120	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics". We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming.	
Course Contents	The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. After taking this course students will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non--trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	Written reports + project presentation + oral exam Overall, the final grade is determined by: 1) Two intermediate reports (10% of grade each, 2 pages each) 2) Final report (10 % of grade, 4 pages) 3) Final project demonstration (70% of grade) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. There is no resit for this course. repair option:A new but related project will be given for the Report (W), Oral(0) and Demonstration (IP). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

CS4035	Cyber Data Analytics	5
Responsible Instructor	Ir. S.E. Verwer	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course provides theoretical and practical background for applying data analytics in the field of cyber security. Cyber data analytics is a huge field with a great diversity of techniques and applications. The course is centered on a selection of seven such techniques: learning from imbalanced data; behavioral profiling and anomaly detection; sequential data mining; data stream mining; learning from software data; adversarial machine learning; and privacy-aware data mining; Anomaly detection is one of the main topics in cyber security. Specific difficulties that the student will learn to handle are the huge amounts of data and the large number of false positives. Behavioral profiling applies to both people and software processes. Different techniques will be taught to construct profiles from software logs. While building such profiles, care should be taken to not infringe upon the privacy of individuals the data is collected from. Finally, attackers will modify their behavior in order to avoid being detected, a cyber data analytics engineer tries to make their models/profiles robust against such modifications.	
Study Goals	The student will be able to: Apply machine learning to real data Understand and modify machine learning algorithms Learn models from time series Detect anomalies in multidimensional time-series Use distributed processing to speed up machine learning Learn models from data streams with limited memory Learn sequential models Use machine learning for fingerprinting and profiling Preserve the privacy of data owners while learning models Learn robust models that can detect evasive attackers Use machine learning to detect fraud, attacks, and botnets	
Education Method	There will be two lectures for each of the seven topics, and 3 large lab assignments on fraud detection, anomaly detection, and behavioral profiling, and 1 smaller lab on adversarial robustness. There is no exam. Teams of two students will work on these assignments which contain both individual and collaborative components. Deadlines are strict as peer-review will be used to both learn of other possible solutions, provide feedback, and get initial estimates on the obtained grade.	
Assessment	The final grade of the course consists of the following components: Lab assignment fraud detection (weighting 30%) Lab assignment anomaly detection (weighting 30%) Lab assignment behavioral profiling (weighting 30%) Lab assignment adversarial robustness (weighting 10%) Final grade calculation: 0.3 * Lab assignment + 0.3 * Lab assignment + 0.3 * Lab assignment + 0.1 * Lab assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignment fraud detection: Re-submit deliverable Lab assignment anomaly detection: Re-submit deliverable Lab assignment behavioral profiling: Re-submit deliverable Lab assignment adversarial robustness: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4125	Seminar Research Methodology for Data Science	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Instructor	Dr. K.A. Hildebrandt	
Instructor	J. Urbano Merino	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	basic knowledge in mathematics (linear algebra, calculus, probability and statistics)	
Course Contents	The course focuses on research methods for data science. It looks at underlying principles and concepts for data collection, analysis and data processing, as well as the use of tools to do this. The main topics of study are: Conceptualizing research questions and experimental design Frequentist and Bayesian data analysis Generalized linear models for statistical analysis Multilevel modelling for hierarchical and longitudinal data analysis Measuring and sampling, validity and reliability Linear and nonlinear dimensional reduction Principles of statistical testing In the course, students will be using software tools such as R, and Matlab	
Study Goals	The main aims of this module for the student is to achieve an understanding of research methods for data science and obtain practical experience with data analysis and data processing methods. This module provides students with the opportunity to develop and demonstrate their understanding, knowledge, and competence. The learning outcomes for the module are that students will be able to: 1. Appreciate and comprehend strategies for collecting and processing data to answer research questions based on data 2. Understand and reproduce key principles underlying statistical data and data processing analysis 3. Learn to identify and avoid typical biases, paradoxes and misunderstandings in data-driven research 4. Apply and select appropriate data modelling techniques to analyse data and data processing	
Education Method	Lectures/Assignments Some lectures have a flipped classroom set-up, where it is expected that students prepare themselves by watching a series of videos. In the lecture, students work together on a set of classroom challenges under the guidance of the lecturer. In some others, students work on preliminary solutions to some exercises that will be discussed during the lecture. Expected Workload Lectures: 26 hours (13 × 2 hours lectures) Reading time: 39 hours Preparation basis tool use: 25 hours (5 × 5 hours for each tool) Coursework project, including writing reports and preparing for presentation: 50 hours (10 × 5 hours) Total = 140 hours	
Literature and Study Materials	Will be provided online	
Assessment	The final grade of the course consists of the following components: Analysis of experimental research data (weighting 40%) Exploration of real-world data set (weighting 30%) Linear and nonlinear dimensional reduction (weighting 30%) Students work in small groups on the 3 assignments. For the assignment, the student group submits reports and gives a presentation, including a question and answer round where individual group members are assessed on the coursework. Final grade calculation: 0.4 * Analysis of experimental research data + 0.3 * Exploration of real-world data set + 0.3 * Linear and nonlinear dimensional reduction A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Analysis of experimental research data: re-submit deliverable Exploration of real-world data set: re-submit deliverable Linear and nonlinear dimensional reduction: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	NA	

CS4145	Crowd Computing	5
Responsible Instructor	U.K. Gadiraju	
Instructor	A. Bozzon	
Instructor	J. Yang	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic knowledge of artificial intelligence and/or human computer interaction is advised. Proficiency in at least one programming language.	
Course Contents	Crowd Computing is an emerging field that sits at the intersection of computer science and data science. Crowd computing studies how large groups of people can solve complex tasks that are currently beyond the capabilities of artificial intelligence algorithms, and that cannot be solved by a single person alone. It involves the algorithmic engagement and coordination of people by means of Web-enabled platforms. These complex tasks are mainly focused on the creation, enrichment, and interpretation of data, making crowd computing a building block of data science. Examples of such tasks include the coordinated creation of data about real world events when electronic sensors are not available; the annotation of existing data sets to create ground truth data for the training of machine learning algorithms; and the analysis and interpretation of Web data to spot identify inappropriate content (e.g.,hate speech, or fake news). Crowd computing is an essential tool for any data-driven company: from Facebook to Microsoft, from Google to IBM, from Spotify to Pandora, all major companies employ crowd computing to fulfil their data needs, both by involving employees, and by reaching out to anonymous crowds through online marketplaces like Amazon Mechanical Turk or Appen. Crowd computing methods therefore play an important role in the design, development and evaluation of a variety of products, services, and systems in a variety of domains. The objective of the Crowd Computing course is to introduce the scientific and technical underpinnings of crowd computing, and to investigate how it can be used for computer science applications (e.g., information retrieval, machine learning, next-generation interfaces, and data mining) and for real world applications (e.g., cultural heritage preservation, online knowledge creation, smart cities, etc.) The course is designed around one key challenge, the creation and consumption of (high quality) data, and will be organized around three themes: 1) Establishing data needs; 2) Fulfilling data needs with crowd computing; and 3) Evaluating the quality of the retrieved data with respect to the original data need. Covered topics include: 1) Establishing Data Needs: - Requirement Elicitation - Requirement Analysis - User Modelling Properties 2) Fulfilling Data Needs with Crowd Computation: - Systems for/with collective intelligence (e.g., recommendation, semiautonomous systems, citizen science, crowdsourcing, and human computation systems) - Multi-modal Interaction (e.g., conversational systems) - Human Computation (e.g., worker modelling, task modelling, incentives, task assignment, recruitment) - Games with a purpose - Algorithms for Crowd Computing - Computational Methods for User Modelling - Interfaces for Crowd Computing Systems 3) Evaluating Retrieved Data: - Expert Evaluation - User Evaluation - Explanation of the output of Crowd Computing Systems 4) Study of Application Domains When applicable, the course will also feature invited lectures from selected academics and professionals in the field. Since instructors of this course are also directing the Design@Scale Delft AI lab, students of this course will have the opportunity to engage with cutting-edge research projects relevant to this lab. This Crowd Computing course is an elective for students following the Data Science and Technology Track and the Software Technology Track. It adds to the master education offer by addressing topics that are complementary to courses like IN4325 Information Retrieval, IN4252 Web Science & Engineering, CS4065 Multimedia Search and Recommendation, and IN4010 Artificial Intelligence Techniques.	
Study Goals	After this course, students will be able to: - Identify the requirements for a Crowd Computing system [LO1] - Design and develop Crowd Computing systems. Support and defend the relevance and correctness of his/her choices [LO2] - Describe and compare several Crowd Computing techniques. [LO3] - Describe and compare design decisions in the context of Crowd Computing interaction paradigms [LO4] - Determine which Crowd Computing technique(s) is most appropriate for being used in a certain problem domain [LO5] - Apply the appropriate Crowd Computing technique to an application domain and evaluate the obtained results. [LO6] - Analyse the performance of a Crowd Computing system by applying the proper evaluation measures. [LO7]	
Education Method	This course consists of 16 2-hour lectures. Each week, a 30-minute assignment tests the knowledge acquired on the discussed topics. Starting from Week 1, students form groups and work on a project, to be presented in week 9. Students are expected to work 6 hours per week (each) on the project assignment. Expected workload is 32 hours for attending lectures, 24 hours of reading study material and preparing lectures, 55 hours for weekly assignments and group assignment, 24 hours for preparing final survey, and 5 hours for exam and plenary presentations (total 140 hours).	
Literature and Study Materials	Books: - Human Computation. Author(s): Edith Law and Luis von Ahn. Synthesis Lectures on Artificial Intelligence and Machine	

	Learning, June 2011, Vol. 5, No. 3. http://www.morganclaypool.com/doi/abs/10.2200/S00371ED1V01Y201107AIM013 - A. Marcus and A. Parameswaran. Crowdsourced Data Management: Industry and Academic Perspectives. Foundations and TrendsR in Databases, vol. 6, no. 1-2, pp. 1161, 2013. DOI: 10.1561/1900000044. https://people.eecs.berkeley.edu/~adityagp/papers/crowd-book.pdf - An Introduction to Hybrid Human-Machine Information Systems. Demartini, G., Difallah, D.E., Gadiraju, U. and Catasta, M., 2017. Foundations and Trends in Web Science, 7(1), pp.1-87. https://edu.nl/np4th Slides: available on Brightspace Articles: available on Brightspace Recommended reading: - Interaction Design: Beyond Human-Computer Interaction (4th Ed, 2015). Authors: Jenny Preece, Helen Sharp, Yvonne Rogers
Assessment	The final grade of the course consists of the following components: Weekly Individual assignment (weighting 15%) Group Assignment (weighting 55%). The group assignment is performed collectively, but graded individually. Final Individual Assignment (weighting 30%) Final grade calculation: 0.15 * Weekly Individual assignment + 0.55 * Group Assignment + 0.3 * Final Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly Individual assignment: re-submit deliverable Group Assignment: re-submit deliverable Final Individual Assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Tags	Algorithmics Artificial intelligence Design Programming Research Methods Software Technology

CS4205	Evolutionary Algorithms	5

Responsible Instructor	P.A.N. Bosman
Instructor	Dr.ir. P.A. Bouter
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4 5
Course Language	English
Expected prior knowledge	Must have the ability to program, predominantly in Python (but also Java and C/C++ are very useful) so that you can create your own (extensions of) evolutionary algorithms. Further, a basic understanding of key aspects in linear algebra (e.g., matrix multiplications, inversions, decompositions), probability theory and statistics (e.g., normal distribution, statistical hypothesis testing), and complexity theory (e.g., big-O notation), will be very helpful.
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, and real-world applications. The course is planned to have 10 lectures, multiple small practical assignemnts and 1 larger practical assignment. The first practical assignments pertains to experimenting with already implemented EAs on predefined problems. The second practical assignment offers more freedom, allowing you, in a group, to build your own EA (this may vary depending on student numbers and other circumstances).
Study Goals	Upon successful completion of this course, students will be able to: 1) Explain and apply key concepts underlying the main streams in Evolutionary Algorithm (EA) research, in particular genetic algorithms, evolution strategies, genetic programming, evolutionary multi-objective optimization, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2) Explain and apply key ingredients underlying the rationale of when these algorithms work and when they do not work and several ways to overcome this. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3) Name and explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4) Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5) Properly (scientifically) experiment with EAs as well as program your own.
Education Method	10 Lectures Several smaller, individual, lab projects One larger lab project in groups of several students
Literature and Study Materials	Papers and slides that will be made available.
Assessment	The final grade of the course consists of the following components: Written exam (weighting 60%) Lab practical work (weighting 40%) Final grade calculation: 0.6 * written exam + 0.4 * lab practical work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab practical work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Permitted Materials during Tests	None
Tags	Algorithmics Artificial intelligence Optimalisation

CS4210-B	Intelligent Decision Making Project	5
Responsible Instructor	Dr. M.T.J. Spaan	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Dr. V. Robu	
Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. J.W. Böhmer	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Instructor	Dr. A. Lukina	
Contact Hours / Week x/x/x/x	0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Theoretical knowledge regarding algorithms for decision making in Artificial Intelligence, obtained for instance by passing one of the following courses: - CS4210-A Algorithms for Intelligent Decision Making - CS4400 Deep Reinforcement Learning - CS4375 Artificial Intelligence Techniques - IN4344 Advanced Algorithms.	
Course Contents	Intelligent decision making is a key skill of computational agents. Research on this topic focuses on building models and algorithms that enable AI systems to take appropriate decisions. Building upon theoretical knowledge gained in other courses, students collaborate in small groups on a distinct research project per group, for instance on decision-making problems in transport, logistics or smart energy grids. Purely algorithmic challenges will also be provided. The research projects provide a good opportunity to learn about topics suitable for Masters projects.	
Study Goals	After completing the Intelligent Decision Making Project course, the student is able to: 1. Apply algorithms for decision making to problem domains, and can compare and evaluate them. 2. Design and implement an extension of a decision-making algorithm. 3. Identify and discuss relevant topics in the research field of algorithms for intelligent decision making. 4. Describe and apply the appropriate research methodology. 5. Communicate his/her findings effectively.	
Education Method	A research project in a small group.	
Literature and Study Materials	Mainly survey papers and book chapters. Details are provided via Brightspace.	
Assessment	The final grade of the course consists of the following components: Quality of work of the research project (weighting 40%) A scientific report of the research project (including peer review of a report) (weighting 20%) Performance during the project (weighting 30%) Oral presentation of the research project (weighting 10%) Final grade calculation: 0.4 * Quality of work of the research project + 0.2 * A scientific report of the research project + 0.3 * Performance during the project + 0.1 * Oral presentation of the research project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Quality of work of the research project: re-submit deliverable A scientific report of the research project: re-submit deliverable Oral presentation of the research project: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Only a limited number of students can participate in this course. In order to be admitted, please submit a short motivation letter (max 200 words) via Brightspace. Attending the first lecture is compulsory.	
Tags	Artificial intelligence	
maximum aantal deelnemers	40	

CS4245	Seminar Computer Vision by Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	N. Tömen	
Instructor	Dr. X. Zhang	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	MSc thesis in the Computer Vision lab	
Expected prior knowledge	Deep Learning (CS4240)	

Course Contents

The recent boom in computer vision and automatic image understanding represents an inflection point in human productivity, permeating wide aspects of the economy and society. Examples of visual tasks which are repetitive or require expert knowledge include medical diagnosis, industrial inspection, autonomous vehicles, etc. When machines can meaningfully assist or even completely take-over such tasks it will change the world as we know it.

The breakthrough in the 2012 ImageNet automatic image recognition competition shows all previously existing methods decisively defeated by a deep neural network. Deep learning replaces feature engineering methods and is able to successfully learn image features from huge annotated datasets.

This course is on automatically understanding visual content such as images and videos by deep learning.

The range of topics (although, not limited to): Fundamentals in Vision, object detection, per-pixel labelings, video recognition, image similarity learning, efficiency, self-supervision, 3D computer vision, adverserial attacks, explainability, generative models, etc.

The course will have lectures, a seminar and a lab practical:

- The lectures will be on established topics based on the current literature.

- The seminar will have students read, critique, and present relevant computer vision research papers. You will have to read 2 papers per week, for 7 weeks (14 papers).

- The lab will have students apply and design their own (small) computer vision project.

The course build on top of the Deep Learning course (CS4240) and follows a similar setup.

Study Goals

Upon successful completion of the course, students will be able to:

[LO1]. Describe the deep learning techniques reviewed in the course for computer vision applications such as image classification, object detection, per-pixel labelings, video recognition, image similarity learning.

[LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers

[LO3]. Debate upon positive and negative aspects of techniques and research papers

[L04]. Quickly identify the core contributions of a research paper

[LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (we focus on Pytorch)

[LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain.

[LO7]. Apply the appropriate technique to a (simple) Computer Vision problem.

[LO8]. Write clearly and concisely about your code, method, results, and analyis.

Education Method

Lectures

Lab project: design and execute your own Computer Vision project.

Seminar: paper reading, critiquing, and presenting.

Assessment

The final grade of the course consists of the following components:
Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once.
Exam (weighting 60%). Exam about all material.
Project (weighting 30%). in a small group of students you work on a computer vision project.

Final grade calculation: 0.1 * Presentation + 0.3 * project + 0.6 * Exam
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

Resit/ Repair opportunities:
Exam: resit
In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:
Presentation: resit
Project: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

CS4280	Language-Based Software Security	5
Responsible Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	This course has no formal prerequisites. However, for the homework assignments you will have to implement several program analysis techniques using the Scala programming language. If you have not used Scala before, you are thus expected to learn the basics of the language through self-study.	
Course Contents	Security vulnerabilities often arise due to programming errors in the source code of an application. Recent programming errors with severe security implications include Heartbleed (buffer over-read), Shellshock (code injection), and goto-fail (ill-formated code). Rather than hunt for individual vulnerabilities in programs, a more structural approach to improve security is to improve the programming language. This is the goal of language-based security: to rule out whole classes of potential security vulnerabilities in one go. This course studies various security properties and program analysis techniques for enforcing these properties at the level of the programming language to improve software security. In particular, we will study the following properties: - Memory safety: prevent buffer overflows and overreads - Type safety: prevent undefined behaviour - Information flow control: prevent data leaks and code injection attacks We will study techniques to address these problems at the language level through dynamic analysis, static analysis, and language design. To facilitate a precise study and comparison, we will define the above techniques formally in class. To facilitate student experimentation and exploration of trade-offs, students will implement the above techniques in homework assignments.	
Study Goals	After taking this course, students should be able to: 1. Describe the nature and causes of security vulnerabilities in software systems, and give concrete examples of how these security vulnerabilities can be exploited. 2. Explain the properties that can be enforced at the level of the programming language to rule out security vulnerabilities, such as memory safety, type safety, and non-interference. 3. Formally define the semantics of a simple programming language. 4. Formally define dynamic and static analysis techniques for enforcing these security properties. 5. Implement these techniques for a small programming language. 6. Discuss and evaluate the importance of soundness and precision of a given program analysis. 7. Contrast programming languages based on the set of countermeasures they provide, and give an appropriate recommendation for a specific application. 8. Analyse and apply results from scientific literature in the area of language based security.	
Education Method	The course work consists of the following activities: 1 or 2 instruction sessions per week. Weekly homework assignments consisting of theoretical questions, programming assignments, and reading assignments	
Assessment	The final grade of the course consists of the following components: Weekly homework assignments (weighting 40%). The weekly homework assignments will test your ability to design and implement (variants of) the techniques discussed in the lectures (study goals 3-5). Written or oral exam (weighting 60%). The final written or oral exam will test your theoretical understanding of the security vulnerabilities and their countermeasures discussed in class (study goals 1-2) and your ability to discuss and contrast the different aspects of these techniques (study goals 6-8). Final grade calculation: 0.4 * Weekly homework assignments + 0.6 * Written or oral exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written or oral exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4290	Seminar on Distributed Machine Learning Systems	5
Responsible Instructor	Dr. Y. Chen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Machine learning systems are often conventionally designed for centralized processing in that they first collect data from distributed sources and then execute algorithms on a single server. Due to the limited scalability of processing large amount of data and the long latency delay, there is a strong demand for a paradigm shift to distributed or decentralized ML systems which execute ML algorithms on multiple and in some cases even geographically dispersed nodes. The aim of this seminar course is to let students learn how to design and build distributed ML systems via paper reading, presentation, discussion, and project prototyping. We provide a broad overview on the design of the state-of-the-art distributed ML systems, with a strong focus on the scalability, resource efficiency, data requirements, and robustness of the solutions. We will present an array of methodologies and techniques that can efficiently scale ML analysis to a large number of distributed nodes against all operation conditions, e.g., system failures and malicious attacks. The specific course topics are listed below. The course materials will be based on a mixture of classic and recently published papers. For each topic, the basic concepts and technology landscape will be first provided and then two state-of-the art of papers will be presented and discussed by students. We offer a testbed of a distributed (deep) ML system in which students are encouraged to apply different techniques to jointly improve its scalability and resilience. Course topics include Overview of distributed machine learning systems Performance and scalability of state-of-the-art systems Acceleration of ML workloads Slim distributed ML systems on small data Robust deep learning systems Federated machine learning systems	
Study Goals	Students are able to argue and reason about distributed ML from a systems perspective. Students understand the behavior and tradeoffs of distributed ML in terms of performance and scalability. Students can estimate the importance of data inputs via different techniques, i.e., core set and decomposition methods, for distributed ML systems. Students understand data poison attacks and design defense strategy for distributed ML systems. Students can analyze the state-of-the art federated machine learning systems and design the failure-resilient communication protocols. Students are able to design and implement methods and techniques for making distributed ML systems more efficient.	
Education Method	Lectures: 7 weeks X 2h Papers: one paper presentation, two paper reviews, and paper discussion. Practical: apply system and algorithmic optimization techniques learned in the lecture to improve the performance of distributed machine learning systems, e.g., image recognition on CIFAR 10. The testbed environment, learning algorithms, and dataset will be given. Deliverables include git commit of functioning code and a report summarizing the contribution	
Assessment	The final grade of the course consists of the following components: Paper presentation by group (10%): each group of 2 to 3 students needs to choose from a given set a papers to present (15 minutes) and lead the discussion (10 minutes). Paper reviews by individual (30%): each student needs to write three reviews of papers assigned from a given set and those two papers have to be different from the paper for the presented. Each review will account for 10% of the grade. Questionnaires by individual (0%): each student needs to hand in a list of questions at the beginning of the lectures that have paper presentation from other students. This is not graded. Individual project (60%): The project is collaborative among the entire class and competitive as a whole group. The objective is to continuously improve the performance of the given distributed ML system. The students need to hand in a final project report in style of a short scientific paper, stating their individual contribution to the overall system performance. Final grade calculation: 0.1 * Paper presentation group + 0.3 * Paper presentation individual + 0.6 * Individual project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper presentation by group: resit Paper reviews by individual: resit Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4295	Release Engineering for Machine Learning Applications	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Course Contents

The world of Software Engineering has been revolutionized in the last decade. Instead of releasing software updates yearly, companies can now release multiple times per week, sometimes even per day, to their customers. This allows quick reactions to market demands, software failures, and it is crucial for increasing the business value of software. These improvements have been mostly enabled by advances in release engineering and, in this course, we will learn about the techniques and technologies that build the foundation for modern release engineering.

This course is addressing two main target audiences: 1) software engineers that would like to learn how to systematically release their applications and make them scalable in an orchestrated environment and 2) machine learning experts that would like to learn how to make their proof-of-concept applications production ready.

Together, we will go on a journey that starts at continuous integration and then moves on to continuous delivery, continuous deployment, and continuous experimentation. We will discuss the theory and the current research on various related subjects like containerization, testing, or monitoring and will put the learned theory into practice. As a running example, we will build a pipeline for a machine learning application, which -compared to traditional release engineering- poses additional challenges, like data versioning or model deployment.

Important: Make sure to register for a group in Brightspace AT THE LATEST during the first lecture.

Study Goals

After following this course, you will be able to...

Explain the concepts of DevOps and MLOps

Automate the development and operation process
... Automate packaging, versioning, and releasing in modern build environments like GitLab or GitHub
... Release artifacts in public registries like the GitHub Maven Registry.
... Provision servers with Vagrant and Ansible
... Automate the model training pipeline using DVC

Use modern containerization technology to build a scalable micro-service architecture
... Use Docker to isolate a complex, web-based application in a container
... Use Kubernetes to orchestrate a non-trivial, distributed application
... Create an Istio service mesh in Kubernetes that enables custom routing strategies
... Use Prometheus and Grafana to monitor the state of a Kubernetes-based application

Deploy and maintain machine learning models in production
... Embed a typical ML model in an HTTP micro-service
... Apply version-control techniques to data and models in a machine learning project
... Implement quality-control techniques in a machine learning application
... Design a testing strategy that covers data, model, infrastructure, and monitoring aspects

Document the state and extensions of release engineering pipeline
... Find related work and inspiration for release engineering tasks
... Identify extension opportunities in existing release pipelines

Education Method

Following this course will require you to...

Follow interactive lectures and pre-recorded material for self-study
Execute a programming project in small teams
Transfer lecture contents into a project through practical assignments
Assess the work of yourself and others through rubric-supported peer-feedback
Actively participate in Q&A sessions
Document a release engineering pipeline in a report
Present your project results and answer questions about the lecture contents.

Assessment

--- Formative Assessment ---

You can check your individual learning progress through weekly rubrics
... You will receive feedback from other groups as a neutral outsider
... You can check your own progress via self-assessment
Joint Q&A sessions will discuss common points of the peer feedback and concrete questions

--- Individual Knock-Out Elements ---

The course enforces several criteria to protect the active members of a team. Students not reaching the following criteria will be removed from the course:

Upholding a professional attitude when interacting with peers and staff.
Actively participating in the team process and contributing to all team deadlines and deliverables
Meaningful and visible project contribution throughout each of the seven project weeks (W2-W4, W6-W9)
... 1+ commit to one of the team repositories
... Create 1+ PR and approve 1+ PR that is not your own
... The changes must contain a meaningful contribution to code, config, or report
Constructive participation in all peer reviews for assignments (one for another team and one for your own)
Ability to explain lecture contents and defend all aspects of the team project in the oral examination
Availability of team results for grading (repositories, report, artifacts)

We offer repair options for the following criteria:

Not having meaningful and visible contributions in one week can be repaired in the following week. Failing to satisfy the criteria in more than one week will lead to removal.
Failing to submit a peer review in time can be repaired in the following week. Missing a review a second time will lead to removal
The inability to explain or justify the project in the oral examination can be repaired in a second oral examination.

--- Summative Assessment ---

Students attending the course will be graded on the following grading components. ALL graded components are rounded to .1 and must at least be "almost passed" (>5.0). Their weighted average will be rounded to .5 to calculate the final team grade.

Project
... Versioning & Releases (10%)
... Containers & Orchestration (10%)
... Kubernetes & Monitoring (10%)
... Provisioning (10%)
... Istio Service Mesh (10%)
... ML Configuration Management (10%)
... ML Testing (10%)
Report (30%)
Presentation & Oral Examination (Pass/Fail)

In case of large performance differences in a group, we may choose to assess the team members individually, based on the individual contributions to the deliverables and the interaction traces that we find in the repository.

--- Please Note ---

All assignment deadlines throughout the course are fixed, we do not accept late submissions. Late submissions in systems that do not enforce due dates will be treated as not submitted (e.g., Brightspace, Buddycheck).
Failing to submit a deliverable cannot be repaired.
Insufficient deliverables (>=4.0 and <6.0) can be repaired, but the grade of the deliverable is then capped at 6.0.
There is NO resit opportunity for this course.
Partial grades are not carried over to the next academic year.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)

Enrolment / Application

In addition to your regular enrollment of the course, you must also register in Brightspace, AT THE LATEST during the first lecture, so you can be assigned to a group. Late registrations after the group creation cannot be considered anymore.

Co-Instructor

Prof.dr. A.E. Zaidman

CS4350	Machine Learning for Graph Data	5
Responsible Instructor	M. Khosla	
Responsible Instructor	E. Isufi	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics Machine Learning: data division in train-validation-test, overfitting, kernels, regularization; Basics Deep Learning: back-propagation, stochastic gradient descent (SGD), multi-layer perceptron. Basics in programming: Python or equivalents, minimal experience with Pytorch / TensorFlow or equivalents is helpful but not strongly expected.	
Course Contents	Graph data are present in a myriad of modern computer sciences systems and applications. Examples include data generated over social, brain, financial, power, water and sensor networks. Because these data have a complicated structure they require different tools conventionally used in machine and deep learning to develop end to end solutions. These solutions falls generally under the umbrella of graph-based machine learning and can be used to perform recommendations, detect anomalies in the brain, predict financial crisis, estimate the sate of a power or water network, and coordinate group of autonomous moving sensor, to name a few. This course deals with the foundations and principles of machine and deep learning for network data. Topics include: unsupervised and semi-supervised learning on graphs; graph representation learning; graph signal processing; graph convolutions; graph neural networks; spatiotemporal learning on graphs; scalable algorithms; explainability.	
Study Goals	By the end of this course, you should be able to: Lo1: Recognise key challenges in learning from graph-structured data. You are able to recognise challenges in real applications involving graph-based data, explain them with a machine learning language, and discuss strategies to solve them. You are able to identify the issues of lack of interpretability in graph-based machine learning models and describe some possible solutions. You are also able to explain a few techniques to improve scalability of existing techniques for large graphs. Lo2: Explain and compare representation learning techniques on graphs. You are able to explain, and compare different graph-based machine learning techniques, identifying advantages and limitations in each of them, and discuss the latter with respect to a specific application. Lo3: Implement and analyse machine and deep learning techniques for graph data. You are able to apply different graph-based machine learning techniques including graph kernels, graph neural networks, graph-time learning solutions to real data. You are also able to analyse the different trade-off of these solutions, compare them with each other and inspects their behaviour. LO4: Design machine learning solutions for real world graph data tasks. You are able to create a novel solution for a practical graph-based challenge, discusses different graph-machine learning strategies for solving the task, test one or more of them, and contrast the performance of the developed approach with alternative baselines.	
Education Method	The course consists of: - 12 lectures in which the main concepts are explained and discussed; - 3 hands-on homework labs, in which practical assignments need to be solved individually; - At least two Q&A and feedback sessions about the final project;	
Literature and Study Materials	The instructors will provide online accessible material for the specific lectures. Also the slides used during the lectures are made available via Brightspace. Practical final project examples will be made available for download via Brightspace. During the course several pointer to on-line video lectures will be given.	
Assessment	The final grade of the course consists of the following components: Lab assignments (Pass/Fail). The labs assignments must be solved individually and consists of only pass/fail grade. Students can solve their assignment at home or during the lab sessions. During the labs instructors and TAs (if available) will provide feedback on the questions regarding the assignment. Students need to pass all the three lab assignments before the final project submission to pass the course. Final project (weighting 100%). The final project will be solved in groups and a group grade 1-10 will be given. The project grade will be evaluated on both the written final document as well as the question-answering session during the final interview. Final grade calculation: P/F * Lab assignments + 1 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignments: re-submit deliverable Final project: resit Q&A Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4360	Natural Language Processing	5
Responsible Instructor	J. Yang	
Instructor	Dr. C. Lofi	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Proficiency in at least one programming language. Knowledge of basic algebra. Knowledge of Web Information Systems and Machine Learning can be helpful.	
Course Contents	Natural Language Processing (NLP) concerns the interaction between computers and human language, in particular how to build systems to process and analyze large amounts of human language data. It has become an important technology in many parts of computer science, facilitating key applications of large-scale text understanding, automatic question answering, knowledge base construction, information retrieval, text or speech-driven dialogue agents, translation, text summarization, etc. This course aims to introduce the principles and techniques for Natural Language Processing. Through the lectures, assignments, and the group project, students will gain an understanding of classic and modern techniques for language understanding and generation and learn the skills to design, develop, and evaluate their own Natural Language Processing systems and applications. Covered topics include: = Lexical, Syntactic and semantic analysis = Language models = Word, and subword embedding techniques = Attention mechanism and Transformers = Transfer Learning = Knowledge-driven language processing = NLP applications (e.g., QA, summarization, machine translation) = Responsible NLP (explainability, fairness) = Evaluation of NLP systems	
Study Goals	At the completion of this course, students will be able to: [LO1] Recognize typical NLP tasks and components of an NLP system [LO2] Describe lexical analysis techniques such as language modeling and smoothing [LO3] Describe syntactic and semantic analysis techniques [LO4] Describe and compare different token embedding techniques, common neural architectures (including attention mechanisms) for text processing and their implementation in neural language models (e.g., GPT-3 and BERT). [LO5] Describe information extraction techniques and its application to NLP tasks [LO6] Apply NLP techniques to information retrieval, question answering, text summarization [LO7] Explain the implications of data-driven models in terms of transparency, accountability, and fairness [LO8] Design, develop, and evaluate NLP systems, possibly using neural methods with semantic enrichment and user adaptation functionalities. Support and defend the relevance and correctness of the design choices.	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 65 hours for the group project and 30 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books, authentic blog posts - all resources are available on Brightspace.	
Books	Dan Jurafsky and James H. Martin. 2014. Speech and language processing. Pearson. Yoav Goldberg. 2015. A Primer on Neural Network Models for Natural Language Processing. Steven Bird, Ewan Klein, and Edward Loper. 2009. Natural language processing with Python: analyzing text with the natural language toolkit. " O'Reilly Media, Inc.".	
Assessment	The final grade of the course consists of the following components: Weekly individual assignment (weighting 20%) Group project (weighting 80%). The group assignment is performed collectively, but graded individually. Final grade calculation: 0.2 * Weekly individual assignment + 0.8 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly individual assignment: re-submit deliverable Group project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor anaylsis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: 0.5 * Oral discussion 1 + 0.5 * Oral discussion 2. Both partial grades must be sufficient, i.e., .a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case the take-home assignments are insufficient, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4030	Error Correcting Codes	4
Responsible Instructor	Dr.ir. J.H. Weber	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic linear algebra and probability theory, as being treated in B.Sc. programmes like Electrical Engineering, Computer Science, and Mathematics	
Course Contents	Introduction into error-correcting codes; mathematical basics; block codes fundamentals; cyclic codes; co-operating codes; soft-decision decoding; convolutional codes; iterative decoding (turbo codes, LDPC codes); applications.	
Study Goals	The general overall objectives of this course are that the students can - reproduce the error control concepts as presented in the lecture notes; - explain the meaning and usefulness of these concepts; - apply the concepts to transmission and storage systems, including basic calculations, at the level of the exercises in the lecture notes and previous exams. More specifically, the students are able to - discuss the basic trade-offs between efficiency, reliability, and complexity in designing coding schemes for transmission and storage systems; - apply finite fields theory in order to construct (cyclic) codes; - encode data into code sequences that are resistant to a certain number of random and/or burst errors; - decode received sequences using (iterative) hard or soft decision techniques.	
Education Method	Lectures; expected workload is 20 hours attending lectures, 60 hours preparing for the lectures, studying the lecture notes, and making suggested exercises, and 32 hours for preparing and making the exam.	
Literature and Study Materials	Lecture notes "Error-Correcting Codes" by J.H. Weber, available on Brightspace. Also previous exams as well as demos are available via Brightspace.	
Assessment	The final grade will be fully determined by a scheduled written exam, which will be held at the end of Q4. The resit will take place as an oral exam in Q5, on appointment with the lecturer. The exam format is closed-book in any case. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	Actual course information available on Brightspace.	

IN4255	Geometric Data Processing	5
Responsible Instructor	Dr. K.A. Hildebrandt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic knowledge in mathematics (linear algebra, calculus): CSE1205, CSE1200 or comparable courses. Students who haven't followed any of these courses can follow the course, but should be willing to invest more time.	
Course Contents	Geometric data processing is concerned with the representation, analysis, manipulation, and optimization of digital shapes. Thanks to the advances in 3D acquisition and manufacturing technologies (like 3D-Scanning and 3D-printing), the usage of geometric data is continuously increasing and an efficient processing of digital shapes plays an important role for a variety of applications in areas such as computer graphics, computer-aided design and engineering, medical imaging and surgery planning, architecture, and entertainment. In this course, we will study concepts and algorithms for creating, analyzing, editing and optimizing digital geometric shapes.	
Study Goals	After successfully completing this course, the student is able to: - describe the fundamental techniques used for representing, analyzing, processing and modeling digital 3D-shapes treated in the course and to explain the mathematical and algorithmic concepts associated with them - apply the learned mathematical concepts to solve basic geometric problems arising in geometric modeling applications - design algorithms that can solve simple geometric processing tasks and evaluate the drawbacks, benefits and limitations of the proposed algorithms - implement the designed algorithms in a geometric processing software framework	
Education Method	The course combines lectures, tutorials, practical project work, and homework assignments.	
Literature and Study Materials	References to textbooks and recent research and survey papers are given in the lectures.	
Assessment	The final grade of the course consists of the following components: Practical projects (weighting 60%) Theoretical assignment (weighting 40%) Final grade calculation: 0.6 * Practical projects + 0.4 * Theoretical assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Practical projects: re-submit deliverable Theoretical assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4331	Web-scale Data Management	5
Responsible Instructor	Dr. A. Katsifodimos	
Instructor	Dr. C. Lofi	
Instructor	Dr. R. Hai	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor level courses in database management systems and operating systems. A prior course in distributed systems or middleware would be helpful but is not required. Programming skills are important for the final assignment.	
Course Contents	This course addresses the challenges of Data Management at Web-scale. Especially, it covers the need for large-scale distributed data storage systems. The lecture therefore introduces step-by-step increasingly complex distributed storage systems, leading up to modern implementations of different NoSQL data storage systems. The challenges arising from such systems are presented and discussed, especially focusing on the CAP theorem and the resulting trade-offs with respect to data models, transactional power, query expressivity, and replication consistency. These discussions lead to different variants of NoSQL database systems, like Key-Value Stores, Document Stores, Wide-Columnar stores, and Graph Databases. The advantages, disadvantages, and general properties of these systems are discusses in more detail. There is special focus on distributed transactions and consistency guarantees of different data management systems and methods.	
Study Goals	At the end of this course the student can - assess the nature of a given storage problem, and can select a suitable technology for solving it - understands the different data models encountered in Web Data Management, and their impact on modelling and querying - understands the issues arising from distributing and replicating data, especially with respect to the CAP theorem -understands the trade-offs which can be chosen within the design space of the CAP theorem -categorize and explain modern NoSQL databases within the framework of the previously mentioned trade-offs	
Education Method	Lectures and assignments	
Literature and Study Materials	Course slides and Lecture Videos	
Books	Literature mentioned in the lecture, mostly research papers.	
Assessment	There is a group assignment (80%) and an individual exam (20%). In order to pass the class, a student has to have a grade of at least 5.0 for the group assignment as well as the individual exam, with a weighted average of 5.8 overall. The exam can be resit in the following academic quarter. If the student achieves a 4.0 - 6.0 in the group assignment, a repair opportunity can also be given (e.g., extra work, corrections on the group assignment, etc.).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Seminar Courses CS & Literature Survey 2023

CESE4065	Advanced Practical I.o.T. and Seminar	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.	
Parts	Each seminar: 2x 45 minutes (2 parts) + 10 minute break	
Summary	his course is an involved hands on course for self motivated students. Students work in a group of two (max 3) usually. Students are expected to have sufficient programming and hardware development skills. The course is project oriented thus students are expected to deliver a working model with demonstrations. Preference will be given to Q5 students. Q1 students are discouraged unless otherwise explicitly allowed. --- IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority]. --- This course is done in a group of two (max 3). Instructor/TA will help connecting a potential groupmate in the first class using Brightspace/Feedback Fruits. Activities: How to in Groups? ----------------------------- Part 1: Seminar We guide the groups to select a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar all the students MUST be present. Absenting without permission will be taken seriously. Part 2: Project (1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID19 related instructions from the university/government. (2). Students may exchange the components/boards within the group but carefully adhering to 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their own responsibility. Please note safety is first. For Students not in NL or not able to be in Delft: (3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/raspberry pi, etc. They should try to use the method as in (1) above. (4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki based networks or Open testbeds projects. In these cases, online group activities are possible. NOTE: In some extreme cases, Instructor/TA may allow one person project after assessing the situation and after discussing with the student. Supply of Hardware: ------------------- 1. Instructor/TA is able to provide take home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules. 2. Students may also use their own components.	
Course Contents	Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers.	
Study Goals	To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature.	
Education Method	Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed. Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea. Presentation skills, thinking and reflection abilities are looked into carefully. The teams are composed of two (or three) students generally. NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment. In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs. However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s).	
Literature and Study Materials	This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful. For papers to read and present, we expect students to look into: Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers. Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc. ACM Trans on CPS, etc.	
Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc.	

	Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project. Part 1: 30% of the whole marks; Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment. Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference. Report will not have individual component, however, Q/A may have individual component. Please NOTE: There would be no resit since this is a hands-on course being executed in a team. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Tags	Circuits Group work Programming Project Software
maximum aantal deelnemers	Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

CS4120	**Seminar Science and Methods in Cyber security**	**5**
Responsible Instructor	Prof.dr. M. Conti	
Instructor	Dr. M.P.M. Franssen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This seminar course Cyber Security covers the following topics: (i) an introduction to the philosophy of (classical and design) science, (ii) the art of writing a scientific research proposal, (iii) an overview of useful and relevant scientific methods, (iv) introduction to scientific writing (of a paper and of a MSc thesis), (v) an overview of some representative research areas in the Cyber Security domain (e.g. Cyber Physical Systems Security and IoT Security; Machine Learning and Security; Network Security), possibly illustrated through guest lectures from experts on these topics.	
Study Goals	1. Getting a basic knowledge and understanding of what science entails and how scientific knowledge is being created 2. Getting knowledge and understanding of relevant scientific methods applicable in the field of Cyber Security 3. Getting knowledge, understanding and skills for writing a research proposal related to the creation of a MSc thesis 4. Getting knowledge and understanding on how to execute a scientific article and MSc thesis 5. Getting knowledge and understanding of how to execute a literature review. 6. Getting familiar with recent results and challenges on some key representative research areas in Cyber Security	
Education Method	Lecturers supported by the execution of mostly individual assignments. Attendance of participants in this course is mandatory.	
Assessment	The final grade of the course consists of the following components: Presence and level of participation (weighting 10%) Quality of the research proposal/essay, with preliminary contribution of idea/protocol/algorithm, preliminary results and discussion, to be written and presented (weighting 60%) Assignments during the course (weighting 30%) Final grade calculation: 0.1 * Presence and level of participation + 0.6 * Quality of the research proposal/essay, written and presented + 0.3 * Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Further detail about evaluation of the research proposal/essay: At the end of the course, each student must write a 5-page long proposal/essay. The topic and structure of the proposal/essay must be agreed with the lecturer. The essay might include some implementation prototype or experiments/simulations to evaluate/support the claim in the paper (in case this is a significant part of the essay, two students can agree with the lecturer to work together). During the presentation of the proposal/essay, the student is asked to give a 15-minutes presentation (with slides) to the lecturer about the essay The presentation is graded according to the following components: (30%) Style (30%) Originality (40%) Organization (clarity in your argumentation, coherence between assumptions and conclusions, logical organisation, evidence to support claims) Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Quality of the research proposal/essay, written and presented: re-submit deliverable and/or resit presentation Assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Elective	Yes	
Tags	Research Methods	

CS4125	Seminar Research Methodology for Data Science	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Instructor	Dr. K.A. Hildebrandt	
Instructor	J. Urbano Merino	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	basic knowledge in mathematics (linear algebra, calculus, probability and statistics)	
Course Contents	The course focuses on research methods for data science. It looks at underlying principles and concepts for data collection, analysis and data processing, as well as the use of tools to do this. The main topics of study are: Conceptualizing research questions and experimental design Frequentist and Bayesian data analysis Generalized linear models for statistical analysis Multilevel modelling for hierarchical and longitudinal data analysis Measuring and sampling, validity and reliability Linear and nonlinear dimensional reduction Principles of statistical testing In the course, students will be using software tools such as R, and Matlab	
Study Goals	The main aims of this module for the student is to achieve an understanding of research methods for data science and obtain practical experience with data analysis and data processing methods. This module provides students with the opportunity to develop and demonstrate their understanding, knowledge, and competence. The learning outcomes for the module are that students will be able to: 1. Appreciate and comprehend strategies for collecting and processing data to answer research questions based on data 2. Understand and reproduce key principles underlying statistical data and data processing analysis 3. Learn to identify and avoid typical biases, paradoxes and misunderstandings in data-driven research 4. Apply and select appropriate data modelling techniques to analyse data and data processing	
Education Method	Lectures/Assignments Some lectures have a flipped classroom set-up, where it is expected that students prepare themselves by watching a series of videos. In the lecture, students work together on a set of classroom challenges under the guidance of the lecturer. In some others, students work on preliminary solutions to some exercises that will be discussed during the lecture. Expected Workload Lectures: 26 hours (13 × 2 hours lectures) Reading time: 39 hours Preparation basis tool use: 25 hours (5 × 5 hours for each tool) Coursework project, including writing reports and preparing for presentation: 50 hours (10 × 5 hours) Total = 140 hours	
Literature and Study Materials	Will be provided online	
Assessment	The final grade of the course consists of the following components: Analysis of experimental research data (weighting 40%) Exploration of real-world data set (weighting 30%) Linear and nonlinear dimensional reduction (weighting 30%) Students work in small groups on the 3 assignments. For the assignment, the student group submits reports and gives a presentation, including a question and answer round where individual group members are assessed on the coursework. Final grade calculation: 0.4 * Analysis of experimental research data + 0.3 * Exploration of real-world data set + 0.3 * Linear and nonlinear dimensional reduction A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Analysis of experimental research data: re-submit deliverable Exploration of real-world data set: re-submit deliverable Linear and nonlinear dimensional reduction: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	NA	

CS4130	Seminar Programming Languages	5
Responsible Instructor	Dr. S.S. Chakraborty	
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	C.B. Poulsen	
Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	4/0/0/0 0/0/0/4	
Education Period	1 4	
Start Education	1 4	
Exam Period	1 4	
Course Language	English	
Expected prior knowledge	Followed at least one other Programming Languages master course	
Course Contents	Programming languages is a core field in computer science that studies the design, theory and applications of both new and existing programming languages. Topics in programming languages include compiler construction, program analysis, program transformations, meta programming, parsing, formal semantics, program verification, and type systems. In this course, we will read scientific journal and conference articles in the field of programming languages to get a deeper understanding of programming languages. If you wish to do a MSc thesis in the programming languages group, we highly recommend taking this course.	
Study Goals	The student will acquire: - Skills to read and discuss scientific articles. - Understanding of the topics in the research field of programming languages. - Understanding of the research methodology in the research field programming languages.	
Education Method	We will run this seminar as a discussion seminar with meetings twice a week. In each meeting, we discuss a scientific article that has been studied by the participants in advance. The following activities are required for each meeting: - Reading a scientific article - Writing and submitting a short summary of the article (max 0.5 pages) - Active participation in the discussion of the article Expected Workload: - 4h Discussion sessions - 6h Reading paper at home - 2h Writing summary at home	
Literature and Study Materials	Papers from the programming languages literature will be assigned at the start of the course	
Books	No books	
Assessment	The final grade of the course consists of the following components: Students get a grade for each meeting based on the participation in the discussion (weighting 100%) Final grade calculation: The final grade is the average of the grades for the meetings. A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. There are no repair opportunities for this course in accordance with TER Implementation Regulations Art 5, sub 5. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	not applicable	
Enrolment / Application	The number of participants for this course is limited. Students in the second year of the master and students that follow the Programming Languages specialization have priority.	

CS4165	Seminar Social Signal Processing	5
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Instructor	Dr. H.S. Hung	
Contact Hours / Week x/x/x/x	4/4/0/0 + Project	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	Your background should consist of a combination of at least two of these topics or related topics: Signal Processing, Speech/Audio Processing, Computer Vision, AI, Machine Learning, Pattern Recognition, Reinforcement Learning, Deep Learning/ Neural Networks, Cognitive Modelling. These can be topics that you learned about at either Bachelor or Master level.	
Course Contents	The core of social intelligence is our ability to understand and interpret social signals of a person we are communicating with is. Social intelligence is a facet of human intelligence that has been argued to be indispensable and perhaps the most important for success in life. Social Signal Processing (SSP), the new, emerging, domain aimed at understanding social interactions through machine analysis and production of nonverbal behavior. In this course you will learn how next-generation computing can make use of such social signals by giving it the ability to recognize and produce human social signals and social behaviors. Think about turn taking, politeness, disagreement, emotions, rapport. You will learn about relevant findings in social psychology, and you will learn computational techniques that allow systems to make use of social signals to become more effective and more efficient by being able to detect but also simulate (e.g. in virtual agents) blinks, smiles, crossed arms, laughter. Socially aware computing. These techniques can be used in robots, virtual agents, smart homes, crowd monitoring, etc.	
Study Goals	Know what social signals are. Be able to apply computational methods to detect and simulate such signals. Position the field of social signal processing in computer science and psychology, and identify its major goals and angles of study. Define and explain social signals in humans and know about major psychological theories of social interaction. Explain major social signal recognition, simulation and expression techniques in computational systems. Develop (in groups) a research project that uses social signals in a non-trivial manner with hypotheses, research questions, and a supporting literature survey, and together evaluate the resulting system. These are important skills that prepare students towards their own masters thesis study later on.	
Education Method	This course is run as is a two quarter course running in Q1 and Q2. The course has been historically open to students from Leiden University as we see that mixed university groups leads to better quality projects and peer learning, which is a key part of the course. The course has therefore been designed to block off Monday afternoons where lecture times are scheduled to allow travel between the two institutions as part of the timetabled course. In light of the move to have campus education again, the lectures will be on campus and not virtual. The course has historically been run also as a Q1 only course. However, we have found that the 2 quarter model allows students more time to learn and absorb the course learning objectives leading to higher quality projects. In practice most of the contact hours are in Q1 with more time being devoted to the project work in Q2 with occasional progress meetings. Seminar: 2 hours of lectures per week for most of Q1. Self-study of papers. The papers will be made available at the start of each lecture. Project: 2 hours of class contact hours every 2-3 weeks for latter part of Q1 and Q2. Perform a piece of research (survey, research question, programming, testing) and write a paper about it. Students will work in teams of about 3-4 persons! Depending on the total number of students enrolled, teams will either work on a topic of their own or we will all together work on one big topic.	
Literature and Study Materials	Selected papers made available before the course.	
Assessment	The final grade of the course consists of the following components: Mini exam (weighting 10%). This multiple choice assessment helps to establish basic knowledge of the course material before the project work starts. Project proposal quality (midterm presentation + survey) (weighting 40%) Project execution (weighting 50%) Final grade calculation: 0.1 * mini exam + 0.4 * Project proposal quality + 0.5 Project execution A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mini exam: re-submit deliverable Project proposal quality: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Maximum number of participants	60	

CS4210-B	Intelligent Decision Making Project	5
Responsible Instructor	Dr. M.T.J. Spaan	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Dr. V. Robu	
Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. J.W. Böhmer	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Instructor	Dr. A. Lukina	
Contact Hours / Week x/x/x/x	0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Theoretical knowledge regarding algorithms for decision making in Artificial Intelligence, obtained for instance by passing one of the following courses: - CS4210-A Algorithms for Intelligent Decision Making - CS4400 Deep Reinforcement Learning - CS4375 Artificial Intelligence Techniques - IN4344 Advanced Algorithms.	
Course Contents	Intelligent decision making is a key skill of computational agents. Research on this topic focuses on building models and algorithms that enable AI systems to take appropriate decisions. Building upon theoretical knowledge gained in other courses, students collaborate in small groups on a distinct research project per group, for instance on decision-making problems in transport, logistics or smart energy grids. Purely algorithmic challenges will also be provided. The research projects provide a good opportunity to learn about topics suitable for Masters projects.	
Study Goals	After completing the Intelligent Decision Making Project course, the student is able to: 1. Apply algorithms for decision making to problem domains, and can compare and evaluate them. 2. Design and implement an extension of a decision-making algorithm. 3. Identify and discuss relevant topics in the research field of algorithms for intelligent decision making. 4. Describe and apply the appropriate research methodology. 5. Communicate his/her findings effectively.	
Education Method	A research project in a small group.	
Literature and Study Materials	Mainly survey papers and book chapters. Details are provided via Brightspace.	
Assessment	The final grade of the course consists of the following components: Quality of work of the research project (weighting 40%) A scientific report of the research project (including peer review of a report) (weighting 20%) Performance during the project (weighting 30%) Oral presentation of the research project (weighting 10%) Final grade calculation: 0.4 * Quality of work of the research project + 0.2 * A scientific report of the research project + 0.3 * Performance during the project + 0.1 * Oral presentation of the research project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Quality of work of the research project: re-submit deliverable A scientific report of the research project: re-submit deliverable Oral presentation of the research project: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Only a limited number of students can participate in this course. In order to be admitted, please submit a short motivation letter (max 200 words) via Brightspace. Attending the first lecture is compulsory.	
Tags	Artificial intelligence	
maximum aantal deelnemers	40	

CS4245	Seminar Computer Vision by Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	N. Tömen	
Instructor	Dr. X. Zhang	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	MSc thesis in the Computer Vision lab	
Expected prior knowledge	Deep Learning (CS4240)	
Course Contents	The recent boom in computer vision and automatic image understanding represents an inflection point in human productivity, permeating wide aspects of the economy and society. Examples of visual tasks which are repetitive or require expert knowledge include medical diagnosis, industrial inspection, autonomous vehicles, etc. When machines can meaningfully assist or even completely take-over such tasks it will change the world as we know it. The breakthrough in the 2012 ImageNet automatic image recognition competition shows all previously existing methods decisively defeated by a deep neural network. Deep learning replaces feature engineering methods and is able to successfully learn image features from huge annotated datasets. This course is on automatically understanding visual content such as images and videos by deep learning. The range of topics (although, not limited to): Fundamentals in Vision, object detection, per-pixel labelings, video recognition, image similarity learning, efficiency, self-supervision, 3D computer vision, adverserial attacks, explainability, generative models, etc. The course will have lectures, a seminar and a lab practical: - The lectures will be on established topics based on the current literature. - The seminar will have students read, critique, and present relevant computer vision research papers. You will have to read 2 papers per week, for 7 weeks (14 papers). - The lab will have students apply and design their own (small) computer vision project. The course build on top of the Deep Learning course (CS4240) and follows a similar setup.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the deep learning techniques reviewed in the course for computer vision applications such as image classification, object detection, per-pixel labelings, video recognition, image similarity learning. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [L04]. Quickly identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (we focus on Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) Computer Vision problem. [LO8]. Write clearly and concisely about your code, method, results, and analyis.	
Education Method	Lectures Lab project: design and execute your own Computer Vision project. Seminar: paper reading, critiquing, and presenting.	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Exam (weighting 60%). Exam about all material. Project (weighting 30%). in a small group of students you work on a computer vision project. Final grade calculation: 0.1 * Presentation + 0.3 * project + 0.6 * Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4290	Seminar on Distributed Machine Learning Systems	5
Responsible Instructor	Dr. Y. Chen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Machine learning systems are often conventionally designed for centralized processing in that they first collect data from distributed sources and then execute algorithms on a single server. Due to the limited scalability of processing large amount of data and the long latency delay, there is a strong demand for a paradigm shift to distributed or decentralized ML systems which execute ML algorithms on multiple and in some cases even geographically dispersed nodes. The aim of this seminar course is to let students learn how to design and build distributed ML systems via paper reading, presentation, discussion, and project prototyping. We provide a broad overview on the design of the state-of-the-art distributed ML systems, with a strong focus on the scalability, resource efficiency, data requirements, and robustness of the solutions. We will present an array of methodologies and techniques that can efficiently scale ML analysis to a large number of distributed nodes against all operation conditions, e.g., system failures and malicious attacks. The specific course topics are listed below. The course materials will be based on a mixture of classic and recently published papers. For each topic, the basic concepts and technology landscape will be first provided and then two state-of-the art of papers will be presented and discussed by students. We offer a testbed of a distributed (deep) ML system in which students are encouraged to apply different techniques to jointly improve its scalability and resilience. Course topics include Overview of distributed machine learning systems Performance and scalability of state-of-the-art systems Acceleration of ML workloads Slim distributed ML systems on small data Robust deep learning systems Federated machine learning systems	
Study Goals	Students are able to argue and reason about distributed ML from a systems perspective. Students understand the behavior and tradeoffs of distributed ML in terms of performance and scalability. Students can estimate the importance of data inputs via different techniques, i.e., core set and decomposition methods, for distributed ML systems. Students understand data poison attacks and design defense strategy for distributed ML systems. Students can analyze the state-of-the art federated machine learning systems and design the failure-resilient communication protocols. Students are able to design and implement methods and techniques for making distributed ML systems more efficient.	
Education Method	Lectures: 7 weeks X 2h Papers: one paper presentation, two paper reviews, and paper discussion. Practical: apply system and algorithmic optimization techniques learned in the lecture to improve the performance of distributed machine learning systems, e.g., image recognition on CIFAR 10. The testbed environment, learning algorithms, and dataset will be given. Deliverables include git commit of functioning code and a report summarizing the contribution	
Assessment	The final grade of the course consists of the following components: Paper presentation by group (10%): each group of 2 to 3 students needs to choose from a given set a papers to present (15 minutes) and lead the discussion (10 minutes). Paper reviews by individual (30%): each student needs to write three reviews of papers assigned from a given set and those two papers have to be different from the paper for the presented. Each review will account for 10% of the grade. Questionnaires by individual (0%): each student needs to hand in a list of questions at the beginning of the lectures that have paper presentation from other students. This is not graded. Individual project (60%): The project is collaborative among the entire class and competitive as a whole group. The objective is to continuously improve the performance of the given distributed ML system. The students need to hand in a final project report in style of a short scientific paper, stating their individual contribution to the overall system performance. Final grade calculation: 0.1 * Paper presentation group + 0.3 * Paper presentation individual + 0.6 * Individual project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper presentation by group: resit Paper reviews by individual: resit Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4340	Seminar Probabilistic Programming	5
Responsible Instructor	S. Dumani	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Probabilistic reasoning is one of the core aspects of artificial intelligence. This course will look into probabilistic programming, the most powerful paradigm of probabilistic reasoning, which specifies probabilistic models as computer programs. Because of the representational power of programming languages, any computable process can be modelled as a probabilistic program. The course covers the topics of probabilistic modelling and inference, but the main focus will be on developing a working probabilistic programming language. The topics include: Probabilistic modelling techniques, Importance sampling, Particle filters, Metropolis-Hastings algorithm, Hamiltonian Monte Carlo and automated differentiation, Variational inference, Discrete and logic probabilistic programming languages, Probabilistic circuits, Compiled/Amortised inference, Deep generative models	
Study Goals	By the end of the course, you will be able to explain the standard probabilistic inference procedure, solve real-world problems using probabilistic programs, identify the strengths and weaknesses in recent probabilistic programming literature, and design a research proposal to address an open problem in the field	
Education Method	This is a seminar course: we will meet twice a week to discuss a research paper. The paper should be read before the lecture and the participants need to submit a short summary (max 0.5 - 1 page). The students are expected to study every discussed paper (but not to the same amount of detail), prepare a presentation and lead a discussion for a few papers, and actively participate in discussions. Aside from paper discussions, the students are expected to do a small practical project (e.g., validating, stress testing or extending the existing method)	
Assessment	The final grade of the course consists of the following components: Research proposal (weighting 65%) Presentation (weighting 25%) Class participation (weighting 10%) Final grade calculation: 0.65 * Research proposal + 0.25 * Presentation + 0.1 * Class participation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Research proposal: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Artificial intelligence	

CS4345	Seminar Formal Methods for Learned Systems	5
Responsible Instructor	Dr. A. Lukina	
Instructor	Dr. M.T.J. Spaan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course will cover several topics at the intersection of formal methods and machine learning, for instance, abstraction refinement for analyzing deep neural networks, algorithms for reliable learning-based control, interpretable model design, and formal methods for verification and monitoring of learned systems. The course should be useful for students of both formal methods and machine learning, and lies at the intersection of these areas. Since the area is relatively new, the course material will be primarily based on research papers.	
Study Goals	After this course students will acquire formal methods knowledge and skills such as: - modeling complex systems, - formally defining critical properties, - applying verification tools to learned systems, - synthesizing controllers for learned systems, - providing interpretability for learned models.	
Education Method	The course will consist of one lecture and one discussion session per week. The students will be assigned projects in groups of two, where they will need to apply the material from the lectures, e.g., train a learned system, define a property and verify it using one of the open-source tools from state-of-the-art research papers. Discussion sessions will be used for questions about the projects.	
Assessment	The final grade of the course consists of the following components: Participation (weighting 15%) Presenting Papers (weighting 15%) Assignment #1 (weighting 15%) Assignment #2 (weighting 15%) Final Report (weighting 40%) Final grade calculation: 0.15 * Participation + 0.15 * Presenting Papers + 0.15 * Assignment #1 + 0.15 * Assignment #2 + 0.4 * Final report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presenting Papers: resit Assignment #1: re-submit deliverable Assignment #1: re-submit deliverable Final report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4306	Literature Survey	10
Responsible Instructor	L. Miranda da Cruz	
Contact Hours / Week x/x/x/x	Not applicable	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Literature Survey is an individual assignment carried out under the supervision of a Computer Science staff member, i.e. an assistant, associate or full professor. For this assignment the student reads a broad range of papers in the chosen specialisation field and writes a report in which the ideas found in the papers are discussed and compared. It is not allowed to merge this assignment with the thesis project.	
Study Goals	After the course, the student will be able to: LO1 - Formulate a research question that is relevant, meaningful, and scientifically sound. LO2 - Find high-quality contemporary scientific literature in the chosen research field. LO3 - Explain the scientific contribution of the reviewed papers within the context of the research question.	
Education Method	Individual assignment and individual guidance by a scientific staff member.	
Assessment	Writing a scientific report, individually and under supervision of a staff member. This staff member will also mark the report. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	The Literature Survey may be part of an individual exam programme of a student, which has to be approved by the Board of Examiners (BoE). To apply for a literature study the student should contact a staff member of the research group of her/his chosen specialisation after having received approval of his or her individual exam programme. The staff member and the student make arrangements regarding content and scope of the survey. The abovementioned staff member will supervise the student during this course.	
Co-Instructor	C.B. Poulsen	
Co-Instructor	Dr. J.C. van Gemert	
Co-Instructor	Dr. K.A. Hildebrandt	
Co-Instructor	T.E.P.M.F. Abeel	
Co-Instructor	Dr.ir. W.P. Brinkman	
Co-Instructor	Dr.ir. A.R. Bidarra	

IN4310	Seminar Computer Graphics	5
Responsible Instructor	P. Kellnhofer	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Strongly advised: One of the CS core courses (IN4089 Data Visualization, and IN4152 3D Computer Graphics and Animation), and at least one of the Computer Graphics specialization courses (IN4255 Geometric Modeling, IN4302TU Building Serious Games, CS4365 Applied Image Processing, and IN4307 Medical Visualization) are expected as prior knowledge.	
Course Contents	In this seminar you work on a selection of recent topics and results in one of the areas of Computer Graphics.	
Study Goals	To obtain in-depth knowledge about an advanced topic within Computer Graphics, in particular in rendering, game technology, visualization, appearance capture, animation or geometric modeling. The seminar may be used as a preparation for an MSc thesis topic. The student will acquire practical skills in reading, presenting, explaining, and discussing scientific papers, as well as writing scientific papers.	
Education Method	This course has the format of a student seminar. Students will prepare a scientific presentation of a recent research paper. The presentation goes in-depth and covers the papers strengths and weaknesses. After each presentation, a research discussions takes place and will be held in the plenary colloquium sessions. Students will attend the paper presentations and participate in a scientific discussion. Finally, each student will realize a particular aspect of the chosen paper in form of an implementation, which will be presented in a dedicated session and described in a short document (1 page + figures) following the structure of a scientific article.	
Literature and Study Materials	Recent research papers about the selected topic.	
Assessment	The final grade of the course consists of the following components: - Presentation of a recent scientific paper (50%) - Implementation component reproducing an aspect of the paper (40%, including short report and presentation) - Preparation of questions for the presented research papers (10%) Final grade calculation: (0.5 * Presentation + 0.4 * Implementation + 0.1 * Questions) The course is passed if all components have at least a grade of 5.0 and the weighted average is a 6 (rounded on 0.5). Repair possibilities (under conditions described in TER Implementation Regulations Art 5, sub 5): - Presentation of a recent scientific paper can be repaired by an individual video presentation of a state-of-the-art report + an oral examination. The content of each presented paper must be understood by the student. - Implementation component can be repaired by a new implementation that illustrates the difference of two algorithms for the same task against each other. - Preparation of each question for the assigned paper can be repaired by a short video presentation of the same paper. The maximum obtainable result for each repair option is a 6.0 (see: TER Art 5, sub 5). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. M. Weinmann	
Co-Instructor	Dr. M. Skrodzki	
Co-Instructor	Dr. R. Marroquim	
Co-Instructor	T. Höllt	
Co-Instructor	Dr. K.A. Hildebrandt	
Co-Instructor	Prof.dr. E. Eisemann	
Co-Instructor	Dr.ir. A.R. Bidarra	

IN4314	Seminar Selected Topics in Multimedia Computing	5
Responsible Instructor	P.S. Cesar Garcia	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	signal (image, audio) processing, pattern recognition, networking and distributed systems	
Course Contents	Through all the exciting recent advances in digital media technology and the rapid growth of social media platforms, multimedia content is increasingly embedded in our daily lives, gaining enormous potential in improving the traditional educational, professional, business, communication and entertainment processes. To be able to use this potential for transferring these processes into user-centric interactive multimedia applications, technology is required that can help us access, deliver, enrich and share rich-media content. This course provides insight into the state-of-the-art cross-disciplinary research efforts related to the development of such technology. The topics covered by the course include, but are not limited to, multimedia systems (transport and delivery, telepresence and VR, mobile), multimedia experiences (Quality of Experience, Collaboration), and multimedia engagement (emotional and social signals and social multimedia).	
Study Goals	To become acquainted with the state-of-the-art research and development activities in the field of Multimedia Computing, and to become an expert in one particular "hot topic", such that they are able to identify the "knowledge gap" (i.e., the place in which more research is needed in order to advance the state of the art).	
Education Method	readings, seminar discussions, presentations, survey paper	
Literature and Study Materials	Readings, possibly including video lectures.	
Assessment	The final grade of the course consists of the following components: Presentation on a pre-existing survey (weighting 10%) Writing own survey on a new topic (weighting 65%) Presentation on own survey topic (weighting 25%) Final grade calculation: 0.1 * Presentation pre-existing survey + 0.65 * Writing own survey + 0.25 * Presentation own survey A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation on a pre-existing survey: resit Writing own survey on a new topic: re-submit deliverable Presentation on own survey topic: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4326	Seminar Web Information Systems	5
Responsible Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Standard bachelor-level computer science or equivalent.	
Course Contents	In this course we discuss recent developments in the area of web information systems. We select topics to discuss from the areas of: - web technology (e.g. web engineering, hypertext, adaptive web), - web data management (e.g. web data interoperability, system and data integration), - web data and semantics (e.g. ontologies, semantic web, metadata), - web data analytics (e.g. user modeling, web personalization, web information filtering and retrieval), - social web (e.g. social web data analytics, social networking, human computing), - web science (e.g. crowdsourcing, trust, data science). We discuss this content while learning about the role of scientific communication and about the scientific methodologies and approaches for conducting research in the area. In this seminar the students will have to prepare and give scientific presentations on the basis of research papers about selected topics - the topics are selected in the first session together with the students. The students will also have to attend the presentations and participate in discussions on the presented papers. In addition, students will have to write a short survey about a topic in the area of web information systems of their choice.	
Study Goals	- to expose the student to current developments in research on web information systems and be aware of the methodologies and approaches to conduct research in the area; - to familiarise the student with reading, presenting and discussing scientific literature in the area and be aware of the most important academic journals and conferences in the area (and their review processes); - to help the student in reading and writing scientific papers and choosing a topic for her/his thesis in the area.	
Education Method	Student seminar.	
Literature and Study Materials	Is provided in the seminar, depending on the chosen subjects.	
Assessment	The final grade of the course consists of the following components: Quality of presentation of the scientific paper studied (weighting 15%) Participation in the seminar discussions (weighting 10%) Quality of paper written (weighting 75%) Final grade calculation: 0.15 * Quality of presentation of the scientific paper studied + 0.1 * Participation in the seminar discussions + 0.75 * Quality of paper written A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Quality of presentation of the scientific paper studied: resit Quality of paper written: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Students are asked to register/enrol on Brightspace beforehand. Students are also asked to be present and active in the first seminar session, to facilitate the proper planning of the seminar.	
Remarks	The expected workload of 5ects is distributed uniformly over the quarter. The seminar asks for active participation and therefore can only be completed as part of the first quarter edition; there is no re-sit.	
Tags	Artificial intelligence Databases	
Maximum number of participants	This course has a maximum capacity of 50 students. Students of 1) Web Information Systems group and 2) EEMCS have priority for other students.	

IN4334	Analytics and Machine Learning for Software Engineering	5
Responsible Instructor	M. Izadi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	- Experience with programming is required - Experience with statistics/machine and specifically deep learning (Transformer-based language models) is nice to have - Experience with research methods is nice to have	
Course Contents	Software repositories archive valuable software engineering data, such as source code, execution traces, historical code changes, mailing lists, and bug reports. This data contains a wealth of information about a projects status and history. Advances in Machine Learning and AI technologies, as demonstrated by the successful application of Deep Neural Networks in various domains did not go unnoticed in the field of Software Engineering; researchers have applied machine learning to tackle different software engineering tasks. IN4334 is a seminar course that aims to give students a deep understanding of and a hands-on approach to how deep neural networks and NLP techniques are used to represent knowledge and solve existing SE problems in novel ways.	
Study Goals	At the end of the course, the students will be able to: - Understand current literature in the area of software analytics and machine learning for software engineering - Apply software analytics techniques to extract actionable software engineering insights - Apply machine learning / deep learning algorithms to solve software engineering tasks	
Education Method	The course is a seminar, which means that we will be studying the literature in the area of software analytics and machine learning for software engineering. The course consists of the following education methods: - Self-study and presentation of papers - Development of software analytics tools and machine learning models to solve known or new problems To finish the course, students (in groups) will have to: - Study at least 10 research papers on ML4SE, which will be discussed during the lectures (reading and discussion) - Present a research paper and lead the discussion as a group (presentation) - Replicate existing work or propose new work as a group (project) The course also features guest lectures from top researchers in the area.	
Literature and Study Materials	Research papers are the main literature used in this course. We will share the resources with the students once the course starts.	
Prerequisites	The following are not strict prerequisites, however, if you have followed these courses you will be better prepared for the course: * CS4240 - Deep Learning * CS4360 - Natural Language Processing * IN4325 - Information Retrieval	
Assessment	The final grade of the course consists of the following components: Project results in the form of a research paper (weighting 80%) Final presentation/discussion of results (weighting 20%) One Research paper presentation during the course (Pass/Fail) Mid-term project report (Pass/Fail) Peer feedback on project reports (Pass/Fail) Final grade calculation: (P/F * One Research paper presentation + Mid-term project report + Peer feedback on project reports) + 0.8 * Project results in the form of a research paper + 0.2 * Final presentation/discussion A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project results in the form of a research paper: re-submit deliverable Final presentation/discussion of results: resit Mid-term project report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Each student who wants to take part in this course is *required* to: - Register/enroll on Brightspace before the start of the course - Participate in the first lecture of the course	
Tags	Project Research Methods Software Software Engineering	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Free Elective Space 2023

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Quantum Computing 2023

AP3432	Quantum Hardware 1 - Theoretical Concepts	4
Responsible Instructor	C.K. Andersen	
Instructor	M.F. Rimbach-Russ	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Undergraduate electricity and magnetism; AP3421 Fundamentals of quantum information; Bachelor Quantum Mechanics	
Course Contents	Quantum hardware is what turns the novel concepts of quantum computation and communication into reality. The key challenge is to control, couple, transmit and read out the fragile state of quantum systems with great precision, and in a technologically viable way. Quantum Hardware I is focused on teaching theoretical physics concepts for understanding this Hamiltonian engineering challenge in various quantum hardware platforms. The material will be taught using example systems such as superconducting qubits, atomic qubits, defect centers, and trapped-ion qubits.	
Study Goals	This course should enable you to: L01: Discuss single-qubit and two-qubit gate dynamics, qubit measurement, Rabi oscillations, dephasing & relaxation times on an abstract level. L02: Analyse quantum hardware systems by obtaining effective dynamics using approximations to the full Lindbladian of the system. L03: Evaluate performance of quantum hardware by identifying sources of noise and quantifying the effect of these. L04: Establish various quantum hardware systems relative to each other by identifying challenges and opportunities of the systems for quantum information processing.	
Education Method	The course will consist of weekly lectures and tutorial exercise sessions. Every week there will be hand-in assignments, which are factored in the final evaluation of the course.	
Course Relations	This course forms part of the curriculum on Quantum Technologies offered at TU Delft, which at present consists of: AP3421 Fundamentals of quantum information CS4090 Quantum communication and cryptography AP3432 Quantum Hardware 1 - Theoretical Concepts AP3442 Quantum Hardware 2 - Experimental State of the Art EE4575 Electronics for quantum computation Other Related Courses: AP3112 Quantum Optics and Lasers	
Literature and Study Materials	In this course, we will make use of: 1) 0Lecture notes And some selected material from the textbooks: a) Nielsen and Chuang, Quantum computation and information, Cambridge University Press. b) Gerry and Knight, "Introductory Quantum Optics", Cambridge University Press	
Assessment	Both hand-in assignments during the course and a final written exam will contribute to the final assessment of the course. Specifically, hand-in assignments will contribute 30% and the final written exam 70% of the final grade. Feedback on the hand-in assignments will be provided on a weekly basis and should help prepare students for the final exam.	
Continuing Courses	AP3442 Quantum Hardware 2 - Experimental State of the Art AP3472 Modeling of Superconducting Devices AP3662 Special Topics in Quantum Technology	

AP3442	Quantum Hardware 2 - Experimental State of the Art	4
Responsible Instructor	Prof.dr.ir. L.M.K. Vandersypen	
Instructor	Prof.dr.ir. R. Hanson	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	This course requires an understanding of undergraduate electricity and magnetism and the following concepts from quantum information theory: - Hamiltonian of a spin-1/2 particle and of a harmonic oscillator, and tensor products - quantum states (pure and mixed states, including density matrices) - unitary evolution (generated from the above Hamiltonians) - standard quantum operations (X, Y, Z rotations, Hadamard, CNOT, CPhase, SWAP, ...) - decoherence and non-unitary operations (e.g. through the product operator formalism) - simple quantum circuits and algorithms (e.g. teleportation, Deutsch-Jozsa algorithm, Grover's algorithm) - basics of quantum error correction (3-qubit bit flip/phase flip code, notion of fault-tolerance threshold) - simple quantum communication protocols (e.g. BB84, entanglement swapping for quantum repeaters) - quantum measurement (projective measurement in different bases)	
Course Contents	Quantum hardware is what turns the novel concepts of quantum computation and communication into reality. The key challenge is to initialize, control, couple, transmit and read out the fragile stage of quantum systems with great precision and in a technologically viable way. While Quantum Hardware I is focused on teaching the underpinning theoretical tools, Quantum Hardware II will give you an overview of the experimental state-of-the-art. You will learn about several promising approaches for realizing quantum hardware, and critically assess the strengths and weaknesses of each approach. You will also get insight in the conceptual similarities and differences between the various technologies. Specifically, the course will cover general concepts and considerations of qubit hardware, trapped ions, superconducting circuits, quantum dots, impurities, cold atoms, photonic circuits, single-photon sources, single-photon detectors and quantum repeaters.	
Study Goals	- To acquire a good understanding of the requirements of quantum hardware for computing, communication and sensing, both at the conceptual level and at the practical level. - To acquire conceptual insight in the operation, opportunities, and challenges of various qubit realisations. - To obtain a good overview of the state-of-the-art.	
Education Method	In this course, lectures are combined with homework assignments as well as student presentations of recent research papers.	
Literature and Study Materials	Review papers to accompany the lecture slides.	
Assessment	Grades will be established based on homework assignments (40%), paper presentations (25%), and a final exam (35%). The lowest homework grade does not count. This means you can miss one homework without having points deducted. Two missed homeworks will affect your grade, regardless of why you missed them. In unusual cases, a retake exam--which will be oral--will be offered by appointment.	
Continuing Courses	This course forms part of the QuTech Academy curriculum on Quantum Technologies offered at TU Delft, which at present consists of: AP3421 Fundamentals of quantum information CS4090 Quantum communication and cryptography AP3432 Quantum Hardware 1 - Theoretical Concepts AP3442 Quantum Hardware 2 - Experimental State of the Art EE4575 Electronics for quantum computation	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: A good understanding of the fundamental concepts of quantum communication Insight into the differences between classical and quantum communication and cryptography Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)	
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session	
Literature and Study Materials	Primary: Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: Homework (weighting 70%) Final Project (weighting 30%) Final grade calculation: 0.7 * Homework + 0.3 * Final Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Homework: re-submit deliverable Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. M. Blaauboer	
Responsible Instructor	Dr. L. di Carlo	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	30% homework assignments, 10% online quizzes, 60% final exam. A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course.	

QIST4400	Quantum Computer Architecture	5
Responsible Instructor	Dr. S. Feld	
Responsible Instructor	F. Sebastiano	
Instructor	Dr.ir. T.H. Oosterkamp	
Instructor	Dr. R. Ishihara	
Contact Hours / Week x/x/x/x	0/0/4/0/	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The course will assume knowledge of fundamentals of quantum information. Basic knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce the students the overall system of a quantum computer, focusing on the classical hardware and software infrastructure required to build a quantum computer together with the quantum hardware. Covered topics include: - Quantum computer architecture; quantum languages; compilers; QISA; microarchitecture. - Quantum simulator (QX simulator); - Quantum error correction: quantum error correction codes; encoding; logical operations. - Mapping of quantum circuits; placement of qubits; scheduling of operations; routing of qubits; - Classical hardware for quantum computing: classical controller and quantum processor; qubit hardware and the need for cryogenic electronics; - Receiver and transmitter architectures for controlling qubits: frequency up/down-conversion, modulation schemes, frequency generation. - Hardware building blocks: amplifiers, analog-to-digital converters, digital-to-analog converters, digital processing, FPGA - Cryo-CMOS: device characteristics, state-of-the-art, design principles.	
Study Goals	By the end of this course, students are able to: - Explain the different layers that are required to build a quantum computer. - Discuss the basic functioning of quantum error correction (QEC). - Carry out the mapping step of simple quantum circuits onto specific physical device layouts. - Interpret the results of different quantum circuits written in language QASM and executed using simulator QX, including the following: Basic quantum algorithms; Fundamental QEC codes; Surface code, especially for a Ninja star. - Identify and describe the different hardware blocks in the classical controller in a quantum computer. - Derive the specifications of the individual components in a classical controller form the requirements of the whole controller. - Compare different hardware blocks based on their specifications and can select the appropriate component for target functionality and specifications. - Describe the main features and advantages/disadvantages of the adoption of cryogenic electronics in general and cryo-CMOS specifically.	
Education Method	- Slide decks including small exercises - Videos including quizzes - Interactive lectures including activities - Homework assignments - Paper reading including processing its content	
Assessment	The final grade consists of two parts: - Part 1: Weekly homework, will be submitted electronically. Individual submission. - Part 2: Final assignment, preparation of presentation of 20 minutes with 10 minutes Q&A. Assignment will be handled by pairs of two, there will be individual grades. The global course grade is an average of each parts grade (50% part 1 + 50% part 2). Students are allowed to omit one homework. There is no resit.	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Language Courses & Skills 2023

TPM018A	English Grammar for the University	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/2/0/2	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The course is designed for students who want to improve their English grammar with the aim of enhancing their writing skills. We use a university grammar that addresses the limitations many students face when interacting in English. The focus of this course is on improving your grammar range and accuracy to help you express complex thoughts and ideas more clearly.	
Study Goals	During this course, students will: 1. expand their basic understanding of sentence structure in academic discourse 2. learn how to use punctuation in academic writing 2. demonstrate an understanding of more complex grammatical structures 3. write a variety of sentence types needed to write an academic text 4. write paragraphs to practise targeted grammar structures 5. demonstrate an understanding of grammar through text analysis of academic articles	
Education Method	This is a 7-week course with a weekly 90-minute class. Homework typically involves grammar theory as well as text and written assignments. In-class activities tend to focus on text analysis and writing assignments.	
Literature and Study Materials	Grammar for Academic Purposes 2 - Steve Marshall This textbook is available in e-format (approx. 20). Details of how to obtain a copy will be given after enrolment.	
Practical Guide		
Prerequisites		
Assessment	Assignments and a Final Test in week 7. Students have to complete weekly assignments with set deadlines. Students who submit their weekly assignments late or not at all may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). In the event of a missed class, the assignments for that particular week still need to be completed. Students receive their course credits when they complete four out of five weekly assignments at pass level and pass the Final Test in week 7.	
Enrolment / Application	Enrolment is through Brightspace.	
Remarks	The course will be taught if there is a minimum enrolment of 12 students.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	Bachelor's and Master's students. No Placement Test is required to take the course.	
Category	BSc and MSc level	

TPM303A	Intermediate Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on the acquisition of necessary language skills in the areas of grammar and usage, as well as reading and writing at a level appropriate to university study. Students will develop skills that develop an idea with relevant support for that idea by structuring the text according to current writing conventions, using various transitions to link ideas and demonstrating a knowledge of grammar and punctuation. Students will edit their own texts and work with other students in peer-editing groups.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at the Intermediate level.	
Books	Course material will be announced on Brightspace	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at intermediate level. After receiving your Placement Test results you can enrol via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q1 and Q3. However, the course will only be offered if there are sufficient participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM304A	Advanced Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/2/0/2	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on writing effective, well-structured and coherent academic texts in English. It looks at the writing process, including planning and structuring the text. Students will then develop paragraphs into longer academic texts and work on text coherence and cohesion. The course also pays attention to advanced academic vocabulary, sentence structure, punctuation and grammar.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who already have a good understanding of academic writing, but who wish to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at Advanced level.	
Books	Course material will be announced on Brightspace	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at Advanced level. After receiving your Placement Test results you can enroll via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q2 and Q4. However, the course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure an essay, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English. You may not submit texts that were previously submitted for any other course. Students who submit their weekly assignments late, repeatedly miss deadlines or lack commitment may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). Students will receive their course credits when their work shows sufficient progress and weekly deadlines and all other conditions are met.	
Special Information	Enrolment is through Brightspace.	
Remarks	The course will only be offered when there is sufficient demand. If there are not enough enrollments, the course may be cancelled.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl.	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filing tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will self-study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final exam + a speaking test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

WM1116TU	Dutch Elementary 2	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 1 (course code WM1117TU)	
Expected prior knowledge	A completed course Elementary 1 (WM1115TU). It is also possible to enrol if you have a similar level: CEF A2. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl.	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filing tests.	
Course Contents	Dutch Elementary 2 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final test + a speaking test.. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1117TU	Dutch Intermediate 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 2. This is a paid course, also for students and will not give you ECTS credits.	
Expected prior knowledge	A completed course Elementary 2 (WM1116TU). It is also possible to enrol if you have a similar level: CEF A2+. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl.	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filing tests.	
Course Contents	Dutch Intermediate 1 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Wesdijk en Blom Sciarone e.a., "Tweede Ronde, herziene editie. Nederlands voor buitenlanders. De Delftse Methode". Text book.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Intermediate 1 (WM1117TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test + speaking test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1135TU	Advanced English for the University	3
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	This course is given at C1 CEFR (Common European Framework of Reference) level. Advanced English for the University is the highest level integrated skills course offered by the English Unit within the ITAV institute at TPM. It is designed for those who already have a thorough working knowledge of English grammar and vocabulary but who still wish to further perfect their proficiency in order to become more adept in both writing and speaking. Time will be devoted to: Academic texts and writing conventions Academic vocabulary and collocations Grammar and punctuation Presentations and discussions Language learning strategies Advanced English for the University is a 14-week course with 1 lesson a week (2 hours) plus approx. 4 hours of homework assignments per week. The groups tend to be small (max. 18 participants) and the sessions are highly interactive. Attendance is mandatory. Students can miss a maximum of two lessons. Courses begin in the first week of September and in the first or second week of February . Please see the website for Placement Test dates and for any further course details: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit	
Study Goals	By the end of the course students should be familiar with academic writing conventions academic vocabulary with a specific focus on collocations common grammatical errors punctuation conventions structural devices to improve coherence various language resources such as AI tools	
Education Method	Students will do in-class exercises completed either in small groups or in pairs. They will also do a group PowerPoint presentation and a discussion led in pairs. The feedback on course components will be given both by the lecturer and by fellow course participants.	
Literature and Study Materials	Course book:Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine Publisher: Cambridge University Press, 2012 ISBN: 978-0-521-16521-1	
Prerequisites	To qualify for the course students need to have received an Advanced English for the University recommendation following the completion of the Placement Test. Please see in this connection: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test	
Assessment	This is a pass/fail course. There is no final exam. Assessment is based on the six written assignments and on the in-class discussions and presentations.	
Enrolment / Application	After receiving their Placement Test results students can enrol via Brightspace by first registering for the course and then selecting the group of their choice.	
Remarks	More information can be found on Brightspace WM1135TU	
Elective	Yes	
Tags	English proficiency	
Targetgroup	BSc, MSc and PhD students are welcome	
Category	BSc and MSc level	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Projects 2023

IFEEMCS520200	Interdisciplinary Advanced Artificial Intelligence Project	15
Responsible Instructor	Dr. P.K. Murukannaiah	
Responsible Instructor	J. Zatarain Salazar	
Contact Hours / Week x/x/x/x	10/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	This course is aimed at students who have already obtained at least 20EC computer science/artificial intelligence background. Having a solid shared pool of knowledge on AI enables the students embarking on this course to work in a team on an AI challenge at a master level. Prospective students, for example, have completed an AI minor, or have a good amount of AI knowledge from their bachelor (e.g., Mechanical Engineering).	
Course Contents	Artificial Intelligence (AI) is an interdisciplinary challenge. This course aims to provide students with a unique experience to learn, develop and apply AI to a real-world challenge, working in an interdisciplinary project team. During the course, the project team executes AI research, and/or produces AI solutions, that address real-world problems. Students collaborate in a group of 4-6 students, ideally, of diverse backgrounds (including Computer Science). Mini-tutorials will be organized during the course to provide students with knowledge and skill-based input required to successfully complete their project. The topics of the mini-tutorials will be based on the initiative from the project supervisors and other AI researchers, e.g., from the TU Delft AI Labs. The course will include a kick-off, mini-tutorials, and lab sessions with project supervisors.	
Study Goals	Upon finishing this course, the student should be able to: 1. position an AI challenge in the broader literature, e.g., by performing a literature review on the AI challenge in a domain; 2. set-up a research strategy for an interdisciplinary AI project; 3. develop and deploy AI solutions according to a research strategy; 4. effectively collaborate with the team and supervisor; 5. evaluate and reflect on their research / developed AI solution; 6. assess the impact of deploying AI-based solutions and interventions on individuals, organizations, and society; and 7. report and present their research on a scientific level.	
Education Method	The students may work and discuss in groups. Each student is responsible for gathering, sharing and implementing information within the overall project. This allows for collaboration and communication as well as self-directed learning. At least once a week, the students meet with their supervisor. Peer feedback as well as feedback from the supervisor is organized on essential steps in the research project, which include: 1. Project proposal 2. Mid-term review, and 3. A scientific report, describing the problem, methods, results, and conclusion 4. A scientific presentation The mini-tutorials are organized as electives, i.e., students identify which tutorials contribute to their assigned task and field of expertise within the group project.	
Assessment	The scientific report and the project presentation are assessed by the project supervisor and an external examiner. The assessment based on the clarity of presentation, the soundness of the methods used, and the conclusions drawn. Resit next quarter: A repair option is allowed only for students that received a fail mark (<5.8). Overall, the resit mark will be capped to 5.8.	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning; The ability to integrate (scientific and practical technological)knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

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Education	**Master Computer Science**

Thesis Project 2023

IN5000	Final Project	45
Responsible Instructor	Dr. J.C. van Gemert	
Responsible Instructor	C.B. Poulsen	
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	L. Cruz	
Responsible Instructor	Dr.ir. A.R. Bidarra	
Responsible Instructor	Dr.ir. W.P. Brinkman	
Responsible Instructor	Dr. K.A. Hildebrandt	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	IN5000 Final Project is the final part of the Master's degree programme. During this project, you will be required to demonstrate your ability to solve a research or engineering problem. The project must be carried out using the techniques of project management. You will begin by making a project plan in cooperation with your Masters thesis advisor. Several aspects of the project are defined within the plan, including the assignment, the frequency of interaction with the advisors, the milestones of the project and the resources and facilities offered by the faculty. You will be required to adhere to your plan throughout the project. It is obviously possible to adjust your plan under certain circumstances and after discussion with your daily supervisor. At the end of the project, you will submit your Masters thesis, which must be written in English, and make an oral presentation of your work to the Thesis Committee. The Thesis Committee will announce the final mark, which is based on the project performance, the thesis, the presentation and the subsequent discussion. More information about the graduation process: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-policy-msc/ More information on the thesis project on Brightspace: MSc CS Thesis project EEMCS https://brightspace.tudelft.nl/d2l/home/492868	
Study Goals	1. The student is able to design a research project: - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. 2. The student is able to execute a research project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. 3. The student is able to write a research report: - The student is able to write a research report that shows sufficient coherence of content; - The students is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; 4. The student is able to present and defend the research project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.	
Education Method	Project	
Prerequisites	Before starting the project, students must have completed at least 60 EC of the Master's degree programme as shown by a Declaration of Obtained Credits (DCO). To be able to get this DCO, the individual exam program (IEP) should have been approved by the Board of Examiners. Note: In some cases, the thesis supervisor may impose additional conditions for starting this project.	
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-policy-msc/assessment/ The voting members of the thesis committee determine the final grade. The grade should reflect a weighted average of the four scores above, but need not to be an exact arithmetical mean. The final mark starts from 5 up to and 10. Marks ending in .5 may also be used. If the student shows excellence (is nominated for a 10) the chair of the thesis committee should consult the chair of the Board of Examiners, at least five working days in advance of the defense. The chair may advice to add an extra member to the thesis committee. The motivation for the grade at each of the four criteria as listed above is summarized on a form and signed by the chairman of the thesis committee. The candidate is given a short account of the assessment, either in private or in front of the audience. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
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Research Groups 2023	
Introduction 1	Students do the Master Thesis project IN5000 under supervision of a CS research group from the INSY or ST department. INSY - Intelligent Systems Department research groups: 1. CGV - Computer Graphics and Visualization 2. CYS - Cyber Security 3. II - Interactive Intelligence 4. MMC - Multimedia Computing 5. PRB - Pattern Recognition and Bioinformatics 6. SDM - Sequential Decision Making ST - Software Technology Department research groups: 1. ALG - Algorithmics 2. DS - Distributed Systems 3. ES - Embedded Systems 4. NS - Networked Systems 5. PL - Programming Languages 6. SE - Software Engineering 7. QCS - Quantum Computer Science 8. WIS - Web Information Systems

Year	**2023/2024**
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Education	**Master Computer Science**

Algorithmics 2023	
Introduction 1	The Algorithms group designs and evaluates algorithms to solve problems in complex systems where decentralization, uncertainty, conflicting interests, and time constraints are major issues. Think of real-time balancing of energy supply and demand in communities of producers and consumers, or of coordinating schedules for service providers at airports in order to ensure that planes are cleaned at the right time and provided with fuel and food services. To design such algorithmic solutions we build upon fascinating fundamental scientific findings in computer science. In our group you learn how to design and use advanced algorithms using methods from planning and scheduling, algorithmic game theory, and sequential decision making under uncertainty. Also you will learn to implement these algorithms efficiently and to use the right methods to evaluate their performance. Once you have obtained this algorithmic expertise, you can participate in our groups research projects on topics such as smart grids, transportation systems, surveillance or maintenance. Quite a number of these projects involve industrial partners such as Alliander, NedTrain and Thales. A good example of a recent master thesis is Frits de Nijs thesis Project Scheduling: The Impact of Instance Structure on Heuristic Performance, available at our website (http://www.alg.ewi.tudelft.nl). This website also contains information about possible master thesis projects. To participate in a master thesis project you need to have completed at least our Advanced Algorithms course and two of our specialization courses.

CS4205	Evolutionary Algorithms	5
Responsible Instructor	P.A.N. Bosman	
Instructor	Dr.ir. P.A. Bouter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Must have the ability to program, predominantly in Python (but also Java and C/C++ are very useful) so that you can create your own (extensions of) evolutionary algorithms. Further, a basic understanding of key aspects in linear algebra (e.g., matrix multiplications, inversions, decompositions), probability theory and statistics (e.g., normal distribution, statistical hypothesis testing), and complexity theory (e.g., big-O notation), will be very helpful.	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, and real-world applications. The course is planned to have 10 lectures, multiple small practical assignemnts and 1 larger practical assignment. The first practical assignments pertains to experimenting with already implemented EAs on predefined problems. The second practical assignment offers more freedom, allowing you, in a group, to build your own EA (this may vary depending on student numbers and other circumstances).	
Study Goals	Upon successful completion of this course, students will be able to: 1) Explain and apply key concepts underlying the main streams in Evolutionary Algorithm (EA) research, in particular genetic algorithms, evolution strategies, genetic programming, evolutionary multi-objective optimization, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2) Explain and apply key ingredients underlying the rationale of when these algorithms work and when they do not work and several ways to overcome this. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3) Name and explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4) Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5) Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	10 Lectures Several smaller, individual, lab projects One larger lab project in groups of several students	
Literature and Study Materials	Papers and slides that will be made available.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 60%) Lab practical work (weighting 40%) Final grade calculation: 0.6 * written exam + 0.4 * lab practical work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab practical work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Tags	Algorithmics Artificial intelligence Optimalisation	

CS4210-A	Algorithms for Intelligent Decision Making	5
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. V. Robu	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: CS4375: Artificial Intelligence Techniques, or equivalent; and/or IN4344: Advanced Algorithms, or equivalent Required: basic course(s) in algorithm design and analysis, logic and probability; basic programming (in Python)	
Course Contents	Decision making is at the centre of artificial intelligence. This course gives you practical skills on a solid theoretical base. The course looks at solving mathematical models of NP-hard discrete optimisation problems. These kinds of problems lie at the heart of AI techniques such as planning, machine learning and mechanism design, and more generally combinatorial optimisation. You will learn about a range of modelling techniques from boolean satisfiability to constraint programming, and how advanced solvers for these models work. The course has plenty of real-world case studies as well as theoretical results. Apply the skills you learn in this course by taking CS4210-B: Intelligent Decision Making Project in quarter 4!	
Study Goals	By the end of this course, you will be able to identify features of real-world combinatorial decision problems, and be able to model and design systems for simplified instances of these problems using boolean satisfiability, mixed integer programming, and constraint programming over finite and real domains. You will be able to explain how SAT, CP and LCG solvers work in some detail, and how MIP solvers work at a high level. You will also be able to explain the basics of program synthesis and combinatorial game theory.	
Education Method	Lectures, homework exercises (optional, and not counted towards the course grade), and programming assignment(s) (mandatory, but not counted towards the course grade). The expected workload is: 30% lectures (including preparation for the exams) 30% homework exercises (optional) 40% programming assignments	
Literature and Study Materials	Provided on Brightspace	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Besides the summative assessment, formative coursework will be offered. This includes a mandatory assignment, performed in pairs. Students must submit a non-trivial attempt at the assignment in order to pass the course. The assignment receives formative feedback, but is not counted in computing the course grade. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Elective	Yes	
Tags	Algorithmics Artificial intelligence Group work Modelling Optimalisation Programming Projects Small groups	

CS4210-B	Intelligent Decision Making Project	5
Responsible Instructor	Dr. M.T.J. Spaan	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Dr. V. Robu	
Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. J.W. Böhmer	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumani	
Instructor	Dr. A. Lukina	
Contact Hours / Week x/x/x/x	0/0/0/1	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Theoretical knowledge regarding algorithms for decision making in Artificial Intelligence, obtained for instance by passing one of the following courses: - CS4210-A Algorithms for Intelligent Decision Making - CS4400 Deep Reinforcement Learning - CS4375 Artificial Intelligence Techniques - IN4344 Advanced Algorithms.	
Course Contents	Intelligent decision making is a key skill of computational agents. Research on this topic focuses on building models and algorithms that enable AI systems to take appropriate decisions. Building upon theoretical knowledge gained in other courses, students collaborate in small groups on a distinct research project per group, for instance on decision-making problems in transport, logistics or smart energy grids. Purely algorithmic challenges will also be provided. The research projects provide a good opportunity to learn about topics suitable for Masters projects.	
Study Goals	After completing the Intelligent Decision Making Project course, the student is able to: 1. Apply algorithms for decision making to problem domains, and can compare and evaluate them. 2. Design and implement an extension of a decision-making algorithm. 3. Identify and discuss relevant topics in the research field of algorithms for intelligent decision making. 4. Describe and apply the appropriate research methodology. 5. Communicate his/her findings effectively.	
Education Method	A research project in a small group.	
Literature and Study Materials	Mainly survey papers and book chapters. Details are provided via Brightspace.	
Assessment	The final grade of the course consists of the following components: Quality of work of the research project (weighting 40%) A scientific report of the research project (including peer review of a report) (weighting 20%) Performance during the project (weighting 30%) Oral presentation of the research project (weighting 10%) Final grade calculation: 0.4 * Quality of work of the research project + 0.2 * A scientific report of the research project + 0.3 * Performance during the project + 0.1 * Oral presentation of the research project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Quality of work of the research project: re-submit deliverable A scientific report of the research project: re-submit deliverable Oral presentation of the research project: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Only a limited number of students can participate in this course. In order to be admitted, please submit a short motivation letter (max 200 words) via Brightspace. Attending the first lecture is compulsory.	
Tags	Artificial intelligence	
maximum aantal deelnemers	40	

CS4340	Seminar Probabilistic Programming	5
Responsible Instructor	S. Dumani	
Instructor	Prof.dr. M.M. de Weerdt	
Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Probabilistic reasoning is one of the core aspects of artificial intelligence. This course will look into probabilistic programming, the most powerful paradigm of probabilistic reasoning, which specifies probabilistic models as computer programs. Because of the representational power of programming languages, any computable process can be modelled as a probabilistic program. The course covers the topics of probabilistic modelling and inference, but the main focus will be on developing a working probabilistic programming language. The topics include: Probabilistic modelling techniques, Importance sampling, Particle filters, Metropolis-Hastings algorithm, Hamiltonian Monte Carlo and automated differentiation, Variational inference, Discrete and logic probabilistic programming languages, Probabilistic circuits, Compiled/Amortised inference, Deep generative models	
Study Goals	By the end of the course, you will be able to explain the standard probabilistic inference procedure, solve real-world problems using probabilistic programs, identify the strengths and weaknesses in recent probabilistic programming literature, and design a research proposal to address an open problem in the field	
Education Method	This is a seminar course: we will meet twice a week to discuss a research paper. The paper should be read before the lecture and the participants need to submit a short summary (max 0.5 - 1 page). The students are expected to study every discussed paper (but not to the same amount of detail), prepare a presentation and lead a discussion for a few papers, and actively participate in discussions. Aside from paper discussions, the students are expected to do a small practical project (e.g., validating, stress testing or extending the existing method)	
Assessment	The final grade of the course consists of the following components: Research proposal (weighting 65%) Presentation (weighting 25%) Class participation (weighting 10%) Final grade calculation: 0.65 * Research proposal + 0.25 * Presentation + 0.1 * Class participation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Research proposal: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Artificial intelligence	

CS4345	Seminar Formal Methods for Learned Systems	5
Responsible Instructor	Dr. A. Lukina	
Instructor	Dr. M.T.J. Spaan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course will cover several topics at the intersection of formal methods and machine learning, for instance, abstraction refinement for analyzing deep neural networks, algorithms for reliable learning-based control, interpretable model design, and formal methods for verification and monitoring of learned systems. The course should be useful for students of both formal methods and machine learning, and lies at the intersection of these areas. Since the area is relatively new, the course material will be primarily based on research papers.	
Study Goals	After this course students will acquire formal methods knowledge and skills such as: - modeling complex systems, - formally defining critical properties, - applying verification tools to learned systems, - synthesizing controllers for learned systems, - providing interpretability for learned models.	
Education Method	The course will consist of one lecture and one discussion session per week. The students will be assigned projects in groups of two, where they will need to apply the material from the lectures, e.g., train a learned system, define a property and verify it using one of the open-source tools from state-of-the-art research papers. Discussion sessions will be used for questions about the projects.	
Assessment	The final grade of the course consists of the following components: Participation (weighting 15%) Presenting Papers (weighting 15%) Assignment #1 (weighting 15%) Assignment #2 (weighting 15%) Final Report (weighting 40%) Final grade calculation: 0.15 * Participation + 0.15 * Presenting Papers + 0.15 * Assignment #1 + 0.15 * Assignment #2 + 0.4 * Final report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presenting Papers: resit Assignment #1: re-submit deliverable Assignment #1: re-submit deliverable Final report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4400	Deep Reinforcement Learning	5
Responsible Instructor	Dr. J.W. Böhmer	
Instructor	Dr. M.T.J. Spaan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students must have passed CS4375 "Artificial Intelligence Techniques", or have acquired equivalent knowledge about: - basic probability theory, analysis and algebra - general machine learning methodology, e.g. regression - fully and partially observable Markov decision processes - tabular reinforcement learning methods, e.g. Q-learning - the exploration/exploitation trade-off, e.g. RMAX or UCB - multi-agent learning, e.g. centralized training and decentralized execution The course will require a lot of math and implementations of deep-learning algorithms. Students are encouraged to close any gaps in the above knowledge and to familiarize themselves with the Python/PyTorch deep-learning framework before the start of the course.	
Course Contents	This course will cover the breadth of modern model-free RL methods, discuss their limitations and introduce a variety of current research topics. In particular, we expect to cover the following: - deep learning methodology and architectures - stabilization of approximated value estimation - modern actor-critic methods - planning as inference - exploration with deep networks - offline reinforcement learning - deep multi-agent reinforcement learning - multi-task and meta learning	
Study Goals	After successful completion of this course, students - can list the strengths and limitations of modern deep RL approaches, - explain the underlying concepts of the discussed methods, and how they differ from each other, - derive the objectives and constraints of selected algorithms, - can implement selected algorithms/architectures, and - can analyze a new task to decide which algorithms/architectures to apply.	
Education Method	The course will be taught in lectures and the content will be solidified in homework submissions, which will be presented in tutorials.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Exercises (Pass/Fail) To be eligible for the exam, students must hand in homework exercises. Homework will not be individually graded, but at least 75% of the answers must be of sufficient quality (in terms of time commitment, not necessarily correctness) to be eligible to take the exam. Final grade calculation: P/F * Exercises + 1 * Witten exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Exercises: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4344	Advanced Algorithms	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic courses in Algorithmics and Complexity Theory	
Course Contents	The course is on solving (abstract models of) complex real-world problems, with a focus on solving intractable problems exactly. The course consists of two main parts: * modeling and solving using linear programming * exact algorithms using search trees, dynamic programming, and/or decision diagrams	
Study Goals	By the end of this course, students will have knowledge of and experience with the following advanced algorithmic techniques: (Part 1:) - linear programming (LP) and LP modelling - duality and simplex algorithm - integer LP and integer modelling - LP relaxation and branch and bound (Part 2:) - complete and bounded search trees - tree decomposition - dynamic programming - preprocessing - decision diagrams Furthermore, they will be able to * recognize situations where they can use these techniques and adapt them to different applications. construct new algorithms that are similar to these techniques. construct proofs that are similar to a selected set of proofs from the material. analyze the run time of algorithms. implement an algorithm that is given in pseudo-code. experimentally evaluate the quality and the runtime of an algorithm on a set of instances.	
Education Method	Interactive lectures (usually 2x2 per week), optional homework exercises, programming assignments (2 hours lab support per week) The expected workload is 30% studying the written material and the recorded lectures, and participating in the interactive sessions 30% making the homework exercises and preparation for the exams 40% working on the programming assignments	
Literature and Study Materials	Part 1 of the course will be mainly based on chapters 1-8, some of 9-11 (most of 9-11 is assumed to be known), and 12-14 of the syllabus "Optimization" by Karen Aardal, Leo van Iersel and Remie Janssen, which can be ordered via https://www.webedu.nl/bestellen/tudelft/ Part 2 will use chapter 10 of the following textbook: J. Kleinberg and E. Tardos, Algorithm Design, Pearson Education, 2006. ISBN 0-321-37291-3 As well as some supplemental study material, which will be provided via BrightSpace.	
Assessment	The final mark depends on the marks obtained for (a) written reports and evaluation of programming assignments, PA (weight 40%) (b) a written exam, EX (2 parts, weight 60%) Each programming assignment is graded on a scale from 0 to 10. The final mark for the programming assignment (PA) is the average of the mark obtained for the assignments. Programming exercises can be completed by 2 students working together. If the average grade for the four programming assignments (PA) is between 4.0 and 6.0 (excluding 6.0), a repair is possible by making one extra assignment to replace the lowest grade of the four assignments. This leads to a repaired PA score, maximized at 6. (E.g. if grades were 3.5, 4, 4, 4.5, the 3.5 can be replaced. If the grade for the replacement assignment is a 7.5, the repaired PA score is a 5. If grades were 5, 5.5, 5.5, 6, the 5 can be replaced. If the replacement is a 9, the repaired PA score is not 6.25 but 6.) The exam consists of two parts. Each part will be examined after the lectures about that part have been delivered and will be graded on a scale from 1 to 10. The final mark for the exam (EX) is the average of the marks for the parts. Each exam part contains one or more challenging assignments; during each part of the course two representative homework assignments will be made available to prepare for this. There is a resit for the exam where any of the two parts can be redone. The result for a part after the resits is determined by the maximum score obtained for the original exam and the resit. The final mark for the course is determined as follows: - if the PA and the EX mark are above 5, the final mark is the weighted average of these three marks: 60% EX, 40% PA - if at least one of PA, EX is less than or equal to 5, the final mark is the minimum of the results obtained for PA or EX. Partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	For content questions / of general interest, please use StackOverflow. For personal questions, please use the course email address: aa-cs-ewi@tudelft.nl	
Tags	Algorithmics Artificial intelligence Mathematics	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Computer Graphics and Visualisation 2023	
Introduction 1	The focus of Computer Graphics and Visualization is the creation of visual content. This research field has many important applications in domains, such as architecture, health, geosciences, simulations, games, and movies. We work on many different topics in rendering, visualization of (scientific) information and modelling of 3D objects. Our goal is to develop new algorithms to generate models, represent and interpret data, and to find efficient solutions for display and interaction. One particularly important aspect are complex and large data sets, as they play an increasingly important role in many scientific, medical, entertainment, and engineering applications. In recent years, we offered several Master projects around topics related to our core fields: Game Technology, Geometry Processing, Interactive Visualization and Virtual Reality, Medical Visualization, Image Synthesis, and Rendering Techniques. We maintain an internal Master project page with example topics, but are also open to discuss additional possibilities. Currently, these options include work on scene compositing, physics-based deformations, stylization and artistic rendering, image processing, analysis of brain shape bio markers, visual analytics methods, geometric model fitting, stereo/multi-view rendering, and animation systems.

CS4365	Applied Image Processing	5
Responsible Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Weinmann	
Instructor	P. Kellnhofer	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Basic knowledge of programming concepts, linear algebra, and calculus. Prior experience with Python and C++ is useful but it is not required.	
Course Contents	The course discusses image processing techniques and their application to visual data problems in computer graphics and computer vision including neural image processing techniques. The practical part of the course focuses on the efficient implementation of image processing algorithms on parallel computer hardware. The final project guides students to work with literature sources and state-of-the-art surveys in an area. This course can serve as an introduction to the field of visual computing, which spans the domains of Computer Graphics, Computer Vision, Computational Photography and more. During the course, you will receive an overview of image processing with a deeper focus on applications for visual data. This will include: image representations, color models, linear filters, image sampling, digital photography (including HDR and tone mapping), image retargeting, image abstraction and non-photorealism, multiscale image processing (pyramids, Fourier transform), geometrical image transformations (warping, morphing), stereoscopic imaging, and neural image processing (concept, texture synthesis, style transfer, super-resolution, denoising, editing, depth map estimation). The course will cover classical signal processing and advanced image techniques, as well as approaches utilizing neural networks and machine learning. During lab sessions, you will learn about efficient implementations of the relevant algorithms with a focus on parallel data processing. This will include use of C++ with OpenMP and the Python machine-learning library Pytorch. Introduction to these technologies will be provided before their use. Assignments are solved by using the information presented in the lectures and lab sessions. Formative feedback will be provided during the lab sessions.	
Study Goals	By the end of this course, you should be able to: - Explain the process in which a digital image is formed. - Efficiently conceive, apply, and implement algorithms for processing digital images in C++ or Pytorch. - Analyze algorithms for image processing based on their computational costs, qualitative performance, and application areas.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of image processing, common problems and their solutions. Various algorithms and their properties will be discussed. During the guided lab sessions (4 hours each), you will be introduced to guidelines for efficient implementation of selected algorithms, and you will receive formative feedback on your approach to practical assignments. The final project will focus on survey and assessment of state-of-the-art approaches to a specified image processing problem.	
Literature and Study Materials	Required literature: - Lecture slides with additional references. Recommended literature: - Szeliski, Richard. Computer vision: algorithms and applications. Springer Science & Business Media, 2010. - Chapters 3, 4 and 10. - Burger, Wilhelm, and Mark J. Burge. Digital image processing: an algorithmic introduction using Java. Springer, 2016. - Chapters 22-26. - Sundararajan, Duraisamy. Digital image processing: a signal processing and algorithmic approach. Springer, 2017. - Chapters 2-7. - Goodfellow, I., Bengio, Y. and Courville, A.. Deep Learning. MIT Press, 2016. - Chapters 6-9 and 14.	
Assessment	The final grade of the course consists of the following components: - 3 assignments during the quarter (weighting 55%) - a final project (weighting 45%) Final grade calculation: 0.55 * Assignments + 0.45 * Final project The course is passed if all components have at least a grade of 5.0 and the weighted average is a 6 (rounded on 0.5). Repair possibilities (under conditions described in TER Implementation Regulations Art 5, sub 5): - Assignments: An individual replacement assignment with a maximal grade 6.0 in the following quarter. - Final project: An individual replacement project with a maximal grade 6.0 in the following quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

IN4089	Data Visualization	5
Responsible Instructor	T. Höllt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/2/0/0 & lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Master course MKE/ST/DS	
Expected prior knowledge	IN2905-A/IN4152/CSE2215 Computer Graphics (recommended, not required). The practicals will be implemented in HTML/Javascript/D3 (InfoVis) and C++ (VolVis). We consider programming skills as a requirement but not necessarily in the mentioned languages and no advanced skills are needed. Relevant topics will be introduced and experience in other programming languages should make adaption feasible.	
Course Contents	Data visualization is the visual representation of data by computer generated images. The data sets can be results of numerical simulations or measurements (scientific visualization), or other data collections such as databases (information visualization). The goal is to improve insight, understanding and/or communication of data. Data visualizations use a combination of methods from a very diverse variety of disciplines: perception, computer graphics, human computer interaction, algorithmics, image processing, machine learning, numerical analysis, optimization, The course has two main parts; information and scientific visualization that will involve knowledge of diverse disciplines. As a computer science course, affinity to algorithmic thinking and programing skills will be needed. Topics covered: models of the visualization process; colour models and use of colour; information visualization; representation and processing of data; volume visualization; interactive visual data analysis; visualization of vector fields and flows. Guest lectures might be given on selected topics.	
Study Goals	The goal of the course is to get knowledge on the fundamentals that are part of data visualization. The main principles and techniques that are the basis of generating effective visual representations of data. Techniques and cases of data visualization are discussed. There are several applications for the techniques, such as medical, engineering, finances, economics, game analytics, and more. By the end of the course, you should be able to LO1: Discuss a large range of visualization techniques. LO2: Discuss a perception principle of visualization. LO3: Explain mathematical principles and algorithms of visualization techniques. LO4: Design suitable visualization systems for a given practical data analysis problem. LO5: Implement visualization systems for a given practical data analysis problem.	
Education Method	Lectures, practical assignments, self-study, and projects.	
Literature and Study Materials	Course slides, instructions for projects, and selected literature via Brightspace. Chapters from: Visualization Analysis and Design Author: Tamara Munzner CRC Press Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Books available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 30%). You are allowed to have the slides and material of the course during the exam. No computer or laptop is allowed. Information visualization project (weighting 35%) Volume visualization project (weighting 35%) The projects will be developed in groups of 3 and are evaluated based on the reasoning/justification of the techniques used based on the material given at the course, effectiveness of the results, technical contribution or implementation, quality of the documentation and presentation Final grade calculation: 0.3 * Written exam + 0.35 * Information visualization project + 0.35 * Volume visualization project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Information visualization project: re-submit deliverable Volume visualization project: re-submit deliverable The project resits are not automatic and must be requested within 2 weeks after publishing the grades. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

IN4152	3D Computer Graphics and Animation	5
Responsible Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. R. Marroquim	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Students that haven't followed any previous Computer Graphics courses (like CSE2215) will be able to participate, but might have to invest some more time to catch up in the first lectures.	
Course Contents	Have you ever wondered how Toy Story was made, why the game Last of Us 2 looks so beautiful, or have you ever wanted to create your own graphics application or game? Then you should consider following this course. If not, then you should still follow it... maybe, you will become interested! In this course, you will get a good idea of Computer Graphics in general. The topic is of very high relevance for the industry and the research community and has numerous applications in different domains, such as scientific visualization, video games, simulators, special effects, animated movies and many more. Here, you will learn about basic algorithms, as well as modern techniques. We will address several topics: the principles of image synthesis, object representations, geometric and hierarchical transformations, graphics cards and the graphics pipeline, realistic rendering (including global illumination and effects, such as reflections), expressive rendering, physics simulations, rendering control (including previsualization systems used by professionals in the movie industry), and perceptual rendering, which relies on properties of the human visual system to enhance the quality of the images. Besides course sessions on the theory of Computer Graphics, some of the algorithms will also be reproduced in practice, and deepened during the final project.	
Study Goals	The course teaches computer graphics techniques on an advanced level. After the course the student is able to classify the different modeling, shading, and display techniques. The student can reproduce the basic mathematical and algorithmic notions associated with these concepts, can comment on the weak and strong points of these techniques, and can apply the core concepts within a graphics program in practice.	
Education Method	lectures, instructions, research papers, lab work	
Literature and Study Materials	Research Papers in domain of selected topics, lecture sheets, online sources, optional books (see below)	
Books	Fundamentals of Computer Graphics by Shirley et al. - CRC Press Real-time Rendering by Tomas Akenine-Möller, Eric Haines, Naty Hoffman - Peters, Wellesley Real-Time Shadows by Elmar Eisemann, Michael Schwarz, Ulf Assarsson, Michael Wimmer - Taylor & Francis Computer Graphics. Principles and Practice by James D. Foley, Andries VanDam, Steven K. Feiner - Addison Wesley Advanced Material: Physically Based Rendering, From Theory to Implementation by Matt Pharr, Wenzel Jakob and Greg Humphreys - MIT Press	
Assessment	The final grade of the course consists of the following components (details of both elements will be presented during the lecture): Project (weighting 60%). The project grade is the result of a project and its presentation that is building upon the assignments that are handed out (roughly) weekly during the duration of this course. Paper (weighting 40%). The paper grade is the result of the presentation of a scientific paper and the development of an associated practical implementation. Final grade calculation: 0.6 * Project + 0.4 * Paper A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Paper: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4255	Geometric Data Processing	5
Responsible Instructor	Dr. K.A. Hildebrandt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic knowledge in mathematics (linear algebra, calculus): CSE1205, CSE1200 or comparable courses. Students who haven't followed any of these courses can follow the course, but should be willing to invest more time.	
Course Contents	Geometric data processing is concerned with the representation, analysis, manipulation, and optimization of digital shapes. Thanks to the advances in 3D acquisition and manufacturing technologies (like 3D-Scanning and 3D-printing), the usage of geometric data is continuously increasing and an efficient processing of digital shapes plays an important role for a variety of applications in areas such as computer graphics, computer-aided design and engineering, medical imaging and surgery planning, architecture, and entertainment. In this course, we will study concepts and algorithms for creating, analyzing, editing and optimizing digital geometric shapes.	
Study Goals	After successfully completing this course, the student is able to: - describe the fundamental techniques used for representing, analyzing, processing and modeling digital 3D-shapes treated in the course and to explain the mathematical and algorithmic concepts associated with them - apply the learned mathematical concepts to solve basic geometric problems arising in geometric modeling applications - design algorithms that can solve simple geometric processing tasks and evaluate the drawbacks, benefits and limitations of the proposed algorithms - implement the designed algorithms in a geometric processing software framework	
Education Method	The course combines lectures, tutorials, practical project work, and homework assignments.	
Literature and Study Materials	References to textbooks and recent research and survey papers are given in the lectures.	
Assessment	The final grade of the course consists of the following components: Practical projects (weighting 60%) Theoretical assignment (weighting 40%) Final grade calculation: 0.6 * Practical projects + 0.4 * Theoretical assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Practical projects: re-submit deliverable Theoretical assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4302TU	Building Serious Games	5
Responsible Instructor	Dr.ir. A.R. Bidarra	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	For CS students: programming experience with some object-oriented language; experience with graphics, AI and/or some game engine(s) is a plus. For all students: though not compulsory, it may be convenient to have followed the course SEN9235 (Game Design Project), which is taught in the first quarter.	
Course Contents	Project-based interdisciplinary course, open to MSc students of all faculties. The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, training, health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose. Assignments for this course will be provided by real-world end-users (e.g. companies or the Science Centre Delft), to whom the group will be reporting throughout the term of the project. Project-based interdisciplinary course, open to MSc students of all faculties (of LDE universities). The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, professional- and health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose, solving a variety of possibly conflicting, technical challenges. Assignments for this course are provided by real-world commissioners (from startups to NGOs to research institutes), to whom the group will be reporting throughout the project. In previous years, various serious game prototypes originated in this course were actually developed and eventually deployed for their intended application; and many more were published and presented by students in international research venues (incl. IEEE CoG, FDG, ICIDS, GALA, ISAGA).	
Study Goals	At the end of the project, the student will demonstrate proficiency in the following aspects: o identifying and valuing the soft skills necessary to work effectively in interdisciplinary teams o responsibly interacting in a team, integrating its members' talents and expertise in a variety of team roles o adapting with flexibility to the dynamic requirements of a complex external assignment o translating feedback received into proactive personal development steps o formulating research questions, identifying research contributions, and describe them in a scientific publication Additionally, the CS student will demonstrate proficiency in the following specific aspects: o identifying, selecting and deploying the most adequate game technologies for the given serious game domain and constraints o deepening programming skills while building a complex and large software system in an agile context	
Education Method	Project: teams work intensively as a small game studio, according to a tight planning. Also a few plenary sessions and/or lectures.	
Assessment	The final grade of the course consists of the following components: Product grade (weighting 50%). Unique for the whole group, based on both the game itself and the required documentation. Process grade (individual) (weighting 45%). Including personal contribution, performance, attitude, and peer evaluation. Final presentation (weighting 5%) The commissioner will be involved both as advisor and as assessor. The final documentation includes writing a scientific paper and preparing it for actual submission to a conference on serious games and/or their application. Final grade calculation: 0.5 * Product grade + 0.45 * Process grade + 0.05 * Final presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final presentation: resit There is no repair possibility for the other components (product and process), since their assessment necessarily requires an extended development period. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. R. Marroquim	

IN4307	Medical Visualization	5
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge of linear algebra, calculus and programming is needed. The project will be implemented in MeVisLab and Python. Learning MeVisLab will be part of the practice assignments.	
Course Contents	Theory and practice (Notice project extends to Q2) of medical visualization. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measuremen; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for medical data; advanced applications.	
Study Goals	By the end of the course, you should be able to LO1: Explain medical visualization algorithms and their applicability to medical problems. LO2: Discuss the advantages and disadvantage of medical visualization algorithms. LO3: Build a medical visualization system for a given problem: a. Discuss a suitable visualization for a given medical problem. b. Implement the most suitable solution. c. Judge the performance of the implemented solution.	
Education Method	The course will be based on a combination of lectures and practical assignments. A final project will be developed in Q2	
Literature and Study Materials	Course slides, instructions for project and assignments, and selected literature via Brightspace. Chapters from the following book can be used as alternate view on the topics but are not mandatory: Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Book available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam Q1 (weighting 40%) Final project Q2 (weighting 60%) The final project will be the design and implementation of a visualization system for a given medical problem. The final project will be carried out in teams of three. The deliverables for the final project will be a report (paper), the results (e.g., code) and a short video presenting the project (i.e. screencast). Final grade calculation: 0.4 * Written exam + 0.6 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Q2 In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final project: re-submit deliverable (the project resit is not automatic and must be requested within 2 weeks after publishing the grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Notes and written material. No computers.	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	
Co-Instructor	Prof.dr. E. Eisemann	

IN4310	Seminar Computer Graphics 5
Responsible Instructor	P. Kellnhofer
Contact Hours / Week x/x/x/x	2/0/0/0
Education Period	1
Start Education	1
Exam Period	1
Course Language	English
Expected prior knowledge	Strongly advised: One of the CS core courses (IN4089 Data Visualization, and IN4152 3D Computer Graphics and Animation), and at least one of the Computer Graphics specialization courses (IN4255 Geometric Modeling, IN4302TU Building Serious Games, CS4365 Applied Image Processing, and IN4307 Medical Visualization) are expected as prior knowledge.
Course Contents	In this seminar you work on a selection of recent topics and results in one of the areas of Computer Graphics.
Study Goals	To obtain in-depth knowledge about an advanced topic within Computer Graphics, in particular in rendering, game technology, visualization, appearance capture, animation or geometric modeling. The seminar may be used as a preparation for an MSc thesis topic. The student will acquire practical skills in reading, presenting, explaining, and discussing scientific papers, as well as writing scientific papers.
Education Method	This course has the format of a student seminar. Students will prepare a scientific presentation of a recent research paper. The presentation goes in-depth and covers the papers strengths and weaknesses. After each presentation, a research discussions takes place and will be held in the plenary colloquium sessions. Students will attend the paper presentations and participate in a scientific discussion. Finally, each student will realize a particular aspect of the chosen paper in form of an implementation, which will be presented in a dedicated session and described in a short document (1 page + figures) following the structure of a scientific article.
Literature and Study Materials	Recent research papers about the selected topic.
Assessment	The final grade of the course consists of the following components: - Presentation of a recent scientific paper (50%) - Implementation component reproducing an aspect of the paper (40%, including short report and presentation) - Preparation of questions for the presented research papers (10%) Final grade calculation: (0.5 * Presentation + 0.4 * Implementation + 0.1 * Questions) The course is passed if all components have at least a grade of 5.0 and the weighted average is a 6 (rounded on 0.5). Repair possibilities (under conditions described in TER Implementation Regulations Art 5, sub 5): - Presentation of a recent scientific paper can be repaired by an individual video presentation of a state-of-the-art report + an oral examination. The content of each presented paper must be understood by the student. - Implementation component can be repaired by a new implementation that illustrates the difference of two algorithms for the same task against each other. - Preparation of each question for the assigned paper can be repaired by a short video presentation of the same paper. The maximum obtainable result for each repair option is a 6.0 (see: TER Art 5, sub 5). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Co-Instructor	Dr. M. Weinmann
Co-Instructor	Dr. M. Skrodzki
Co-Instructor	Dr. R. Marroquim
Co-Instructor	T. Höllt
Co-Instructor	Dr. K.A. Hildebrandt
Co-Instructor	Prof.dr. E. Eisemann
Co-Instructor	Dr.ir. A.R. Bidarra

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Cyber Security 2023	
Introduction 1	Our society critically depends on cyber space for almost everything, including banking, transport & logistics, air travel, energy, telecommunications, flood defences, health care, email, social networks, and even warfare. The consequences of cyber security failures could be disastrous and the demand for cyber security specialists is therefore high and rising. The Cyber Security (CYS) Section is within the Intelligent Systems Department in the Faculty of Electrical Engineering, Computer Science and Mathematics (EEMCS). It has strong connections with the ICT Section in the Faculty of Technology, Policy, and Management (TPM), with the 3TU Federation, with the Hague Security Delta, and with companies like Fox-IT. Hence, there are plenty of opportunities to explore the multi-disciplinary aspects of cyber security and to contribute to cutting-edge research. The research focus of the CYS Section is on the following topics. Computing with Encrypted Data - Secure Information Sharing - Homomorphic Encryption - Lightweight Cryptography Data Analytics, Machine Learning - Privacy Preserving Data Mining - Automated Reverse Engineering - Botnet Detection - Monitoring and Analytics Applications of (Quantum) Information Theory - Information Theoretic Security and Privacy - Quantum Information, Computation, Crypto, Error Correction (within QuTech)

CS4035	Cyber Data Analytics	5
Responsible Instructor	Ir. S.E. Verwer	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course provides theoretical and practical background for applying data analytics in the field of cyber security. Cyber data analytics is a huge field with a great diversity of techniques and applications. The course is centered on a selection of seven such techniques: learning from imbalanced data; behavioral profiling and anomaly detection; sequential data mining; data stream mining; learning from software data; adversarial machine learning; and privacy-aware data mining; Anomaly detection is one of the main topics in cyber security. Specific difficulties that the student will learn to handle are the huge amounts of data and the large number of false positives. Behavioral profiling applies to both people and software processes. Different techniques will be taught to construct profiles from software logs. While building such profiles, care should be taken to not infringe upon the privacy of individuals the data is collected from. Finally, attackers will modify their behavior in order to avoid being detected, a cyber data analytics engineer tries to make their models/profiles robust against such modifications.	
Study Goals	The student will be able to: Apply machine learning to real data Understand and modify machine learning algorithms Learn models from time series Detect anomalies in multidimensional time-series Use distributed processing to speed up machine learning Learn models from data streams with limited memory Learn sequential models Use machine learning for fingerprinting and profiling Preserve the privacy of data owners while learning models Learn robust models that can detect evasive attackers Use machine learning to detect fraud, attacks, and botnets	
Education Method	There will be two lectures for each of the seven topics, and 3 large lab assignments on fraud detection, anomaly detection, and behavioral profiling, and 1 smaller lab on adversarial robustness. There is no exam. Teams of two students will work on these assignments which contain both individual and collaborative components. Deadlines are strict as peer-review will be used to both learn of other possible solutions, provide feedback, and get initial estimates on the obtained grade.	
Assessment	The final grade of the course consists of the following components: Lab assignment fraud detection (weighting 30%) Lab assignment anomaly detection (weighting 30%) Lab assignment behavioral profiling (weighting 30%) Lab assignment adversarial robustness (weighting 10%) Final grade calculation: 0.3 * Lab assignment + 0.3 * Lab assignment + 0.3 * Lab assignment + 0.1 * Lab assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignment fraud detection: Re-submit deliverable Lab assignment anomaly detection: Re-submit deliverable Lab assignment behavioral profiling: Re-submit deliverable Lab assignment adversarial robustness: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: A good understanding of the fundamental concepts of quantum communication Insight into the differences between classical and quantum communication and cryptography Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)	
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session	
Literature and Study Materials	Primary: Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: Homework (weighting 70%) Final Project (weighting 30%) Final grade calculation: 0.7 * Homework + 0.3 * Final Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Homework: re-submit deliverable Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4120	Seminar Science and Methods in Cyber security	5
Responsible Instructor	Prof.dr. M. Conti	
Instructor	Dr. M.P.M. Franssen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This seminar course Cyber Security covers the following topics: (i) an introduction to the philosophy of (classical and design) science, (ii) the art of writing a scientific research proposal, (iii) an overview of useful and relevant scientific methods, (iv) introduction to scientific writing (of a paper and of a MSc thesis), (v) an overview of some representative research areas in the Cyber Security domain (e.g. Cyber Physical Systems Security and IoT Security; Machine Learning and Security; Network Security), possibly illustrated through guest lectures from experts on these topics.	
Study Goals	1. Getting a basic knowledge and understanding of what science entails and how scientific knowledge is being created 2. Getting knowledge and understanding of relevant scientific methods applicable in the field of Cyber Security 3. Getting knowledge, understanding and skills for writing a research proposal related to the creation of a MSc thesis 4. Getting knowledge and understanding on how to execute a scientific article and MSc thesis 5. Getting knowledge and understanding of how to execute a literature review. 6. Getting familiar with recent results and challenges on some key representative research areas in Cyber Security	
Education Method	Lecturers supported by the execution of mostly individual assignments. Attendance of participants in this course is mandatory.	
Assessment	The final grade of the course consists of the following components: Presence and level of participation (weighting 10%) Quality of the research proposal/essay, with preliminary contribution of idea/protocol/algorithm, preliminary results and discussion, to be written and presented (weighting 60%) Assignments during the course (weighting 30%) Final grade calculation: 0.1 * Presence and level of participation + 0.6 * Quality of the research proposal/essay, written and presented + 0.3 * Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Further detail about evaluation of the research proposal/essay: At the end of the course, each student must write a 5-page long proposal/essay. The topic and structure of the proposal/essay must be agreed with the lecturer. The essay might include some implementation prototype or experiments/simulations to evaluate/support the claim in the paper (in case this is a significant part of the essay, two students can agree with the lecturer to work together). During the presentation of the proposal/essay, the student is asked to give a 15-minutes presentation (with slides) to the lecturer about the essay The presentation is graded according to the following components: (30%) Style (30%) Originality (40%) Organization (clarity in your argumentation, coherence between assumptions and conclusions, logical organisation, evidence to support claims) Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Quality of the research proposal/essay, written and presented: re-submit deliverable and/or resit presentation Assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Elective	Yes	
Tags	Research Methods	

CS4150	Systems Security	5
Responsible Instructor	Dr. S. Picek	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Security and Cryptography (IN4191) and Network Security (CS4430) and a Bachelor level Operating Systems course. The topics below should be covered to a good bachelor level. Basic machine learning knowledge (e.g., being able to run classification algorithms). To allow students to assess their level of knowledge, they can have a short oral interview.	
Course Contents	IoT security, Hardware, Countermeasures, Covert channels, Secure System Engineering	
Study Goals	The student will acquire: An appreciation of the security architecture of computer systems Detailed knowledge of the security of a specific operating system Skills in exploiting vulnerabilities of computer systems Skills in developing counter measures against exploits	
Education Method	2 hours per week lectures, 4 hours per week lab	
Assessment	The final grade of the course consists of the following components: Lab work (weighting 50%) Written exam (weighting 50%) Final grade calculation: 0.5 * Lab work + 0.5 * Written exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4380	Privacy Enhancing Technologies	5
Responsible Instructor	Z. Erkin	
Instructor	Dr. S. Roos	
Instructor	Dr. K. Liang	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	IN4191 Security and Cryptography Course	
Course Contents	anonymous communication; identity management; anonymous credentials; anonymity systems; mix networks; onion routing; database privacy; k-anonymity; differential privacy; other probabilistic approaches; private data processing; secure multiparty computation; (fully/somewhat) homomorphic encryption; garbled circuits; secret sharing; privacy-preserving clustering; private recommender systems; private smart metering; privacy-preserving biometrics.	
Study Goals	Concepts like the Internet of Things or Big Data inherently utilize massive amounts of data containing private information collected and stored by websites, sensors, monitoring systems, auditing systems, and so on. Examples include electronic records in health care systems and location information in ubiquitous computing applications. But how can we protect the privacy of participating users while at the same time enable effective sharing and utilization of the distributed data? There are several dimensions in the area of privacy, ranging from technical and juridical to societal and economical. While we will touch upon all these different aspects in the course, we will focus on the technical dimension. We will explore potential techniques for building new platforms, services, and tools that protect users' privacy. The study of promising component technologies ranging from advances in anonymous communication and identity management to theoretic tools like differential privacy and cryptography will be the core of this course. Learning objectives: Good understanding of privacy in the Internet of Things Ability to analyse and evaluate anonymity mechanisms, both for anonymous communication and for database privacy Ability to apply and analyse the concept of secure multiparty computation to protect privacy in different application domains Gain hands-on experience with different privacy-enhancing technologies	
Education Method	Lectures (joint lecturing), lectures slides and articles.	
Literature and Study Materials	Lecture Slides, essential reading material provided with the lectures (articles and tutorials)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%). Closed book. Assignments (weighting 30%) Final grade calculation: 0.7 * Written exam + 0.3 * Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4430	Network Security	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best practices in computer and network security. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course's primary focus will be on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design, review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics, as well as conducting measurements on the effectiveness of attack and defense schemes.	
Study Goals	See course contents.	
Education Method	Lectures, Labs, and Project.	
Assessment	The final grade of the course consists of the following components: Assignments (Pass/Fail) Final project (Report + Presentation) (weighting 100%) Final grade calculation: P/F * Assignments + 1* Final project A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable, resit presentation Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4191	Security and Cryptography	5
Responsible Instructor	Z. Erkin	
Instructor	J.A. Martinez Castaneda	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	UT-201500042 Privacy Enhancing Technologies (Q4)	
Expected prior knowledge	Basic understanding on the following is suggested. -Probability and statistics -Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Learning outcomes: The goal is to make students familiar with the basic concepts applied cryptography, including classical cryptography and modern secret key and public key cryptography. In particular, the students will acquire A sound understanding of the notion of security An understanding of the confidentiality, integrity and authenticity needs of the society Understand the role of cryptographic primitives including the differences between symmetric and asymmetric cryptography, the role of hash functions, digital signatures and PKI Understand the advanced topics in cryptography needed for the modern society with untrustworthy entities Among others things, the following topics are covered: -Classical systems -Information theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures, -*Secret Sharing. (if time permits) -*Zero Knowledge proofs (if time permits)	
Study Goals	It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security, as well as is familiar with present applications. Through assignments, students are expected to have the chance to work on the topics covered in the lectures.	
Education Method	Lectures, assignments and weekly exercises. Attention: This course requires full effort of 140 hours. Even more, if you lack the background (probability and modular arithmetic) Planned Workload: Lectures: 28 x 45minutes sessions, total 22 hours Practice session: 7 x 90 min. total 12 hours Assignments: 3 x 20 hour, total 60 hours Weekly study: 7 x 4 hours, total 28 hours Exam preparation: 20 hours Exam: 3 hours	
Literature and Study Materials	-Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition -Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) Handouts of lectures	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%) Mandatory assignments (weighting 30%) Final grade calculation: 0.7 * Written exam + 0.3 * Mandatory assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mandatory assignments: re-submit deliverable	

	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	In case of in person examination at campus: The exam is closed book. A cheat sheet of size A4, hand written is allowed for the written exam. Name and student number has to be present on each side. In case of remote exam from home: open book: textbook, slides and self-made notes only. randomised/customised exam.
Permitted Materials during Tests	Only non-scientific calculators.

IN4253ET	**"Hacking Lab"-Applied Security Analysis**	**5**
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Necessary background differs per student project, see first lecture or contact instructors for details	
Course Contents	The security of computer and telecommunication systems is becoming an increasing concern. In this course, we will review the current state of the art on security research and gain practical experience in assessing the security and vulnerabilities of communication systems. Engineers are typically taught to focus on performance, correctness, scalability, and maintainability when building communication and information processing systems. However, an additional set of design principles are required to achieve security. In this course, we discuss security principles, common pitfalls and vulnerabilities. The weekly lectures provide an introduction into security research, with a focus on real-world security, privacy-enhancing technology and common security pitfalls. Each student participates in a "Hack Project", with a group of one to four students. Students can select between a wide range of available Hack Project outlines within the first week. The goal may be to evaluate the security of a real-world IT system, developing a proof-of-concept exposing a vulnerability or focussed on preserving privacy in a post-Snowden world. Students may propose their own Hack Project based on their background knowledge and skills. Such Hack Projects need to be approved and shaped together with the instructor. Example of possible outlined hardware-oriented projects are: development of a wifi tracker, programing an FPGA system to break passwords, assess the security of RFID cards, or to transparently intercept Ethernet traffic. Concrete software projects are: hacking Bitcoin, improving the TOR anonymity protocol and create Android-based tools for human rights activists in Iran, Egypt and Russia, reprogramming neural networks attacks. Each Hack Project is documented with a written report. This can be in the form of a 6-8 page IEEE-style scientific article or a traditional more lengthy report. All results, experiences and findings are presented to the entire class in the last week of the course. Hack Projects also report their progress several times during the course, after the weekly lectures.	
Study Goals	After this course, the student will have a thorough knowledge of security in real-world systems, and will be able to explore the literature on this topic independently. The student will be aware of the poor state of security in real-world computer systems. The student can explain the common pitfalls, why these known failures still occur and reasons behind the poor state of security in general.	
Education Method	Lectures, student presentations, written final report and active participation. Attendance and active participation during lectures is mandatory. This sadly means telelecturing is not possible.	
Literature and Study Materials	Customize literature lists and study materials are provided per project topic	
Assessment	The final grade of the course consists of the following components: Written Hack Project report (weighting 60%) Final presentation of result (weighting 10%) Presentation of ongoing project progress (weighting 20%) Participation in discussions, overall quality of the practical work and class attendance (weighting 10%) Attendance to lectures is mandatory. Final grade calculation: 0.6 * Written Hack Project report + 0.1 * Final presentation of result + 0.2 * Presentation of ongoing project progress + 0.1 * Participation in discussions, overall quality of the practical work and class attendance A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Written Hack Project report: re-submit deliverable Final presentation of result: resit Presentation of ongoing project progress: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
maximum aantal deelnemers	If there is an unexpected high demand for this course, then enrollment will be based on past performance in relevant courses.	
Co-Instructor	Dr. S. Picek	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Cyber Security/SERG 2023

CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5
Responsible Instructor	Ir. S.E. Verwer	
Responsible Instructor	Dr. A. Panichella	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Software is one of the most complex artifacts of mankind has ever created, but complexity is the enemy of correctness. Modern software testing and validation tools use a multitude of techniques geared toward correct computer code, most of these are base on artificial intelligence. In this course, we study these techniques in details, specifically we will understand and implement: Execution monitoring and taint analysis Branch distance computation Hill-climbing and genetic algorithms Concrete and symbolic (concolic) execution Active state machine learning Genetic programming The goal is to better understand and test software using artificial intelligence. Using the taught techniques you will be able to automatically: Discover which code is reachable Find (security) bugs in software Write tests that cover all reachable code Reverse engineer a code's functionality Patch code to remove bugs and failing tests	
Study Goals	The student will: Understand modern AI techniques for software testing. Be able to implement several such techniques from scratch: - smart fuzzing (probing software with input to find crashes/bugs), - symbolic execution (using logic to construct inputs that trigger specific code branches), - fault localization (given that a program fails, find the line of code responsible for the failure), and - automated program repair (using a patch library and genetic programming to improve code) Be able to apply this technology to locate bugs in real-world software implementations.	
Education Method	The main part of the course will consist of 3 lab assignments covering the theory (fuzzing&tainting, symbolic execution, automated program repair), and one lab assignment for the application to real software. The students will implement the taught techniques from scratch in the first 3 assignments, which will be scored with a pass/fail. All three assignments need to be passed to complete the course. The final lab will contain a recap from the first three assignments and an application of a state-of-the-art tool on real software. The final lab will be graded and be the final course grade. There will be instruction sessions where students can work on their assignment and ask the teachers for assistance.	
Assessment	The final grade of the course consists of the following components: Three lab assignments (pass/fail) Final lab (weighting 100%) Final grade calculation: P/F * 3 lab assignments + 1 * Final lab A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: 3 lab assignments: re-submit deliverable Final lab: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Artificial intelligence Software	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Distributed Systems 2023	
Introduction 1	The Distributed Systems group concentrates on the modeling, the design, the implementation, and the analysis of parallel and distributed systems and algorithms. This research is fundamental in that we aim at the development and evaluation of generic methods and techniques, and application-driven in that the research is motivated by application areas. Most of our research is experimental: we try to build prototypes of systems, preferably used in the real world, to demonstrate the quality of the proposed solutions. The main research areas of the group are P2P systems and online social networks, large-scale reputation systems and crowdsourcing systems, grids and clouds, and multicore architectures and parallel programming. The teaching of the group includes MSc courses on High-Performance Computing, Parallel Algorithms, Distributed Algorithms, and the Hacking Lab. We have excellent computer and lab facilities for its experimental research. We have a 32-node cluster computer (DAS-4; 32x2x4 cores, 18TB storage, 10 Gbps networking). Additionally, we have a Bitcoin-driven infrastructure for anonymous online purchases of equipment. A good example of a recent thesis is the work of Risto J.H. Tanaskoski titled Anonymous HD video streaming. This project created the first scalable privacy-enhancing technology capable of supporting video streaming. An operational and Internet-deployed prototype evolved in various stages. First, basic onion routing technology was implemented. Next, the codebase was extended with the Tor routing protocol for relaying. Finally, it was extended with support for Libtorrent and crypto-readyness was implemented. Performance measurements showed that the download speed exceeded 5 Mbyte per second. Therefore, provided there are sufficient user-donations of proxy bandwidth this thesis created an initial solution for anonymous streaming from existing Bittorrent swarms.

CS4160	Blockchain Engineering	5
Responsible Instructor	Dr.ir. J.A. Pouwelse	
Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	In this course you will learn all aspects of blockchain technology, including tamper-proof data structures, digital identities, transitive trust, fault tolerance, distributed consensus, smart contracts and applications. Ledgers and blockchains are an emerging technology with the potential to radically improve financial transactions, supply-chain flows, transactions in general, and distributed databases. The first three weeks of the course will provide a fast-paced introduction to Bitcoin, Ethereum, and TrustChain developed by TUDelft itself. The main component in this course is a team-based complex engineering project. This course is designed for computer scientists to understand blockchain technology and to produce significant hands-on experience. To provide a deep understanding of blockchain technology and understand why it is special you need to experience first-hand how it operates at a detailed technical level. Students design, implement, and test their own independent project in teams of 3-5 students. Students can choose from a pool of possible project ideas. This course requires you to like software engineering. Topics covered: -Blockchain basics and evolution Bitcoin 1st generation, smart contract generation, future 3rd generation (trust or trust in math) -identity and transitive trust Authentication and security primitives, tamper-proof identities, trust models, MITM attacks, Sybil attacks, and TrustChain by TUDelft -Consensus models Proof-of-work, permissioned, Proof-of-stake, Corda no-global-consensus, TUDelft bottom-up fast consensus model -Smart Contract pro/con encrypted data, Bitcoin scripts, Etherium execution model, Hyperledger + Docker issues, Corda Jar file approach, Tezos difficult to use, powerful technology, vision of the future: trusted verified execution -Markets and exchanges Disruption by open markets, winner-takes-all, and multi-sided market platforms, Uber, Airbnb, 22 years of eBay, Silk Road, honesty among drug dealers, the role of trust in markets, P2P exchange markets -Decentralized Autonomous Organization, novel method to collaborate and organise any economic activity -Web3: how the Internet can be re-decentralised using DAOs Within this 2023 edition "the Delft DAO" will be prominently featured. TUDelft achieved a world-first in DAO research. We devised a full end-to-end proof-of-principle of a DAO which is capable of 0) near unbounded scalability 1) controlling money 2) democratic decision making and 3) continuous sustained self-evolution. This course provides you with the knowledge to work with this advanced technology. After this course you will have a firm grasp on the current operational blockchain-based systems, realistic view of real-world applications that may be built on top of ledger technology. You will be able to reason and discuss the open challenges and questions that still need to be resolved. This course is a key course for distributed systems students.	
Study Goals	After this course students are able to design and engineer complex blockchain-based systems. Students are able to describe blockchain technology, the various consensus model, smart contracts, markets, and relation to existing database technology. Student are able to setup a new architecture for blockchain applications.	
Education Method	This course consists of four 2-hour lectures. Each lecture is followed by a 4-hour homework period in the same week focused on understanding the background material. In week 1 you will form teams and initiate work on your blockchain engineering project. A list of projects to select from will be provided at the start of this course.	
Literature and Study Materials	Online course textbook: Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction by Narayanan, Bonneau, Felten, Miller and Goldfeder.	
Prerequisites	It is highly recommended to follow this course (see remarks): Security and Cryptography (Q1) Distributed Algorithms (Q2)	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) You will write down your project results in a single-page report, markdown format. You will be graded on your open source efforts located on Github and single-page report. Your grade will be expressed on a scale of 0 to 10. Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This class has a limited capacity (50). If there is a larger number of enrollments than the capacity of the class, students will be assigned to their preferred blockchain engineering project based on their background, engineering experience level, and match to the course goals. Students who followed Security and Cryptography (Q1) and are also enrolled in Distributed Algorithms (Q2) will have priority for placement. Mathematics students are exempts from this, if they can show some minimal software development experience (e.g. Github profile). Finally, students with a Grade Point Average of 8.0 or higher are eligible for the challenging scientific projects, resulting in a research paper. These project receive intense guidance, but have no capacity limits.	

CS4290	Seminar on Distributed Machine Learning Systems	5

Responsible Instructor	Dr. Y. Chen
Contact Hours / Week x/x/x/x	0/0/0/2
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	Machine learning systems are often conventionally designed for centralized processing in that they first collect data from distributed sources and then execute algorithms on a single server. Due to the limited scalability of processing large amount of data and the long latency delay, there is a strong demand for a paradigm shift to distributed or decentralized ML systems which execute ML algorithms on multiple and in some cases even geographically dispersed nodes. The aim of this seminar course is to let students learn how to design and build distributed ML systems via paper reading, presentation, discussion, and project prototyping. We provide a broad overview on the design of the state-of-the-art distributed ML systems, with a strong focus on the scalability, resource efficiency, data requirements, and robustness of the solutions. We will present an array of methodologies and techniques that can efficiently scale ML analysis to a large number of distributed nodes against all operation conditions, e.g., system failures and malicious attacks. The specific course topics are listed below. The course materials will be based on a mixture of classic and recently published papers. For each topic, the basic concepts and technology landscape will be first provided and then two state-of-the art of papers will be presented and discussed by students. We offer a testbed of a distributed (deep) ML system in which students are encouraged to apply different techniques to jointly improve its scalability and resilience. Course topics include Overview of distributed machine learning systems Performance and scalability of state-of-the-art systems Acceleration of ML workloads Slim distributed ML systems on small data Robust deep learning systems Federated machine learning systems
Study Goals	Students are able to argue and reason about distributed ML from a systems perspective. Students understand the behavior and tradeoffs of distributed ML in terms of performance and scalability. Students can estimate the importance of data inputs via different techniques, i.e., core set and decomposition methods, for distributed ML systems. Students understand data poison attacks and design defense strategy for distributed ML systems. Students can analyze the state-of-the art federated machine learning systems and design the failure-resilient communication protocols. Students are able to design and implement methods and techniques for making distributed ML systems more efficient.
Education Method	Lectures: 7 weeks X 2h Papers: one paper presentation, two paper reviews, and paper discussion. Practical: apply system and algorithmic optimization techniques learned in the lecture to improve the performance of distributed machine learning systems, e.g., image recognition on CIFAR 10. The testbed environment, learning algorithms, and dataset will be given. Deliverables include git commit of functioning code and a report summarizing the contribution
Assessment	The final grade of the course consists of the following components: Paper presentation by group (10%): each group of 2 to 3 students needs to choose from a given set a papers to present (15 minutes) and lead the discussion (10 minutes). Paper reviews by individual (30%): each student needs to write three reviews of papers assigned from a given set and those two papers have to be different from the paper for the presented. Each review will account for 10% of the grade. Questionnaires by individual (0%): each student needs to hand in a list of questions at the beginning of the lectures that have paper presentation from other students. This is not graded. Individual project (60%): The project is collaborative among the entire class and competitive as a whole group. The objective is to continuously improve the performance of the given distributed ML system. The students need to hand in a final project report in style of a short scientific paper, stating their individual contribution to the overall system performance. Final grade calculation: 0.1 * Paper presentation group + 0.3 * Paper presentation individual + 0.6 * Individual project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper presentation by group: resit Paper reviews by individual: resit Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

IN4150	Distributed Algorithms	6
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for network overlays, message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, broadcast, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	Understand the fundamental problems that distributed systems face Understand the most important distributed algorithms that solve these problems Reason about the execution of distributed algorithms Program distributed algorithms Select and summarize relevant literature on distributed algorithms	
Education Method	Lab work executed in groups of two students	
Literature and Study Materials	Lecture notes and lecture slides (available on Brightspace)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 75%). In case in-person, on-campus written exams are not possible because of unforeseen circumstances or measures, the written exam will be replaced by individual online oral exams. Paper summary (weighting 25%) Final grade calculation: 0.75 * Written exam + 0.25 * Paper summary A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper summary: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None except the list of algorithms	
Special Information	Email: in4150-ewi@tudelft.nl. Please send any emails related to the course to this address.	
Remarks	Lab work is 40 hrs.	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Embedded and Networked Systems 2023

CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	- basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	The course intends to bring the student into the position to: - Explain the fundamental concepts and terminology of real-time systems - Construct task schedules using different scheduling policies under a given set of realistic system constraints - Analyze the timing behavior of a system for a given system model and scheduling policy - Discuss advantages and disadvantages of different scheduling policies for a given platform or system - Discuss the effect of hardware and software interferences on the timing behavior of a given system - Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system - Derive (reverse engineer) the system specification from a given implementation (in the lab) - Evaluate the scheduling overheads of a given implementation (in the lab) - Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	Written exam (grade) + lab work; the exam has a resit Repair option for the Individual Assignment = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CESE4030	Embedded Systems Laboratory	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. G. Lan	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This highly multi-disciplinary course comes with a lab project where teams of 4 students each will have to develop an embedded control unit for a tethered electrical model quad rotor aerial vehicle (the Quadrupel drone), in order to provide stabilization such that it can hover and (ideally!) fly, with only limited user control (one joystick). The control algorithm (which is given) must be mapped onto a home-brew PCB holding a modern RF SoC interfacing a sensor module and the motor controllers. The students will be exposed to simple physics, signal processing, sensors (gyros, accelerometers), actuators (motors, servos), and basic control principles. The embedded software running on the flight controller board (FCB) driving the drone must be written in the Rust programming language. The project work (including written report) covers the entire duration of the course period, and will take approximately 128 hours, of which 32 hours are spent at the lab facilities.	
Study Goals	Student is acquainted with real-time programming in an embedded context, along with a basic understanding of embedded systems, real-time communication, sensor data processing, actuator control, control theory, and simulation. Moreover, the student has had exposure to integrating the various multidisciplinary aspects at the system level.	
Education Method	Lectures (6*2hrs), lab work (8*4hrs), coding@home (8*12hrs), report (8hrs), so on average 2 days per week	
Prerequisites	Literacy in Rust (CESE4000 or equivalent, and CSE4015)	
Assessment	Lab. project (75%) + written report (25%), no exam, no resit. Failed student has to do an additional assignment/report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for CESE students- they get preference over other MSc students.	

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet has become of critical importance to society. However, the large size of networks and abundance of protocols have made network management very complex. The novel concept of network programmability addresses this complexity and has resulted in a paradigm shift in how networks are (or can be) operated. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network programmability and which treats fundamental networking concepts like Quality of Service and network resilience.	
Study Goals	The learning objectives of this course are twofold: (1) The student should gain knowledge of the treated networking technologies. (2) The student should be able to apply and work with the programmable network technologies in a network emulator (Mininet).	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final assessment (and hence 100% of the grade) will be based on an exam that covers both the theory from the slides as well as the content from the reader. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4050	Measuring and Simulating the Internet	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/2/0 (Lectures(2x2h), Pract)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Strongly advised: (Advanced) Networking course (e.g., CESE4045) and Programming skills.	
Course Contents	The Internet is a complex network without a fixed structure. Hence, measuring the Internet is crucial to acquire knowledge about the Internet infrastructure (topology), traffic, and performance (e.g., loss, delay, bandwidth, etc.). This course will discuss the design requirements and challenges in measuring and simulating the Internet, and the existing measurement methodologies (how/where/when to measure). Knowledge of how to conduct and evaluate Internet measurements enables the design and enhancement of a large set of applications, including: capacity planning and traffic engineering, network management and trouble-shooting, detecting network abuse and intrusions, etc.	
Study Goals	The goal of this course is to introduce the students to basic Internet measurement tools, as well as the state-of-the-art in Internet measurements research. The students will learn several Internet measurement techniques (e.g., active vs. passive measurements), and different software tools. Through a measurement assignment, the students will learn how to define/formulate a research problem, choose a specific approach, and complete a measurements-related research project.	
Education Method	Lectures + weekly instructions (discussion of papers and progress) + independent project work.	
Literature and Study Materials	Papers	
Assessment	Groups of students will be assigned a project that requires the students to put the theory on measuring and simulating the Internet into practice. The students have approximately 1 month to complete their assignment. The final assessment is based on the presentation (via report and/or demonstration) of the project assignment results and on the individual contribution and level of participation. Students within a group may thus receive different grades. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Maximum number of participants	Because this is a project-based course, we can only admit a limited number of students (typically around 30, but the actual number depends on the number of TAs involved).	

CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	

Course Contents

IMPORTANT NOTICE

Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students.
The above applies to all the fields in this coursebase for this course.

The course will be offered in PHYSICAL format.

Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure.

Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network.

The contents of the course are as follows:

Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5)
o Definition of ad-hoc networks
o Comparison with infrastructure based systems
o Typical applications
o Advantages and challenges
o Radio technologies for ad-hoc networks
o Wi-Fi, Zigbee, Bluetooth

Modelling ad-hoc networks
o Propagation models
o Topology models based on graph theory
o Degree and hopcount
o Connectivity theorems

MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook)
o Introduction to MAC protocols
o Issues and design goals
o Classification
o Directional, muli-channel MAC protocols
o Energy efficiency in MAC protocols
o Quality of service

Self organisation and Routing (Chapter 7, 8, 11 of the textbook)
o Flooding
o Node discovery, neighbour discovery
o Route establishment
o Topology maintenance, localisation
o Proactive, reactive and hybrid routing
o Typical protocols
o Energy efficiency in routing
o Broadcast and multicast
o Effects of mobility on connectivity and capacity
o Effect of nodes joining and leaving the network

Advanced issues in ad hoc networks
o Wireless sensor networks (Chapter 12 of the textbook and papers)
o Cooperation (Reference papers)
o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET)
o Energy Harvesting

Project presentations by students

Study Goals

By the end of this course students should be able to:
- Model the ad-hoc networks using Graphs.
- Describe the working principles of medium access control protocols for ad-hoc networks
- Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks
- Choose various components to form a coherent ad hoc networking architecture
- Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks
- Assess the suitability of ad-hoc networks for different communication needs and scenarios

Education Method

The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.

Literature and Study Materials

1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004.

2. Lecture notes consisting of slides presented at the lectures (Slides are only teaching aid and they are not substitute for

	textbooks, research papers, etc). 3. Some recent journal papers 4. Optional Reference Books 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms) 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar , C Puttamadappa , and T. G Basavaraju, Auerbach Publications, 2008. This book is avaliable online in the library. 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is avaliable online in the library. 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer. 5. OPNET/ns-2 web pages, tutorials and video lectures
Books	Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.
Assessment	1. There will be totally three written/oral tests/examinations for this course (including the final exam). 2. The students will carry out a project in a group and submit a short report. 3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis. Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class. Breakup: 2 Tests + Final Exam = 55% Project 40% Self assessment + Reflection 3% Activities on Feedback Fruits or any Online platform 2% Resit: written/oral exam [+ carry over from projects]. == Changes due to COVID19 in 2021 is below; in case there is a similar situation, we use the following methods. == There are three chapters Chapter 1: Network modelling Chapter 2: MAC protocol Chapter 3: Routing + Lab: Simple experiments using laptops (individual) + simulations (group-wise) Marks Breakup -------------- Homework/Assignments/Group works Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks Part B1: Individual experiment + report Part B2: Group-wise simulations + report + demo Part B1 + Part B2 - 50 Marks Part C: Self-assessment + peer activities -- 5 marks Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit. == (More information will be given in the first class) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Permitted Materials during Tests	Different conditions for different test/exams. Conditions will be informed 1 week before the exams/test.

CESE4060	Wireless IoT and Local Area Networks	5
Responsible Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students are advised to follow the course Wireless Communications (ET4358) before taking this Wireless Networking course. An advantage is to have entry-level programming skills (Matlab, Python, C/C++). Nonetheless, students with little knowledge of programming will be helped.	
Course Contents	The following modules will be discussed during the lectures: Introduction (example topics): - What is wireless networking - Where to search for (academic) wireless network literature and resources Medium Access Control (example topics): - WiFi: hidden/exposed terminal problem, Carrier Sense Multiple Access - Bluetooth standard: in-depth look into the channel hopping, protocol specifications WiFi (example topics): - Review of IEEE 802.11 standards - Protocol format - ISM band regulation - Adaptive Modulation and Coding - WiFi Matlab class (assignment) IoT networking standards (example topics): - LoRa: protocol specifications, energy consumption, modulation format, network design Review of wireless tools (example topics): - Introduction to wireless packet sniffing and analysis using Wireshark (assignment) - Simple simulations of WiFi network with NS3 RFID networking (example topics): - Principles of backscatter - Protocol formats: EPC C1G2 - RFID hackathon (assignment) Cognitive radio (example topics): - Basics of spectrum management - White Space Databases - Theory of spectrum sensing	
Study Goals	At the end of the course students will be able to: (i) to understand how practical wireless systems work and get a deeper understanding of how the theoretical concepts of wireless communications apply to practice; (ii) employ their own analysis methodology to assess new wireless network systems (especially at the physical layer); (iii) understand rapid prototyping of new wireless systems (for instance, with software defined radio).	
Education Method	Lecture presentations, mini-project assignments, assigned paper reading and its critical analysis and presentation.	
Computer Use	Each student should have its own laptop (preferably with a Linux distribution, where Linux must not be installed on a virtual machine). We will be using Matlab, and/or NS3 and/or GNURadio and/or Wireshark for the assignments.	
Literature and Study Materials	WiFi Matlab WLAN toolbox: https://nl.mathworks.com/help/wlan/; Wireshark learn page: https://www.wireshark.org/#learnWS; tutorial on NS3 network simulator: https://www.nsnam.org/documentation/; specific chapters from books provided at the beginning of each lecture.	
Prerequisites	Background in programming (Matlab, Python, Bash)	
Assessment	Points from the mini-project assignments. A research paper analysis from conferences such as IEEE INFOCOM, ACM MobiCom, ACM SIGCOMM will be required to pass the course. Repair option: Individual Assignment = Re-submit deliverable Individual Report = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4065	Advanced Practical I.o.T. and Seminar	5

Responsible Instructor	R.R. Venkatesha Prasad
Contact Hours / Week x/x/x/x	2/0/0/0
Education Period	1
Start Education	1
Exam Period	1
Course Language	English
Expected prior knowledge	fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.
Parts	Each seminar: 2x 45 minutes (2 parts) + 10 minute break
Summary	his course is an involved hands on course for self motivated students. Students work in a group of two (max 3) usually. Students are expected to have sufficient programming and hardware development skills. The course is project oriented thus students are expected to deliver a working model with demonstrations. Preference will be given to Q5 students. Q1 students are discouraged unless otherwise explicitly allowed. -- IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority]. -- This course is done in a group of two (max 3). Instructor/TA will help connecting a potential groupmate in the first class using Brightspace/Feedback Fruits. Activities: How to in Groups? ----------------------------- Part 1: Seminar We guide the groups to select a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar all the students MUST be present. Absenting without permission will be taken seriously. Part 2: Project (1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID19 related instructions from the university/government. (2). Students may exchange the components/boards within the group but carefully adhering to 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their own responsibility. Please note safety is first. For Students not in NL or not able to be in Delft: (3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/raspberry pi, etc. They should try to use the method as in (1) above. (4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki based networks or Open testbeds projects. In these cases, online group activities are possible. NOTE: In some extreme cases, Instructor/TA may allow one person project after assessing the situation and after discussing with the student. Supply of Hardware: ------------------- 1. Instructor/TA is able to provide take home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules. 2. Students may also use their own components.
Course Contents	Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers.
Study Goals	To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature.
Education Method	Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed. Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea. Presentation skills, thinking and reflection abilities are looked into carefully. The teams are composed of two (or three) students generally. NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment. In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs. However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s).
Literature and Study Materials	This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful. For papers to read and present, we expect students to look into: Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers. Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc. ACM Trans on CPS, etc.
Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc.

	Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project. Part 1: 30% of the whole marks; Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment. Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference. Report will not have individual component, however, Q/A may have individual component. Please NOTE: There would be no resit since this is a hands-on course being executed in a team. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Tags	Circuits Group work Programming Project Software
maximum aantal deelnemers	Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

CESE4110	Visible Light Communication and Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Instructor	Q. Wang	
Contact Hours / Week x/x/x/x	0,0,0,0,4 Lecture & 0,0,0,0,4 Lab	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement: Programming experience This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object Oriented Programming". If you have not taken this course, experience with programming in C, Python and/or Java is sufficient. Some basic knowledge in wireless communication would help (modulation and demodulation), but it is not required because these concepts will be refreshed during the lectures.	
Course Contents	Nowadays, half of the worlds population (plus billions of things) connect to the Internet using mainly wireless technologies. Our ever-increasing demand for wireless is crowding the radio frequency spectrum: we need more bandwidth. To overcome this challenge, a new generation of devices have started to use the visible light spectrum to transmit data. This new paradigm is promising because the visible light spectrum is empty and free. In this course, you will be introduced to the general field of visible light communication and sensing (VLCS). You will learn about its history, current developments and future opportunities. The course will cover theory and practice. First, you will learn the main modulation techniques used to transmit data with light, and then, you will use that knowledge to create your own VLCS system. The course will also cover the latest research done on this topic at various universities around the world. A wonderful introduction about the potential of Visible Light Communication is given in this TED talk: https://www.ted.com/talks/harald_haas_wireless_data_from_every_light_bulb?language=en#t-4148 This course includes 8 lectures, two lab projects, and one final main project. There are two lectures per week during the first four weeks. After that, there will be only lab sessions to monitor the progress of the main project. The 8 lectures are: Lecture 1: Introduction & Hardware Lecture 2: Physical Layer: Pulse-Based Modulation Lecture 3: Physical Layer: Advanced Modulation Lecture 4: Network Layer Lecture 5: Camera-based communication Lecture 6: Backscattering Communication Lecture 7: Sensing Applications Lecture 8: SoA, Standardization & Commercialization	
Study Goals	By the end of this course, you will achieve the following Learning Objectives (LOs): LO1: Illustrate the principles of Visible Light Communication (VLC) systems: a. Understand the characteristics of VLC devices such as LEDs and photodiodes. b. Formulate the principles of line-of-sight and non-line-of-sight VLC links. c. Elaborate on pulse-based and advanced VLC modulation schemes. d. Explain and innovate upon the mechanisms behind VLC networking systems. LO2: Employ the theoretical and experimental training on VLC systems to: a. Analyze LED-to-camera and backscattering VLC systems. b. Investigate novel visible light systems. LO3: Interpret the standardization, commercialization, and state-of-the-art of VLC. LO4: Build and demonstrate a VLCS system with off-the-shelf devices.	
Education Method	Lectures + Labs + Final Project The lectures, labs and final project cover the entire duration of the course period and will take approximately 120 hours, of which 16 hours are spent on lectures, 44 hours on the two labs (22 hours each) and 80 hours on the project.	
Literature and Study Materials	The course will be based on research papers and web tutorials. An optional book is: Optical Wireless Communications: System and Channel Modelling with MATLAB® by Z. Ghassemlooy, W. Popoola, S. Rajbhandari	
Assessment	There will be two labs and one project. Each lab accounts for 20% of the grade (40% in total) and the final project accounts for the remaining 60%. The labs will consist of guided sessions (4 hours each) and independent work (18 hours). For the guided sessions, we will provide instructions, the HW platforms and a basic SW structure. Lab 1: 20% of grade Goal: Build a basic LED-to-PD link Timeline: Starts after Lecture 2 and is due after Lecture 4 Deliverables: Demo, short report, and Q&A Lab 2: 20% of grade Goal: Build a basic LED-to-Camera link Timeline: Starts after Lecture 5 and is due after Lecture 7 Deliverables: Demo, short report, and Q&A Final Project: 60% of grade Open. You will build an application on top of one of the two labs. Timeline: Starts after the second Lab and is due before the exam week. Deliverables: Demo (30%), Report (15%) & Oral exam (15%) Repair option: A repair option will be given for the labs. A new but related topic wil be given for the final Project, Report and Oral Exam.	

	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Enrolment / Application	2. The first lecture will be compulsory 3. This course can only accommodate 30 students, with ES students having a preference when demand exceeds capacity.

CESE4120	**Smart Phone Sensing**	**5**
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics". We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming.	
Course Contents	The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. After taking this course students will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non--trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	Written reports + project presentation + oral exam Overall, the final grade is determined by: 1) Two intermediate reports (10% of grade each, 2 pages each) 2) Final report (10 % of grade, 4 pages) 3) Final project demonstration (70% of grade) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. There is no resit for this course. repair option:A new but related project will be given for the Report (W), Oral(0) and Demonstration (IP). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Interactive Intelligence 2023	
Introduction 1	CANCELLED: CS4170 Seminar Intimate Computing in 2018-2019. Please note that due to the unforeseen circumstances course CS4170 Seminar Intimate Computing (Q4) will NOT take place this academic year 2018-2019. Therefore this course is not listed.

CS4015	Behaviour Change Support Systems	5

Responsible Instructor
Dr.ir. W.P. Brinkman

Contact Hours / Week x/x/x/x
0/4/0/0

Education Period
2

Start Education
2

Exam Period
2
3

Course Language
English

Expected prior knowledge
none

Course Contents
Behavior change support systems (BCSS) are computer-based systems that support individuals to form, alter or reinforce cognitions, attitudes or behaviors without using coercion or deception. They can serve individuals throughout the various stages of a change process, such as awareness development, contemplation, action strategy development, development of new behaviors, and maintaining these new behaviors. Virtual healthcare coaches, negotiation support systems, and applications that provide individuals with personalised financial guidance are three examples of these systems. To establish, modify or maintain change BCSS can deploy computerized persuasive strategies (e.g. reducing effort to establish target behavior, or argumentation reflection strategies, and also algorithmic nudge, including data-, expert-, and theory-driven), simulations (e.g. serious gaming, virtual reality), relational software agents (e.g. ePartners, virtual coaches), and personalization based on longitudinal user data. BCSSs are found in many domains, including education, sales, negotiation, management, and particularly in the health domain.

The main topics are:

Understanding the behaviour; strategies for behaviour change, action funnel, human decision making
Identifying change support strategies; Behaviour change wheel, product vision, target outcome, target user, population diversity
Conceptual design of BCSS; Behaviour plan, decision-making environment, story-editing technique
Behaviour change and technology acceptance; Motivation, Trans-theoretical model, Theory of reasoned action, theory of planned behaviour, Technology acceptance model, Unified Theory of Acceptance and Use of Technology, technology adaption lifecycle, Fogg behaviour model, Self-Determination theory
Persuasive technology; functional triad, levels of persuasion, Nudge Theory, principles of influence, triggers
Algorithmic nudge/Persuasion and support Algorithms; concept, key characteristics, and feasibility
Learning principles; Behaviouristic learning principles, cognitive learning principles, constructivism based learning principles, social cognitive (learning) theory
Communication and framing; Behavioural economic principles, elaboration likelihood model (ELM), health belief model, protection motivation theory, persuasive communication
Adherence and Gamification; Implementation Intentions, system 1 and system 2, treatment adherence, gamification
Application domains eHealth and Cyber security; Self-monitoring, self-management, virtual health agents, virtual reality therapy, internet cognitive behavioural therapy (iCBT)

Study Goals
This course provides students the opportunity to develop and demonstrate understanding, knowledge and competence in the field of BCSS. The learning outcomes for the course are that students will be able to:
1. Apply psychological theories, principles, and concepts in the design of BCSS systems.
2. Apply BCSS practices, principles and techniques.
3. Construct an implementation of an algorithmic nudge/boost/persuasion for a specific real-world application domain and objective
4. Design a BCSS system using BCSS principles, concepts, theories, and design methods.

Education Method
The course uses a flipped-classroom approach, where students watch the pre-recorded lectures in their own time and complete online quizzes. In the on-campus lectures, students and teachers discuss small challenges related to the lecture topic. In the pre-recorded video material, theories, principles and methods are presented, discussed and illustrated with examples from the field. The video material is supported by online self-tests.

In their own time, students work in small groups on coursework assignments to develop a product design for a BCSS with a computer simulation of supporting algorithmic nudge/boost/persuasion. In the on-campus session, student groups present the progress of their coursework and receive formative feedback. The presentations are done three times, following topic as the project progress over time:
1. The targeted behaviour
2. Behaviour plan
3. Conceptual design and the algorithmic nudge/boost/persuasion

Literature and Study Materials
Available on brightspace

Books
Wendel, S. (2020). Designing for Behavior Change: Applying Psychology and Behavioral Economics. O'Reilly Media, Inc,. or the older edition Wendel, S. (2013). Designing for Behavior Change: Applying Psychology and Behavioral Economics. " O'Reilly Media, Inc.

Assessment
The final grade of the course consists of the following components:
Computerised examination (or oral exam) (weighting 60%) If the expected number of students registering for the exam or resit is small, the teacher might decide to replace the computerized examination with an oral examination.
Design BCSS Coursework Project (resulting in presentation slides and demonstration of computer simulation of a supporting algorithmic nudge/boost/persuasion, including question and answer round where individual group members are assessed on the coursework) (weighting 40%)

Final grade calculation: 0.6 * Computerised examination (or oral exam) + 0.4 * Design BCSS Coursework Project
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

Resit/ Repair opportunities:
Computerised examination (or oral exam): Resit

In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:
Design BCSS Coursework Project: Resubmit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

Exam Hours
3

Co-Instructor
M.L. Tielman

CS4125	Seminar Research Methodology for Data Science	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Instructor	Dr. K.A. Hildebrandt	
Instructor	J. Urbano Merino	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	basic knowledge in mathematics (linear algebra, calculus, probability and statistics)	
Course Contents	The course focuses on research methods for data science. It looks at underlying principles and concepts for data collection, analysis and data processing, as well as the use of tools to do this. The main topics of study are: Conceptualizing research questions and experimental design Frequentist and Bayesian data analysis Generalized linear models for statistical analysis Multilevel modelling for hierarchical and longitudinal data analysis Measuring and sampling, validity and reliability Linear and nonlinear dimensional reduction Principles of statistical testing In the course, students will be using software tools such as R, and Matlab	
Study Goals	The main aims of this module for the student is to achieve an understanding of research methods for data science and obtain practical experience with data analysis and data processing methods. This module provides students with the opportunity to develop and demonstrate their understanding, knowledge, and competence. The learning outcomes for the module are that students will be able to: 1. Appreciate and comprehend strategies for collecting and processing data to answer research questions based on data 2. Understand and reproduce key principles underlying statistical data and data processing analysis 3. Learn to identify and avoid typical biases, paradoxes and misunderstandings in data-driven research 4. Apply and select appropriate data modelling techniques to analyse data and data processing	
Education Method	Lectures/Assignments Some lectures have a flipped classroom set-up, where it is expected that students prepare themselves by watching a series of videos. In the lecture, students work together on a set of classroom challenges under the guidance of the lecturer. In some others, students work on preliminary solutions to some exercises that will be discussed during the lecture. Expected Workload Lectures: 26 hours (13 × 2 hours lectures) Reading time: 39 hours Preparation basis tool use: 25 hours (5 × 5 hours for each tool) Coursework project, including writing reports and preparing for presentation: 50 hours (10 × 5 hours) Total = 140 hours	
Literature and Study Materials	Will be provided online	
Assessment	The final grade of the course consists of the following components: Analysis of experimental research data (weighting 40%) Exploration of real-world data set (weighting 30%) Linear and nonlinear dimensional reduction (weighting 30%) Students work in small groups on the 3 assignments. For the assignment, the student group submits reports and gives a presentation, including a question and answer round where individual group members are assessed on the coursework. Final grade calculation: 0.4 * Analysis of experimental research data + 0.3 * Exploration of real-world data set + 0.3 * Linear and nonlinear dimensional reduction A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Analysis of experimental research data: re-submit deliverable Exploration of real-world data set: re-submit deliverable Linear and nonlinear dimensional reduction: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	NA	

CS4165	Seminar Social Signal Processing	5
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Instructor	Dr. H.S. Hung	
Contact Hours / Week x/x/x/x	4/4/0/0 + Project	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Expected prior knowledge	Your background should consist of a combination of at least two of these topics or related topics: Signal Processing, Speech/Audio Processing, Computer Vision, AI, Machine Learning, Pattern Recognition, Reinforcement Learning, Deep Learning/ Neural Networks, Cognitive Modelling. These can be topics that you learned about at either Bachelor or Master level.	
Course Contents	The core of social intelligence is our ability to understand and interpret social signals of a person we are communicating with is. Social intelligence is a facet of human intelligence that has been argued to be indispensable and perhaps the most important for success in life. Social Signal Processing (SSP), the new, emerging, domain aimed at understanding social interactions through machine analysis and production of nonverbal behavior. In this course you will learn how next-generation computing can make use of such social signals by giving it the ability to recognize and produce human social signals and social behaviors. Think about turn taking, politeness, disagreement, emotions, rapport. You will learn about relevant findings in social psychology, and you will learn computational techniques that allow systems to make use of social signals to become more effective and more efficient by being able to detect but also simulate (e.g. in virtual agents) blinks, smiles, crossed arms, laughter. Socially aware computing. These techniques can be used in robots, virtual agents, smart homes, crowd monitoring, etc.	
Study Goals	Know what social signals are. Be able to apply computational methods to detect and simulate such signals. Position the field of social signal processing in computer science and psychology, and identify its major goals and angles of study. Define and explain social signals in humans and know about major psychological theories of social interaction. Explain major social signal recognition, simulation and expression techniques in computational systems. Develop (in groups) a research project that uses social signals in a non-trivial manner with hypotheses, research questions, and a supporting literature survey, and together evaluate the resulting system. These are important skills that prepare students towards their own masters thesis study later on.	
Education Method	This course is run as is a two quarter course running in Q1 and Q2. The course has been historically open to students from Leiden University as we see that mixed university groups leads to better quality projects and peer learning, which is a key part of the course. The course has therefore been designed to block off Monday afternoons where lecture times are scheduled to allow travel between the two institutions as part of the timetabled course. In light of the move to have campus education again, the lectures will be on campus and not virtual. The course has historically been run also as a Q1 only course. However, we have found that the 2 quarter model allows students more time to learn and absorb the course learning objectives leading to higher quality projects. In practice most of the contact hours are in Q1 with more time being devoted to the project work in Q2 with occasional progress meetings. Seminar: 2 hours of lectures per week for most of Q1. Self-study of papers. The papers will be made available at the start of each lecture. Project: 2 hours of class contact hours every 2-3 weeks for latter part of Q1 and Q2. Perform a piece of research (survey, research question, programming, testing) and write a paper about it. Students will work in teams of about 3-4 persons! Depending on the total number of students enrolled, teams will either work on a topic of their own or we will all together work on one big topic.	
Literature and Study Materials	Selected papers made available before the course.	
Assessment	The final grade of the course consists of the following components: Mini exam (weighting 10%). This multiple choice assessment helps to establish basic knowledge of the course material before the project work starts. Project proposal quality (midterm presentation + survey) (weighting 40%) Project execution (weighting 50%) Final grade calculation: 0.1 * mini exam + 0.4 * Project proposal quality + 0.5 Project execution A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mini exam: re-submit deliverable Project proposal quality: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Maximum number of participants	60	

CS4235	Socio-Cognitive Engineering	5
Responsible Instructor	Prof.dr. M.A. Neerincx	
Instructor	F. Broz	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic prior knowledge on human-computer interaction is helpful, but not required.	
Course Contents	Whether you are playing a game in virtual reality, driving a semi-autonomous car, educating yourself with a mobile app, or harmonizing your health and lifestyle with a robot assistant; nowadays intelligent networked information and communication technology is omnipresent. This course focuses on the design of human-aware intelligence into such environments, to support joint human-technology performances that bring about positive human experiences. The focus will be on social robots in general, with practical design and test assignments on robots that support people with dementia, http://rejam.tudelft.nl). In the Socio-Cognitive Engineering (SCE) course (MSc level), you will become acquainted with the application of a coherent set of methods for the design and evaluation of human--agent collaboration. Based on the SCE-method, we will elaborate on the state of the art of intelligent user interfaces (ePartners), such as virtual assistants, robotic teammates, eCoaches, and companion agents. The main topics of study are: - Design methods: Cognitive Engineering, Value Sensitive Design, Scenario-based Design, Claims Analysis, Design Rationale, Design Patterns. - Design for collective intelligence: Knowledge Representation, Ontology Engineering, Mental Models, Theory of Mind, ePartners, Adaptive Automation, Social Robots. - Design Evaluation: Prototyping, Test Methods, Measures, Questionnaires, Ethics. - Human Factors Theories and Models: Human Cognition & Learning, Memory, Emotion, Task Load, Human-Agent Teamwork, Behavior Change and Persuasive Technology.	
Study Goals	At the end of this course, students will be able to: 1. Explain the essential concepts of the design methods addressed in the course. 2. Explain the (dis)advantages of various design methods and their complementarity. 3. Apply the design methods addressed in the course in their research and design projects. 4. Explain what a design rationale is. 5. Construct a design rationale. 6. Create design specifications that are grounded in a design rationale. 7. Evaluate the strengths and weaknesses of a design rationale, e.g. using human-centered evaluations that test the design rationale. 8. Explain some of the state of the art human factors theories, models, and methods relevant to intelligent user interfaces, human-robot collaboration, and ePartner technology. 9. Write a structured report about a design-test cycle, with sufficient detail for a new group of researchers to continue the research. 10. Present work on a design project to an academic audience. 11. Work in a group on collaborative assignments.	
Education Method	LECTURES During the lectures, the teachers will present a range of theories, models, and methods relevant to socio-cognitive engineering. Students are required to read a number of scientific papers which are made available on Brightspace, along with the sheets/slides of the lectures. Together, the sheets/slides and the papers provide the students with the required theoretical knowledge to work on the practical project, and to learn about relevant design methods, human factors theories, conceptual solutions, and design principles. Most of the lectures include practical assignments and discussions stimulating the students to apply the contents of the lecture to their project (also see Project). PROJECT In the project, students work in groups to apply the knowledge acquired during the lectures. Students are required to plan, execute, present, and report on a complete design cycle (i.e. design, prototype, and evaluation) for a given design problem. This year (like the past years), the design problem is a social robot for older adults with dementia, and their social environment (https://rejam.tudelft.nl). The objective of the social robot is to improve humans physical, social, cognitive, and emotional well-being. The students will use the Wiki Socio-Cognitive Engineering (WiSCE) tool to specify the design rationale and its evaluation, step-by-step (see also https://xwiki.ewi.tudelft.nl/). Throughout the course, students will give presentations about their progress, on the design and evaluation of their prototype. Literature and study material consist of: - Papers from scientific journals on Brightspace. - Lecture notes on Brightspace	
Literature and Study Materials	Papers from scientific journals on Brightspace. Lecture notes on Brightspace.	
Assessment	The module assessment concerns the processing and application of the theory and methods; the construction of the design (rationale) and the evaluation; and the provision of the resulting concise and coherent report (including the lessons learned): The final grade of the course consists of the following components: Presentations (weighting 15%) Prototype (weighting 15%) Project report according to the prescribed format (weighting 60%) Individual reflection (weighting 10%) Final grade calculation: 0.15 * Presentations + 0.15 * Prototype + 0.6 * Project report + 0.1 * Individual reflection A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentations: resit Prototype: re-submit deliverable Project report according to the prescribed format: re-submit deliverable Individual reflection: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Exam Hours	There is no exam. The assessment is based on a paper, presentation and report. During the course, students will receive feedback on interim work. There is no resit after the end of the course.

CS4270	Conversational Agents	5
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Responsible Instructor	F. Broz	
Instructor	M.L. Tielman	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming skills (e.g. Python and Java) Probability theory and statistics	
Course Contents	Chatbots, embodied and conversational virtual agents, and social robots are becoming more and more popular. Many people are owning an Alexa, Cortana or Echo or are talking to their virtual assistant on their phone. Indeed, such technologies have the potential of making our lives easier and relieve people from the more repetitive tasks. For example, it is imaginable that such systems are being used for financial applications by helping customers with frequently asked questions but also to advise them on in the long term more impactful decisions such as their pension plans. Further applications can be imagined in the area of healthcare and education, some of which are already in existence today. In this course, attention will be given to different verbal and nonverbal behavioral characteristics, like speech, intonation, gaze and gestures that humans show when communicating with both other people and machines. This behavior is then related to different dialogue functions, including turn-taking, addressing others, and backchanneling, that give shape to the communication process. This course introduces conversational agent technology. We cover agent related technologies which can be grouped into: Dialog Management NLP speech synthesis social robotics	
Study Goals	After this course you have learned to: 1) Apply relevant linguistic and psychological theory to conversational agent systems 2) Design the conversational memory of your agent focusing on a perception/generational model 3) Analyse human-human conversational data to better design ML models for adding generation or perception capabilities to conversational agents 4) Develop the conversational memory in your agent 5) Evaluate the effects of affect and embodiment on human-agent interaction 6) Discuss your results and the limitations of your implementation	
Education Method	There are 2 lectures scheduled per week. Students work in groups of 4 on a group project. Lectures: 26 hours (13 × 2 hours lectures) Reading time: 39 hours Preparation basis tool use: 25 hours Coursework project, including writing report and prepare a video demonstration: 50 hours (10 × 5 hours)	
Literature and Study Materials	We use the book "The conversational interface " by Michael McTear, Zoraida Callejas, David Griol. This book is freely available through the TU Delft library. https://link-springer-com.tudelft.idm.oclc.org/book/10.1007%2F978-3-319-32967-3 Other relevant material will be provided on Brightspace.	
Assessment	The final grade of the course consists of the following components: Online Examination (weighting 30%) Group Assignment (weighting 70%). This assignment will result in a group report and a video demonstration. 3 Lab assignments (pass/fail) Final grade calculation: 0.3 * online examination + 0.7 * Group assignments + P/F * Lab assignments A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Online examination: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Group Assignment: re-submit deliverable Lab assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4375	Artificial Intelligence Techniques	5
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. M.T.J. Spaan	
Instructor	L. Cavalcante Siebert	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	We expect students to have programming expertise at the Bachelor level of Computer Science, in particular knowledge is expected of algorithms (e.g. search algorithms), and probability theory. Programming skills are required for the practical assignments which can be done in Python.	
Course Contents	The course focuses on concepts and techniques from probabilistic Artificial Intelligence for building intelligent agents, and decision making and -support systems. The specific topics include Bayesian networks, inference, belief monitoring, (partially observable) Markov decision processes, reinforcement learning and multiagent systems.	
Study Goals	After completing the course, student will be able 1. to comprehensively summarize the covered topics within probabilistic artificial intelligence and discuss their main issues; 2. to explain the main concepts and techniques; 3. to relate practical problems to appropriate AI techniques and apply the formal models covered in the course to these problems; 4. to implement and evaluate algorithms for complex decision making.	
Education Method	Lectures, lab work (programming assignments).	
Literature and Study Materials	Stuart J. Russel and Peter Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Prentice-Hall. ISBN 9781292401133 Global edition. See website http://aima.cs.berkeley.edu/global-index.html for additional information that goes with the book.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	The practical assignments will be done in groups of 4 to 5 students. The ethical standards of working are expected of all students. Work can be divided over students, but all students are responsible for the overall quality and originality of the work. Students that do not do their share of the work have to be reported by the group to the supervisor so that action can be taken.	
Tags	Artificial intelligence	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Multimedia Computing 2023

CS4195	Modeling and Data Analysis in Complex Networks	5
Responsible Instructor	H. Wang	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The assignment and final project require basic programming skill.	
Course Contents	Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	After this course, students could construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies, model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. Students could obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 20%) Final project (weighting 80%) Final grade calculation: 0.2 * Assignments + 0.8 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4350	Machine Learning for Graph Data	5
Responsible Instructor	M. Khosla	
Responsible Instructor	E. Isufi	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics Machine Learning: data division in train-validation-test, overfitting, kernels, regularization; Basics Deep Learning: back-propagation, stochastic gradient descent (SGD), multi-layer perceptron. Basics in programming: Python or equivalents, minimal experience with Pytorch / TensorFlow or equivalents is helpful but not strongly expected.	
Course Contents	Graph data are present in a myriad of modern computer sciences systems and applications. Examples include data generated over social, brain, financial, power, water and sensor networks. Because these data have a complicated structure they require different tools conventionally used in machine and deep learning to develop end to end solutions. These solutions falls generally under the umbrella of graph-based machine learning and can be used to perform recommendations, detect anomalies in the brain, predict financial crisis, estimate the sate of a power or water network, and coordinate group of autonomous moving sensor, to name a few. This course deals with the foundations and principles of machine and deep learning for network data. Topics include: unsupervised and semi-supervised learning on graphs; graph representation learning; graph signal processing; graph convolutions; graph neural networks; spatiotemporal learning on graphs; scalable algorithms; explainability.	
Study Goals	By the end of this course, you should be able to: Lo1: Recognise key challenges in learning from graph-structured data. You are able to recognise challenges in real applications involving graph-based data, explain them with a machine learning language, and discuss strategies to solve them. You are able to identify the issues of lack of interpretability in graph-based machine learning models and describe some possible solutions. You are also able to explain a few techniques to improve scalability of existing techniques for large graphs. Lo2: Explain and compare representation learning techniques on graphs. You are able to explain, and compare different graph-based machine learning techniques, identifying advantages and limitations in each of them, and discuss the latter with respect to a specific application. Lo3: Implement and analyse machine and deep learning techniques for graph data. You are able to apply different graph-based machine learning techniques including graph kernels, graph neural networks, graph-time learning solutions to real data. You are also able to analyse the different trade-off of these solutions, compare them with each other and inspects their behaviour. LO4: Design machine learning solutions for real world graph data tasks. You are able to create a novel solution for a practical graph-based challenge, discusses different graph-machine learning strategies for solving the task, test one or more of them, and contrast the performance of the developed approach with alternative baselines.	
Education Method	The course consists of: - 12 lectures in which the main concepts are explained and discussed; - 3 hands-on homework labs, in which practical assignments need to be solved individually; - At least two Q&A and feedback sessions about the final project;	
Literature and Study Materials	The instructors will provide online accessible material for the specific lectures. Also the slides used during the lectures are made available via Brightspace. Practical final project examples will be made available for download via Brightspace. During the course several pointer to on-line video lectures will be given.	
Assessment	The final grade of the course consists of the following components: Lab assignments (Pass/Fail). The labs assignments must be solved individually and consists of only pass/fail grade. Students can solve their assignment at home or during the lab sessions. During the labs instructors and TAs (if available) will provide feedback on the questions regarding the assignment. Students need to pass all the three lab assignments before the final project submission to pass the course. Final project (weighting 100%). The final project will be solved in groups and a group grade 1-10 will be given. The project grade will be evaluated on both the written final document as well as the question-answering session during the final interview. Final grade calculation: P/F * Lab assignments + 1 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignments: re-submit deliverable Final project: resit Q&A Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4370	Testing and Validation for AI-intensive Systems	5
Responsible Instructor	C.C.S. Liem	
Responsible Instructor	Dr. A. Panichella	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of machine learning. Knowledge of software testing is preferred but not required.	
Summary	How do you know and assess whether an AI-intensive system is working as intended? On the one hand, quality assurance for such systems can be considered to be a software testing challenge; in parallel, the machine learning components within such systems are normally validated according to machine learning-specific evaluation methodology. In this course, you will learn more about these two complementary takes to evaluation and quality assurance, while learning about (and getting hands-on experience in) state-of-the-art advances in combining these two takes.	
Course Contents	- Basics of testing (assertions, white-box vs. black-box testing) - Basics of machine learning evaluation (validation & validity) - Adversarial attacks - Testing techniques (e.g. fuzzing, metamorphic testing, combinatorial testing, differential testing) - Data labels (oracle) validation	
Study Goals	After completing this course, students are able to: 1. Apply state-of-the-art techniques to validate machine learning models and training routines 2. Explain, analyze and improve robustness considerations in AI-intensive systems 3. Design and implement techniques to generate adversarial attacks 4. Apply fuzzing, metamorphic and combinatorial testing techniques to machine learning models 5. Connect the oracle problem in software testing to questions of validity in datasets 6. Complement classical evaluation reporting of machine learning models with insights obtained through testing	
Education Method	2 hours lecture, 2 hours project	
Literature and Study Materials	The study material will be provided online by the lecturers	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4314	Seminar Selected Topics in Multimedia Computing	5
Responsible Instructor	P.S. Cesar Garcia	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	signal (image, audio) processing, pattern recognition, networking and distributed systems	
Course Contents	Through all the exciting recent advances in digital media technology and the rapid growth of social media platforms, multimedia content is increasingly embedded in our daily lives, gaining enormous potential in improving the traditional educational, professional, business, communication and entertainment processes. To be able to use this potential for transferring these processes into user-centric interactive multimedia applications, technology is required that can help us access, deliver, enrich and share rich-media content. This course provides insight into the state-of-the-art cross-disciplinary research efforts related to the development of such technology. The topics covered by the course include, but are not limited to, multimedia systems (transport and delivery, telepresence and VR, mobile), multimedia experiences (Quality of Experience, Collaboration), and multimedia engagement (emotional and social signals and social multimedia).	
Study Goals	To become acquainted with the state-of-the-art research and development activities in the field of Multimedia Computing, and to become an expert in one particular "hot topic", such that they are able to identify the "knowledge gap" (i.e., the place in which more research is needed in order to advance the state of the art).	
Education Method	readings, seminar discussions, presentations, survey paper	
Literature and Study Materials	Readings, possibly including video lectures.	
Assessment	The final grade of the course consists of the following components: Presentation on a pre-existing survey (weighting 10%) Writing own survey on a new topic (weighting 65%) Presentation on own survey topic (weighting 25%) Final grade calculation: 0.1 * Presentation pre-existing survey + 0.65 * Writing own survey + 0.25 * Presentation own survey A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation on a pre-existing survey: resit Writing own survey on a new topic: re-submit deliverable Presentation on own survey topic: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Pattern Recognition & Bioinformatics 2023

CS4070	Multivariate Data Analysis	5
Responsible Instructor	Dr. J. Söhl	
Responsible Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Required: * Introduction Probability Theory and Statistics: see for instance A Modern Introduction to Probability and Statistics Understanding Why and How Series: Springer Texts in Statistics Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover ISBN: 1-85233-896-2 * Basic calculus * Linear Algebra: matrix multiplication, the inverse of a matrix, the transpose of a matrix, least square solution. see: David C. Lay: Linear Algebra and Its Applications ISBN-10: 0321385179 ISBN-13: 9780321385178 ©2012 Pearson)	
Course Contents	PART I: Stochastic models will be developed on the basis of probability theory. Probability theory describes the behavior of certain phenomena in terms of how likely it is that certain values will occur. Central features of the models will be discussed are random variables, probability density functions, and the expected value operator. In describing random processes and signals, the correlation function and conditional probabilities play a central role. It addresses the following subjects: 1. Random variables. Matlab exercise on estimation of PDF, expected value and variance. 2. Refresher correlation. Calculating with correlation functions. 3. Random processes, correlation function, stationarity, wide sense stationarity, estimation of correlation function (Matlab exercise). 4. Random signal processing, power spectral density function, white noise. 5. AR processes, linear prediction: theory and computer exercise. 6. Markov chains. PART II: A course in advanced statistics about linear models, Bayesian inference, classification problems, Gaussian processes and Markov Chain Monte Carlo.	
Study Goals	PART I: 1. Probability Theory - Conditional) probabilities, the law of total probability, and Bayes rule. - Solve probability problems that require the use of axioms of probability. 2. Definition and Description of Random Variables and Processes PDF, PMF, CDF, Covariance, Correlation- Determine if a given PDF, PMF, CDF, variance, (auto/cross-)correlation(-function), (auto/cross-)covariance(-function), power spectral density complies with (theoretical and analytical) requirements. - Convert the description of a probabilistic problem into a probabilistic model using PDF, PMF, or CDF. 3. PDF/PMF and Expected Value Calculate the various forms of expected value of (combinations of) random variables and random processes - For a given (amplitude continuous/discrete and time continuous/discrete) probability model calculate the following probabilistic (marginal, joint and conditional) characterizations: PDF, PMF, CDF, probability of an event, expected value, variance, covariance, correlation, correlation coefficient, auto/crosscorrelation function, auto/crosscovariance function, (cross) power spectral density. - Calculate the PDF, PMF, expected value and variance of a derived random variable. 4. Properties of Random Processes - Independence, orthogonality, uncorrelated, whiteness, IID- Determine if random variables/processes have the following properties: independent, orthogonal, uncorrelated, white, Poisson, Gaussian, Bernoulli, Markov, IID, stationary, WSS, ergodic. - Calculate the expected value, variance, auto/crosscorrelation(function), auto/crosscovariance(function), power spectral density of a linear combination of random variables and of a linearly filtered (WSS, amplitude discrete/continuous, time discrete/continuous) random process. 5. Large NumbersCentral limit theorem, law of large numbers - Solve problems that require the use of the central limit theorem in an engineering context - Explain the law of the large numbers in an engineering context. 6. Statistical Estimators - Estimated mean, variance, and correlation function - Given a set of outcomes, sample functions or realizations, calculate estimators for expected value, variance, and (auto-)correlation function. 7. Application to Engineering Problems and Simulations - Select and translate a simple electrical engineering or computer science problem into mathematical probability model. The emphasis is on problems in signal and image processing, telecommunication, and media and knowledge technology. The class of probability models encompasses the following random variables/processes: Bernoulli, exponential, binomial, Poisson, Gaussian, uniform. - Justify and reflect on the approach taken in calculating or simulating (MatLab) the following probabilistic properties: PDF,	

	PMF, expected value, variance, autocorrelation function, autocovariance function. PART II: By the end of the course, you should be able to: - compute properties of frequentist and Bayesian estimators in the linear model - derive posterior distributions and their properties - apply hyperparameter optimization in Bayesian models - apply computational methods for posterior approximation - derive properties of frequentist and Bayesian classification methods - derive properties of MCMC methods for sampling from the posterior distributions - derive properties of Gaussian process regression
Education Method	PART I: Lectures, working groups (problem solving), laboratory work (a Matlab exercise) Workload is around 15 hours for attending lectures, 5 hours of reading study material and preparing lectures, 15 hours for the lab course, 20 hours for preparing the exam, 3 hours for the exam, and 8 hours for a final report (66 hours in total). PART II: Classes and weekly exercises.
Books	PART I: R.D. Yates and D.J. Goodman, "Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers", ISBN 0-471-17837-3, John Wiley and Sons, New York, 2005, Second Edition. PART II: Simon Rogers and Mark Girolami "A first coruse in machine learning, 2nd edition" Chapman & Hall It is planned to cover chapters 1--4, 8 and 9 from this book. The precise material will be determined during the course.
Assessment	The final grade of the course consists of the following components: Part I (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Part II (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Final grade calculation: 0.5 * Part I + 0.5 * Part II A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Part I written exam: Resit (written or oral exam, depending on the number of students) Part II written exam: Resit (written or oral exam, depending on the number of students) In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Part I lab assignments: Re-submit deliverable Part II lab assignments: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	PART I: Exam of 3 hours. PART II: Exam of 3 hours.
Permitted Materials during Tests	PART I: Self made notes on a two-sided written A4 sheet. Calculator. PART II: none
Remarks	PART II: This course is particularly interesting for students that are interested in statistical exploratory and quantitative techniques to analyse multivariate data.

CS4176	Algorithms for network-based bioinformatics	5
Responsible Instructor	Prof.dr.ir. M.J.T. Reinders	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The student is expected to have a basic knowledge of molecular biology, statistics and linear algebra. It is advisable to have followed CS4220 Machine Learning 1 (old code: IN4085 Pattern Recognition).	
Course Contents	Molecular biology is concerned with the study of the presence of and interactions between molecules, at the cellular and sub-cellular level. In bioinformatics and systems biology, algorithms and tools are developed to model these interactions, with various goals: predicting yet unobserved interactions, assigning functions to yet unknown molecules through their relations with known molecules; predicting certain phenotypes such as diseases; or just to build up biological knowledge in a structured way. Such interaction models are often best modelled as networks or graphs, which opens up the possibility of using a large number of readily available algorithms for inferring networks, performing simulations of biology, optimising paths or flows through networks, graph-based data integration and graph mining. Many of these algorithms can be applied (sometimes with slight alterations) to solve a particular biological problem, such as modeling transcriptional regulation or predicting protein interaction/complex formation, but also to derive systems behaviour by breaking down networks into modules or motifs with certain characteristics. In this course, we will first give a brief overview of molecular biology, the advent of high-throughput measurement techniques and large databases containing biological knowledge, and the importance of networks to model all this. We will highlight a number of peculiar features of biological networks. Next, a number of basic network models (linear, Boolean, Bayesian) will be discussed, as well as methods of inferring these from observed measurement data. Building on the network inference methods, a number of ways of integrating various data sources and databases to refine biological networks will be discussed, with specific attention to the use of sequence information to refine transcription regulation networks. Finally, we will give some examples of algorithms exploiting the networks found to learn about biology, specifically for inspecting protein interaction networks and for finding active subnetworks.	
Study Goals	After succesfully completing this course, a student is able to: list the basic elements of a living cell and their interactions, and describe how these can be measured; explain what type of mathematical model is applicable to what measurement(s), at what level(s), in a given systems biology problem; read and comment upon recent network-based computational biology literature; discuss the state-of-the-art in systems biology and integrative bioinformatics, and future challenges	
Education Method	The course consists of a mixture of lectures by the teachers and paper presentations by one or more of the students. Each paper presentation will be followed by a in-depth discussion. There will also be a practical session allowing students to get hands-on experience with network models.	
Literature and Study Materials	Slides, collection of papers and lab course manual (Brightspace).	
Assessment	The final grade of the course consists of the following components: Proposal for a research project (weighting 80%). Students are required to write a proposal for a research project in which they clearly state the biological problem to be solved, the necessary data, the computational approach as well as the innovative parts of the approach. Paper presentations/discussions (weighting 20%) Final grade calculation: 0.8 * Proposal for a research project + 0.2 * Paper presentations/discussions A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Proposal for a research project: re-submit deliverable Paper presentations/discussions: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	As students depend on each other (to present the material to the class), a commitment to follow the course through to the end is required.	

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4230	Machine Learning 2	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Instructor	Dr. D.M.J. Tax	
Instructor	T.J. Viering	
Instructor	Dr. F.A. Oliehoek	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Required: This course is the more advanced and research oriented follow-up to Machine Learning 1 (CS4220). CS4220 or an equivalent machine learning course is therefore expected as prior knowledge.	
Course Contents	The course will treat a number of machine learning settings and theories spanning the breadth of machine learning research in detail and on an advanced level. Topics that are covered include (subject to change): learning theory, learning curves, latent variable models, causal inference and discovery, adversarial learning, online learning and neuromorphic computing.	
Study Goals	After completing the course, you should be able to: 1. Identify which setting beyond (un)supervised learning a problem belongs to 2. Explain the essential problem that (adversarial/causal/online/PAC/neuromorphic/...) learning addresses and how it differs from the classical ML setting 3. Evaluate methods that are proposed in these learning scenarios, both theoretically and empirically 4. Mathematically prove basic properties of methods in these scenarios 5. Apply common methods in these learning scenarios on real-world datasets 6. Develop machine learning techniques adapted for a particular learning task	
Education Method	Each week: 1 lecture and 1 Q&A session where the weekly exercise set will be discussed.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) 4 combined assignments (pass/fail). There will be multiple small assignments (in weeks 2,4,6,8). Final grade calculation: P/F * assignments + 1 * Written exam A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: 4 combined assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4240	Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Seminar Computer Vision by Deep Learning	
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy). Experience in programming and installing programming languages/libraries is required. Without affinity/experience with python (and numpy) programming this course will become too time consuming as the project will become difficult.	
Course Contents	In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.	
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.	
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ - https://udlbook.github.io/udlbook/ Research papers will be made available through Brightspace. Assignments are based on PyTorch: https://pytorch.org/	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Digital Exam (weighting 90%). Exam about the papers, lab assignments, and the theory (lectures). Project (Pass/Fail). In a small group of students you work on a deep learning paper reproducibility project. Final grade calculation: (pass project) * (0.1 * Presentation + 0.9 * Exam) A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4245	Seminar Computer Vision by Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	N. Tömen	
Instructor	Dr. X. Zhang	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	MSc thesis in the Computer Vision lab	
Expected prior knowledge	Deep Learning (CS4240)	
Course Contents	The recent boom in computer vision and automatic image understanding represents an inflection point in human productivity, permeating wide aspects of the economy and society. Examples of visual tasks which are repetitive or require expert knowledge include medical diagnosis, industrial inspection, autonomous vehicles, etc. When machines can meaningfully assist or even completely take-over such tasks it will change the world as we know it. The breakthrough in the 2012 ImageNet automatic image recognition competition shows all previously existing methods decisively defeated by a deep neural network. Deep learning replaces feature engineering methods and is able to successfully learn image features from huge annotated datasets. This course is on automatically understanding visual content such as images and videos by deep learning. The range of topics (although, not limited to): Fundamentals in Vision, object detection, per-pixel labelings, video recognition, image similarity learning, efficiency, self-supervision, 3D computer vision, adverserial attacks, explainability, generative models, etc. The course will have lectures, a seminar and a lab practical: - The lectures will be on established topics based on the current literature. - The seminar will have students read, critique, and present relevant computer vision research papers. You will have to read 2 papers per week, for 7 weeks (14 papers). - The lab will have students apply and design their own (small) computer vision project. The course build on top of the Deep Learning course (CS4240) and follows a similar setup.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the deep learning techniques reviewed in the course for computer vision applications such as image classification, object detection, per-pixel labelings, video recognition, image similarity learning. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [L04]. Quickly identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (we focus on Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) Computer Vision problem. [LO8]. Write clearly and concisely about your code, method, results, and analyis.	
Education Method	Lectures Lab project: design and execute your own Computer Vision project. Seminar: paper reading, critiquing, and presenting.	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Exam (weighting 60%). Exam about all material. Project (weighting 30%). in a small group of students you work on a computer vision project. Final grade calculation: 0.1 * Presentation + 0.3 * project + 0.6 * Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4250	Selected Topics in Molecular Biology	5
Responsible Instructor	T.E.P.M.F. Abeel	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course covers a broad range of essential topics in molecular biology necessary to start critically working with bioinformatics algorithms. The choice and implementation of algorithms critically depend on a proper understanding of the application domain. In this case the application domain is biology and in this course we review key biological concepts and topics that have impact on the implementation and selection of various computer algorithms and models. We use the following book: B. Alberts et al., 'Molecular Biology of the Cell', 6th edition Publication Date: November 18, 2014 ISBN-13: 978-0815344322 ISBN-10: 0815344325 A detailed overview of the actual reading material will be provided through the online learning platform Brightspace and will include material from Chapters 1,3,4,5,6,7 and 8. Extra material to aid in learning are available from the publisher's website.	
Study Goals	The goal of this course is to learn about the basic concepts in molecular biology required for bioinformaticians. At the end of this course: students will be able to reproduce, discuss and reflect on information about basic molecular biological processes, concepts and ideas. This includes: - cell structure, organization, information encoding and transfer - core concepts around DNA, chromosomes, genomes - DNA replication, repair and recombination - central dogma molecular, including replication, transcription, translation - regulation of the processes above	
Education Method	Self-study with weekly Q&A sessions.	
Books	B. Alberts et al., 'Molecular Biology of the Cell', 6th edition Publication Date: November 18, 2014 ISBN-13: 978-0815344322 ISBN-10: 0815344325	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 100%) Final grade calculation: 1 * Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4255	Algorithms for sequence-based Bioinformatics	5
Responsible Instructor	T.E.P.M.F. Abeel	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Bioinformatics analyses in genomics aim to compare large sets of genomes in order to understand and explain differences in traits of an organism. Contemporary methods are powered by fundamental algorithms and data structures, which are efficient and scale to large data sets. A thorough understanding of these algorithms and data structures is necessary for advanced users and developers in this area. In addition, understanding how comparative genomics is developing is important to shape your own research. In this course, we will cover genome analysis, variant analysis, and pangenomics. Core concepts, applications, and future trends will be discussed, with a focus on the algorithms and data structures underlying state-of-the-art methods.	
Study Goals	After having followed this course, the student has a good understanding of algorithms and data structures in genomics used for DNA sequence analysis. The student is able to implement algorithms in python, and can translate methods described in scientific literature into a working implementation.	
Education Method	The course is offered as a mix of lectures, exercises and a project	
Books	http://bioinformaticsalgorithms.com	
Assessment	The final grade of the course consists of the following components: Exercises (weighting 60%) Project report (weighting 40%) Final grade calculation: 0.6 * Exercises + 0.4 * Project report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Exercises: re-submit deliverable Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4260	Machine Learning in Bioinformatics	5

Responsible Instructor	Dr. J.S. de Pinho Gonçalves
Contact Hours / Week x/x/x/x	0/0/4/0 Lectures + 0/0/4/0 lab
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Expected prior knowledge	Recommended for fundamental knowledge in machine learning and multivariate data analysis: CS4220 Machine Learning 1 CS4070 Multivariate Data Analysis Recommended for basic concepts of molecular biology: CS4250 Selected Topics in Molecular Biology Required: CSE1200/TI1106M Calculus (or similar) CSE1205/TI1206M Linear Algebra (or similar) CSE1210/TI2216M Probability Theory and Statistics (or similar) The project requires basic programming skills.
Course Contents	Learning from patterns in molecular biology data plays an important role in diagnosing disease, discovering new targets for therapy, and more generally in answering biological questions that lead to an improved understanding of biological systems with relevance to human health and industrial areas including biotechnology and agriculture. This course focuses on methodology for the analysis of high-dimensional data in molecular biology, naturally addressing challenges that commonly arise in the field such as learning from unlabelled data or from small numbers of samples. The methodology is introduced in the context of meaningful applications in molecular biology with examples using real data. Covered topics will be (a selection of) the following: - Discrete probability models and statistical modeling: statistical models for experiments with categorical outcomes, goodness of fit, maximum likelihood and Bayesian estimation. - Mixture models: finite mixture models of normals and the EM algorithm, common infinite mixture models such as beta-binomial and gamma-Poisson, ECDF and bootstrapping. - Clustering: comparing observations, iterative partitioning, density-based clustering, hierarchical clustering, clustering validation, choosing the number of clusters. - Statistical hypothesis testing: p-values, single and multiple hypothesis testing, family-wise error rate, (local) false discovery rate. - Testing using high-throughput count data: multifactorial designs, linear models, analysis of variance, generalized linear models, robustness and outlier detection, shrinkage estimation. - Linear dimensionality reduction (PCA): preprocessing data for multivariate analysis, projecting onto lower dimensions, matrix decomposition (SVD), biplot representations, projecting additional variables for interpretation). - Multivariate methods for heterogenous data: orderings, gradients and latent variables (ordination); multidimensional scaling (MDS), robust (non-metric) MDS, batch effect removal, correspondence analysis, finding gradients and trajectories (local non-linear methods such as tSNE and UMAP), canonical correlation. - Supervised learning: discrimination, performance measures, curse of dimensionality, generalizability and model complexity, regularization and penalization, cross-validation, supervised learning methods (SVM, decision trees, ...), method hacking. - Design of high-throughput experiments and their analyses: types of variability (error, noise, bias), confounding, dependencies, batch effects, statistical power, mean-variance relationships and data transformations, workflow design, data representation, efficient computation.
Study Goals	After successfully completing this course, the student should be able to: - Recognise, characterise, and interpret different kinds of high-throughput molecular biology data and their statistical properties. - Recognise, categorise and compare common statistical techniques and machine learning algorithms and the data analysis problems that they address. - Recognise typical research questions that can arise when analysing such kinds of data, reason about and select appropriate methodology to address them. - Understand, reason about, and discuss the different steps of a data analysis workflow: from experiment design to the interpretation of results. - Design a research plan, execute it, and write a scientific paper about it.
Education Method	The course is run in a flipped classroom setting. Lectures: students take turns explaining the material from the different chapters of the course book. All students are required to read and prepare the course material before every lecture, so that they can contribute to the discussion of the material in the classroom. Project/labs: students will work in small groups on a research project throughout the course. Each group will present and discuss the status of the project weekly or bi-weekly, and deliver a written report at the end. There will be one or two lectures (90 min. each) and one project discussion lab (45 to 90 min.) per week. The frequency and duration of the sessions will depend on the number of students following the course. The exact schedule will be determined after the first lecture.
Books	The material of the course follows the book: "Modern Statistics for Modern Biology" Authors: Susan Holmes, Stanford University, California Wolfgang Huber, European Molecular Biology Laboratory Published: February 2019 ISBN: 9781108705295 The complete book can be browsed online at: http://web.stanford.edu/class/bios221/book/. This website also contains the R code and data accompanying the examples in the book.

Assessment	The final grade of the course consists of the following components: Book chapter presentation (weighting 25%) Participation (weighting 15%) Project report/paper (weighting 60%) Students are graded individually. Final grade calculation: 0.25 * Book chapter presentation + 0.15 * Participation + 0.6 * Project report/paper A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Book chapter presentation: re-submit deliverable Project report/paper: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Tags	Artificial intelligence Research Methods

CS4329	**Recent topics in bioinformatics**	**5**
Responsible Instructor	Prof.dr.ir. M.J.T. Reinders	
Instructor	Dr. J.S. de Pinho Gonçalves	
Instructor	T.E.P.M.F. Abeel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Bioinformatics is at the heart of many modern systems biology analyses, and encompasses the application of statistics and computer science to (large-scale) biomolecular datasets. In essence, bioinformatics is about smart ways of extracting knowledge from the enormous amounts of data that can be generated using modern measurement techniques. For instance, it plays an important role in finding the genetic origins of various diseases, such as cancer, diabetes or alzheimer.	
Study Goals	After successfully completing this course, the student is able to: explain several high-throughput data acquisition experiments, such as DNA/RNA sequencing, and discuss the benefits and limitations of these methods comprehend the statistical and computer science issues in analyzing high-throughput data discuss the basic systems biology approach, and the role of high-throughput measurements, gene selection and classification therein explain bioinformatics methods, algorithms and models to a non-expert audience implement or execute basic algorithms from descriptions provided in scientific literature read and comprehend a current scientific paper and reflect on the bioinformatics methods used in such paper	
Education Method	In this course we will study some key examples of bioinformatics analyses by reading a set of selected papers that present some significant biological conclusions.The course is run a flipped class-room students take turns guiding us through the selected papers. Teachers moderate the discussion and fill in any gaps in understanding.All students are required to read, and prepare the course material before every lecture to effectively participate. Each week there are one or two lectures, each of 90 minutes In each lecture one paper (the course material) will be discussed in detail. One or more students will present and explain the details of this paper. It is essential that you highlight the (bioinformatics) methodology of the paper. The schedule for this will be prepared in the first lecture. This presentation is graded. All students are expected to have read the paper and should have an active role in the discussion about the paper. Each student should be prepared to raise at least two critical remarks or questions during the discussion. The quantity and quality of participation of each student is graded. The student will perform a practical project in groups where they gain practical experience in state-of-the-art bioinformatics methods.	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 20%) Participation during discussions (weighting 20%) Reporting about unseen paper with oral presentation (weighting 20%) Final grade calculation: 0.2 * Presentations + 0.2 * Participation + 0.2 * Reporting about unseen paper with oral presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Reporting about unseen paper with oral presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). en circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Programming Languages 2023

CS4130	Seminar Programming Languages	5
Responsible Instructor	Dr. S.S. Chakraborty	
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	C.B. Poulsen	
Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	4/0/0/0 0/0/0/4	
Education Period	1 4	
Start Education	1 4	
Exam Period	1 4	
Course Language	English	
Expected prior knowledge	Followed at least one other Programming Languages master course	
Course Contents	Programming languages is a core field in computer science that studies the design, theory and applications of both new and existing programming languages. Topics in programming languages include compiler construction, program analysis, program transformations, meta programming, parsing, formal semantics, program verification, and type systems. In this course, we will read scientific journal and conference articles in the field of programming languages to get a deeper understanding of programming languages. If you wish to do a MSc thesis in the programming languages group, we highly recommend taking this course.	
Study Goals	The student will acquire: - Skills to read and discuss scientific articles. - Understanding of the topics in the research field of programming languages. - Understanding of the research methodology in the research field programming languages.	
Education Method	We will run this seminar as a discussion seminar with meetings twice a week. In each meeting, we discuss a scientific article that has been studied by the participants in advance. The following activities are required for each meeting: - Reading a scientific article - Writing and submitting a short summary of the article (max 0.5 pages) - Active participation in the discussion of the article Expected Workload: - 4h Discussion sessions - 6h Reading paper at home - 2h Writing summary at home	
Literature and Study Materials	Papers from the programming languages literature will be assigned at the start of the course	
Books	No books	
Assessment	The final grade of the course consists of the following components: Students get a grade for each meeting based on the participation in the discussion (weighting 100%) Final grade calculation: The final grade is the average of the grades for the meetings. A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. There are no repair opportunities for this course in accordance with TER Implementation Regulations Art 5, sub 5. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	not applicable	
Enrolment / Application	The number of participants for this course is limited. Students in the second year of the master and students that follow the Programming Languages specialization have priority.	

CS4135	Software Verification	5
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	Ir. K.F. Wullaert	
Contact Hours / Week x/x/x/x	0/2/0/0 + 0/4/0/0 practicum	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	How can we ensure that software cannot crash and is guaranteed to be correct? In this course we tackle this question by viewing programs and programming languages as mathematical objects. That way we can use logic to prove properties about programs and thereby guarantee that software is correct. To make reasoning about actual programs and programming languages feasible, we will not be doing these proofs by hand, but instead use a tool called a proof assistant to build proofs that can be checked by a computer. As we will show during this course, proof assistants turn the activity of doing proofs into programming. This course is a specialization course for programming languages and software engineering Prior knowledge: - Required: - Strongly advised: some familiarity with functional programming and propositional and predicate logic.	
Study Goals	Learning Objectives: - State and prove properties of functional programs in logic. - Specify the semantics of a programming language in logic. - State and prove the correctness of imperative programs. - Use a proof assistant to perform a mechanized proof.	
Education Method	This course is being taught in a "flipped classroom" style. It consists of 2 hours of studying (by reading or watching lecture videos, for greater flexibility and avoiding clashes with other courses - material will be provided) and two lab sessions of 2 hours each. During the lab sessions students will work on stating and proving theorems. Towards the end of the course students will carry out research projects that apply the ideas of the course.	
Literature and Study Materials	Supplementary material: Free online text book "Logic and Proof": https://leanprover.github.io/logic_and_proof/ Free online text book "The Hitchhikers Guide to Logical Verification": https://github.com/blanchette/logical_verification_2021/raw/main/hitchhikers_guide.pdf	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the individual project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4200-A	Compiler Construction	5
Responsible Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	- programming (required) - software engineering (recommended) - functional programming (recommended) - formal languages and automata (recommended)	
Course Contents	Compilers translate the source code of programs in a high-level programming language into executable (virtual) machine code. In this course you will study the architecture of compilers, and important concepts and techniques underlying the components of that architecture. You will study these techniques by applying them in practice to implement your own compiler for a small language. The course covers the following topics: - Syntax and parsing - Static semantics and type checking - Virtual machines, assemby code, byte code - Interpretation vs. compilation - Code generation - Memory management, garbage collection - Data-flow analysis - Optimization	
Study Goals	At the end of this course, you will be able to: - Explain the architecture of a compiler and the role of each component of that architecture - Explain the algorithms and techniques for the implementation of compiler components and apply these techniques to examples - Implement a code generator that translates source language abstract syntax trees to target language instructions - Implement transformations on abstract syntax terms to simplify programs - Implement simple type checkers - Implement simple data-flow analyzers - Integrate these components into a working compiler	
Education Method	Attending the course involves: - Attending weekly lectures - Attending weekly lab session (which may start with a group tutorial) - Reading lecture material and papers - Making homework assignments (building a compiler)	
Literature and Study Materials	Required reading: - We will use selected chapters from the book "Essentials of Compilation" by Jeremy Siek, available online: https://github.com/IUCompilerCourse/Essentials-of-Compilation - Lecture slides and selected papers from the literature Optional reading: - We will use Scala to implement our compilers. The following book is optional reading which may be helpful in getting up to speed with programming in Scala: https://www.artima.com/shop/programming_in_scala_5ed (Additional) reading materials will be made available on Brightspace. We will use WebLab (https://weblab.tudelft.nl/cs4200/) for the submission of homework assignments.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Homework assignments (weighting 50%) Final grade calculation: 0.5 * Written exam + 0.5 * Homework assignments	

	A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Permitted Materials during Tests	No materials are permitted during the exam.
Co-Instructor	Dr. S.S. Chakraborty

CS4280	**Language-Based Software Security**	**5**
Responsible Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	This course has no formal prerequisites. However, for the homework assignments you will have to implement several program analysis techniques using the Scala programming language. If you have not used Scala before, you are thus expected to learn the basics of the language through self-study.	
Course Contents	Security vulnerabilities often arise due to programming errors in the source code of an application. Recent programming errors with severe security implications include Heartbleed (buffer over-read), Shellshock (code injection), and goto-fail (ill-formated code). Rather than hunt for individual vulnerabilities in programs, a more structural approach to improve security is to improve the programming language. This is the goal of language-based security: to rule out whole classes of potential security vulnerabilities in one go. This course studies various security properties and program analysis techniques for enforcing these properties at the level of the programming language to improve software security. In particular, we will study the following properties: - Memory safety: prevent buffer overflows and overreads - Type safety: prevent undefined behaviour - Information flow control: prevent data leaks and code injection attacks We will study techniques to address these problems at the language level through dynamic analysis, static analysis, and language design. To facilitate a precise study and comparison, we will define the above techniques formally in class. To facilitate student experimentation and exploration of trade-offs, students will implement the above techniques in homework assignments.	
Study Goals	After taking this course, students should be able to: 1. Describe the nature and causes of security vulnerabilities in software systems, and give concrete examples of how these security vulnerabilities can be exploited. 2. Explain the properties that can be enforced at the level of the programming language to rule out security vulnerabilities, such as memory safety, type safety, and non-interference. 3. Formally define the semantics of a simple programming language. 4. Formally define dynamic and static analysis techniques for enforcing these security properties. 5. Implement these techniques for a small programming language. 6. Discuss and evaluate the importance of soundness and precision of a given program analysis. 7. Contrast programming languages based on the set of countermeasures they provide, and give an appropriate recommendation for a specific application. 8. Analyse and apply results from scientific literature in the area of language based security.	
Education Method	The course work consists of the following activities: 1 or 2 instruction sessions per week. Weekly homework assignments consisting of theoretical questions, programming assignments, and reading assignments	
Assessment	The final grade of the course consists of the following components: Weekly homework assignments (weighting 40%). The weekly homework assignments will test your ability to design and implement (variants of) the techniques discussed in the lectures (study goals 3-5). Written or oral exam (weighting 60%). The final written or oral exam will test your theoretical understanding of the security vulnerabilities and their countermeasures discussed in class (study goals 1-2) and your ability to discuss and contrast the different aspects of these techniques (study goals 6-8). Final grade calculation: 0.4 * Weekly homework assignments + 0.6 * Written or oral exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written or oral exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4405	Analysis of Concurrent and Distributed Programs	5
Responsible Instructor	Dr. B. Özkan	
Responsible Instructor	Dr. S.S. Chakraborty	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	

Course Contents

Software systems are becoming highly concurrent and distributed to utilize modern multicore architectures and increasing speed and bandwidth in networks. Shared-memory concurrency in multicore programs and message-passing concurrency in distributed programs share many common abstractions and problems.

In the multicore era, all performance-critical software employs some form of concurrent programming; typically shared memory concurrency. In this setting,
programmers use a number of primitives to develop efficient and correct concurrent programs. To do so the programmers have to understand the behaviors of the primitives and reason about them. It is also important to match the programming paradigms and underlying architectures. For instance, traditionally programmers have assumed that a multithreaded program executed simply by interleaving the executions of its threadsa model known as sequential consistency (SC). This assumption is, however, invalidated both by mainstream multicore architectures, which often execute instructions out of order, and by compilers, whose optimizations affect the outcomes of concurrent programs. As a result, concurrent programs have more outcomes than SC allows.

In the distributed setting, the units of concurrency are independent processes that do not share memory but communicate by exchanging asynchronous messages. The execution of such a system involves two main sources of nondeterminism: concurrency and partial failures. As the processes run concurrently, the exchanged messages can be delivered and processed in many different orderings. The distributed set of processes is also prone to network of process failures. The trade-off between the systems availability in the existence of failures and the consistency between the processes gives rise to a spectrum of weak consistency notions. It is important to reason about concurrency, possible failures, and consistency guarantees to implement distributed programs correctly and understand their behavior.

This course aims to explore analysis techniques for concurrent and distributed programs.

Outline of Lectures:

Shared memory concurrency:
- Abstractions for shared memory concurrency
- Relaxed memory concurrency
- Correctness of concurrent programs

Distributed concurrency:
- Distributed system components, models and assumptions
- Fundamental abstractions for distributed systems

Study Goals

This course aims to give students a deep understanding of concurrency and distribution in modern systems and hands-on experience for analyzing these systems.
By the end of this course, you should be able to:

- Describe the fundamental concurrency models in multicore and distributed systems

- Explain the concurrency nondeterminism in the executions of concurrent and distributed programs

- Discover concurrency bugs in multicore and distributed programs by program analysis and testing techniques

- Evaluate the pros and cons of program analysis and testing techniques for specific multicore and distributed programs

Education Method

The course consists of the following education methods:

- Lectures for reviewing concurrency and distribution concepts
- Homeworks/assignments
- Developing a course project, writing a report, and presenting it (course project)

To finish the course, students (in teams) will have to:

- Study several papers which will be discussed during the lectures
- Deliver their assignments
- Deliver and present their implementation project

Assessment

The final grade of the course consists of the following components:
Research project implementation (weighting 40%)
Research project report (weighting 20%)
Research project presentation (weighting 20%)
Homework assignments (weighting 20%)

Final grade calculation: 0.4 * Research project implementation + 0.2 * Research project report + 0.2 * Research project presentation + 0.2 * Homework assignments
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

Resit/ Repair opportunities:
In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:
Research project report: re-submit deliverable
Research project presentation: resit
Homework assignments: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

CS4410	Category Theory for Programmers	5
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	Ir. K.F. Wullaert	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Categorical structures occur in programming languages on different levels: (1) within programming languages, providing design principles and guidance on how to write modular and correct-by-design programmes (as demonstrated in the practical programming language Haskell) and (2) in the design and study of programming languages, as a guiding meta-theory. In particular, category theory provides a mathematical justification for recursion schemes for inductive datatypes. This course aims to provide solid foundations on both (1) and (2). Lecture notes for the course are available at https://github.com/benediktahrens/CT4P. This course is suitable for people with an interest in functional programming, abstract mathematics, and logical reasoning. Prior knowledge: - Required: None - Strongly advised: None	
Study Goals	Learning Objectives: - Use categorical constructions (e.g., monads) in the design and structuring of computer programmes - Specify datatypes as initial algebras - Prove properties of computer programmes, guided by categorical intuition - Use categorical fusion laws to optimize functions - Use the theory of infinite data structures to define functions on infinite data	
Education Method	Learning in this course is achieved through lectures, problem sessions, and guided self-study.	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Software Engineering 2023	
Introduction 1	The Software Engineering Research Group aims at developing a deep understanding of how people build and evolve software systems and, with that knowledge, develops novel methods, techniques and tools that advance the way in which software is built and modified. This means that we study how software is developed (e.g., by mining collaborative platforms like GitHub or setting up experiments with developers) in areas such as software testing, code reviewing, end-user development or continuous integration. At the same time, we also build tools and prototypes that alleviate some of the pains that we observe (tools like TestNForce, Cloneboard, etc.). In addition, we are also working in the area of software language design and engineering, which aims to effectively design, implement, and apply domain-specific software languages. Main areas that we are working in here are: the automatic derivation of efficient, scalable, incremental compilers and effective IDEs from high-level, declarative language definitions. At the same time, we investigate the systematic design of DSLs with an optimal tradeoff between expressivity, completeness, portability, coverage, and maintainability. A good example of a recent master thesis is Alex Nederlofs thesis Analyzing Web Applications: An Empirical Study in which he fully automatically tests 3,422 randomly selected web sites for errors and accessibility standard violations. This thesis can be found at the groups website: www.se.ewi.tudelft.nl. This website also contains information about possible master thesis projects.

CS4295	Release Engineering for Machine Learning Applications	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The world of Software Engineering has been revolutionized in the last decade. Instead of releasing software updates yearly, companies can now release multiple times per week, sometimes even per day, to their customers. This allows quick reactions to market demands, software failures, and it is crucial for increasing the business value of software. These improvements have been mostly enabled by advances in release engineering and, in this course, we will learn about the techniques and technologies that build the foundation for modern release engineering. This course is addressing two main target audiences: 1) software engineers that would like to learn how to systematically release their applications and make them scalable in an orchestrated environment and 2) machine learning experts that would like to learn how to make their proof-of-concept applications production ready. Together, we will go on a journey that starts at continuous integration and then moves on to continuous delivery, continuous deployment, and continuous experimentation. We will discuss the theory and the current research on various related subjects like containerization, testing, or monitoring and will put the learned theory into practice. As a running example, we will build a pipeline for a machine learning application, which -compared to traditional release engineering- poses additional challenges, like data versioning or model deployment. *Important:* Make sure to register for a group in Brightspace AT THE LATEST during the first lecture.	
Study Goals	After following this course, you will be able to... Explain the concepts of DevOps and MLOps Automate the development and operation process ... Automate packaging, versioning, and releasing in modern build environments like GitLab or GitHub ... Release artifacts in public registries like the GitHub Maven Registry. ... Provision servers with Vagrant and Ansible ... Automate the model training pipeline using DVC Use modern containerization technology to build a scalable micro-service architecture ... Use Docker to isolate a complex, web-based application in a container ... Use Kubernetes to orchestrate a non-trivial, distributed application ... Create an Istio service mesh in Kubernetes that enables custom routing strategies ... Use Prometheus and Grafana to monitor the state of a Kubernetes-based application Deploy and maintain machine learning models in production ... Embed a typical ML model in an HTTP micro-service ... Apply version-control techniques to data and models in a machine learning project ... Implement quality-control techniques in a machine learning application ... Design a testing strategy that covers data, model, infrastructure, and monitoring aspects Document the state and extensions of release engineering pipeline ... Find related work and inspiration for release engineering tasks ... Identify extension opportunities in existing release pipelines	
Education Method	Following this course will require you to... Follow interactive lectures and pre-recorded material for self-study Execute a programming project in small teams Transfer lecture contents into a project through practical assignments Assess the work of yourself and others through rubric-supported peer-feedback Actively participate in Q&A sessions Document a release engineering pipeline in a report Present your project results and answer questions about the lecture contents.	
Assessment	--- Formative Assessment --- You can check your individual learning progress through weekly rubrics ... You will receive feedback from other groups as a neutral outsider ... You can check your own progress via self-assessment Joint Q&A sessions will discuss common points of the peer feedback and concrete questions --- Individual Knock-Out Elements --- The course enforces several criteria to protect the active members of a team. Students not reaching the following criteria will be removed from the course: Upholding a professional attitude when interacting with peers and staff. Actively participating in the team process and contributing to all team deadlines and deliverables Meaningful and visible project contribution throughout each of the seven project weeks (W2-W4, W6-W9) ... 1+ commit to one of the team repositories ... Create 1+ PR and approve 1+ PR that is not your own ... The changes must contain a meaningful contribution to code, config, or report Constructive participation in all peer reviews for assignments (one for another team and one for your own) Ability to explain lecture contents and defend all aspects of the team project in the oral examination Availability of team results for grading (repositories, report, artifacts) We offer repair options for the following criteria: Not having meaningful and visible contributions in one week can be repaired in the following week. Failing to satisfy the criteria in more than one week will lead to removal. Failing to submit a peer review in time can be repaired in the following week. Missing a review a second time will lead to removal The inability to explain or justify the project in the oral examination can be repaired in a second oral examination.	

	--- Summative Assessment --- Students attending the course will be graded on the following grading components. ALL graded components are rounded to .1 and must at least be "almost passed" (>5.0). Their weighted average will be rounded to .5 to calculate the final team grade. Project ... Versioning & Releases (10%) ... Containers & Orchestration (10%) ... Kubernetes & Monitoring (10%) ... Provisioning (10%) ... Istio Service Mesh (10%) ... ML Configuration Management (10%) ... ML Testing (10%) Report (30%) Presentation & Oral Examination (Pass/Fail) In case of large performance differences in a group, we may choose to assess the team members individually, based on the individual contributions to the deliverables and the interaction traces that we find in the repository. --- Please Note --- All assignment deadlines throughout the course are fixed, we do not accept late submissions. Late submissions in systems that do not enforce due dates will be treated as not submitted (e.g., Brightspace, Buddycheck). Failing to submit a deliverable cannot be repaired. Insufficient deliverables (>=4.0 and <6.0) can be repaired, but the grade of the deliverable is then capped at 6.0. There is NO resit opportunity for this course. Partial grades are not carried over to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Enrolment / Application	In addition to your regular enrollment of the course, you must also register in Brightspace, AT THE LATEST during the first lecture, so you can be assigned to a group. Late registrations after the group creation cannot be considered anymore.
Co-Instructor	Prof.dr. A.E. Zaidman

CS4415	**Sustainable Software Engineering**	**5**
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Sustainable Software Engineering is an overarching discipline that addresses the long-term consequences of designing, building, and releasing a software project. By definition, sustainability covers five main perspectives: environmental, social, individual, economic, technical. This course mainly focuses on the first, also known as Green Software Engineering. On top of that, we also cover fundamental aspects of social and individual sustainability of software projects. Software Engineering (SE) has long addressed sustainability by narrowing it down to economic and technical sustainability. However, our society is facing major sustainability challenges that can no longer be overlooked by software engineers and computer scientists. It is estimated that, by 2040, the ICT sector will contribute to 14% of the global carbon footprint. Hence, the environmental, social, and individual perspectives ought to be part of the equation when it comes to design, build, and release software systems. The problem is far from simple and we need expert computer scientists to bring sustainability into the core values of the next generation of tech-leading organisations.	
Study Goals	After attending this course, you will be able to: LO1. Explain the fundamental sustainability principles and their relevance to software engineering. LO2. Assess the energy consumption of software systems using measurement and testing techniques. LO3. Create actionable strategies to address sustainability issues within software organisations. LO4. Evaluate and compare different solutions for sustainable software systems based on their effectiveness in achieving sustainability goals and their potential impact on various stakeholders.	
Education Method	To meet these objectives, you will be involved in a broad set of learning activities: lectures, paper reading, software analysis, software development, essay writing and presentation. These heterogenous set of activities aim at building a strong set of hard skills for energy-efficient code development combined with a strong set of soft-skills and critical thinking. You will work on ideas that will help real-world software projects embrace a green software culture.	
Assessment	The final grade of the course consists of the following components: Software project (weighting 45%) Essay (weighting 45%) Presentation (weighting 10%) Final grade calculation: 0.45 * Software project+ 0.45 * Essay + 0.1 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Software project: re-submit deliverable Essay: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4315	Software Architecture	5
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software engineering	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four, in which the actual architectures of existing open source systems are analyzed in considerable detail. These systems are taken from github, and student teams are challenged to actually contribute to the systems under analysis in the course.	
Study Goals	Bring students into the position that they can (1) explain the key architectural concepts and methods for modeling software architectures; (2) apply viewpoints and perspectives to model software architectures; (3) discuss the benefits of architecting and the role of the software architect; (4) evaluate and validate software architectures; (5) explain and discuss the concepts of component-based and plugin architectures, service-oriented architectures, and software product lines; (6) explain and recognize technical debt and have an understanding of possible refactorings.	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books; Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; and Coplien and Bjørnvig, "Lean Architecture", Wiley, 2010. Additional reading material will be announced in the lectures.	
Assessment	Grades will be based on lab assignment including essay writing, coding, (video) presentation, peer reviewing, participation. The final grade of the course consists of the following components: Essays (weighting 60%) Coding (weighting 20%) Presentation (weighting 20%) Final grade calculation: 0.6 * Essays + 0.2 * Coding + 0.2 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Essays: re-submit deliverable Coding: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Course web site: https://se.ewi.tudelft.nl/delftswa/index.html	
Co-Instructor	M. Finavaro Aniche	

IN4334	Analytics and Machine Learning for Software Engineering	5
Responsible Instructor	M. Izadi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	- Experience with programming is required - Experience with statistics/machine and specifically deep learning (Transformer-based language models) is nice to have - Experience with research methods is nice to have	
Course Contents	Software repositories archive valuable software engineering data, such as source code, execution traces, historical code changes, mailing lists, and bug reports. This data contains a wealth of information about a projects status and history. Advances in Machine Learning and AI technologies, as demonstrated by the successful application of Deep Neural Networks in various domains did not go unnoticed in the field of Software Engineering; researchers have applied machine learning to tackle different software engineering tasks. IN4334 is a seminar course that aims to give students a deep understanding of and a hands-on approach to how deep neural networks and NLP techniques are used to represent knowledge and solve existing SE problems in novel ways.	
Study Goals	At the end of the course, the students will be able to: - Understand current literature in the area of software analytics and machine learning for software engineering - Apply software analytics techniques to extract actionable software engineering insights - Apply machine learning / deep learning algorithms to solve software engineering tasks	
Education Method	The course is a seminar, which means that we will be studying the literature in the area of software analytics and machine learning for software engineering. The course consists of the following education methods: - Self-study and presentation of papers - Development of software analytics tools and machine learning models to solve known or new problems To finish the course, students (in groups) will have to: - Study at least 10 research papers on ML4SE, which will be discussed during the lectures (reading and discussion) - Present a research paper and lead the discussion as a group (presentation) - Replicate existing work or propose new work as a group (project) The course also features guest lectures from top researchers in the area.	
Literature and Study Materials	Research papers are the main literature used in this course. We will share the resources with the students once the course starts.	
Prerequisites	The following are not strict prerequisites, however, if you have followed these courses you will be better prepared for the course: * CS4240 - Deep Learning * CS4360 - Natural Language Processing * IN4325 - Information Retrieval	
Assessment	The final grade of the course consists of the following components: Project results in the form of a research paper (weighting 80%) Final presentation/discussion of results (weighting 20%) One Research paper presentation during the course (Pass/Fail) Mid-term project report (Pass/Fail) Peer feedback on project reports (Pass/Fail) Final grade calculation: (P/F * One Research paper presentation + Mid-term project report + Peer feedback on project reports) + 0.8 * Project results in the form of a research paper + 0.2 * Final presentation/discussion A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project results in the form of a research paper: re-submit deliverable Final presentation/discussion of results: resit Mid-term project report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Each student who wants to take part in this course is *required* to: - Register/enroll on Brightspace before the start of the course - Participate in the first lecture of the course	
Tags	Project Research Methods Software Software Engineering	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Web Information Systems 2023	
Introduction 1	The WIS research group concentrates its research on engineering and science of the Web. The research specifically considers the role of Web data in the engineering of Web-based information systems. The group's research is aimed at improving the understanding of people's actions, interests, motivations, and behaviors on the Web, and subsequently leveraging that knowledge to build Web applications that are semantic, personalized and adaptive. This includes topics in harvesting, integrating, transforming, analyzing, and retrieving Web data, with focus on the special properties of Web data. A large portion of Web data is human-made, e.g. in social networks or Twitter streams, and this brings scientific challenges in how to effectively attribute meaning to Web data. The size of the Web brings challenges in how to efficiently store, index, and analyze data at Web scale. WIS researchers and students strive to advance the state-of-the-art in relevant disciplines like user modeling, Web science, information retrieval, Web engineering, Web data management, and crowdsourcing. A good example of a recent thesis is Catalin Stanculescus thesis with the title "Driving engagement and online social behavior of employees in an enterprise environment". This thesis contributes to the studies on enterprise gamification with a study performed at a large multinational enterprise. The student designed and implemented a modular and extensible framework for studying gamification and instantiated it as a Q&A game combined with news sharing and social connections capabilities. The framework was used to test the effectiveness of several game mechanics for promoting several types of desirable behavior concerning news sharing and knowledge acquisition.

CS4145	Crowd Computing	5
Responsible Instructor	U.K. Gadiraju	
Instructor	A. Bozzon	
Instructor	J. Yang	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic knowledge of artificial intelligence and/or human computer interaction is advised. Proficiency in at least one programming language.	
Course Contents	Crowd Computing is an emerging field that sits at the intersection of computer science and data science. Crowd computing studies how large groups of people can solve complex tasks that are currently beyond the capabilities of artificial intelligence algorithms, and that cannot be solved by a single person alone. It involves the algorithmic engagement and coordination of people by means of Web-enabled platforms. These complex tasks are mainly focused on the creation, enrichment, and interpretation of data, making crowd computing a building block of data science. Examples of such tasks include the coordinated creation of data about real world events when electronic sensors are not available; the annotation of existing data sets to create ground truth data for the training of machine learning algorithms; and the analysis and interpretation of Web data to spot identify inappropriate content (e.g.,hate speech, or fake news). Crowd computing is an essential tool for any data-driven company: from Facebook to Microsoft, from Google to IBM, from Spotify to Pandora, all major companies employ crowd computing to fulfil their data needs, both by involving employees, and by reaching out to anonymous crowds through online marketplaces like Amazon Mechanical Turk or Appen. Crowd computing methods therefore play an important role in the design, development and evaluation of a variety of products, services, and systems in a variety of domains. The objective of the Crowd Computing course is to introduce the scientific and technical underpinnings of crowd computing, and to investigate how it can be used for computer science applications (e.g., information retrieval, machine learning, next-generation interfaces, and data mining) and for real world applications (e.g., cultural heritage preservation, online knowledge creation, smart cities, etc.) The course is designed around one key challenge, the creation and consumption of (high quality) data, and will be organized around three themes: 1) Establishing data needs; 2) Fulfilling data needs with crowd computing; and 3) Evaluating the quality of the retrieved data with respect to the original data need. Covered topics include: 1) Establishing Data Needs: - Requirement Elicitation - Requirement Analysis - User Modelling Properties 2) Fulfilling Data Needs with Crowd Computation: - Systems for/with collective intelligence (e.g., recommendation, semiautonomous systems, citizen science, crowdsourcing, and human computation systems) - Multi-modal Interaction (e.g., conversational systems) - Human Computation (e.g., worker modelling, task modelling, incentives, task assignment, recruitment) - Games with a purpose - Algorithms for Crowd Computing - Computational Methods for User Modelling - Interfaces for Crowd Computing Systems 3) Evaluating Retrieved Data: - Expert Evaluation - User Evaluation - Explanation of the output of Crowd Computing Systems 4) Study of Application Domains When applicable, the course will also feature invited lectures from selected academics and professionals in the field. Since instructors of this course are also directing the Design@Scale Delft AI lab, students of this course will have the opportunity to engage with cutting-edge research projects relevant to this lab. This Crowd Computing course is an elective for students following the Data Science and Technology Track and the Software Technology Track. It adds to the master education offer by addressing topics that are complementary to courses like IN4325 Information Retrieval, IN4252 Web Science & Engineering, CS4065 Multimedia Search and Recommendation, and IN4010 Artificial Intelligence Techniques.	
Study Goals	After this course, students will be able to: - Identify the requirements for a Crowd Computing system [LO1] - Design and develop Crowd Computing systems. Support and defend the relevance and correctness of his/her choices [LO2] - Describe and compare several Crowd Computing techniques. [LO3] - Describe and compare design decisions in the context of Crowd Computing interaction paradigms [LO4] - Determine which Crowd Computing technique(s) is most appropriate for being used in a certain problem domain [LO5] - Apply the appropriate Crowd Computing technique to an application domain and evaluate the obtained results. [LO6] - Analyse the performance of a Crowd Computing system by applying the proper evaluation measures. [LO7]	
Education Method	This course consists of 16 2-hour lectures. Each week, a 30-minute assignment tests the knowledge acquired on the discussed topics. Starting from Week 1, students form groups and work on a project, to be presented in week 9. Students are expected to work 6 hours per week (each) on the project assignment. Expected workload is 32 hours for attending lectures, 24 hours of reading study material and preparing lectures, 55 hours for weekly assignments and group assignment, 24 hours for preparing final survey, and 5 hours for exam and plenary presentations (total 140 hours).	
Literature and Study Materials	Books: - Human Computation. Author(s): Edith Law and Luis von Ahn. Synthesis Lectures on Artificial Intelligence and Machine	

	Learning, June 2011, Vol. 5, No. 3. http://www.morganclaypool.com/doi/abs/10.2200/S00371ED1V01Y201107AIM013 - A. Marcus and A. Parameswaran. Crowdsourced Data Management: Industry and Academic Perspectives. Foundations and TrendsR in Databases, vol. 6, no. 1-2, pp. 1161, 2013. DOI: 10.1561/1900000044. https://people.eecs.berkeley.edu/~adityagp/papers/crowd-book.pdf - An Introduction to Hybrid Human-Machine Information Systems. Demartini, G., Difallah, D.E., Gadiraju, U. and Catasta, M., 2017. Foundations and Trends in Web Science, 7(1), pp.1-87. https://edu.nl/np4th Slides: available on Brightspace Articles: available on Brightspace Recommended reading: - Interaction Design: Beyond Human-Computer Interaction (4th Ed, 2015). Authors: Jenny Preece, Helen Sharp, Yvonne Rogers
Assessment	The final grade of the course consists of the following components: Weekly Individual assignment (weighting 15%) Group Assignment (weighting 55%). The group assignment is performed collectively, but graded individually. Final Individual Assignment (weighting 30%) Final grade calculation: 0.15 * Weekly Individual assignment + 0.55 * Group Assignment + 0.3 * Final Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly Individual assignment: re-submit deliverable Group Assignment: re-submit deliverable Final Individual Assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Tags	Algorithmics Artificial intelligence Design Programming Research Methods Software Technology

CS4225	Educational Technologies	5
Responsible Instructor	Prof. M.M. Specht	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	* Theories of Human Information Processing and Learning * Learning Management Systems * Learning Analytics * Personalisation and Adaptive Educational Systems * Mobile and Seamless Learning Technologies * Artificial Intelligence in Education * Realtime Learning Technologies * Project Design * Project Implementation	
Study Goals	The course will enable you to classify, understand, design and implement the core functionalities and systems for supporting human learning processes. As well current practices implemented as also approaches for technology enhanced learning currently researched will be presented. You will learn how educational technologies provide human learning process support, implement guidance and recommendation, create personalised learning support, as also give real-time feedback and support reflection of learners. In the final project you will identify a problem, design a solution based on the presented approaches and implement your own educational technology solution.	
Education Method	Lectures, weekly assignments and quiz questions, final project	
Assessment	The final grade of the course consists of the following components: Weekly assignments (weighting 30%) Final project (weighting 70%) Final grade calculation: 0.3 * weekly assignments + 0.7 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4360	Natural Language Processing	5
Responsible Instructor	J. Yang	
Instructor	Dr. C. Lofi	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Proficiency in at least one programming language. Knowledge of basic algebra. Knowledge of Web Information Systems and Machine Learning can be helpful.	
Course Contents	Natural Language Processing (NLP) concerns the interaction between computers and human language, in particular how to build systems to process and analyze large amounts of human language data. It has become an important technology in many parts of computer science, facilitating key applications of large-scale text understanding, automatic question answering, knowledge base construction, information retrieval, text or speech-driven dialogue agents, translation, text summarization, etc. This course aims to introduce the principles and techniques for Natural Language Processing. Through the lectures, assignments, and the group project, students will gain an understanding of classic and modern techniques for language understanding and generation and learn the skills to design, develop, and evaluate their own Natural Language Processing systems and applications. Covered topics include: = Lexical, Syntactic and semantic analysis = Language models = Word, and subword embedding techniques = Attention mechanism and Transformers = Transfer Learning = Knowledge-driven language processing = NLP applications (e.g., QA, summarization, machine translation) = Responsible NLP (explainability, fairness) = Evaluation of NLP systems	
Study Goals	At the completion of this course, students will be able to: [LO1] Recognize typical NLP tasks and components of an NLP system [LO2] Describe lexical analysis techniques such as language modeling and smoothing [LO3] Describe syntactic and semantic analysis techniques [LO4] Describe and compare different token embedding techniques, common neural architectures (including attention mechanisms) for text processing and their implementation in neural language models (e.g., GPT-3 and BERT). [LO5] Describe information extraction techniques and its application to NLP tasks [LO6] Apply NLP techniques to information retrieval, question answering, text summarization [LO7] Explain the implications of data-driven models in terms of transparency, accountability, and fairness [LO8] Design, develop, and evaluate NLP systems, possibly using neural methods with semantic enrichment and user adaptation functionalities. Support and defend the relevance and correctness of the design choices.	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 65 hours for the group project and 30 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books, authentic blog posts - all resources are available on Brightspace.	
Books	Dan Jurafsky and James H. Martin. 2014. Speech and language processing. Pearson. Yoav Goldberg. 2015. A Primer on Neural Network Models for Natural Language Processing. Steven Bird, Ewan Klein, and Edward Loper. 2009. Natural language processing with Python: analyzing text with the natural language toolkit. " O'Reilly Media, Inc.".	
Assessment	The final grade of the course consists of the following components: Weekly individual assignment (weighting 20%) Group project (weighting 80%). The group assignment is performed collectively, but graded individually. Final grade calculation: 0.2 * Weekly individual assignment + 0.8 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly individual assignment: re-submit deliverable Group project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4252	Web Science & Engineering	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	J. Yang	
Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The expected entry level is (equivalent to) standard bachelor-level computer science.	
Course Contents	The main subject of the course is the Web, and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g. Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan / research paper draft in collaboration with others in a small group to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies, and verbally present and defend this plan / draft.	
Education Method	The education methods include: - Lectures, before which and after which students study material by themselves, to get an understanding of the relevant material; - Small assignments and hands-on exercises, to apply the understanding of relevant material; - One large assignment, with a number of feedback moments, to learn how to write a web science paper / research project plan. Lectures will not be in each week in the class period (1+2): in between lectures there is time reserved for studying before and after lectures, for small assignments and exercises, and for writing the large assignment paper. The writing of the large assignment paper / project plan happens throughout the class period (1+2) to enable frequent feedback.	
Literature and Study Materials	Will be provided in class, depending on the topics chosen for the assignments and final paper.	
Assessment	Assessments are based on two components: the "small assignments" (accompanying the lectures) which provide a total of 20% of the grade (these are a mix of literature and programming homework), and the "final paper" which provides 80% of the final grade. To pass, each of the components ("small assignments" and "final paper") need to be graded with a minimum of 5.0. The final grade must be at least a 5.8. If these criteria are not met, the "small assignments" and the "final paper" can be repaired provided that the to-be-repaired component was at least graded with a 4 in accordance with OER/TER article 17. This repair is not a resit, and is limited to a maximum of 6.0 as a result. Components that have been graded with less than a 4.0 cannot be repaired. If the course is still not passed after the repair or cannot be repaired, all components need to be redone if the course is retaken in a later year. Both the small assignments and the final paper are group works of groups of preferably 2 members.	
Special Information	Students are asked to register/enrol on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	The expected workload is 5ects and that is principally distributed uniformly over the two quarters.	

IN4325	Information Retrieval	5
Responsible Instructor	J. Yang	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language. Knowledge of Web information systems and software engineering can be helpful.	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. The World Wide Web is now the primary source of information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, analysed, and exploited so that each user can quickly find information that is both relevant and comprehensive for their needs. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation of Web and Enterprise search engines. Covered topics include: = Information Retrieval Models; = Indexing Techniques; = Web Search; = Information Seeking Paradigms; = Evaluation of information retrieval systems;	
Study Goals	At the completion of this course, students will be able to: = Describe the different information retrieval models, and compare their strenghts and weaknesses. [Learning Objective 1] = Describe and implement different indexing techniques. [Learning Objective 2] = Describe and analyze querying techniques with respect to their most suited application domains. [Learning Objective 3] = Analyse the effectiveness of an information retrieval system through proper use of evaluation metrics. [Learning Objective 4] = Design and implement (Web) Information Retrieval systems, possibly using advanced social and semantic search functionalities. Support and defend the relevance and correctness the choices with regards to the adopted information retrieval model, indexing technique, and querying technique. [Learning Objective 5]	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 80 hours for the group project and 15 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books - all resources are available on Brightspace.	
Books	Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. Introduction to Information Retrieval. Cambridge University Press, New York, NY, USA. Stefan Büttcher, Charles Clarke, and Gordon V. Cormack. 2010. Information Retrieval: Implementing and Evaluating Search Engines. The MIT Press.	
Assessment	The final grade of the course consists of the following components: Individual assignment (weighting 10%) Weekly individual assignment (weighting 20%) Group project (weighting 70%) Final grade calculation: 0.1 * Individual assignment + 0.2 * Weekly individual assignment + 0.7 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual assignment: re-submit deliverable Weekly individual assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4326	Seminar Web Information Systems	5
Responsible Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Standard bachelor-level computer science or equivalent.	
Course Contents	In this course we discuss recent developments in the area of web information systems. We select topics to discuss from the areas of: - web technology (e.g. web engineering, hypertext, adaptive web), - web data management (e.g. web data interoperability, system and data integration), - web data and semantics (e.g. ontologies, semantic web, metadata), - web data analytics (e.g. user modeling, web personalization, web information filtering and retrieval), - social web (e.g. social web data analytics, social networking, human computing), - web science (e.g. crowdsourcing, trust, data science). We discuss this content while learning about the role of scientific communication and about the scientific methodologies and approaches for conducting research in the area. In this seminar the students will have to prepare and give scientific presentations on the basis of research papers about selected topics - the topics are selected in the first session together with the students. The students will also have to attend the presentations and participate in discussions on the presented papers. In addition, students will have to write a short survey about a topic in the area of web information systems of their choice.	
Study Goals	- to expose the student to current developments in research on web information systems and be aware of the methodologies and approaches to conduct research in the area; - to familiarise the student with reading, presenting and discussing scientific literature in the area and be aware of the most important academic journals and conferences in the area (and their review processes); - to help the student in reading and writing scientific papers and choosing a topic for her/his thesis in the area.	
Education Method	Student seminar.	
Literature and Study Materials	Is provided in the seminar, depending on the chosen subjects.	
Assessment	The final grade of the course consists of the following components: Quality of presentation of the scientific paper studied (weighting 15%) Participation in the seminar discussions (weighting 10%) Quality of paper written (weighting 75%) Final grade calculation: 0.15 * Quality of presentation of the scientific paper studied + 0.1 * Participation in the seminar discussions + 0.75 * Quality of paper written A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Quality of presentation of the scientific paper studied: resit Quality of paper written: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Students are asked to register/enrol on Brightspace beforehand. Students are also asked to be present and active in the first seminar session, to facilitate the proper planning of the seminar.	
Remarks	The expected workload of 5ects is distributed uniformly over the quarter. The seminar asks for active participation and therefore can only be completed as part of the first quarter edition; there is no re-sit.	
Tags	Artificial intelligence Databases	
Maximum number of participants	This course has a maximum capacity of 50 students. Students of 1) Web Information Systems group and 2) EEMCS have priority for other students.	

IN4331	Web-scale Data Management	5
Responsible Instructor	Dr. A. Katsifodimos	
Instructor	Dr. C. Lofi	
Instructor	Dr. R. Hai	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor level courses in database management systems and operating systems. A prior course in distributed systems or middleware would be helpful but is not required. Programming skills are important for the final assignment.	
Course Contents	This course addresses the challenges of Data Management at Web-scale. Especially, it covers the need for large-scale distributed data storage systems. The lecture therefore introduces step-by-step increasingly complex distributed storage systems, leading up to modern implementations of different NoSQL data storage systems. The challenges arising from such systems are presented and discussed, especially focusing on the CAP theorem and the resulting trade-offs with respect to data models, transactional power, query expressivity, and replication consistency. These discussions lead to different variants of NoSQL database systems, like Key-Value Stores, Document Stores, Wide-Columnar stores, and Graph Databases. The advantages, disadvantages, and general properties of these systems are discusses in more detail. There is special focus on distributed transactions and consistency guarantees of different data management systems and methods.	
Study Goals	At the end of this course the student can - assess the nature of a given storage problem, and can select a suitable technology for solving it - understands the different data models encountered in Web Data Management, and their impact on modelling and querying - understands the issues arising from distributing and replicating data, especially with respect to the CAP theorem -understands the trade-offs which can be chosen within the design space of the CAP theorem -categorize and explain modern NoSQL databases within the framework of the previously mentioned trade-offs	
Education Method	Lectures and assignments	
Literature and Study Materials	Course slides and Lecture Videos	
Books	Literature mentioned in the lecture, mostly research papers.	
Assessment	There is a group assignment (80%) and an individual exam (20%). In order to pass the class, a student has to have a grade of at least 5.0 for the group assignment as well as the individual exam, with a weighted average of 5.8 overall. The exam can be resit in the following academic quarter. If the student achieves a 4.0 - 6.0 in the group assignment, a repair opportunity can also be given (e.g., extra work, corrections on the group assignment, etc.).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

QCE/ Network Architectures and Services 2023

EE4396	Mobile Networks	5
Responsible Instructor	Dr. R. Litjens	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Not formally required, but recommended to have the basic knowledge taught in 'Fundamentals of Wireless Communications' (ET4358). If you didn't take this course, key aspects will be repeated during the class. In case you are lacking a comfortable background, talk to me in the first week of class and I'll help you catch up.	
Course Contents	The course addresses the technology and management of contemporary and future mobile cellular communication networks. With a focus on radio access, as it dominates both the performance and cost aspects of mobile networking, the course addresses 5G, 4G (LTE/LTE-A), 3G (UMTS/HSPA) and 2G (GSM/GPRS/EDGE) technologies, all of which are operational network technologies. An outlook to 6G will also be given. Fundamental aspects of cellular networking are explained and discussed, including e.g. the intrinsic tradeoffs between coverage, capacity and quality and key aspects that influence performance (traffic, propagation, technology, deployment, spectrum. Cellular network planning, dimensioning and performance analysis are treated, as well as radio resource management mechanisms (adaptive beamforming, channel-aware single/multi-user scheduling, handover control, link adaptation, admission, congestion control,) and the current trend towards (potentially AI/ML-based) self-management. Cutting-edge radio features including single/multi-user massive MIMO beamforming, which is at the heart of 5G, are also covered. Considering the on-going technological developments in this exciting domain, the course material is updated every year. The principal lecturer has a 25+-year track record of research and consultancy in the field. The course is enriched with notes about actual deployments, historical anecdotes, popular misconceptions, fascinating factoids, illustrative demonstrations and (potentially) guest appearances.	
Study Goals	The objective of the course is to provide the student with a thorough understanding of contemporary mobile cellular networking technologies, their fundamental properties and tradeoffs, as well as the key challenges and approaches related to network dimensioning, planning and (self-)optimization towards efficient provisioning of coverage, capacity and quality.	
Education Method	Lectures, demonstrations, in-class discussions, assignment(s).	
Literature and Study Materials	Primarily: the lecture slides! For the enthusiastic student, supplementary background materials are recommended in the form of e.g. (white/scientific) papers and book chapters. A number of books on wireless/mobile communications in general or about 2/3/4/5G technologies specifically are recommended as background material; specifics will be given in class / in the lecture slides. As said, this is merely recommended for the most eager students. In principle, the lecture slides cover all relevant content.	
Assessment	Closed-book exam, either written or oral (to be determined). And one (may be two) assignments. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	

ET4034	Telecom Business Architectures and Models	4
Responsible Instructor	Dr. E.F.M. van Boven	
Responsible for assignments	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	none	
Summary	Edgar van Boven, (KPN, TUD EEMCS)-- Past, present & future of communications -- Assignment 1 Eric Smeitink (KPN, TUD EEMCS)--Strategy, Spectrum & 5G Ton van der Knaap, (Stedin)-- What, why and how of business architecture; experiences from a grid operator and a telecom operator compared Samuel Pronk, (Krauthammer)-- Real Options-- Assignment 2 Ramin Hekmat, (KPN)-- Security and privacy in contemporary communication networks-- Assignment 3 Wouter de Vries (KPN)-- Artificial Intelligent Processes for People Tim Daeleman, (Prodapt Consulting)and Erik Lemmens (KPN)-- Digital Telecom Platforms Jose Almodovar, (TNO)--Standardization of Mobile Communications Frank Mertz (KPN)--5G Networks Developments and Opportunities Remco Helwerda (KPN) -- Digitalization Marco van der Pal, (Cap Gemini)--The IoT landscape architecture Ruud van de Bovenkamp (KPN)-- Fiber-to-the-Home Access Networks -- Assignment 4 Mark Dirksen (KPN)-- Business Continuity Management Pieter Veenstra, (Titanium)-- Service Architecture and 5G Roaming Security Edgar van Boven (KPN, TUD EEMCS)-- Value Case pitch practicing -- Assignment 5	
Course Contents	The essence of the course The ET4034 course Telecom, Architectures and Business Models was designed in 2001 and scheduled since 2002. The aim of the course is giving students from any faculty of the Delft University of Technology the opportunity to learn about a large variety of aspects of the telecom domain, not only its technologies, but also the diverse and challenging daily practice of organisations active in the communications sector. The course is provided by a team of 15 lecturers interactively sharing their knowledge about communications infrastructure in the context of overarching societal trends, service & network architecture, long term developments and new technology evolving from standardization initiatives. As a consequence of continuous innovation and societal developments, the course content is being updated each year including a well-considered choice of central themes that enhance and connect the course lectures. In 2022, the themes Internet of Things, Artificial Intelligence and 5G were highlighted, to be updated in 2023.	
Course Contents Continuation	EE5010 Internships or ET4399 projects in telecom related industries.	
Study Goals	Overview and understanding the Communications Sector, Services, Telecom infrastructures and Value Cases serving society	
Education Method	Online lectures, one powerpoint pitch presentation given by student teams and four off-line written assignments in teams of 2 or 3 students	
Literature and Study Materials	See information on Brightspace.tudelft.nl	
Reader	Course lectures, background information and video recordings available on ET4034 Brightspace	
Prerequisites	Finalised BSc as ET4034 is meant for Master students. PhD students are also welcome and can obtain credits when successfully passing the course assignments via the Graduate School.	
Assessment	The examination of the course Together forming the basis for the examination, the ET4034 course contains five assignments of which four written documents and one opportunity to strengthen ones pitching skills presenting a self-developed Value Case to a live audience. Students are invited to work in couples or trio's on the five assignments. The four written assignments each have a weight factor 1 and the Value Case pitch presentation has a weight factor 0,25 in the end mark of the ET4034 course. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	Not applicable, the course assignments are four documents written offline together with a course partner. The Value Case powerpoint presentation is a pitch that takes circa 7 minutes.	
Maximum number of participants	No limitation	
Self test	Not applicable	
maximum aantal deelnemers	No limitation	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Literature and Study Materials	We follow the book Performance Analysis of Complex Networks and Systems, by P. Van Mieghem, Cambridge University Press (2014). See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Written and closed book. A formularium is provided that can be consulted at the examination. Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Special Programmes 2023	
Introduction 1	There are three Special programmes: Bioinformatics (BI) Information Architecture (AI) Cyber Security

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Information Architecture 2023

Introduction 1

Compulsory Courses CS-IA:
IN4252 Web Science & Engineering, 5EC
IN4325 Information Retrieval, 5EC
IN4331 Web-scale Data Management, 5EC
SEN1622 I&C Services Design, 5EC
SEN1141 Managing Multi Actor Decision Making, 5EC
SEN1611 I&C Architecture Design, 5EC
SEN1121 Complex Systems engineering, 5EC

And in addition to said compulsory Information Architecture courses:

- Students of the Data Science & Technology track need to take 3 additional common core courses.
- Students of the Software Technology track need to take 4 additional common core courses.
- Students of the Artificial Technology track need to take 4 additional common core courses.

Thesis project in Information Architecture (45 EC)
IEP approved by Information Architecture Coordinator

IN4252	Web Science & Engineering	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	J. Yang	
Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The expected entry level is (equivalent to) standard bachelor-level computer science.	
Course Contents	The main subject of the course is the Web, and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g. Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan / research paper draft in collaboration with others in a small group to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies, and verbally present and defend this plan / draft.	
Education Method	The education methods include: - Lectures, before which and after which students study material by themselves, to get an understanding of the relevant material; - Small assignments and hands-on exercises, to apply the understanding of relevant material; - One large assignment, with a number of feedback moments, to learn how to write a web science paper / research project plan. Lectures will not be in each week in the class period (1+2): in between lectures there is time reserved for studying before and after lectures, for small assignments and exercises, and for writing the large assignment paper. The writing of the large assignment paper / project plan happens throughout the class period (1+2) to enable frequent feedback.	
Literature and Study Materials	Will be provided in class, depending on the topics chosen for the assignments and final paper.	
Assessment	Assessments are based on two components: the "small assignments" (accompanying the lectures) which provide a total of 20% of the grade (these are a mix of literature and programming homework), and the "final paper" which provides 80% of the final grade. To pass, each of the components ("small assignments" and "final paper") need to be graded with a minimum of 5.0. The final grade must be at least a 5.8. If these criteria are not met, the "small assignments" and the "final paper" can be repaired provided that the to-be-repaired component was at least graded with a 4 in accordance with OER/TER article 17. This repair is not a resit, and is limited to a maximum of 6.0 as a result. Components that have been graded with less than a 4.0 cannot be repaired. If the course is still not passed after the repair or cannot be repaired, all components need to be redone if the course is retaken in a later year. Both the small assignments and the final paper are group works of groups of preferably 2 members.	
Special Information	Students are asked to register/enrol on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	The expected workload is 5ects and that is principally distributed uniformly over the two quarters.	

IN4325	Information Retrieval	5
Responsible Instructor	J. Yang	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language. Knowledge of Web information systems and software engineering can be helpful.	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. The World Wide Web is now the primary source of information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, analysed, and exploited so that each user can quickly find information that is both relevant and comprehensive for their needs. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation of Web and Enterprise search engines. Covered topics include: = Information Retrieval Models; = Indexing Techniques; = Web Search; = Information Seeking Paradigms; = Evaluation of information retrieval systems;	
Study Goals	At the completion of this course, students will be able to: = Describe the different information retrieval models, and compare their strenghts and weaknesses. [Learning Objective 1] = Describe and implement different indexing techniques. [Learning Objective 2] = Describe and analyze querying techniques with respect to their most suited application domains. [Learning Objective 3] = Analyse the effectiveness of an information retrieval system through proper use of evaluation metrics. [Learning Objective 4] = Design and implement (Web) Information Retrieval systems, possibly using advanced social and semantic search functionalities. Support and defend the relevance and correctness the choices with regards to the adopted information retrieval model, indexing technique, and querying technique. [Learning Objective 5]	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 80 hours for the group project and 15 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books - all resources are available on Brightspace.	
Books	Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. Introduction to Information Retrieval. Cambridge University Press, New York, NY, USA. Stefan Büttcher, Charles Clarke, and Gordon V. Cormack. 2010. Information Retrieval: Implementing and Evaluating Search Engines. The MIT Press.	
Assessment	The final grade of the course consists of the following components: Individual assignment (weighting 10%) Weekly individual assignment (weighting 20%) Group project (weighting 70%) Final grade calculation: 0.1 * Individual assignment + 0.2 * Weekly individual assignment + 0.7 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual assignment: re-submit deliverable Weekly individual assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4331	Web-scale Data Management	5
Responsible Instructor	Dr. A. Katsifodimos	
Instructor	Dr. C. Lofi	
Instructor	Dr. R. Hai	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor level courses in database management systems and operating systems. A prior course in distributed systems or middleware would be helpful but is not required. Programming skills are important for the final assignment.	
Course Contents	This course addresses the challenges of Data Management at Web-scale. Especially, it covers the need for large-scale distributed data storage systems. The lecture therefore introduces step-by-step increasingly complex distributed storage systems, leading up to modern implementations of different NoSQL data storage systems. The challenges arising from such systems are presented and discussed, especially focusing on the CAP theorem and the resulting trade-offs with respect to data models, transactional power, query expressivity, and replication consistency. These discussions lead to different variants of NoSQL database systems, like Key-Value Stores, Document Stores, Wide-Columnar stores, and Graph Databases. The advantages, disadvantages, and general properties of these systems are discusses in more detail. There is special focus on distributed transactions and consistency guarantees of different data management systems and methods.	
Study Goals	At the end of this course the student can - assess the nature of a given storage problem, and can select a suitable technology for solving it - understands the different data models encountered in Web Data Management, and their impact on modelling and querying - understands the issues arising from distributing and replicating data, especially with respect to the CAP theorem -understands the trade-offs which can be chosen within the design space of the CAP theorem -categorize and explain modern NoSQL databases within the framework of the previously mentioned trade-offs	
Education Method	Lectures and assignments	
Literature and Study Materials	Course slides and Lecture Videos	
Books	Literature mentioned in the lecture, mostly research papers.	
Assessment	There is a group assignment (80%) and an individual exam (20%). In order to pass the class, a student has to have a grade of at least 5.0 for the group assignment as well as the individual exam, with a weighted average of 5.8 overall. The exam can be resit in the following academic quarter. If the student achieves a 4.0 - 6.0 in the group assignment, a repair opportunity can also be given (e.g., extra work, corrections on the group assignment, etc.).	

SEN1121	Complex Systems Engineering	5
Module Manager	Prof.dr. F.M. Brazier	
Instructor	Y. Huang	
Instructor	Prof.dr. F.M. Brazier	
Instructor	Dr. P.H.G. van Langen	
Responsible for assignments	Prof.dr. F.M. Brazier	
Co-responsible for assignments	Y. Huang	
Contact Hours / Week x/x/x/x	4/0/0/0 and interactive lab sessions	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course introduces design thinking in complex sociotechnical systems in line with the core orientation of the CoSem MSc programme. Students learn about designing complex technological systems in multi-actor environments. Confronted with a challenge students explore the context, values and requirements of the different stakeholders involved (both directly and indirectly). Students learn to structure and coordinate a design process: creating, exploring and evaluating potential designs in relation to requirements given a mission statement. The course provides the foundation for further design-oriented courses. Students learn to - explore the context of a challenge in a multi-actor setting - understand stakeholder values and requirements - formulate a mission statement for a complex sociotechnical system - structure a design process, creating and evaluating potential design options - manage their design project Students participate in online interactive lectures and work together in groups (and with each other) on a specific challenge to put the knowledge into practice: learning by experience.	
Study Goals	After completion of the course, a student is able to - use and differentiate between concepts and terminology related to the design of complex sociotechnical systems - analyze challenges in complex sociotechnical systems in multi-actor environments - structure and coordinate a design process for a complex sociotechnical system - use and combine methods for requirements acquisition and analysis - use and combine approaches for conceptual design - use and compare methods for systems design and engineering - use and compare methods for project and risk management	
Education Method	Learning by participation: online interactive lectures (for which participation is essential), and labs on campus in which groups work together and with each other supported by TAs and teaching staff.	
Literature and Study Materials	The main content of this course is provided on the slides used during the lecture, together with papers, articles and book chapters as appropriate. These reading materials are made available on Brightspace.	
Assessment	The overall grade for this course consists of 2 sub grades: 1) Exam: The exam focuses both on the theoretical knowledge and the ability to apply and reflect on the theory taught. The digital exam is open book and contributes to 70% of the final grade.. 2) Assignments: During the labs, theory from the lectures is applied to a practical case. The assignments contribute to 30% of the final grade. Both parts must be completed with a passing grade (at least 5.75).	
Tags	Analysis Challenging Design Group Dynamics/Project Organisation Group work Modelling Projects Sustainability	

SEN1141	Managing Multi-actor Decision-making	5
Module Manager	Dr. M.L.C. de Bruijne	
Instructor	Prof.mr.dr. J.A. de Bruijn	
Instructor	Dr. M. Leijten	
Instructor	Dr.ir. E. Minkman	
Instructor	F. Hirschhorn Zonana	
Instructor	Dr. F.S. Gürses	
Instructor	B. Wagner	
Responsible for assignments	Dr. M.L.C. de Bruijne	
Co-responsible for assignments	B. Wagner	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Previous courses have focused on, first, diagnosing the complexity of socio-technical systems and identifying problems and, then, designing for solutions (which oftentimes have a large engineering component) to be introduced and their accompanying institutions in the midst of this complexity. This course on multi-actor decision-making goes one step further: (reflecting on) realizing change. We focus on the actors in the system. How do they behave, individually and collectively? What does the system look like from an actor perspective? Why does intentional change in socio-technical systems often appear so hard? What limits the designability of systems? By studying characteristics of sociotechnical systems and identifying the complexity of the challenges they pose for engineering (e.g. the increased importance of IT for design, the increased interdependencies between systems) as well as the specific actor characteristics (e.g. the global character of the manufacturers and supply chains, issues of plurality and inclusiveness) we identify key factors that affect intentional change (i.e. the wickedness of problems, public values, strategic behaviour of actors and dynamics of the system). How do these factors affect decision-making, design and change of complex engineering systems? How can we assess the consequences of this decision-making complexity and how does this affect the potential for interventions for (re)design in these systems in deliberate and responsible ways? In this course, we identify key characteristics which affect multi-actor decision making in socio-technical systems. Furthermore, we assess how and under which conditions process managerial tactics may help to manage multi-actor decision-making with its plural rationalities and dynamics of negotiation and design.	
Study Goals	At the end of this course, students will be able to explain why actors in networks behave as they do, and how their behaviour evolves describe different network structures and their practical implications assess the designability of institutions, systems, policies and technologies argue why certain process managerial tactics might work under which conditions create case-specific process managerial tactics to anticipate complex decision-making. assess the quality of process designs reflect on multi-actor decision-making in light of broader scientific and public policy trends and debates	
Education Method	Lectures Assignments (group and individual) Readings	
Assessment	Assessments group assignment analysing a multi-actor decision making process (1/2) individual assignments which enable students to reflect on prescribed literature (1/2)	

SEN1611	I&C Architecture Design	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	N. Bharosa	
Instructor	Dr. F. Kleiman	
Instructor	Dr.ir. J. De Stefani	
Responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Co-responsible for assignments	G.A. De Reuver	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	· Basic knowledge of databases, including Entity-Relationship Diagram (ERD) · Basic programming skills (such as JavaScript) · Basic knowledge of Unified Modelling Language (UML) and software engineering (agile development) · Basic understanding of multi-actor system and stakeholder analyses methods This prior knowledge are part of the linkage program.	
Course Contents	More and more data is available using various sources like databases, social media or the Internet of Things (IoT). Information sharing requires the integration of these systems. The ability to execute and process data is a key capability required by public and private organizations. The large variety of heterogeneous data, applications and business processes results in a complex and fragmented landscape that need to be dealt with when architecting and engineering new systems. Path dependencies and legacy block the easy integration, whereas at the same time new technologies appear that need to be integrated in the complex systems landscape. The purpose of this course is to teach the architectural design of innovative and large-scale ICT infrastructures and services in the light of the challenges imposed by the requirements from the systems physical, economic and social environment. Emphasis will be put on the concepts and role of ICT-architecture and modelling in order to properly design ICT solutions within a multi-actor context. For this purpose application and data integration technologies and information and system quality theories will be addressed.	
Study Goals	-The student is familiar with the state-of-the-art knowledge of ICT-architecting, design and governance within the field of large-scale ICT-systems within a multi-actor context -The student is able to describe basic concepts related to architecting and designing large and complex ICT-infrastructures and service systems within a multi-actor context. -The student should master architecture theories, methods and tools with the ability to combine and switch between them when dealing with complex ICT-problems. -The student is able to structure and analyze problems and identify dilemmas arising during the design process with regard to designing large ICT-systems within a multi-actor context. -The student is able to apply system engineering and architecture-based approaches, methods and tools to deal with problems with regard to designing large ICT-systems. -The student is able to report about the use of architectural concepts, methods and tools for translating business needs into ICT-designs within constellation of public and private actors.	
Education Method	· Lectures. The lectures are aimed at providing an overview of the knowledge of this course. During the lectures exercises are given to practice to internalize the knowledge. · Guest lectures (obliged). The guest lectures are aimed at giving an overview of the state-of-the art in practice and provide insight into practical challenges when using ICT in a multi-actor domain. · Exercises During some of the lectures assignments are given to practice what is learned. · Assignments. Various assignments are given which together create the final report. This includes developing various architectural models. · Presentation. Students should present their results during a lectures. · Report. The results should be reported in a concise report including the argumentation of design choices, the resulting architecture and an evaluation	
Computer Use	Architecture modelling in Archimate tool (http://www.archimatetool.com/) BPMN modelling using Eunomia process builder (http://www.eunomia-process.com/), Bizagi (http://www.bizagi.com/) or LucidChart (https://www.lucidchart.com/)	
Literature and Study Materials	slides book (open access e-book) Nitesh Bharosa, Remco van Wijk, Niels de Winne, Marijn F.W.H.A. Janssen (2015). Challenging the chain. Governing the Automated Exchange and Processing of Business Information. ISBN: 978-1-61499-497-8 (open access). This book covers a range from governance to technical issues and integrates them. Open access book: http://ebooks.iospress.nl/book/challenging-the-chain-governing-the-automated-exchange-and-processing-of-business-information Reader providing an outline of the area and focus on some aspects which is complemented by papers which goes in more depth.	
Assessment	Group assignment is 60% of the grade. The students work together in groups of 4-6 persons. The assignment is assessed by a report which should adhere to the APA reference standards. The report cannot be updated after handing it in. Groups that will not pass will be given the opportunity to write a new report. The exam is 40% of the grade. The exam will be digital and held in a computer room at the university. Each grade should be 5.75 or higher If one part is not passed and the other part is then the other part will remain valid for the next year.	
Elective	Yes	
Tags	Information & Communication Modelling Prototyping Technology	
Targetgroup	SEN I&C track Information Architecture (IA) students Electives for MOT, EPA, Computer science students, and non-TBM students having the required knowledge in this field are welcome	

SEN1622	I&C Service Design	5
Module Manager	Dr. Y. Ding	
Instructor	Dr.ir. G.A. de Reuver	
Instructor	Dr. Y. Ding	
Co-responsible for assignments	Dr.ir. G.A. de Reuver	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Mobile apps, Internet-of-Things, cloud edge architectures and sensors are enabling a range of new I&C services. These services are typically offered in ecosystems of interdependent actors. However, designing services that add value for users as well as ecosystem stakeholders is challenging. In this course, you learn how to design practical ICT services and develop a valorization plan. The course contains two main parts. In the first part, it is all about designing innovative services that meet the needs of users. You will design service mockups and interview users to test your ideas. In the second part, it is about bringing that service idea to reality. You will design a valorization plan that specifies how to deal with external technologies, stakeholders and revenues. The course is practical and hands-on: you will create, test and plan your service ideas. At the same time, all is theory-informed and you will learn how to support design choices using relevant kernel and design theories.	
Study Goals	After the course, you are able to: - Analyze practical ICT services by applying concepts of design science research, action design research, agile development and service engineering - Evaluate practical requirements and make informed choices on supporting technologies and platforms, with specific attention to autonomous driving, medical IoT, AI, XR, and edge computing technologies - Evaluate an ICT-enabled service concept through semi-structured interviews - Design and illustrate a value-adding service concept driven by ICT - Design a valorization plan that explicitly covers how and when to involve external stakeholders in a design process and integrate the ICT service with value network	
Education Method	Lectures, design project, peer review, presentations, guest lecture.	
Assessment	Group essay = 3 EC; Group mockup = 1.25 EC, Individual reflection document = 0.75 EC All three marks should be >5.75. Participation in peer reviewing, guest lecture and presentations is mandatory.	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Bioinformatics 2023

Introduction 1

General Setup 2022-2023
I. DST common core courses (>20EC: choose 4 out 7) AIT common core courses (>20EC: choose 4 out of 8) or ST Common Core courses (>25EC: choose 5 out of 10)
II. BI core courses, 25EC
III. BI specialization courses, >15EC
IV. Literature study, 10EC
V. Free electives, >10EC
VI. Thesis project, 45EC
The IEP must be approved by the Bioinformatics MSc Coordinator

I DST common core courses (>20EC: choose 4 out 7) AIT common core courses (>20EC: choose 4 out of 8) or ST Common Core courses (>25EC: choose 5 out of 10)

II Compulsory Bioinformatics courses (25EC)
1. CS4250 Selected topics in molecular biology, 5EC
2. CS4255 Algorithms for sequence-based bioinformatics, 5EC
3. CS4176 Algorithms for network-based bioinformatics, 5EC
4. CS4260 Machine learning in bioinformatics, 5EC
5. CS4329 Recent topics in bioinformatics, 5EC

III Specialization courses: choose >15EC

Bioinformatics specialization Courses Q1
1. CS4070 Multivariate Data Analysis, 5EC
2. EE4C06 Networking, 5EC
3. IN4049TU Introduction to High Performance Computing, 6EC
4. IN4252 Web Science & Engineering, 5EC
5. IN4344 Advanced Algorithms, 5EC
6. CS4375 Artificial Intelligence Techniques, 5EC
7. IN4307 Medical Visualization, 5EC

Bioinformatics specialization Courses Q2
1. IN4089 Data Visualization, 5EC
2. IN4150 Distributed Algorithms, 6EC
3. CS4220 Machine Learning 1, 5EC

Bioinformatics specialization Courses Q3
1. CS4240 Deep Learning, 5EC
2. CS4195 Modeling and Data Analysis in Complex Networks, 5EC
3. CS4230 Machine Learning 2, 5EC
4. IN4325 Information Retrieval, 5EC
5. IN4315 Software Architecture, 5EC

Bioinformatics specialization Courses Q4
1. CS4205 Evolutionary Algorithms, 5EC
2. IN4331 Web Data Management, 5EC
3. CS4290 Seminar Distributed Machine Learning Systems, 5EC
4. CS4245 Seminar Computer Vision by Deep Learning, 5EC

IV Literature study (10EC)
IN4306 Literature Survey, 10EC

V Free Electives (10EC)

VI Thesis project (45EC) with Bioinformatics group
IN5000 Final Project, 45EC

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

DST & AIT Common Core courses 2023

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Compulsory Bioinformatics courses 2023

CS4176	Algorithms for network-based bioinformatics	5
Responsible Instructor	Prof.dr.ir. M.J.T. Reinders	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The student is expected to have a basic knowledge of molecular biology, statistics and linear algebra. It is advisable to have followed CS4220 Machine Learning 1 (old code: IN4085 Pattern Recognition).	
Course Contents	Molecular biology is concerned with the study of the presence of and interactions between molecules, at the cellular and sub-cellular level. In bioinformatics and systems biology, algorithms and tools are developed to model these interactions, with various goals: predicting yet unobserved interactions, assigning functions to yet unknown molecules through their relations with known molecules; predicting certain phenotypes such as diseases; or just to build up biological knowledge in a structured way. Such interaction models are often best modelled as networks or graphs, which opens up the possibility of using a large number of readily available algorithms for inferring networks, performing simulations of biology, optimising paths or flows through networks, graph-based data integration and graph mining. Many of these algorithms can be applied (sometimes with slight alterations) to solve a particular biological problem, such as modeling transcriptional regulation or predicting protein interaction/complex formation, but also to derive systems behaviour by breaking down networks into modules or motifs with certain characteristics. In this course, we will first give a brief overview of molecular biology, the advent of high-throughput measurement techniques and large databases containing biological knowledge, and the importance of networks to model all this. We will highlight a number of peculiar features of biological networks. Next, a number of basic network models (linear, Boolean, Bayesian) will be discussed, as well as methods of inferring these from observed measurement data. Building on the network inference methods, a number of ways of integrating various data sources and databases to refine biological networks will be discussed, with specific attention to the use of sequence information to refine transcription regulation networks. Finally, we will give some examples of algorithms exploiting the networks found to learn about biology, specifically for inspecting protein interaction networks and for finding active subnetworks.	
Study Goals	After succesfully completing this course, a student is able to: list the basic elements of a living cell and their interactions, and describe how these can be measured; explain what type of mathematical model is applicable to what measurement(s), at what level(s), in a given systems biology problem; read and comment upon recent network-based computational biology literature; discuss the state-of-the-art in systems biology and integrative bioinformatics, and future challenges	
Education Method	The course consists of a mixture of lectures by the teachers and paper presentations by one or more of the students. Each paper presentation will be followed by a in-depth discussion. There will also be a practical session allowing students to get hands-on experience with network models.	
Literature and Study Materials	Slides, collection of papers and lab course manual (Brightspace).	
Assessment	The final grade of the course consists of the following components: Proposal for a research project (weighting 80%). Students are required to write a proposal for a research project in which they clearly state the biological problem to be solved, the necessary data, the computational approach as well as the innovative parts of the approach. Paper presentations/discussions (weighting 20%) Final grade calculation: 0.8 * Proposal for a research project + 0.2 * Paper presentations/discussions A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Proposal for a research project: re-submit deliverable Paper presentations/discussions: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	As students depend on each other (to present the material to the class), a commitment to follow the course through to the end is required.	

CS4250	Selected Topics in Molecular Biology	5
Responsible Instructor	T.E.P.M.F. Abeel	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course covers a broad range of essential topics in molecular biology necessary to start critically working with bioinformatics algorithms. The choice and implementation of algorithms critically depend on a proper understanding of the application domain. In this case the application domain is biology and in this course we review key biological concepts and topics that have impact on the implementation and selection of various computer algorithms and models. We use the following book: B. Alberts et al., 'Molecular Biology of the Cell', 6th edition Publication Date: November 18, 2014 ISBN-13: 978-0815344322 ISBN-10: 0815344325 A detailed overview of the actual reading material will be provided through the online learning platform Brightspace and will include material from Chapters 1,3,4,5,6,7 and 8. Extra material to aid in learning are available from the publisher's website.	
Study Goals	The goal of this course is to learn about the basic concepts in molecular biology required for bioinformaticians. At the end of this course: students will be able to reproduce, discuss and reflect on information about basic molecular biological processes, concepts and ideas. This includes: - cell structure, organization, information encoding and transfer - core concepts around DNA, chromosomes, genomes - DNA replication, repair and recombination - central dogma molecular, including replication, transcription, translation - regulation of the processes above	
Education Method	Self-study with weekly Q&A sessions.	
Books	B. Alberts et al., 'Molecular Biology of the Cell', 6th edition Publication Date: November 18, 2014 ISBN-13: 978-0815344322 ISBN-10: 0815344325	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 100%) Final grade calculation: 1 * Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4255	Algorithms for sequence-based Bioinformatics	5
Responsible Instructor	T.E.P.M.F. Abeel	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Bioinformatics analyses in genomics aim to compare large sets of genomes in order to understand and explain differences in traits of an organism. Contemporary methods are powered by fundamental algorithms and data structures, which are efficient and scale to large data sets. A thorough understanding of these algorithms and data structures is necessary for advanced users and developers in this area. In addition, understanding how comparative genomics is developing is important to shape your own research. In this course, we will cover genome analysis, variant analysis, and pangenomics. Core concepts, applications, and future trends will be discussed, with a focus on the algorithms and data structures underlying state-of-the-art methods.	
Study Goals	After having followed this course, the student has a good understanding of algorithms and data structures in genomics used for DNA sequence analysis. The student is able to implement algorithms in python, and can translate methods described in scientific literature into a working implementation.	
Education Method	The course is offered as a mix of lectures, exercises and a project	
Books	http://bioinformaticsalgorithms.com	
Assessment	The final grade of the course consists of the following components: Exercises (weighting 60%) Project report (weighting 40%) Final grade calculation: 0.6 * Exercises + 0.4 * Project report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Exercises: re-submit deliverable Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4260	Machine Learning in Bioinformatics	5
Responsible Instructor	Dr. J.S. de Pinho Gonçalves	
Contact Hours / Week x/x/x/x	0/0/4/0 Lectures + 0/0/4/0 lab	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Recommended for fundamental knowledge in machine learning and multivariate data analysis: CS4220 Machine Learning 1 CS4070 Multivariate Data Analysis Recommended for basic concepts of molecular biology: CS4250 Selected Topics in Molecular Biology Required: CSE1200/TI1106M Calculus (or similar) CSE1205/TI1206M Linear Algebra (or similar) CSE1210/TI2216M Probability Theory and Statistics (or similar) The project requires basic programming skills.	
Course Contents	Learning from patterns in molecular biology data plays an important role in diagnosing disease, discovering new targets for therapy, and more generally in answering biological questions that lead to an improved understanding of biological systems with relevance to human health and industrial areas including biotechnology and agriculture. This course focuses on methodology for the analysis of high-dimensional data in molecular biology, naturally addressing challenges that commonly arise in the field such as learning from unlabelled data or from small numbers of samples. The methodology is introduced in the context of meaningful applications in molecular biology with examples using real data. Covered topics will be (a selection of) the following: - Discrete probability models and statistical modeling: statistical models for experiments with categorical outcomes, goodness of fit, maximum likelihood and Bayesian estimation. - Mixture models: finite mixture models of normals and the EM algorithm, common infinite mixture models such as beta-binomial and gamma-Poisson, ECDF and bootstrapping. - Clustering: comparing observations, iterative partitioning, density-based clustering, hierarchical clustering, clustering validation, choosing the number of clusters. - Statistical hypothesis testing: p-values, single and multiple hypothesis testing, family-wise error rate, (local) false discovery rate. - Testing using high-throughput count data: multifactorial designs, linear models, analysis of variance, generalized linear models, robustness and outlier detection, shrinkage estimation. - Linear dimensionality reduction (PCA): preprocessing data for multivariate analysis, projecting onto lower dimensions, matrix decomposition (SVD), biplot representations, projecting additional variables for interpretation). - Multivariate methods for heterogenous data: orderings, gradients and latent variables (ordination); multidimensional scaling (MDS), robust (non-metric) MDS, batch effect removal, correspondence analysis, finding gradients and trajectories (local non-linear methods such as tSNE and UMAP), canonical correlation. - Supervised learning: discrimination, performance measures, curse of dimensionality, generalizability and model complexity, regularization and penalization, cross-validation, supervised learning methods (SVM, decision trees, ...), method hacking. - Design of high-throughput experiments and their analyses: types of variability (error, noise, bias), confounding, dependencies, batch effects, statistical power, mean-variance relationships and data transformations, workflow design, data representation, efficient computation.	
Study Goals	After successfully completing this course, the student should be able to: - Recognise, characterise, and interpret different kinds of high-throughput molecular biology data and their statistical properties. - Recognise, categorise and compare common statistical techniques and machine learning algorithms and the data analysis problems that they address. - Recognise typical research questions that can arise when analysing such kinds of data, reason about and select appropriate methodology to address them. - Understand, reason about, and discuss the different steps of a data analysis workflow: from experiment design to the interpretation of results. - Design a research plan, execute it, and write a scientific paper about it.	
Education Method	The course is run in a flipped classroom setting. Lectures: students take turns explaining the material from the different chapters of the course book. All students are required to read and prepare the course material before every lecture, so that they can contribute to the discussion of the material in the classroom. Project/labs: students will work in small groups on a research project throughout the course. Each group will present and discuss the status of the project weekly or bi-weekly, and deliver a written report at the end. There will be one or two lectures (90 min. each) and one project discussion lab (45 to 90 min.) per week. The frequency and duration of the sessions will depend on the number of students following the course. The exact schedule will be determined after the first lecture.	
Books	The material of the course follows the book: "Modern Statistics for Modern Biology" Authors: Susan Holmes, Stanford University, California Wolfgang Huber, European Molecular Biology Laboratory Published: February 2019 ISBN: 9781108705295 The complete book can be browsed online at: http://web.stanford.edu/class/bios221/book/. This website also contains the R code and data accompanying the examples in the book.	

Assessment	The final grade of the course consists of the following components: Book chapter presentation (weighting 25%) Participation (weighting 15%) Project report/paper (weighting 60%) Students are graded individually. Final grade calculation: 0.25 * Book chapter presentation + 0.15 * Participation + 0.6 * Project report/paper A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Book chapter presentation: re-submit deliverable Project report/paper: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Tags	Artificial intelligence Research Methods

CS4329	**Recent topics in bioinformatics**	**5**
Responsible Instructor	Prof.dr.ir. M.J.T. Reinders	
Instructor	Dr. J.S. de Pinho Gonçalves	
Instructor	T.E.P.M.F. Abeel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Bioinformatics is at the heart of many modern systems biology analyses, and encompasses the application of statistics and computer science to (large-scale) biomolecular datasets. In essence, bioinformatics is about smart ways of extracting knowledge from the enormous amounts of data that can be generated using modern measurement techniques. For instance, it plays an important role in finding the genetic origins of various diseases, such as cancer, diabetes or alzheimer.	
Study Goals	After successfully completing this course, the student is able to: explain several high-throughput data acquisition experiments, such as DNA/RNA sequencing, and discuss the benefits and limitations of these methods comprehend the statistical and computer science issues in analyzing high-throughput data discuss the basic systems biology approach, and the role of high-throughput measurements, gene selection and classification therein explain bioinformatics methods, algorithms and models to a non-expert audience implement or execute basic algorithms from descriptions provided in scientific literature read and comprehend a current scientific paper and reflect on the bioinformatics methods used in such paper	
Education Method	In this course we will study some key examples of bioinformatics analyses by reading a set of selected papers that present some significant biological conclusions.The course is run a flipped class-room students take turns guiding us through the selected papers. Teachers moderate the discussion and fill in any gaps in understanding.All students are required to read, and prepare the course material before every lecture to effectively participate. Each week there are one or two lectures, each of 90 minutes In each lecture one paper (the course material) will be discussed in detail. One or more students will present and explain the details of this paper. It is essential that you highlight the (bioinformatics) methodology of the paper. The schedule for this will be prepared in the first lecture. This presentation is graded. All students are expected to have read the paper and should have an active role in the discussion about the paper. Each student should be prepared to raise at least two critical remarks or questions during the discussion. The quantity and quality of participation of each student is graded. The student will perform a practical project in groups where they gain practical experience in state-of-the-art bioinformatics methods.	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 20%) Participation during discussions (weighting 20%) Reporting about unseen paper with oral presentation (weighting 20%) Final grade calculation: 0.2 * Presentations + 0.2 * Participation + 0.2 * Reporting about unseen paper with oral presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Reporting about unseen paper with oral presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). en circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Specialization courses 2023

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Bioinformatics specialization Courses Q1 2023

CS4070	Multivariate Data Analysis	5
Responsible Instructor	Dr. J. Söhl	
Responsible Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Required: * Introduction Probability Theory and Statistics: see for instance A Modern Introduction to Probability and Statistics Understanding Why and How Series: Springer Texts in Statistics Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover ISBN: 1-85233-896-2 * Basic calculus * Linear Algebra: matrix multiplication, the inverse of a matrix, the transpose of a matrix, least square solution. see: David C. Lay: Linear Algebra and Its Applications ISBN-10: 0321385179 ISBN-13: 9780321385178 ©2012 Pearson)	
Course Contents	PART I: Stochastic models will be developed on the basis of probability theory. Probability theory describes the behavior of certain phenomena in terms of how likely it is that certain values will occur. Central features of the models will be discussed are random variables, probability density functions, and the expected value operator. In describing random processes and signals, the correlation function and conditional probabilities play a central role. It addresses the following subjects: 1. Random variables. Matlab exercise on estimation of PDF, expected value and variance. 2. Refresher correlation. Calculating with correlation functions. 3. Random processes, correlation function, stationarity, wide sense stationarity, estimation of correlation function (Matlab exercise). 4. Random signal processing, power spectral density function, white noise. 5. AR processes, linear prediction: theory and computer exercise. 6. Markov chains. PART II: A course in advanced statistics about linear models, Bayesian inference, classification problems, Gaussian processes and Markov Chain Monte Carlo.	
Study Goals	PART I: 1. Probability Theory - Conditional) probabilities, the law of total probability, and Bayes rule. - Solve probability problems that require the use of axioms of probability. 2. Definition and Description of Random Variables and Processes PDF, PMF, CDF, Covariance, Correlation- Determine if a given PDF, PMF, CDF, variance, (auto/cross-)correlation(-function), (auto/cross-)covariance(-function), power spectral density complies with (theoretical and analytical) requirements. - Convert the description of a probabilistic problem into a probabilistic model using PDF, PMF, or CDF. 3. PDF/PMF and Expected Value Calculate the various forms of expected value of (combinations of) random variables and random processes - For a given (amplitude continuous/discrete and time continuous/discrete) probability model calculate the following probabilistic (marginal, joint and conditional) characterizations: PDF, PMF, CDF, probability of an event, expected value, variance, covariance, correlation, correlation coefficient, auto/crosscorrelation function, auto/crosscovariance function, (cross) power spectral density. - Calculate the PDF, PMF, expected value and variance of a derived random variable. 4. Properties of Random Processes - Independence, orthogonality, uncorrelated, whiteness, IID- Determine if random variables/processes have the following properties: independent, orthogonal, uncorrelated, white, Poisson, Gaussian, Bernoulli, Markov, IID, stationary, WSS, ergodic. - Calculate the expected value, variance, auto/crosscorrelation(function), auto/crosscovariance(function), power spectral density of a linear combination of random variables and of a linearly filtered (WSS, amplitude discrete/continuous, time discrete/continuous) random process. 5. Large NumbersCentral limit theorem, law of large numbers - Solve problems that require the use of the central limit theorem in an engineering context - Explain the law of the large numbers in an engineering context. 6. Statistical Estimators - Estimated mean, variance, and correlation function - Given a set of outcomes, sample functions or realizations, calculate estimators for expected value, variance, and (auto-)correlation function. 7. Application to Engineering Problems and Simulations - Select and translate a simple electrical engineering or computer science problem into mathematical probability model. The emphasis is on problems in signal and image processing, telecommunication, and media and knowledge technology. The class of probability models encompasses the following random variables/processes: Bernoulli, exponential, binomial, Poisson, Gaussian, uniform. - Justify and reflect on the approach taken in calculating or simulating (MatLab) the following probabilistic properties: PDF,	

	PMF, expected value, variance, autocorrelation function, autocovariance function. PART II: By the end of the course, you should be able to: - compute properties of frequentist and Bayesian estimators in the linear model - derive posterior distributions and their properties - apply hyperparameter optimization in Bayesian models - apply computational methods for posterior approximation - derive properties of frequentist and Bayesian classification methods - derive properties of MCMC methods for sampling from the posterior distributions - derive properties of Gaussian process regression
Education Method	PART I: Lectures, working groups (problem solving), laboratory work (a Matlab exercise) Workload is around 15 hours for attending lectures, 5 hours of reading study material and preparing lectures, 15 hours for the lab course, 20 hours for preparing the exam, 3 hours for the exam, and 8 hours for a final report (66 hours in total). PART II: Classes and weekly exercises.
Books	PART I: R.D. Yates and D.J. Goodman, "Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers", ISBN 0-471-17837-3, John Wiley and Sons, New York, 2005, Second Edition. PART II: Simon Rogers and Mark Girolami "A first coruse in machine learning, 2nd edition" Chapman & Hall It is planned to cover chapters 1--4, 8 and 9 from this book. The precise material will be determined during the course.
Assessment	The final grade of the course consists of the following components: Part I (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Part II (50%) Lab assignments (pass/fail) Written exam (weighting 100%) Final grade calculation: 0.5 * Part I + 0.5 * Part II A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Part I written exam: Resit (written or oral exam, depending on the number of students) Part II written exam: Resit (written or oral exam, depending on the number of students) In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Part I lab assignments: Re-submit deliverable Part II lab assignments: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	PART I: Exam of 3 hours. PART II: Exam of 3 hours.
Permitted Materials during Tests	PART I: Self made notes on a two-sided written A4 sheet. Calculator. PART II: none
Remarks	PART II: This course is particularly interesting for students that are interested in statistical exploratory and quantitative techniques to analyse multivariate data.

CS4375	Artificial Intelligence Techniques	5
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. M.T.J. Spaan	
Instructor	L. Cavalcante Siebert	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	We expect students to have programming expertise at the Bachelor level of Computer Science, in particular knowledge is expected of algorithms (e.g. search algorithms), and probability theory. Programming skills are required for the practical assignments which can be done in Python.	
Course Contents	The course focuses on concepts and techniques from probabilistic Artificial Intelligence for building intelligent agents, and decision making and -support systems. The specific topics include Bayesian networks, inference, belief monitoring, (partially observable) Markov decision processes, reinforcement learning and multiagent systems.	
Study Goals	After completing the course, student will be able 1. to comprehensively summarize the covered topics within probabilistic artificial intelligence and discuss their main issues; 2. to explain the main concepts and techniques; 3. to relate practical problems to appropriate AI techniques and apply the formal models covered in the course to these problems; 4. to implement and evaluate algorithms for complex decision making.	
Education Method	Lectures, lab work (programming assignments).	
Literature and Study Materials	Stuart J. Russel and Peter Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Prentice-Hall. ISBN 9781292401133 Global edition. See website http://aima.cs.berkeley.edu/global-index.html for additional information that goes with the book.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	The practical assignments will be done in groups of 4 to 5 students. The ethical standards of working are expected of all students. Work can be divided over students, but all students are responsible for the overall quality and originality of the work. Students that do not do their share of the work have to be reported by the group to the supervisor so that action can be taken.	
Tags	Artificial intelligence	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IN4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Reuired: Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations Strongly advised: some experience with a programming language (e.g., C) is preferred.	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas are considered. The scaling behavior of these algorithms in case of increasing problem size and/or an increasing number of processors is analyzed. Attention is paid to those aspects of computer architectures that are important to understand the resulting performance, such as the memory hierarchy and the interconnection network. By analyzing a number of case studies (applications) with respect to their computing-intensive character, possible bottlenecks will be determined. Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to get high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and the necessary optimization for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found the Brightspace.	
Course Contents Continuation	High Performance Computing, parallel programming, parallel algorithm	
Study Goals	1. Is able to categorize high-performance computer systems 2. Is able to explain and classify parallel and distributed architectures, and programming models; 3. Is able to explain the concepts of data decomposition and parallel algorithms; 4. Is able to apply various parallelization methods to benefit from the potential parallel computer systems; 5. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 6. Capable to evaluate and analyze the performance of parallel programs.	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): implementing (small) parallel programs with C, MPI and Cuda.	
Literature and Study Materials	Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Lab work (weighting 50%) Quizzes, bonus of max 1 point can be obtained (only valid on the first examination occasion after the lectures) Final grade calculation: 0.5 * Written exam + 0.5 * lab work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Via Brightspace	

IN4252	Web Science & Engineering	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	J. Yang	
Instructor	Prof.dr.ir. G.J.P.M. Houben	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The expected entry level is (equivalent to) standard bachelor-level computer science.	
Course Contents	The main subject of the course is the Web, and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g. Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan / research paper draft in collaboration with others in a small group to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies, and verbally present and defend this plan / draft.	
Education Method	The education methods include: - Lectures, before which and after which students study material by themselves, to get an understanding of the relevant material; - Small assignments and hands-on exercises, to apply the understanding of relevant material; - One large assignment, with a number of feedback moments, to learn how to write a web science paper / research project plan. Lectures will not be in each week in the class period (1+2): in between lectures there is time reserved for studying before and after lectures, for small assignments and exercises, and for writing the large assignment paper. The writing of the large assignment paper / project plan happens throughout the class period (1+2) to enable frequent feedback.	
Literature and Study Materials	Will be provided in class, depending on the topics chosen for the assignments and final paper.	
Assessment	Assessments are based on two components: the "small assignments" (accompanying the lectures) which provide a total of 20% of the grade (these are a mix of literature and programming homework), and the "final paper" which provides 80% of the final grade. To pass, each of the components ("small assignments" and "final paper") need to be graded with a minimum of 5.0. The final grade must be at least a 5.8. If these criteria are not met, the "small assignments" and the "final paper" can be repaired provided that the to-be-repaired component was at least graded with a 4 in accordance with OER/TER article 17. This repair is not a resit, and is limited to a maximum of 6.0 as a result. Components that have been graded with less than a 4.0 cannot be repaired. If the course is still not passed after the repair or cannot be repaired, all components need to be redone if the course is retaken in a later year. Both the small assignments and the final paper are group works of groups of preferably 2 members.	
Special Information	Students are asked to register/enrol on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	The expected workload is 5ects and that is principally distributed uniformly over the two quarters.	

IN4307	Medical Visualization	5
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge of linear algebra, calculus and programming is needed. The project will be implemented in MeVisLab and Python. Learning MeVisLab will be part of the practice assignments.	
Course Contents	Theory and practice (Notice project extends to Q2) of medical visualization. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measuremen; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for medical data; advanced applications.	
Study Goals	By the end of the course, you should be able to LO1: Explain medical visualization algorithms and their applicability to medical problems. LO2: Discuss the advantages and disadvantage of medical visualization algorithms. LO3: Build a medical visualization system for a given problem: a. Discuss a suitable visualization for a given medical problem. b. Implement the most suitable solution. c. Judge the performance of the implemented solution.	
Education Method	The course will be based on a combination of lectures and practical assignments. A final project will be developed in Q2	
Literature and Study Materials	Course slides, instructions for project and assignments, and selected literature via Brightspace. Chapters from the following book can be used as alternate view on the topics but are not mandatory: Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Book available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam Q1 (weighting 40%) Final project Q2 (weighting 60%) The final project will be the design and implementation of a visualization system for a given medical problem. The final project will be carried out in teams of three. The deliverables for the final project will be a report (paper), the results (e.g., code) and a short video presenting the project (i.e. screencast). Final grade calculation: 0.4 * Written exam + 0.6 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Q2 In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final project: re-submit deliverable (the project resit is not automatic and must be requested within 2 weeks after publishing the grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Notes and written material. No computers.	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	
Co-Instructor	Prof.dr. E. Eisemann	

IN4344	Advanced Algorithms	5
Responsible Instructor	Prof.dr. M.M. de Weerdt	
Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic courses in Algorithmics and Complexity Theory	
Course Contents	The course is on solving (abstract models of) complex real-world problems, with a focus on solving intractable problems exactly. The course consists of two main parts: * modeling and solving using linear programming * exact algorithms using search trees, dynamic programming, and/or decision diagrams	
Study Goals	By the end of this course, students will have knowledge of and experience with the following advanced algorithmic techniques: (Part 1:) - linear programming (LP) and LP modelling - duality and simplex algorithm - integer LP and integer modelling - LP relaxation and branch and bound (Part 2:) - complete and bounded search trees - tree decomposition - dynamic programming - preprocessing - decision diagrams Furthermore, they will be able to * recognize situations where they can use these techniques and adapt them to different applications. construct new algorithms that are similar to these techniques. construct proofs that are similar to a selected set of proofs from the material. analyze the run time of algorithms. implement an algorithm that is given in pseudo-code. experimentally evaluate the quality and the runtime of an algorithm on a set of instances.	
Education Method	Interactive lectures (usually 2x2 per week), optional homework exercises, programming assignments (2 hours lab support per week) The expected workload is 30% studying the written material and the recorded lectures, and participating in the interactive sessions 30% making the homework exercises and preparation for the exams 40% working on the programming assignments	
Literature and Study Materials	Part 1 of the course will be mainly based on chapters 1-8, some of 9-11 (most of 9-11 is assumed to be known), and 12-14 of the syllabus "Optimization" by Karen Aardal, Leo van Iersel and Remie Janssen, which can be ordered via https://www.webedu.nl/bestellen/tudelft/ Part 2 will use chapter 10 of the following textbook: J. Kleinberg and E. Tardos, Algorithm Design, Pearson Education, 2006. ISBN 0-321-37291-3 As well as some supplemental study material, which will be provided via BrightSpace.	
Assessment	The final mark depends on the marks obtained for (a) written reports and evaluation of programming assignments, PA (weight 40%) (b) a written exam, EX (2 parts, weight 60%) Each programming assignment is graded on a scale from 0 to 10. The final mark for the programming assignment (PA) is the average of the mark obtained for the assignments. Programming exercises can be completed by 2 students working together. If the average grade for the four programming assignments (PA) is between 4.0 and 6.0 (excluding 6.0), a repair is possible by making one extra assignment to replace the lowest grade of the four assignments. This leads to a repaired PA score, maximized at 6. (E.g. if grades were 3.5, 4, 4, 4.5, the 3.5 can be replaced. If the grade for the replacement assignment is a 7.5, the repaired PA score is a 5. If grades were 5, 5.5, 5.5, 6, the 5 can be replaced. If the replacement is a 9, the repaired PA score is not 6.25 but 6.) The exam consists of two parts. Each part will be examined after the lectures about that part have been delivered and will be graded on a scale from 1 to 10. The final mark for the exam (EX) is the average of the marks for the parts. Each exam part contains one or more challenging assignments; during each part of the course two representative homework assignments will be made available to prepare for this. There is a resit for the exam where any of the two parts can be redone. The result for a part after the resits is determined by the maximum score obtained for the original exam and the resit. The final mark for the course is determined as follows: - if the PA and the EX mark are above 5, the final mark is the weighted average of these three marks: 60% EX, 40% PA - if at least one of PA, EX is less than or equal to 5, the final mark is the minimum of the results obtained for PA or EX. Partial results are valid only in the current academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	For content questions / of general interest, please use StackOverflow. For personal questions, please use the course email address: aa-cs-ewi@tudelft.nl	
Tags	Algorithmics Artificial intelligence Mathematics	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Bioinformatics specialization Courses Q2 2023

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4089	Data Visualization	5
Responsible Instructor	T. Höllt	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/2/0/0 & lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Master course MKE/ST/DS	
Expected prior knowledge	IN2905-A/IN4152/CSE2215 Computer Graphics (recommended, not required). The practicals will be implemented in HTML/Javascript/D3 (InfoVis) and C++ (VolVis). We consider programming skills as a requirement but not necessarily in the mentioned languages and no advanced skills are needed. Relevant topics will be introduced and experience in other programming languages should make adaption feasible.	
Course Contents	Data visualization is the visual representation of data by computer generated images. The data sets can be results of numerical simulations or measurements (scientific visualization), or other data collections such as databases (information visualization). The goal is to improve insight, understanding and/or communication of data. Data visualizations use a combination of methods from a very diverse variety of disciplines: perception, computer graphics, human computer interaction, algorithmics, image processing, machine learning, numerical analysis, optimization, The course has two main parts; information and scientific visualization that will involve knowledge of diverse disciplines. As a computer science course, affinity to algorithmic thinking and programing skills will be needed. Topics covered: models of the visualization process; colour models and use of colour; information visualization; representation and processing of data; volume visualization; interactive visual data analysis; visualization of vector fields and flows. Guest lectures might be given on selected topics.	
Study Goals	The goal of the course is to get knowledge on the fundamentals that are part of data visualization. The main principles and techniques that are the basis of generating effective visual representations of data. Techniques and cases of data visualization are discussed. There are several applications for the techniques, such as medical, engineering, finances, economics, game analytics, and more. By the end of the course, you should be able to LO1: Discuss a large range of visualization techniques. LO2: Discuss a perception principle of visualization. LO3: Explain mathematical principles and algorithms of visualization techniques. LO4: Design suitable visualization systems for a given practical data analysis problem. LO5: Implement visualization systems for a given practical data analysis problem.	
Education Method	Lectures, practical assignments, self-study, and projects.	
Literature and Study Materials	Course slides, instructions for projects, and selected literature via Brightspace. Chapters from: Visualization Analysis and Design Author: Tamara Munzner CRC Press Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Books available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 30%). You are allowed to have the slides and material of the course during the exam. No computer or laptop is allowed. Information visualization project (weighting 35%) Volume visualization project (weighting 35%) The projects will be developed in groups of 3 and are evaluated based on the reasoning/justification of the techniques used based on the material given at the course, effectiveness of the results, technical contribution or implementation, quality of the documentation and presentation Final grade calculation: 0.3 * Written exam + 0.35 * Information visualization project + 0.35 * Volume visualization project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Information visualization project: re-submit deliverable Volume visualization project: re-submit deliverable The project resits are not automatic and must be requested within 2 weeks after publishing the grades. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	

IN4150	Distributed Algorithms	6
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for network overlays, message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, broadcast, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	Understand the fundamental problems that distributed systems face Understand the most important distributed algorithms that solve these problems Reason about the execution of distributed algorithms Program distributed algorithms Select and summarize relevant literature on distributed algorithms	
Education Method	Lab work executed in groups of two students	
Literature and Study Materials	Lecture notes and lecture slides (available on Brightspace)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 75%). In case in-person, on-campus written exams are not possible because of unforeseen circumstances or measures, the written exam will be replaced by individual online oral exams. Paper summary (weighting 25%) Final grade calculation: 0.75 * Written exam + 0.25 * Paper summary A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper summary: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None except the list of algorithms	
Special Information	Email: in4150-ewi@tudelft.nl. Please send any emails related to the course to this address.	
Remarks	Lab work is 40 hrs.	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Bioinformatics specialization Courses Q3 2023

CS4195	Modeling and Data Analysis in Complex Networks	5
Responsible Instructor	H. Wang	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The assignment and final project require basic programming skill.	
Course Contents	Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	After this course, students could construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies, model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. Students could obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 20%) Final project (weighting 80%) Final grade calculation: 0.2 * Assignments + 0.8 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4230	Machine Learning 2	5
Responsible Instructor	Dr.ir. J.H. Krijthe	
Instructor	Dr. D.M.J. Tax	
Instructor	T.J. Viering	
Instructor	Dr. F.A. Oliehoek	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Required: This course is the more advanced and research oriented follow-up to Machine Learning 1 (CS4220). CS4220 or an equivalent machine learning course is therefore expected as prior knowledge.	
Course Contents	The course will treat a number of machine learning settings and theories spanning the breadth of machine learning research in detail and on an advanced level. Topics that are covered include (subject to change): learning theory, learning curves, latent variable models, causal inference and discovery, adversarial learning, online learning and neuromorphic computing.	
Study Goals	After completing the course, you should be able to: 1. Identify which setting beyond (un)supervised learning a problem belongs to 2. Explain the essential problem that (adversarial/causal/online/PAC/neuromorphic/...) learning addresses and how it differs from the classical ML setting 3. Evaluate methods that are proposed in these learning scenarios, both theoretically and empirically 4. Mathematically prove basic properties of methods in these scenarios 5. Apply common methods in these learning scenarios on real-world datasets 6. Develop machine learning techniques adapted for a particular learning task	
Education Method	Each week: 1 lecture and 1 Q&A session where the weekly exercise set will be discussed.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) 4 combined assignments (pass/fail). There will be multiple small assignments (in weeks 2,4,6,8). Final grade calculation: P/F * assignments + 1 * Written exam A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: 4 combined assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4240	Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Seminar Computer Vision by Deep Learning	
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy). Experience in programming and installing programming languages/libraries is required. Without affinity/experience with python (and numpy) programming this course will become too time consuming as the project will become difficult.	
Course Contents	In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.	
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.	
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ - https://udlbook.github.io/udlbook/ Research papers will be made available through Brightspace. Assignments are based on PyTorch: https://pytorch.org/	
Assessment	The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Digital Exam (weighting 90%). Exam about the papers, lab assignments, and the theory (lectures). Project (Pass/Fail). In a small group of students you work on a deep learning paper reproducibility project. Final grade calculation: (pass project) * (0.1 * Presentation + 0.9 * Exam) A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4315	Software Architecture	5
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Responsible Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software engineering	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four, in which the actual architectures of existing open source systems are analyzed in considerable detail. These systems are taken from github, and student teams are challenged to actually contribute to the systems under analysis in the course.	
Study Goals	Bring students into the position that they can (1) explain the key architectural concepts and methods for modeling software architectures; (2) apply viewpoints and perspectives to model software architectures; (3) discuss the benefits of architecting and the role of the software architect; (4) evaluate and validate software architectures; (5) explain and discuss the concepts of component-based and plugin architectures, service-oriented architectures, and software product lines; (6) explain and recognize technical debt and have an understanding of possible refactorings.	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books; Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; and Coplien and Bjørnvig, "Lean Architecture", Wiley, 2010. Additional reading material will be announced in the lectures.	
Assessment	Grades will be based on lab assignment including essay writing, coding, (video) presentation, peer reviewing, participation. The final grade of the course consists of the following components: Essays (weighting 60%) Coding (weighting 20%) Presentation (weighting 20%) Final grade calculation: 0.6 * Essays + 0.2 * Coding + 0.2 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Essays: re-submit deliverable Coding: re-submit deliverable Presentation: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Special Information	Course web site: https://se.ewi.tudelft.nl/delftswa/index.html	
Co-Instructor	M. Finavaro Aniche	

IN4325	Information Retrieval	5
Responsible Instructor	J. Yang	
Instructor	Dr. A. Anand	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Knowledge of basic algebra. Proficiency in at least one programming language. Knowledge of Web information systems and software engineering can be helpful.	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. The World Wide Web is now the primary source of information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, analysed, and exploited so that each user can quickly find information that is both relevant and comprehensive for their needs. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation of Web and Enterprise search engines. Covered topics include: = Information Retrieval Models; = Indexing Techniques; = Web Search; = Information Seeking Paradigms; = Evaluation of information retrieval systems;	
Study Goals	At the completion of this course, students will be able to: = Describe the different information retrieval models, and compare their strenghts and weaknesses. [Learning Objective 1] = Describe and implement different indexing techniques. [Learning Objective 2] = Describe and analyze querying techniques with respect to their most suited application domains. [Learning Objective 3] = Analyse the effectiveness of an information retrieval system through proper use of evaluation metrics. [Learning Objective 4] = Design and implement (Web) Information Retrieval systems, possibly using advanced social and semantic search functionalities. Support and defend the relevance and correctness the choices with regards to the adopted information retrieval model, indexing technique, and querying technique. [Learning Objective 5]	
Education Method	Lectures; course long group project (research and development) as well as an individual literature survey and small weekly assignments. Expected workload is is 140 hours: 45 hours for lectures and lecture preparation plus the weekly assignment, 80 hours for the group project and 15 hours for the literature survey.	
Literature and Study Materials	Scientific papers, course slides, course books - all resources are available on Brightspace.	
Books	Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. Introduction to Information Retrieval. Cambridge University Press, New York, NY, USA. Stefan Büttcher, Charles Clarke, and Gordon V. Cormack. 2010. Information Retrieval: Implementing and Evaluating Search Engines. The MIT Press.	
Assessment	The final grade of the course consists of the following components: Individual assignment (weighting 10%) Weekly individual assignment (weighting 20%) Group project (weighting 70%) Final grade calculation: 0.1 * Individual assignment + 0.2 * Weekly individual assignment + 0.7 * Group project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual assignment: re-submit deliverable Weekly individual assignment: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Bioinformatics specialization Courses Q4 2023

CS4205	Evolutionary Algorithms	5
Responsible Instructor	P.A.N. Bosman	
Instructor	Dr.ir. P.A. Bouter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Must have the ability to program, predominantly in Python (but also Java and C/C++ are very useful) so that you can create your own (extensions of) evolutionary algorithms. Further, a basic understanding of key aspects in linear algebra (e.g., matrix multiplications, inversions, decompositions), probability theory and statistics (e.g., normal distribution, statistical hypothesis testing), and complexity theory (e.g., big-O notation), will be very helpful.	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, and real-world applications. The course is planned to have 10 lectures, multiple small practical assignemnts and 1 larger practical assignment. The first practical assignments pertains to experimenting with already implemented EAs on predefined problems. The second practical assignment offers more freedom, allowing you, in a group, to build your own EA (this may vary depending on student numbers and other circumstances).	
Study Goals	Upon successful completion of this course, students will be able to: 1) Explain and apply key concepts underlying the main streams in Evolutionary Algorithm (EA) research, in particular genetic algorithms, evolution strategies, genetic programming, evolutionary multi-objective optimization, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2) Explain and apply key ingredients underlying the rationale of when these algorithms work and when they do not work and several ways to overcome this. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3) Name and explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4) Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5) Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	10 Lectures Several smaller, individual, lab projects One larger lab project in groups of several students	
Literature and Study Materials	Papers and slides that will be made available.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 60%) Lab practical work (weighting 40%) Final grade calculation: 0.6 * written exam + 0.4 * lab practical work A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab practical work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	None	
Tags	Algorithmics Artificial intelligence Optimalisation	

CS4245	Seminar Computer Vision by Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	N. Tömen	
Instructor	Dr. X. Zhang	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	MSc thesis in the Computer Vision lab	
Expected prior knowledge	Deep Learning (CS4240)	

Course Contents

The recent boom in computer vision and automatic image understanding represents an inflection point in human productivity, permeating wide aspects of the economy and society. Examples of visual tasks which are repetitive or require expert knowledge include medical diagnosis, industrial inspection, autonomous vehicles, etc. When machines can meaningfully assist or even completely take-over such tasks it will change the world as we know it.

The breakthrough in the 2012 ImageNet automatic image recognition competition shows all previously existing methods decisively defeated by a deep neural network. Deep learning replaces feature engineering methods and is able to successfully learn image features from huge annotated datasets.

This course is on automatically understanding visual content such as images and videos by deep learning.

The range of topics (although, not limited to): Fundamentals in Vision, object detection, per-pixel labelings, video recognition, image similarity learning, efficiency, self-supervision, 3D computer vision, adverserial attacks, explainability, generative models, etc.

The course will have lectures, a seminar and a lab practical:

- The lectures will be on established topics based on the current literature.

- The seminar will have students read, critique, and present relevant computer vision research papers. You will have to read 2 papers per week, for 7 weeks (14 papers).

- The lab will have students apply and design their own (small) computer vision project.

The course build on top of the Deep Learning course (CS4240) and follows a similar setup.

Study Goals

Upon successful completion of the course, students will be able to:

[LO1]. Describe the deep learning techniques reviewed in the course for computer vision applications such as image classification, object detection, per-pixel labelings, video recognition, image similarity learning.

[LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers

[LO3]. Debate upon positive and negative aspects of techniques and research papers

[L04]. Quickly identify the core contributions of a research paper

[LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (we focus on Pytorch)

[LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain.

[LO7]. Apply the appropriate technique to a (simple) Computer Vision problem.

[LO8]. Write clearly and concisely about your code, method, results, and analyis.

Education Method

Lectures

Lab project: design and execute your own Computer Vision project.

Seminar: paper reading, critiquing, and presenting.

Assessment

The final grade of the course consists of the following components:
Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once.
Exam (weighting 60%). Exam about all material.
Project (weighting 30%). in a small group of students you work on a computer vision project.

Final grade calculation: 0.1 * Presentation + 0.3 * project + 0.6 * Exam
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

Resit/ Repair opportunities:
Exam: resit
In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for:
Presentation: resit
Project: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

CS4290	Seminar on Distributed Machine Learning Systems	5
Responsible Instructor	Dr. Y. Chen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Machine learning systems are often conventionally designed for centralized processing in that they first collect data from distributed sources and then execute algorithms on a single server. Due to the limited scalability of processing large amount of data and the long latency delay, there is a strong demand for a paradigm shift to distributed or decentralized ML systems which execute ML algorithms on multiple and in some cases even geographically dispersed nodes. The aim of this seminar course is to let students learn how to design and build distributed ML systems via paper reading, presentation, discussion, and project prototyping. We provide a broad overview on the design of the state-of-the-art distributed ML systems, with a strong focus on the scalability, resource efficiency, data requirements, and robustness of the solutions. We will present an array of methodologies and techniques that can efficiently scale ML analysis to a large number of distributed nodes against all operation conditions, e.g., system failures and malicious attacks. The specific course topics are listed below. The course materials will be based on a mixture of classic and recently published papers. For each topic, the basic concepts and technology landscape will be first provided and then two state-of-the art of papers will be presented and discussed by students. We offer a testbed of a distributed (deep) ML system in which students are encouraged to apply different techniques to jointly improve its scalability and resilience. Course topics include Overview of distributed machine learning systems Performance and scalability of state-of-the-art systems Acceleration of ML workloads Slim distributed ML systems on small data Robust deep learning systems Federated machine learning systems	
Study Goals	Students are able to argue and reason about distributed ML from a systems perspective. Students understand the behavior and tradeoffs of distributed ML in terms of performance and scalability. Students can estimate the importance of data inputs via different techniques, i.e., core set and decomposition methods, for distributed ML systems. Students understand data poison attacks and design defense strategy for distributed ML systems. Students can analyze the state-of-the art federated machine learning systems and design the failure-resilient communication protocols. Students are able to design and implement methods and techniques for making distributed ML systems more efficient.	
Education Method	Lectures: 7 weeks X 2h Papers: one paper presentation, two paper reviews, and paper discussion. Practical: apply system and algorithmic optimization techniques learned in the lecture to improve the performance of distributed machine learning systems, e.g., image recognition on CIFAR 10. The testbed environment, learning algorithms, and dataset will be given. Deliverables include git commit of functioning code and a report summarizing the contribution	
Assessment	The final grade of the course consists of the following components: Paper presentation by group (10%): each group of 2 to 3 students needs to choose from a given set a papers to present (15 minutes) and lead the discussion (10 minutes). Paper reviews by individual (30%): each student needs to write three reviews of papers assigned from a given set and those two papers have to be different from the paper for the presented. Each review will account for 10% of the grade. Questionnaires by individual (0%): each student needs to hand in a list of questions at the beginning of the lectures that have paper presentation from other students. This is not graded. Individual project (60%): The project is collaborative among the entire class and competitive as a whole group. The objective is to continuously improve the performance of the given distributed ML system. The students need to hand in a final project report in style of a short scientific paper, stating their individual contribution to the overall system performance. Final grade calculation: 0.1 * Paper presentation group + 0.3 * Paper presentation individual + 0.6 * Individual project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Paper presentation by group: resit Paper reviews by individual: resit Individual project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4331	Web-scale Data Management	5
Responsible Instructor	Dr. A. Katsifodimos	
Instructor	Dr. C. Lofi	
Instructor	Dr. R. Hai	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor level courses in database management systems and operating systems. A prior course in distributed systems or middleware would be helpful but is not required. Programming skills are important for the final assignment.	
Course Contents	This course addresses the challenges of Data Management at Web-scale. Especially, it covers the need for large-scale distributed data storage systems. The lecture therefore introduces step-by-step increasingly complex distributed storage systems, leading up to modern implementations of different NoSQL data storage systems. The challenges arising from such systems are presented and discussed, especially focusing on the CAP theorem and the resulting trade-offs with respect to data models, transactional power, query expressivity, and replication consistency. These discussions lead to different variants of NoSQL database systems, like Key-Value Stores, Document Stores, Wide-Columnar stores, and Graph Databases. The advantages, disadvantages, and general properties of these systems are discusses in more detail. There is special focus on distributed transactions and consistency guarantees of different data management systems and methods.	
Study Goals	At the end of this course the student can - assess the nature of a given storage problem, and can select a suitable technology for solving it - understands the different data models encountered in Web Data Management, and their impact on modelling and querying - understands the issues arising from distributing and replicating data, especially with respect to the CAP theorem -understands the trade-offs which can be chosen within the design space of the CAP theorem -categorize and explain modern NoSQL databases within the framework of the previously mentioned trade-offs	
Education Method	Lectures and assignments	
Literature and Study Materials	Course slides and Lecture Videos	
Books	Literature mentioned in the lecture, mostly research papers.	
Assessment	There is a group assignment (80%) and an individual exam (20%). In order to pass the class, a student has to have a grade of at least 5.0 for the group assignment as well as the individual exam, with a weighted average of 5.8 overall. The exam can be resit in the following academic quarter. If the student achieves a 4.0 - 6.0 in the group assignment, a repair opportunity can also be given (e.g., extra work, corrections on the group assignment, etc.).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Literature study 2023

IN4306	Literature Survey	10
Responsible Instructor	L. Miranda da Cruz	
Contact Hours / Week x/x/x/x	Not applicable	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Literature Survey is an individual assignment carried out under the supervision of a Computer Science staff member, i.e. an assistant, associate or full professor. For this assignment the student reads a broad range of papers in the chosen specialisation field and writes a report in which the ideas found in the papers are discussed and compared. It is not allowed to merge this assignment with the thesis project.	
Study Goals	After the course, the student will be able to: LO1 - Formulate a research question that is relevant, meaningful, and scientifically sound. LO2 - Find high-quality contemporary scientific literature in the chosen research field. LO3 - Explain the scientific contribution of the reviewed papers within the context of the research question.	
Education Method	Individual assignment and individual guidance by a scientific staff member.	
Assessment	Writing a scientific report, individually and under supervision of a staff member. This staff member will also mark the report. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	The Literature Survey may be part of an individual exam programme of a student, which has to be approved by the Board of Examiners (BoE). To apply for a literature study the student should contact a staff member of the research group of her/his chosen specialisation after having received approval of his or her individual exam programme. The staff member and the student make arrangements regarding content and scope of the survey. The abovementioned staff member will supervise the student during this course.	
Co-Instructor	C.B. Poulsen	
Co-Instructor	Dr. J.C. van Gemert	
Co-Instructor	Dr. K.A. Hildebrandt	
Co-Instructor	T.E.P.M.F. Abeel	
Co-Instructor	Dr.ir. W.P. Brinkman	
Co-Instructor	Dr.ir. A.R. Bidarra	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Free Electives 2023

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Thesis project 2023

IN5000	Final Project	45
Responsible Instructor	Dr. J.C. van Gemert	
Responsible Instructor	C.B. Poulsen	
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	L. Cruz	
Responsible Instructor	Dr.ir. A.R. Bidarra	
Responsible Instructor	Dr.ir. W.P. Brinkman	
Responsible Instructor	Dr. K.A. Hildebrandt	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	IN5000 Final Project is the final part of the Master's degree programme. During this project, you will be required to demonstrate your ability to solve a research or engineering problem. The project must be carried out using the techniques of project management. You will begin by making a project plan in cooperation with your Masters thesis advisor. Several aspects of the project are defined within the plan, including the assignment, the frequency of interaction with the advisors, the milestones of the project and the resources and facilities offered by the faculty. You will be required to adhere to your plan throughout the project. It is obviously possible to adjust your plan under certain circumstances and after discussion with your daily supervisor. At the end of the project, you will submit your Masters thesis, which must be written in English, and make an oral presentation of your work to the Thesis Committee. The Thesis Committee will announce the final mark, which is based on the project performance, the thesis, the presentation and the subsequent discussion. More information about the graduation process: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-policy-msc/ More information on the thesis project on Brightspace: MSc CS Thesis project EEMCS https://brightspace.tudelft.nl/d2l/home/492868	
Study Goals	1. The student is able to design a research project: - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. 2. The student is able to execute a research project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. 3. The student is able to write a research report: - The student is able to write a research report that shows sufficient coherence of content; - The students is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; 4. The student is able to present and defend the research project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.	
Education Method	Project	
Prerequisites	Before starting the project, students must have completed at least 60 EC of the Master's degree programme as shown by a Declaration of Obtained Credits (DCO). To be able to get this DCO, the individual exam program (IEP) should have been approved by the Board of Examiners. Note: In some cases, the thesis supervisor may impose additional conditions for starting this project.	
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-policy-msc/assessment/ The voting members of the thesis committee determine the final grade. The grade should reflect a weighted average of the four scores above, but need not to be an exact arithmetical mean. The final mark starts from 5 up to and 10. Marks ending in .5 may also be used. If the student shows excellence (is nominated for a 10) the chair of the thesis committee should consult the chair of the Board of Examiners, at least five working days in advance of the defense. The chair may advice to add an extra member to the thesis committee. The motivation for the grade at each of the four criteria as listed above is summarized on a form and signed by the chairman of the thesis committee. The candidate is given a short account of the assessment, either in private or in front of the audience. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Cyber Security 2023

In association with the Faculty of	TBM
Introduction 1	The special programme in Cyber Security offers a variety of courses enabling students to select an individual course program. Student in the special programme in Cyber Security may choose between the Data Science & Technology, the Software Technology track and the Artificial Intelligence Track. Students do 120 EC in total including 80 EC Cyber Security. I Common Core Courses Special Programme Cyber Security 1. IN4191 Security and Cryptography, 5EC 2. TPM027a Cyber risk management, 5EC 3. CS4430 Network Security, 5EC 4. CS4035 Cyber data analytics, 5EC 5. CS4150 Systems Security, 5EC Students have to complete two additional common courses from their respective tracks II Technical Electives: choose at least 3 courses 1. QIST4310 Fundamentals of quantum information, 4EC 2. CS4380 Privacy Enhancing technologies, 5EC 3. CS4090 Quantum communication and cryptography, 5EC 4. IN4253ET "Hacking Lab"-Applied Security Analysis, 5EC 5. CS4160 Blockchain Engineering, 5EC 6. CS4110 Software Testing and Reverse Engineering, 5EC 7. CS4280 Language Based Software Security, 5EC III Socio-Technical Electives: choose at least 3 courses 1. TPM020b Economics of Security, 5EC 2. TPM025a User-Centred Security, 5EC 3. TPM010a Cyber crime science, 5EC 4. TPM030a Introduction to Cloud as Infrastructure: The effects of the new business of computing on practice, 5EC IV Required Courses for CS Graduation 1. CS4120 Seminar Cyber Security, 5EC 2. IN5000 Master Thesis Project in Cyber Security, 45EC The thesis (45 credits) is performed under supervision of the Cyber Security research group. V Free Electives (Prerequisites and other Courses) The remaining credits to make up the programme are chosen in consultation with the master coordinator.
Administration by the Faculty of	EWI

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Common Core Courses Special Programme Cyber Security (7 courses) 2023

CS4035	Cyber Data Analytics	5
Responsible Instructor	Ir. S.E. Verwer	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course provides theoretical and practical background for applying data analytics in the field of cyber security. Cyber data analytics is a huge field with a great diversity of techniques and applications. The course is centered on a selection of seven such techniques: learning from imbalanced data; behavioral profiling and anomaly detection; sequential data mining; data stream mining; learning from software data; adversarial machine learning; and privacy-aware data mining; Anomaly detection is one of the main topics in cyber security. Specific difficulties that the student will learn to handle are the huge amounts of data and the large number of false positives. Behavioral profiling applies to both people and software processes. Different techniques will be taught to construct profiles from software logs. While building such profiles, care should be taken to not infringe upon the privacy of individuals the data is collected from. Finally, attackers will modify their behavior in order to avoid being detected, a cyber data analytics engineer tries to make their models/profiles robust against such modifications.	
Study Goals	The student will be able to: Apply machine learning to real data Understand and modify machine learning algorithms Learn models from time series Detect anomalies in multidimensional time-series Use distributed processing to speed up machine learning Learn models from data streams with limited memory Learn sequential models Use machine learning for fingerprinting and profiling Preserve the privacy of data owners while learning models Learn robust models that can detect evasive attackers Use machine learning to detect fraud, attacks, and botnets	
Education Method	There will be two lectures for each of the seven topics, and 3 large lab assignments on fraud detection, anomaly detection, and behavioral profiling, and 1 smaller lab on adversarial robustness. There is no exam. Teams of two students will work on these assignments which contain both individual and collaborative components. Deadlines are strict as peer-review will be used to both learn of other possible solutions, provide feedback, and get initial estimates on the obtained grade.	
Assessment	The final grade of the course consists of the following components: Lab assignment fraud detection (weighting 30%) Lab assignment anomaly detection (weighting 30%) Lab assignment behavioral profiling (weighting 30%) Lab assignment adversarial robustness (weighting 10%) Final grade calculation: 0.3 * Lab assignment + 0.3 * Lab assignment + 0.3 * Lab assignment + 0.1 * Lab assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab assignment fraud detection: Re-submit deliverable Lab assignment anomaly detection: Re-submit deliverable Lab assignment behavioral profiling: Re-submit deliverable Lab assignment adversarial robustness: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4150	Systems Security	5
Responsible Instructor	Dr. S. Picek	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Security and Cryptography (IN4191) and Network Security (CS4430) and a Bachelor level Operating Systems course. The topics below should be covered to a good bachelor level. Basic machine learning knowledge (e.g., being able to run classification algorithms). To allow students to assess their level of knowledge, they can have a short oral interview.	
Course Contents	IoT security, Hardware, Countermeasures, Covert channels, Secure System Engineering	
Study Goals	The student will acquire: An appreciation of the security architecture of computer systems Detailed knowledge of the security of a specific operating system Skills in exploiting vulnerabilities of computer systems Skills in developing counter measures against exploits	
Education Method	2 hours per week lectures, 4 hours per week lab	
Assessment	The final grade of the course consists of the following components: Lab work (weighting 50%) Written exam (weighting 50%) Final grade calculation: 0.5 * Lab work + 0.5 * Written exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4430	Network Security	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best practices in computer and network security. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course's primary focus will be on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design, review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics, as well as conducting measurements on the effectiveness of attack and defense schemes.	
Study Goals	See course contents.	
Education Method	Lectures, Labs, and Project.	
Assessment	The final grade of the course consists of the following components: Assignments (Pass/Fail) Final project (Report + Presentation) (weighting 100%) Final grade calculation: P/F * Assignments + 1* Final project A passing final grade for the course can only be earned when for all components at least a 5.0 or pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable, resit presentation Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4191	Security and Cryptography	5
Responsible Instructor	Z. Erkin	
Instructor	J.A. Martinez Castaneda	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	UT-201500042 Privacy Enhancing Technologies (Q4)	
Expected prior knowledge	Basic understanding on the following is suggested. -Probability and statistics -Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Learning outcomes: The goal is to make students familiar with the basic concepts applied cryptography, including classical cryptography and modern secret key and public key cryptography. In particular, the students will acquire A sound understanding of the notion of security An understanding of the confidentiality, integrity and authenticity needs of the society Understand the role of cryptographic primitives including the differences between symmetric and asymmetric cryptography, the role of hash functions, digital signatures and PKI Understand the advanced topics in cryptography needed for the modern society with untrustworthy entities Among others things, the following topics are covered: -Classical systems -Information theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures, -*Secret Sharing. (if time permits) -*Zero Knowledge proofs (if time permits)	
Study Goals	It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security, as well as is familiar with present applications. Through assignments, students are expected to have the chance to work on the topics covered in the lectures.	
Education Method	Lectures, assignments and weekly exercises. Attention: This course requires full effort of 140 hours. Even more, if you lack the background (probability and modular arithmetic) Planned Workload: Lectures: 28 x 45minutes sessions, total 22 hours Practice session: 7 x 90 min. total 12 hours Assignments: 3 x 20 hour, total 60 hours Weekly study: 7 x 4 hours, total 28 hours Exam preparation: 20 hours Exam: 3 hours	
Literature and Study Materials	-Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition -Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) Handouts of lectures	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%) Mandatory assignments (weighting 30%) Final grade calculation: 0.7 * Written exam + 0.3 * Mandatory assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Mandatory assignments: re-submit deliverable	

	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	In case of in person examination at campus: The exam is closed book. A cheat sheet of size A4, hand written is allowed for the written exam. Name and student number has to be present on each side. In case of remote exam from home: open book: textbook, slides and self-made notes only. randomised/customised exam.
Permitted Materials during Tests	Only non-scientific calculators.

TPM027A	Cyber Risk Management	5
Module Manager	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Dr. A.P. Afghari	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	MOTIVATION: The challenge of selecting the optimal technical, organisational, legal, and other preventive and repressive measures to reduce cyber risks to acceptable levels can only be understood in the context of the application of Cyber Risk Management. Risk Management is about analysing the relationships between threats, incidents and risks (here in the complex world of cyberspace), based on which an adequate set of countermeasures can be designed. SYNOPSIS: Risk (= the potential lo losing something of value) can manifest itself in cyberspace in all kinds of ways: values at stake are financial wealth, health, physical condition (of people, materials, goods, infrastructures, etc.), well-being, reputation, privacy, trust, etc. Based on a conceptualisation of cyberspace and its various sub-domains (discussed in the project week of year 1), we introduce risk assessment approaches, both of qualitative and quantitative manner, illustrated with case studies. CONTENT: Cyberspace Risk concepts Probabilistic and adversarial risk System and attacker models Requirements and policies Risk assessment methods Qualitative and quantitative methods Measuring security and effectiveness	
Study Goals	After this course, students are able to: Understand and apply risk concepts to (possible) incidents in cyberspace Understand the theoretical principles of cyber risk management Explain the role of adversarial risk models, system models, and attacker profiles in cyber risk assessment Analyze and apply state-of-the-art cyber risk assessment methodologies to (complex, multi-step) cyber incidents Analyze the (expected) effect of measures that help to prevent the occurrence of cyber incidents Justify investments in cyber security	
Education Method	LECTURES: There are two lecture slots per week EXAM: There is a written exam at the end of the course. LANGUAGE: The course is taught in English. LECTURERS: Prof. dr. ir. Pieter v. Gelder (TUD/TPM) and guest lecturers.	
Assessment	#### Assessment Grading will be based on the individual written exam at the end of the course. Final mark of this course is determined by the grade of the individual exam (100%). The minimum grade for the individual exam to pass this course is 6.0. The grade for the individual exam is 2 years valid. The individual exam is summative. Detailed feedback on the individual exam can be reviewed by students by contacting the teaching staff of this course. #### Resit The students are allowed one re-sit per examination. It is not allowed to re-sit an examination or assignment for which they have received a pass (6,0 or higher). It is allowed to re-sit an examination which they haven't done during the first occasion. The re-sit format needs to be discussed with the teacher of the course in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit has to take place within the study year 2023-2024.	
Elective	Yes	
Targetgroup	MSc	
Category	MSc level	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Technical Electives (choose at least 3 courses) 2023

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: A good understanding of the fundamental concepts of quantum communication Insight into the differences between classical and quantum communication and cryptography Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)	
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session	
Literature and Study Materials	Primary: Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: Homework (weighting 70%) Final Project (weighting 30%) Final grade calculation: 0.7 * Homework + 0.3 * Final Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Homework: re-submit deliverable Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4110	Artificial Intelligence for Software Testing and Reverse Engineering	5

Responsible Instructor	Ir. S.E. Verwer
Responsible Instructor	Dr. A. Panichella
Contact Hours / Week x/x/x/x	0/0/4/0
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Course Contents	Software is one of the most complex artifacts of mankind has ever created, but complexity is the enemy of correctness. Modern software testing and validation tools use a multitude of techniques geared toward correct computer code, most of these are base on artificial intelligence. In this course, we study these techniques in details, specifically we will understand and implement: Execution monitoring and taint analysis Branch distance computation Hill-climbing and genetic algorithms Concrete and symbolic (concolic) execution Active state machine learning Genetic programming The goal is to better understand and test software using artificial intelligence. Using the taught techniques you will be able to automatically: Discover which code is reachable Find (security) bugs in software Write tests that cover all reachable code Reverse engineer a code's functionality Patch code to remove bugs and failing tests
Study Goals	The student will: Understand modern AI techniques for software testing. Be able to implement several such techniques from scratch: - smart fuzzing (probing software with input to find crashes/bugs), - symbolic execution (using logic to construct inputs that trigger specific code branches), - fault localization (given that a program fails, find the line of code responsible for the failure), and - automated program repair (using a patch library and genetic programming to improve code) Be able to apply this technology to locate bugs in real-world software implementations.
Education Method	The main part of the course will consist of 3 lab assignments covering the theory (fuzzing&tainting, symbolic execution, automated program repair), and one lab assignment for the application to real software. The students will implement the taught techniques from scratch in the first 3 assignments, which will be scored with a pass/fail. All three assignments need to be passed to complete the course. The final lab will contain a recap from the first three assignments and an application of a state-of-the-art tool on real software. The final lab will be graded and be the final course grade. There will be instruction sessions where students can work on their assignment and ask the teachers for assistance.
Assessment	The final grade of the course consists of the following components: Three lab assignments (pass/fail) Final lab (weighting 100%) Final grade calculation: P/F * 3 lab assignments + 1 * Final lab A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: 3 lab assignments: re-submit deliverable Final lab: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Tags	Artificial intelligence Software

CS4160	Blockchain Engineering	5
Responsible Instructor	Dr.ir. J.A. Pouwelse	
Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	In this course you will learn all aspects of blockchain technology, including tamper-proof data structures, digital identities, transitive trust, fault tolerance, distributed consensus, smart contracts and applications. Ledgers and blockchains are an emerging technology with the potential to radically improve financial transactions, supply-chain flows, transactions in general, and distributed databases. The first three weeks of the course will provide a fast-paced introduction to Bitcoin, Ethereum, and TrustChain developed by TUDelft itself. The main component in this course is a team-based complex engineering project. This course is designed for computer scientists to understand blockchain technology and to produce significant hands-on experience. To provide a deep understanding of blockchain technology and understand why it is special you need to experience first-hand how it operates at a detailed technical level. Students design, implement, and test their own independent project in teams of 3-5 students. Students can choose from a pool of possible project ideas. This course requires you to like software engineering. Topics covered: -Blockchain basics and evolution Bitcoin 1st generation, smart contract generation, future 3rd generation (trust or trust in math) -identity and transitive trust Authentication and security primitives, tamper-proof identities, trust models, MITM attacks, Sybil attacks, and TrustChain by TUDelft -Consensus models Proof-of-work, permissioned, Proof-of-stake, Corda no-global-consensus, TUDelft bottom-up fast consensus model -Smart Contract pro/con encrypted data, Bitcoin scripts, Etherium execution model, Hyperledger + Docker issues, Corda Jar file approach, Tezos difficult to use, powerful technology, vision of the future: trusted verified execution -Markets and exchanges Disruption by open markets, winner-takes-all, and multi-sided market platforms, Uber, Airbnb, 22 years of eBay, Silk Road, honesty among drug dealers, the role of trust in markets, P2P exchange markets -Decentralized Autonomous Organization, novel method to collaborate and organise any economic activity -Web3: how the Internet can be re-decentralised using DAOs Within this 2023 edition "the Delft DAO" will be prominently featured. TUDelft achieved a world-first in DAO research. We devised a full end-to-end proof-of-principle of a DAO which is capable of 0) near unbounded scalability 1) controlling money 2) democratic decision making and 3) continuous sustained self-evolution. This course provides you with the knowledge to work with this advanced technology. After this course you will have a firm grasp on the current operational blockchain-based systems, realistic view of real-world applications that may be built on top of ledger technology. You will be able to reason and discuss the open challenges and questions that still need to be resolved. This course is a key course for distributed systems students.	
Study Goals	After this course students are able to design and engineer complex blockchain-based systems. Students are able to describe blockchain technology, the various consensus model, smart contracts, markets, and relation to existing database technology. Student are able to setup a new architecture for blockchain applications.	
Education Method	This course consists of four 2-hour lectures. Each lecture is followed by a 4-hour homework period in the same week focused on understanding the background material. In week 1 you will form teams and initiate work on your blockchain engineering project. A list of projects to select from will be provided at the start of this course.	
Literature and Study Materials	Online course textbook: Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction by Narayanan, Bonneau, Felten, Miller and Goldfeder.	
Prerequisites	It is highly recommended to follow this course (see remarks): Security and Cryptography (Q1) Distributed Algorithms (Q2)	
Assessment	The final grade of the course consists of the following components: Project (weighting 100%) You will write down your project results in a single-page report, markdown format. You will be graded on your open source efforts located on Github and single-page report. Your grade will be expressed on a scale of 0 to 10. Final grade calculation: 1 * Project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This class has a limited capacity (50). If there is a larger number of enrollments than the capacity of the class, students will be assigned to their preferred blockchain engineering project based on their background, engineering experience level, and match to the course goals. Students who followed Security and Cryptography (Q1) and are also enrolled in Distributed Algorithms (Q2) will have priority for placement. Mathematics students are exempts from this, if they can show some minimal software development experience (e.g. Github profile). Finally, students with a Grade Point Average of 8.0 or higher are eligible for the challenging scientific projects, resulting in a research paper. These project receive intense guidance, but have no capacity limits.	

CS4280	Language-Based Software Security	5
Responsible Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	This course has no formal prerequisites. However, for the homework assignments you will have to implement several program analysis techniques using the Scala programming language. If you have not used Scala before, you are thus expected to learn the basics of the language through self-study.	
Course Contents	Security vulnerabilities often arise due to programming errors in the source code of an application. Recent programming errors with severe security implications include Heartbleed (buffer over-read), Shellshock (code injection), and goto-fail (ill-formated code). Rather than hunt for individual vulnerabilities in programs, a more structural approach to improve security is to improve the programming language. This is the goal of language-based security: to rule out whole classes of potential security vulnerabilities in one go. This course studies various security properties and program analysis techniques for enforcing these properties at the level of the programming language to improve software security. In particular, we will study the following properties: - Memory safety: prevent buffer overflows and overreads - Type safety: prevent undefined behaviour - Information flow control: prevent data leaks and code injection attacks We will study techniques to address these problems at the language level through dynamic analysis, static analysis, and language design. To facilitate a precise study and comparison, we will define the above techniques formally in class. To facilitate student experimentation and exploration of trade-offs, students will implement the above techniques in homework assignments.	
Study Goals	After taking this course, students should be able to: 1. Describe the nature and causes of security vulnerabilities in software systems, and give concrete examples of how these security vulnerabilities can be exploited. 2. Explain the properties that can be enforced at the level of the programming language to rule out security vulnerabilities, such as memory safety, type safety, and non-interference. 3. Formally define the semantics of a simple programming language. 4. Formally define dynamic and static analysis techniques for enforcing these security properties. 5. Implement these techniques for a small programming language. 6. Discuss and evaluate the importance of soundness and precision of a given program analysis. 7. Contrast programming languages based on the set of countermeasures they provide, and give an appropriate recommendation for a specific application. 8. Analyse and apply results from scientific literature in the area of language based security.	
Education Method	The course work consists of the following activities: 1 or 2 instruction sessions per week. Weekly homework assignments consisting of theoretical questions, programming assignments, and reading assignments	
Assessment	The final grade of the course consists of the following components: Weekly homework assignments (weighting 40%). The weekly homework assignments will test your ability to design and implement (variants of) the techniques discussed in the lectures (study goals 3-5). Written or oral exam (weighting 60%). The final written or oral exam will test your theoretical understanding of the security vulnerabilities and their countermeasures discussed in class (study goals 1-2) and your ability to discuss and contrast the different aspects of these techniques (study goals 6-8). Final grade calculation: 0.4 * Weekly homework assignments + 0.6 * Written or oral exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written or oral exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Weekly homework assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4380	Privacy Enhancing Technologies	5
Responsible Instructor	Z. Erkin	
Instructor	Dr. S. Roos	
Instructor	Dr. K. Liang	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	IN4191 Security and Cryptography Course	
Course Contents	anonymous communication; identity management; anonymous credentials; anonymity systems; mix networks; onion routing; database privacy; k-anonymity; differential privacy; other probabilistic approaches; private data processing; secure multiparty computation; (fully/somewhat) homomorphic encryption; garbled circuits; secret sharing; privacy-preserving clustering; private recommender systems; private smart metering; privacy-preserving biometrics.	
Study Goals	Concepts like the Internet of Things or Big Data inherently utilize massive amounts of data containing private information collected and stored by websites, sensors, monitoring systems, auditing systems, and so on. Examples include electronic records in health care systems and location information in ubiquitous computing applications. But how can we protect the privacy of participating users while at the same time enable effective sharing and utilization of the distributed data? There are several dimensions in the area of privacy, ranging from technical and juridical to societal and economical. While we will touch upon all these different aspects in the course, we will focus on the technical dimension. We will explore potential techniques for building new platforms, services, and tools that protect users' privacy. The study of promising component technologies ranging from advances in anonymous communication and identity management to theoretic tools like differential privacy and cryptography will be the core of this course. Learning objectives: Good understanding of privacy in the Internet of Things Ability to analyse and evaluate anonymity mechanisms, both for anonymous communication and for database privacy Ability to apply and analyse the concept of secure multiparty computation to protect privacy in different application domains Gain hands-on experience with different privacy-enhancing technologies	
Education Method	Lectures (joint lecturing), lectures slides and articles.	
Literature and Study Materials	Lecture Slides, essential reading material provided with the lectures (articles and tutorials)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 70%). Closed book. Assignments (weighting 30%) Final grade calculation: 0.7 * Written exam + 0.3 * Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4253ET	"Hacking Lab"-Applied Security Analysis	5
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Necessary background differs per student project, see first lecture or contact instructors for details	
Course Contents	The security of computer and telecommunication systems is becoming an increasing concern. In this course, we will review the current state of the art on security research and gain practical experience in assessing the security and vulnerabilities of communication systems. Engineers are typically taught to focus on performance, correctness, scalability, and maintainability when building communication and information processing systems. However, an additional set of design principles are required to achieve security. In this course, we discuss security principles, common pitfalls and vulnerabilities. The weekly lectures provide an introduction into security research, with a focus on real-world security, privacy-enhancing technology and common security pitfalls. Each student participates in a "Hack Project", with a group of one to four students. Students can select between a wide range of available Hack Project outlines within the first week. The goal may be to evaluate the security of a real-world IT system, developing a proof-of-concept exposing a vulnerability or focussed on preserving privacy in a post-Snowden world. Students may propose their own Hack Project based on their background knowledge and skills. Such Hack Projects need to be approved and shaped together with the instructor. Example of possible outlined hardware-oriented projects are: development of a wifi tracker, programing an FPGA system to break passwords, assess the security of RFID cards, or to transparently intercept Ethernet traffic. Concrete software projects are: hacking Bitcoin, improving the TOR anonymity protocol and create Android-based tools for human rights activists in Iran, Egypt and Russia, reprogramming neural networks attacks. Each Hack Project is documented with a written report. This can be in the form of a 6-8 page IEEE-style scientific article or a traditional more lengthy report. All results, experiences and findings are presented to the entire class in the last week of the course. Hack Projects also report their progress several times during the course, after the weekly lectures.	
Study Goals	After this course, the student will have a thorough knowledge of security in real-world systems, and will be able to explore the literature on this topic independently. The student will be aware of the poor state of security in real-world computer systems. The student can explain the common pitfalls, why these known failures still occur and reasons behind the poor state of security in general.	
Education Method	Lectures, student presentations, written final report and active participation. Attendance and active participation during lectures is mandatory. This sadly means telelecturing is not possible.	
Literature and Study Materials	Customize literature lists and study materials are provided per project topic	
Assessment	The final grade of the course consists of the following components: Written Hack Project report (weighting 60%) Final presentation of result (weighting 10%) Presentation of ongoing project progress (weighting 20%) Participation in discussions, overall quality of the practical work and class attendance (weighting 10%) Attendance to lectures is mandatory. Final grade calculation: 0.6 * Written Hack Project report + 0.1 * Final presentation of result + 0.2 * Presentation of ongoing project progress + 0.1 * Participation in discussions, overall quality of the practical work and class attendance A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Written Hack Project report: re-submit deliverable Final presentation of result: resit Presentation of ongoing project progress: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
maximum aantal deelnemers	If there is an unexpected high demand for this course, then enrollment will be based on past performance in relevant courses.	
Co-Instructor	Dr. S. Picek	

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. M. Blaauboer	
Responsible Instructor	Dr. L. di Carlo	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	30% homework assignments, 10% online quizzes, 60% final exam. A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course.	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Socio-Technical Electives (choose at least 3 courses) 2023

TPM010A	Cyber Crime Science	5
Module Manager	Dr. R.S. van Wegberg	
Instructor	Dr. R.S. van Wegberg	
Instructor	F.E.G. Miedema	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	- Introduction to crime science (i.e., how do scientists study crime), - Socio-technical research skills (i.e., research design, data collection, data analysis), - Studying cybercrime from a socio-technical perspective (e.g., evaluate the effectiveness of phishing detection).	
Study Goals	The student will acquire: A good understanding of the theoretical principles of crime science, An appreciation of the whole spectrum of different cybercrimes, Skills necessary to empirically research cybercrimes.	
Education Method	The course runs from Q3 till Q4. In Q3 the course consists of both lectures and feedback sessions. In Q4 students are expected to work on their research paper, give and receive (peer)feedback and present their results in a conference.	
Assessment	1. Research proposal (groups of 3 students) pass/no-pass; 2. Team presentations pass/no-pass; 3. Reviews of other papers pass/no-pass; 4. Final conference grade - Research paper - Presentation	

TPM020B	Economics of Cybersecurity	5
Module Manager	S.E. Parkin	
Instructor	Prof.dr. M.J.G. van Eeten	
Responsible for assignments	S.E. Parkin	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	With a significant increase in high-profile data breaches and cybersecurity threats in the last couple years, it is critical for businesses to learn about the costs and investment decisions around securing their online systems. While many businesses think of cybersecurity as a technical problem, this course broadens that view and shows that security failures are caused as often by bad business decisions and incentive systems as by bad technical design. This course provides an introduction to the field of the economics behind cybersecurity. It will provide you with the economic concepts, measurement approaches to make better security decisions, while helping you to understand the forces that shape the security decisions of other businesses, products and services.	
Study Goals	The student will: -Gain a sound understanding of the economics of cybersecurity as a systems discipline, from security policies (modelling what ought to be protected) to mechanisms (how to implement the protection goals). -Obtain foundational skills in identifying and assessing data on complex decisions around information security issues. -Gain insights into the design of effective policies to enhance and maintain cyber security must take into account a complex set of incentives facing not only the providers and users of the Internet and computer software, but also those of potential attackers. -Learn to apply economic analysis and analytic approaches to the open issues and pending activities in cybersecurity.	
Education Method	Structured lectures, background reading, group discussions, and regular non-graded quizzes.	
Assessment	Group report and presentation (60%); written individual assignment (40%). There is not a resit option for the group assignment; the resit option for the individual assignment is an oral exam.	

TPM025A	User-Centred Security	5
Module Manager	S.E. Parkin	
Instructor	S.E. Parkin	
Responsible for assignments	S.E. Parkin	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	0/2x2/0/0 This course is shifted to Q2 and is replaced by TPM030a in Q4	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Cybersecurity attacks on organizations and services increasingly target the people who are involved, such as employees in companies or home computer users. Solutions to address this problem need to be feasible for individuals, alongside other user priorities (such as completing paid work). Without consideration of user skills or needs, solutions such as security training or browser warning pop-ups add effort and burden to the user, and encourage less secure behaviours as workarounds (such as writing down difficult-to-remember passwords to ensure system access). In this course, students will learn about the user perspective of security technologies. This will leverage key human factors concepts, around security usability and its connections to decisions in policy, planning, and technology investment. This will include how to assess a security solution from the perspective of different kinds of technology users and their tasks. By assessing the strengths and weaknesses of particular security mechanisms for users in practice (policies, training, monitoring, etc.), security implementation and management decisions can be made which better fit the context in which mechanisms are used. This can ensure long-term security which better matches the requirements of a particular user organisation or community.	
Study Goals	The student will: -Gain a sound understanding of security usability as a discipline, from assessing the context of use alongside primary tasks, to identifying the time and effort costs to users. -Obtain foundational skills in matching security technologies and processes to user abilities, motivations, and perceptions of security-related technologies. This will include examination of authentication and identity technologies, employee security training, trust in online contexts, privacy-related evaluations of personal data disclosure, etc. -Gain insights into the design of effective security technologies and their deployment from a human-centred perspective, to inform interface and policy design and ensure compliance with security expectations. This will include management of security behaviour change activities, and identifying how to position expert support to aid users, e.g., effective communication of policy, use of persuasive design in security.	
Education Method	Structured lectures, background reading, problem-driven group discussions	
Assessment	Individual assignment on assigned reading material (20%); a final individual assignment is an essay with feedback moment (80%). The resit option for both assignments is an oral exam.	
Elective	Yes	

TPM030A	Introduction to Cloud as Infrastructure: The effects of the new business of computing on practice	5

Module Manager	Dr. F.S. Gürses
Instructor	Dr. F.S. Gürses
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten
Contact Hours / Week x/x/x/x	0/0/0/2x2 This course is shifted to Q4 and is replaced by TPM025a in Q2.
Education Period	4
Start Education	4
Exam Period	4 5
Course Language	English
Course Contents	What if intelligence is not the main driver of AI? What if data is not the primary resource of the tech sector? What if mobile phones are not actually personal devices? What if software as a product no longer exists? And what if the dominance of the major tech companies remains a political problem even after we reduce surveillance and discrimination, or apply privacy or fairness by design? These are some of the questions we will ask in this course as we explore the technical, business, and political implications of a specific form of contemporary computational infrastructure (CI [0]): cloud + mobile. The thesis of the course is that CI is an environment for (experimentation with) software production, i.e., an environment for the creation of value through the engineering of software based services. We will study how CI, developed and controlled by a handful of corporations, is vying to become the default environment for a movement that will transform relationships of production --be it digital or physical production-- across the globe. We begin the course by examining common assumptions about the business of computing. We will consider the reigning belief that cloud is somebody elses computer or another node on the internet. We will also unpack the wildly popular take that tech companies generate revenue from personal data and surveillance through personal devices, with privacy as the obvious antidote to individual harms and power asymmetries resulting from the accumulation of information. The second part of the course presents a hard challenge to these assumptions by comparing the past to the present. We will study concrete shifts in the production of software that gave rise to CI such as the turn from personal computers to (mobile) devices attached to the clouds; waterfall methods to agile/lean methodologies; monolith architectures to (micro)services; instruction-based programming to AI. In part three we explore whether and how the mega-corporations that control CI aim to become the environment for software production. We will also look at how the resulting modes of software production go hand-in-hand with transformations, like flattening of the Internet or the move from general-purpose to specialized chips. The final section of the course explores how software production, passing through CI, promotes what we call agile production. Together, we will explore how agile software production enters other industries to become a more general model of production made possible by the rise of CI. We will end the course by examining the long-term objectives of the companies that currently dominate the business of computing. By the end of the course, you will understand why CI is one of the most significant drivers of (geo)political, economic, and technological developments in our day by looking at the way it shapes European digital policies. We will also unpack why CIs hold over software production places important limits on data protection and privacy engineering. [0] A mainframe, server rack, or even a personal computer can serve as any entitys computational infrastructure. In this course, however, we will use the term CI to refer to the dominant and corporate form of materially organizing computing for the development and global delivery of digital services, for example AWS by Amazon, Azure by Microsoft, or mobile devices offered by Apple or Google. Using popular terms, CI is cloud infrastructures plus end- or mobile devices as their accessories.
Study Goals	In this course you will discover how the way we produce software impacts how society is organized by reaching the following learning-objectives: Demonstrate a basic understanding of the history of computing through exemplary shifts in software production Develop an understanding of how corporate computational infrastructures serve as a powerful environment for software production Analyze how privacy or data protection by design are weakened in their protective role due to the rise of computational infrastructures Speculate and predict possible futures of software production in CI and its ambition to transform global economic production Interpret strengths and weaknesses of current EU digital policy vis a vis computational infrastructures
Education Method	Lectures, work groups, case studies, presentations, self-study, writing.
Prerequisites	It is advantageous to have some understanding of or an interest to learn about software engineering and computing to attend the course.
Assessment	10% engagement 50% 5 assignments 40 % oral exam The total of the assignments as well as the oral exam must be graded with at least 5.0. The final grade consists of the average of the three parts (engagement, assignments and exam), and the rounded grade must be at least 6.0 (5.75 rounded up).
Elective	Yes
Targetgroup	This course is listed as part of the Cybersecurity Specialization and is open to students participating in that program TU Delft and Twente. The course is also open to Master Students at other TBM programs, including specializations listed https://www.tudelft.nl/en/tpm/education. If you are not in any of these programs and would like to attend the course, please contact the course teacher per email.
Category	MSc level

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Required Courses for CS Graduation 2023

CS4120	Seminar Science and Methods in Cyber security	5
Responsible Instructor	Prof.dr. M. Conti	
Instructor	Dr. M.P.M. Franssen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This seminar course Cyber Security covers the following topics: (i) an introduction to the philosophy of (classical and design) science, (ii) the art of writing a scientific research proposal, (iii) an overview of useful and relevant scientific methods, (iv) introduction to scientific writing (of a paper and of a MSc thesis), (v) an overview of some representative research areas in the Cyber Security domain (e.g. Cyber Physical Systems Security and IoT Security; Machine Learning and Security; Network Security), possibly illustrated through guest lectures from experts on these topics.	
Study Goals	1. Getting a basic knowledge and understanding of what science entails and how scientific knowledge is being created 2. Getting knowledge and understanding of relevant scientific methods applicable in the field of Cyber Security 3. Getting knowledge, understanding and skills for writing a research proposal related to the creation of a MSc thesis 4. Getting knowledge and understanding on how to execute a scientific article and MSc thesis 5. Getting knowledge and understanding of how to execute a literature review. 6. Getting familiar with recent results and challenges on some key representative research areas in Cyber Security	
Education Method	Lecturers supported by the execution of mostly individual assignments. Attendance of participants in this course is mandatory.	
Assessment	The final grade of the course consists of the following components: Presence and level of participation (weighting 10%) Quality of the research proposal/essay, with preliminary contribution of idea/protocol/algorithm, preliminary results and discussion, to be written and presented (weighting 60%) Assignments during the course (weighting 30%) Final grade calculation: 0.1 * Presence and level of participation + 0.6 * Quality of the research proposal/essay, written and presented + 0.3 * Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Further detail about evaluation of the research proposal/essay: At the end of the course, each student must write a 5-page long proposal/essay. The topic and structure of the proposal/essay must be agreed with the lecturer. The essay might include some implementation prototype or experiments/simulations to evaluate/support the claim in the paper (in case this is a significant part of the essay, two students can agree with the lecturer to work together). During the presentation of the proposal/essay, the student is asked to give a 15-minutes presentation (with slides) to the lecturer about the essay The presentation is graded according to the following components: (30%) Style (30%) Originality (40%) Organization (clarity in your argumentation, coherence between assumptions and conclusions, logical organisation, evidence to support claims) Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Quality of the research proposal/essay, written and presented: re-submit deliverable and/or resit presentation Assignments: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Elective	Yes	
Tags	Research Methods	

IN5000	Final Project	45
Responsible Instructor	Dr. J.C. van Gemert	
Responsible Instructor	C.B. Poulsen	
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	L. Cruz	
Responsible Instructor	Dr.ir. A.R. Bidarra	
Responsible Instructor	Dr.ir. W.P. Brinkman	
Responsible Instructor	Dr. K.A. Hildebrandt	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	IN5000 Final Project is the final part of the Master's degree programme. During this project, you will be required to demonstrate your ability to solve a research or engineering problem. The project must be carried out using the techniques of project management. You will begin by making a project plan in cooperation with your Masters thesis advisor. Several aspects of the project are defined within the plan, including the assignment, the frequency of interaction with the advisors, the milestones of the project and the resources and facilities offered by the faculty. You will be required to adhere to your plan throughout the project. It is obviously possible to adjust your plan under certain circumstances and after discussion with your daily supervisor. At the end of the project, you will submit your Masters thesis, which must be written in English, and make an oral presentation of your work to the Thesis Committee. The Thesis Committee will announce the final mark, which is based on the project performance, the thesis, the presentation and the subsequent discussion. More information about the graduation process: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-policy-msc/ More information on the thesis project on Brightspace: MSc CS Thesis project EEMCS https://brightspace.tudelft.nl/d2l/home/492868	
Study Goals	1. The student is able to design a research project: - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. 2. The student is able to execute a research project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. 3. The student is able to write a research report: - The student is able to write a research report that shows sufficient coherence of content; - The students is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; 4. The student is able to present and defend the research project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.	
Education Method	Project	
Prerequisites	Before starting the project, students must have completed at least 60 EC of the Master's degree programme as shown by a Declaration of Obtained Credits (DCO). To be able to get this DCO, the individual exam program (IEP) should have been approved by the Board of Examiners. Note: In some cases, the thesis supervisor may impose additional conditions for starting this project.	
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-policy-msc/assessment/ The voting members of the thesis committee determine the final grade. The grade should reflect a weighted average of the four scores above, but need not to be an exact arithmetical mean. The final mark starts from 5 up to and 10. Marks ending in .5 may also be used. If the student shows excellence (is nominated for a 10) the chair of the thesis committee should consult the chair of the Board of Examiners, at least five working days in advance of the defense. The chair may advice to add an extra member to the thesis committee. The motivation for the grade at each of the four criteria as listed above is summarized on a form and signed by the chairman of the thesis committee. The candidate is given a short account of the assessment, either in private or in front of the audience. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

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